具身智能视觉语言动作模型综述

A Survey on Vision-Language-Action Models for Embodied AI

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核心摘要校订

根据原文摘要单独校订。正文仍为逐页机器翻译初稿；原始图表、公式和英文条目请查看紧邻原文页。

具身智能通过控制实体智能体完成物理世界任务，是研究通用人工智能的重要方向。随着 LLM 与 VLM 的进步，视觉语言动作模型（VLA）开始利用生成动作的能力解决语言条件下的机器人任务。本文提出详细分类，覆盖三条研究线：VLA 的单个组成组件、能够预测低层动作的 VLA 控制策略，以及将长时程任务分解为子任务并指导 VLA 执行更一般用户指令的高层规划器。本文还总结数据集、仿真器与基准，讨论 VLA 面临的挑战与未来方向，并维护配套文献资源仓库。

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A Survey on Vision-Language-Action Models for Embodied AI

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***Abstract*—Embodied AI is widely recognized as a cornerstone of artificial general intelligence (AGI) because it involves controlling embodied agents to perform tasks in the physical world. Building on the success of large language models (LLMs) and vision-language models (VLMs), a new category of multimodal models—referred to as vision-language-action (VLA) models—has emerged to address language-conditioned robotic tasks in embodied AI by leveraging their distinct ability to generate actions. The recent proliferation of VLAs necessitates a comprehensive survey to capture the rapidly evolving landscape. To this end, we present the first survey on VLAs for embodied AI. This work provides a detailed taxonomy of VLAs, organized into three major lines of research. The first line focuses on individual components of VLAs. The second line is dedicated to developing VLA-based control policies adept at predicting low-level actions. The third line comprises high-level task planners capable of decomposing long-horizon tasks into a sequence of subtasks, thereby guiding VLAs to follow more general user instructions. Furthermore, we provide an extensive summary of relevant resources, including datasets, simulators, and benchmarks. Finally, we discuss the challenges facing VLAs and outline promising future directions in embodied AI. A curated repository associated with this survey is available at: https://github.com/yueen-ma/Awesome-VLA**

***Index Terms*—Embodied AI, large language model (LLM), multimodality, robotics, vision-language-action (VLA) model, world model**

I. Introduction

Vision-language-action (VLA) models are a class of multimodal models within the field of embodied AI, designed to process information from three modalities: vision, language, and action. Unlike conversational AI, embodied AI requires controlling physical embodiments that interact with the environment, and robotics is the most prominent domain of embodied AI. In language-conditioned robotic tasks, the policy must possess the capability to understand language instructions, visually perceive the environment, and generate appropriate actions, necessitating the multimodal capabilities of VLAs. The term was recently coined by RT-2 [1]. Compared with earlier deep reinforcement learning (RL) approaches, VLAs offer superior versatility, dexterity, and generalizability in complex environments. As a result, they are well-suited not only for controlled settings like factories but also for everyday tasks in household environments.

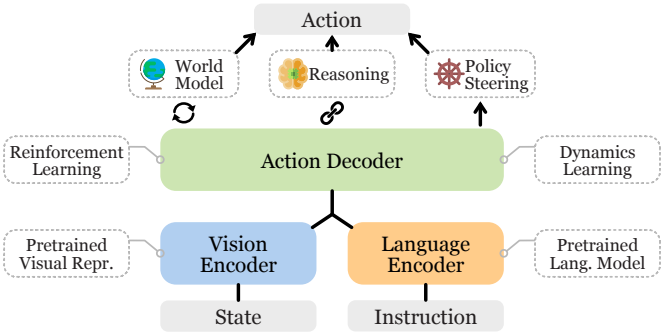


Fig. 1. General architecture of VLA models. Three representative methods for action prediction are shown. Related components are presented in dashed boxes.

Early developments in deep learning primarily consisted of unimodal models. In computer vision (CV), models like AlexNet [2] showcased the potential of artificial neural networks (ANNs). Recurrent neural networks laid the groundwork for numerous natural language processing (NLP) models, but NLP has seen a transition in recent years, with Transformers [3] taking precedence. The deep Q-network (DQN) [4] demonstrated that ANNs can successfully tackle RL problems. Leveraging advancements of unimodal models across diverse machine learning fields, multimodal models now address a wide range of tasks, such as visual question answering (VQA), image captioning, and text-to-video generation.

Conventional robot policies based on RL have largely focused on addressing a limited set of tasks within controlled environments, such as item grasping [5]. However, there is a growing need for more versatile multitask policies, akin to recent trends in large language models (LLMs) and vision-language models (VLMs). Developing a multitask policy is challenging, as it requires learning a broader set of skills and adapting to diverse environments. Furthermore, specifying tasks via language instructions offers a more intuitive user-robot interface, which necessitates the development of language-conditioned robot policies.

Built upon the success of large VLMs, VLA models have demonstrated their potential in addressing these challenges, as illustrated in Fig. 1. Similar to VLMs, VLAs utilize vision foundation models as vision encoders to obtain pretrained

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具身智能视觉语言动作模型综述

1
视觉语言动作（Vision-Language-Action）具身智能（Embodied AI）型号调查

马悦恩、宋子星、庄玉正、郝建业、Irwin King、Fellow, IEEE

摘要 -具身智能（Embodied AI）被广泛认为是通用人工智能 (AGI) 的基石，因为它涉及控制具身智能体在物理世界中执行任务。基于大语言模型 (LLM) 和视觉语言模型 (VLM) 的成功，一类新的多模态模型（称为 视觉语言动作（Vision-Language-Action）(VLA) 模型）已经出现，通过利用其独特的生成动作的能力来解决 具身智能（Embodied AI）中的语言条件机器人任务。最近 VLA 的激增需要进行全面调查，以捕捉快速变化的景观。为此，我们对 具身智能（Embodied AI）的 VLA 进行了首次调查。这项工作提供了 VLA 的详细分类，分为三个主要研究领域。第一行重点关注 VLA 的各个组件。第二条线致力于开发基于 VLA 的控制策略，擅长预测低级行为。第三行包括高级任务规划器，能够将 长时程（Long-Horizon）任务分解为一系列子任务，从而指导 VLA 遵循更通用的用户指令。此外，我们还提供了相关资源的广泛摘要，包括数据集、模拟器和基准测试。最后，我们在 具身智能（Embodied AI）中讨论了 VLA 面临的挑战并概述了有希望的未来方向。与本篇综述相关的精选存储库位于：https://github.com/yueen-ma/Awesome-VLA

索引术语 -具身智能（Embodied AI）、大语言模型 (LLM)、多模态、机器人、视觉语言动作（Vision-Language-Action）(VLA) 模型、世界模型（World Model）
一、简介

视觉语言动作（Vision-Language-Action）(VLA) 模型是 具身智能（Embodied AI）领域内的一类多模态模型，旨在处理来自三种模态的信息：视觉、语言和动作。与对话式人工智能不同，具身智能（Embodied AI）需要控制与环境交互的物理实体，而机器人技术是 具身智能（Embodied AI）最突出的领域。在以语言为条件的机器人任务中，策略必须具备理解语言指令、视觉感知环境并生成适当动作的能力，这就需要 VLA 的多模态能力。该术语最近由 RT-2 [1] 创造。与早期的深度 强化学习（Reinforcement Learning）(RL) 方法相比，

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行动
世界策略
模型推理指导

强化动作解码器动力学
学习学习
预训练视觉语言预训练
视觉再现。编码器编码器语言。模型
状态说明

图 1. VLA 模型的总体架构。显示了三种具有代表性的动作预测方法。相关组件显示在虚线框中。

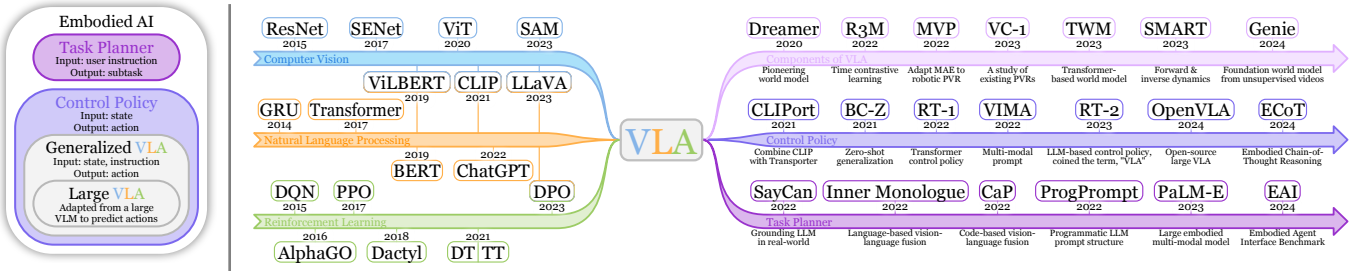
VLA 在复杂环境中提供卓越的多功能性、灵活性和通用性。因此，它们不仅非常适合工厂等受控环境，而且还适合家庭环境中的日常任务。

深度学习的早期发展主要由单峰模型组成。在计算机视觉 (CV) 领域，AlexNet [2] 等模型展示了人工神经网络 (ANN) 的潜力。循环神经网络为众多自然语言处理 (NLP) 模型奠定了基础，但 NLP 近年来发生了转变，其中 Transformers [3] 占据了主导地位。深度 Q 网络（DQN）[4] 证明了 ANN 可以成功解决 RL 问题。利用跨不同机器学习领域的单模态模型的进步，多模态模型现在可以解决广泛的任务，例如视觉问答 (VQA)、图像字幕和文本到视频生成。

基于强化学习的传统机器人策略主要集中于解决受控环境中的一组有限任务，例如物品抓取 [5]。然而，对更通用的多任务策略的需求日益增长，类似于大语言模型 (LLM) 和视觉语言模型 (VLM) 的最新趋势。制定多任务策略具有挑战性，因为它需要学习更广泛的技能并适应不同的环境。此外，通过语言指令指定任务提供了更直观的用户机器人界面，这需要开发以语言为条件的机器人策略。

基于大型 VLM 的成功，VLA 模型已经证明了其在解决这些挑战方面的潜力，如图 1 所示。与 VLM 类似，VLA 利用视觉 基础模型（Foundation Models） 作为视觉编码器来获得预训练的图像

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(a) Concepts (b) Timelines

Fig. 2. (a) Venn diagram that outlines the main concepts in embodied AI discussed in this article. (b) Timelines that trace the evolution from unimodal models to VLA models.

visual representations (PVRs) of the current environmental state, such as object class, pose, and geometry. VLAs encode instructions using the token embeddings of their LLMs and employ various strategies to align vision and language embeddings, including approaches like Flamingo [6], BLIP-2 [7], and LLaVA [8]. By finetuning on robot data, the LLM can function as a decoder to predict actions and perform language-conditioned robotic tasks. These cross-disciplinary innovations drive the advancement of VLAs in embodied AI—a critical building block for AGI.

VLAs are closely related to three lines of work, as depicted by the timelines in Fig. 2(b) and the taxonomy in Fig. 3. Some approaches focus on individual components of VLAs (see Section III-A), such as PVRs, dynamics learning, world models, and reasoning. Meanwhile, a substantial body of research is dedicated to low-level control policies (see Section III-B). In this category, language instructions and visual observations are fed into the control policy, which then generates low-level actions—such as translation and rotation—thereby rendering VLAs an ideal choice for control policies. In contrast, another category of models serves as high-level task planners responsible for task decomposition (see Section IV). These models break down long-horizon tasks into a sequence of subtasks that, in turn, guide VLAs toward the overall goal, as illustrated in Fig. 4. Most current robotic systems adopt such a hierarchical framework [9], [10], [11] because the high-level task planner can leverage models with high capacity while the low-level control policy can focus on speed and precision, similar to hierarchical RL.

To provide a more comprehensive overview of current progress in embodied AI, we propose a generalized definition of "VLA," as illustrated by the Venn diagram in Fig. 2(a). We define a VLA as any model capable of processing multimodal inputs from vision and language to produce robot actions that accomplish embodied tasks, typically following the architecture in Fig. 1. The original concept of VLA referred to a model that adapts VLMs to robotic tasks [1]. Analogous to the distinction between LLMs and more general language models, we designate the original VLAs as "Large VLAs" (LVLAs) because they are based on LLMs or large VLMs.

A. Related Work

To the best of our knowledge, this survey is the first to review the recent progress of VLA models, a rapidly emerging research area. Previous surveys have investigated other facets of embodied AI. Firoozi et al. [12] comprehensively summarized foundation models in robotics up to 2023, while Wang et al. [13] focused on LLMs in robotics. Hu et al. [14] examined more recent vision, language, and robotic foundation models for general-purpose robots. Kawaharazuka et al. [15] concentrated on real-world robot applications. In contrast, our work emphasizes VLA models, thereby complementing and extending the existing literature on embodied AI.

B. Contributions

To the best of our knowledge, this article is the first comprehensive survey of emerging VLA models in the field of embodied AI.

1) *Comprehensive Review:* We present a thorough review of emerging VLA models in embodied AI, covering various aspects including components, architectures, training objectives, and robotic tasks.
2) *Taxonomy:* We introduce a taxonomy of VLAs based on the hierarchical framework of current robotic systems, comprising two hierarchies: a low-level control policy and a high-level task planner. Control policies execute low-level actions based on specified language commands and the perceived environment. Task planners provide guidance to control policies by decomposing long-horizon tasks into subtasks. We also discuss various essential components of VLAs.
3) *Resources:* We summarize the necessary resources for training and evaluating VLAs, including recent datasets and benchmarks in real-world or simulated environments. We also discuss various approaches to address current challenges, such as data scarcity and inconsistency.
4) *Future Directions:* We outline current challenges and promising future opportunities in the field, such as safety, foundation models, and real-world deployment.

II. Background

Embodied AI is a unique form of artificial intelligence that actively interacts with the physical environment. This sets it apart from other AI models, such as conversational AI, which primarily handles textual conversations, or generative AI models that focus on tasks like text-to-video generation.

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具身智能（Embodied AI） ResNet SENet ViT SAM Dreamer R3M MVP VC-1 TWM SMART Genie
任务规划器 2015 2017 2020 2023 2020 2022 2022 2023 2023 2023 2024
输入：用户指令输出：子任务计算机视觉ViLBERT CLIP LLaVA 世界模型 VLA 的开创性组件时间对比学习使 MAE 适应机器人 PVR 现有 PVR 基于世界模型变压器逆动力学的研究前向和无监督视频基础 世界模型（World Model）
控制策略输出：动作输入：状态 GRU2014 Transformer2017 2019 2021 2023 CLIPort2021 BC-Z2021 RT-12022 VIMA2022 RT-22023 OpenVLA2024 ECoT2024
输入：状态、指令输出：动作自然语言处理BERT2019 ChatGPT2022与Transporter组合CLIP控制策略泛化零样本控制策略Transformer多模式提示基于LLM的控制策略，创造了术语“VLA”开源大型VLA体现思想链推理
DQN PPO DPO SayCan Inner Monologue CaP ProgPrompt PaLM-E EAI
改编自大型 2015 2017 2023 2022 2022 2022 2022 2023 2024
VLM 来预测动作 强化学习（Reinforcement Learning） 任务Planner
2016 2018 2021 落地关联（Grounding） 现实世界中的法学硕士 基于语言的视觉-语言融合 基于代码的视觉-语言融合 程序化法学硕士提示结构 多模态模型 大型体现界面 基准 体现智能体
AlphaGO Dactyl DT TT
(a) 概念 (b) 时间线
广义 VLA
大VLA
VLA

图 2. (a) 维恩图概述了本文讨论的 具身智能（Embodied AI） 中的主要概念。 (b) 追踪从单峰模型到 VLA 模型演变的时间表。

当前环境状态的视觉表示 (PVR)，例如对象类、姿势和几何形状。 VLA 使用 LLM 的令牌嵌入对指令进行编码，并采用各种策略来对齐视觉和语言嵌入，包括 Flamingo [6]、BLIP-2 [7] 和 LLaVA [8] 等方法。通过对机器人数据进行微调，法学硕士可以充当解码器来预测动作并执行语言条件的机器人任务。这些跨学科创新推动了 具身智能（Embodied AI） 中 VLA 的进步--AGI 的关键构建模块。

如图 2(b) 中的时间线和图 3 中的分类法所示，VLA 与三项工作密切相关。一些方法侧重于 VLA 的各个组成部分（参见第 III-A 节），例如 PVR、动态学习、世界模型（World Models） 和推理。同时，大量研究致力于低水平控制策略（参见第 III-B 节）。在此类别中，语言指令和视觉观察被输入到控制策略中，然后生成低级操作（例如平移和旋转），从而使 VLA 成为控制策略的理想选择。相比之下，另一类模型充当高级任务规划器，负责任务分解（参见第四节）。这些模型将 长时程（Long-Horizon） 任务分解为一系列子任务，这些子任务反过来引导 VLA 实现总体目标，如图 4 所示

。当前大多数机器人系统都采用这种分层框架[9]、[10]、[11]，因为高级任务规划器可以利用高容量的模型，而低级控制策略可以专注于速度和精度，类似于分层强化学习。

为了更全面地概述 具身智能（Embodied AI） 目前的进展，我们提出了“VLA”的广义定义，如图 2(a) 中的维恩图所示。我们将 VLA 定义为能够处理来自视觉和语言的多模态输入以产生完成具体任务的机器人动作的任何模型，通常遵循图 1 中的架构。VLA 的原始概念是指使 VLM 适应机器人任务的模型 [1]。类似于LLM和更通用的语言模型之间的区别，我们将原始VLA指定为“大型VLA”（LVLA），因为它们基于LLM或大型VLM。

一个。相关工作

据我们所知，这项调查首次回顾了 VLA 模型的最新进展，VLA 模型是一种快速新兴的模型

荷兰国际集团的研究领域。之前的调查已经调查了 具身智能（Embodied AI） 的其他方面。菲鲁齐等人。 [12]全面总结了截至2023年机器人技术中的基础模型（Foundation Models），而Wang等人。 [13] 专注于机器人学领域的法学硕士。胡等人。 [14] 检查了 通用（General-Purpose） 机器人的最新视觉、语言和机器人 基础模型（Foundation Models）。川原冢等人。 [15]专注于现实世界的机器人应用。相比之下，我们的工作强调 VLA 模型，从而补充和扩展了 具身智能（Embodied AI） 的现有文献。

B。贡献

据我们所知，本文是 具身智能（Embodied AI） 领域新兴 VLA 模型的首次全面综述。

- 1)_全面回顾：_我们在 具身智能（Embodied AI） 中对新兴 VLA 模型进行了全面回顾，涵盖组件、架构、训练目标和机器人任务等各个方面。

- 2）_分类法：_我们基于当前机器人系统的分层框架引入了 VLA 分类法，包括两个层次结构：低级控制策略和高级任务规划器。控制策略根据指定的语言命令和感知环境执行低级操作。任务规划器通过将 长时程（Long-Horizon） 任务分解为子任务来提供控制策略的指导。我们还讨论了 VLA 的各种重要组成部分。

- 3)_资源：_我们总结了训练和评估 VLA 所需的资源，包括现实世界或模拟环境中的最新数据集和基准。我们还讨论了解决当前挑战的各种方法，例如数据稀缺和不一致。

- 4) _未来方向：_我们概述了该领域当前的挑战和有希望的未来机遇，例如安全、基础模型（Foundation Models） 和实际部署。

二.背景

具身智能（Embodied AI） 是一种独特的人工智能形式，可以与物理环境主动交互。这使其与其他人工智能模型区分开来，例如主要处理文本对话的对话式人工智能，或专注于文本到视频生成等任务的生成式人工智能模型。

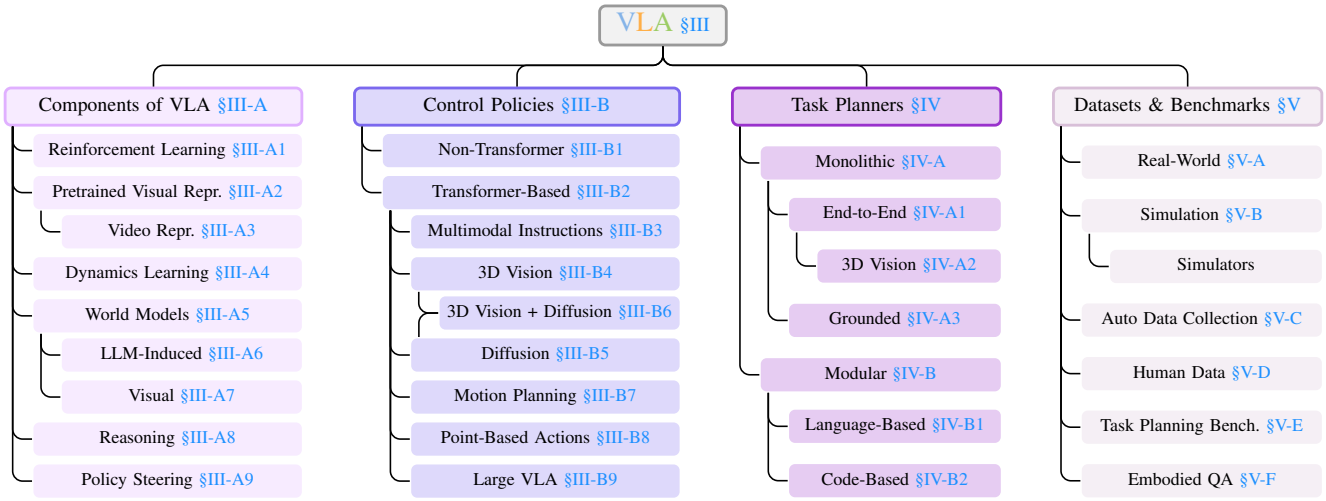


Fig. 3. Taxonomy of VLA models. The organization of this survey follows this taxonomy.

Embodied AI encompasses a broad spectrum of embodiments, including smart appliances, smart glasses, autonomous vehicles, and more. Among them all, robots stand out as one of the most prominent embodiments.

Robot learning is also usually framed as an RL problem, represented as a Markov decision process (MDP) consisting of states (s), actions (a), and rewards (r). An MDP trajectory is denoted by $\tau = (s_1, a_1, r_1, \ldots, s_T, a_T, r_T)$. In certain scenarios, robotic tasks may also be viewed as partially observable Markov decision processes (POMDPs) due to incomplete observations. The primary objective of RL is to train a policy capable of generating optimal actions for the current state $\pi(a_t|s_t)$. Various methods, such as TD learning and policy gradients, can be employed to achieve this. However, in cases where defining the reward function proves challenging, imitation learning is utilized to directly model the action distribution within trajectories devoid of rewards $\tau = (s_1, a_1, \ldots, s_T, a_T)$. Furthermore, many multitask robot models employ language as instructions p to determine which task or skill to execute, leading to the development of language-conditioned robot policies $\pi(a_t|p, s_{\leq t}, a_{<t})$.

III. Vision-Language-Action Models

A. Components of VLA

A body of work focuses on individual components of VLA models, drawing on successes in CV, NLP, and RL. We introduce them through the lens of embodied AI.

1) Reinforcement Learning: RL laid the foundation for embodied AI and continues to help advance VLAs. DQN first demonstrated learning policies directly from high-dimensional pixel inputs, underscoring the need for greater model capacity in end-to-end RL. RL trajectories—sequences of states, actions, and rewards—naturally align with sequence modeling problems, making them well-suited to Transformer architectures. Pioneering efforts in this direction include Decision Transformer (DT) [16] and Trajectory Transformer (TT) [17]. Gato [18] further extended this paradigm to a multimodal, multitask, multiembodiment setting. $\pi^*_{0.6}$ [19] employs RL for VLAs to learn from experience.

Furthermore, a synergy has emerged between RL and LLMs, benefiting embodied AI in multiple ways. RL from human feedback (RLHF) aligns LLMs with human preferences and has also been applied to robot learning. SEED [20] utilizes RLHF alongside skill-based RL to address the sparse reward issue and to improve safety via human evaluation. Conversely, LLMs also enable novel RL methods. Reflexion [21] proposes a verbal RL framework that replaces weight updates in RL models with linguistic feedback, making it naturally applicable to embodied decision-making where planning also occurs in language form. Eureka [22] shows that LLMs can design reward functions for embodied agents that outperform expert human-engineered ones. As LLMs evolve, this synergy accelerates progress in RL and embodied AI.

2) Pretrained Visual Representations: The effectiveness of the vision encoder directly influences the performance of VLAs, as it provides crucial information regarding the current state s_t, such as object categories, positions, and affordances. Consequently, numerous methods are devoted to improving the quality of PVRs. Their technical details are compared in Table I.

CLIP [23] has gained widespread adoption as a vision encoder in robotic models. The training objective of CLIP is to identify the correct text-image pair among all possible combinations in a given batch. CLIP is pretrained on the WIT dataset, comprising 400 million image-text pairs. This large-scale training allows CLIP to develop a rich understanding of the relationship between visual and textual information.

R3M [24] proposes two main pretraining objectives: time contrastive learning (CL) and video-language alignment. The objective of time contrastive learning is to minimize the distance between temporally proximal video frames while simultaneously increasing the separation between temporally distant ones. This objective aims to create PVRs that capture the temporal relationships within the video sequence. On the other hand, video-language alignment aims to learn whether a video corresponds to a language instruction. This objective enriches the semantic relevance embedded in the PVRs. VIP [25] also capitalizes on video temporal relationships.

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VLA §III
VLA 的组件 §III-A 控制策略 §III-B 任务规划器 §IV 数据集和基准 §V
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动力学学习 §III-A4 3D 视觉 §III-B4 3D 视觉 §IV-A2 模拟器
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图 3.VLA 模型的分类。本篇综述的组织遵循这一分类法。

具身智能（Embodied AI） 涵盖了广泛的实施例，包括智能电器、智能眼镜、自动驾驶汽车等。其中，机器人是最突出的体现之一。

机器人学习通常也被视为 RL 问题，表示为由状态（s）、动作（a）和奖励（r）组成的马尔可夫决策过程 (MDP)。 MDP 轨迹由 τ = (s 1 , a 1 , r 1 , ..., sT , aT , rT) 表示。在某些场景中，由于不完整的观察，机器人任务也可能被视为部分可观察的马尔可夫决策过程（POMDP）。 RL 的主要目标是训练能够为当前状态 π (at|st) 生成最佳操作的策略。可以采用多种方法（例如 TD 学习和策略梯度）来实现这一目标。然而，在定义奖励函数具有挑战性的情况下，模仿学习（Imitation Learning） 用于直接对没有奖励的轨迹内的动作分布进行建模 τ = (s 1 , a 1 , . . , sT , aT)。此外，许多多任务机器人模型使用语言作为指令 p 来确定执行哪个任务或技能，从而导致语言条件机器人策略 π (at|p, s≤t, a<t) 的发展。

三． 视觉语言动作（Vision-Language-Action） 型号
_A。 VLA_的组成部分

大量工作重点关注 VLA 模型的各个组成部分，借鉴了 CV、NLP 和 RL 方面的成功经验。我们通过具身智能（Embodied AI）的镜头来介绍它们。

1) 强化学习（Reinforcement Learning）： RL 为 具身智能（Embodied AI） 奠定了基础，并继续帮助推进 VLA。 DQN 首先展示了直接来自高维像素输入的学习策略，强调了端到端强化学习中对更大模型容量的需求。 RL 轨迹（状态、动作和奖励的序列）自然地与序列建模问题保持一致，使它们非常适合 Transformer 架构。这个方向的开创性工作包括决策变压器（DT）[16]和轨迹变压器（TT）[17]。 Gato

[18] 进一步将此范式扩展到多模式、多任务、多实施例设置。 π 0_* ._ 6[19] 使用 RL 来让 VLA 从经验中学习。

此外，RL 和 LLM 之间出现了协同效应，使 具身智能（Embodied AI） 在多个方面受益。来自人类反馈的强化学习 (RLHF) 使法学硕士与人类偏好保持一致，并且也已应用于机器人学习。 SEED [20] 利用 RLHF 和基于技能的 RL 来解决奖励稀疏问题并通过人工评估来提高安全性。相反，法学硕士也支持新颖的强化学习方法。 Reflexion [21] 提出了一种语言强化学习框架，用语言反馈取代强化学习模型中的权重更新，使其自然适用于也以语言形式进行规划的具体决策。 Eureka [22] 表明，法学硕士可以为具体智能体设计奖励函数，其性能优于专家人工设计的智能体。随着法学硕士的发展，这种协同作用加速了 RL 和 具身智能（Embodied AI） 的进步。

2) 预训练视觉表示： 视觉编码器的有效性直接影响 VLA 的性能，因为它提供了有关当前状态 st 的关键信息，例如对象类别、位置和 可供性（Affordances）。因此，许多方法致力于提高 PVR 的质量。它们的技术细节在表一中进行了比较。

CLIP [23] 作为机器人模型中的视觉编码器已得到广泛采用。 CLIP 的训练目标是在给定批次中的所有可能组合中识别正确的文本图像对。 CLIP 在 WIT 数据集上进行预训练，该数据集包含 4 亿个图像文本对。这种大规模的训练使 CLIP 能够对视觉和文本信息之间的关系产生丰富的理解。

R3M [24]提出了两个主要的预训练目标：时间对比学习（CL）和视频语言对齐。时间对比学习的目标是最小化时间上邻近的视频帧之间的距离，同时增加时间上较远的视频帧之间的间隔。该目标旨在创建能够捕获视频序列内的时间关系的 PVR。另一方面，视频语言对齐旨在了解视频是否对应于语言指令。这一目标丰富了 PVR 中嵌入的语义相关性。 VIP [25] 还利用了视频时间关系。

Table I
PVRs. V: VISION. L: LANGUAGE. CL: CONTRASTIVE LEARNING. **SIM/REAL**: SIMULATED/REAL-WORLD. **MANI/NAVI**: MANIPULATION/NAVIGATION. FOR SIMPLICITY, WE ONLY SHOW THE MAIN PART OF THE OBJECTIVE EQUATION, OMITTING ELEMENTS SUCH AS TEMPERATURE, AUXILIARY LOSS. $\mathcal{S}(\cdot)$ DENOTES A SIMILARITY MEASURE

Model	Network	Type	Objective	Notations	Robotic Tasks
CLIP [23]	ViT-B	VL-CL	$\sum_{i=1}^{N} -\log \frac{\exp(\mathcal{S}(x_i,y_i))}{\sum_{j=1}^{N}\exp(\mathcal{S}(x_i,y_j))}$	(x_i, y_i): image-text pair	(Used by CLIPort [29], EmbCLIP [30], and CoW [31])
R3M [24]	ResNet-50	Time-CL	$\sum_{i,j,k} -\log \frac{\exp(\mathcal{S}(x_i,x_j))}{\exp(\mathcal{S}(x_i,x_j))+\exp(\mathcal{S}(x_i,x_k))}$	x_i: video frame at time i; $i<j<k$	**Sim-Mani**: Meta-World, Franka Kitchen, Adroit
MVP [26]	ViT-B/L	MAE	$\sum_{i\in\mathcal{M}} \mathrm{MSE}\left(x_i, f_{\text{mae}}(x_{\notin\mathcal{M}})\right)$	x_i: image patch; $\mathcal{M}$: masked set	**Real-Mani** (xArm7): pick, reach block, push cube, close fridge
VIP [25]	ResNet-50	Time-CL	$\sum_{\tau} -\log \frac{\exp(\mathcal{S}(x_0,x_T))}{\sum_{i,j}\exp(\mathcal{S}(x_i,x_T)-\gamma\mathcal{S}(x_j,x_T))}$	x_i: video frame at time i; $0\le i<j<T$; γ: discount factor; τ: trajectory	**Real-Mani** (Franka): pick and place, close drawer, push bottle, fold towel
VC-1 [32]	ViT-L	MAE, CL	(A combination of previous works, including CLIP, R3M, MVP, and VIP.)	—	**Sim-Mani**: Meta-World, Adroit, DMC, TriFinger, Habitat 2.0; **Sim-Navi**: Habitat
Voltron [33]	ViT-S/B	MAE, Lang-Gen	$\sum_{i\in\mathcal{M}} \mathrm{MSE}\left(x_i, f_{\text{mae}}(x_{\notin\mathcal{M}}, y)\right)$; $\sum_{i=1}^{N} -\log P\left(y_i \mid x, y_{<i}\right)$	x_i: image patch; $\mathcal{M}$: masked set; y: language instruction	**Sim-Mani**: Franka Kitchen; **Real-Mani** (Franka): custom study desk
RPT [28]	ViT	MAE	$\sum_{i\in\mathcal{M}} \mathrm{MSE}\left(x_i, f_{\text{mae}}(x_{\notin\mathcal{M}}, y, z, \dots)\right)$	x, y, z: three distinct modalities; $\mathcal{M}$: masked set	**Real-Mani** (Franka): stack, pick, pick from bin
DINOv2 [34]	ViT	Self-distillation	$\sum_{x}\sum_{x'\neq x} H(P_t(x), P_s(x'))$	x, x': image views; $H()$: cross-entropy; P_t, P_s: teacher, student	(Used by OpenVLA [35], ReKep [36])
I-JEPA [37]	ViT	JEPA	$\sum_{i\in\mathcal{M}} \mathrm{MSE}\left(f'(x_i), g(f(x_{\notin\mathcal{M}}))\right)$	x_i: image block; $\mathcal{M}$: target set; f, f': encoder, EMA encoder; g: predictor	—
Theia [38]	ViT-T/S/B	Distillation	(Distillation of vision foundation models: ViT, CLIP, SAM, DINOv2, Depth-Anything.)	—	**Sim-Mani&Navi**: CortexBench (VC-1); **Real-Mani**: pick, place, open door/drawer

MVP [26] applies a masked autoencoder (MAE) from computer vision to robotic datasets. The MAE involves masking out a portion of input patches to a ViT model and training it to reconstruct the corrupted patches. This approach closely resembles the masked language modeling technique used in BERT [27] and falls within the purview of self-supervised training. RPT [28] is pretrained with a focus not only on reconstructing visual inputs but also on robotic actions and proprioceptive states.

Voltron [33] introduces a pretraining objective by incorporating language conditioning and language generation into the MAE objective. Employing an encoder-decoder Transformer structure, the model alternates between language-conditioned masked image reconstruction and language generation from masked images. This enhances the alignment between language and vision modalities.

VC-1 [32] conducts an in-depth examination of prior PVRs and introduces an enhanced PVR model by systematically exploring optimal ViT configurations across diverse datasets. In addition, they perform a comprehensive comparative analysis of their model against previous methods on various manipulation and navigation datasets, shedding light on the critical factors that contribute to the improvement of PVRs. Another study [39] also compares previous PVRs obtained under supervised learning or self-supervised learning.

DINOv2 [34] proposes a new self-supervised training paradigm for PVRs that achieves performance beyond that of MAE. It employs a self-distillation framework in which the teacher and student networks receive different views of the same image and match their encoded representations. The student network is updated via gradient descent, while the teacher network is maintained as the exponential moving average of the student network.

I-JEPA [37] is motivated by the joint-embedding predictive architectures proposed by LeCun [40]. It constructs a "primitive" internal world model by comparing the embeddings of patches. Unlike DINO, which uses cropped images, I-JEPA employs masked patches. Moreover, it differs from MAE because it is a nongenerative approach.

Theia [38] proposes distilling various vision foundation models into a single model. By fusing diverse visual information, such as segmentation, depth, and semantics, it outperforms previous PVRs while requiring less data and a smaller model size.

3) Video Representations: Videos are simple sequences of images and can be represented by concatenating the usual PVRs of each frame. However, their multiview nature enables a variety of unique representation techniques beyond those mentioned above, such as time CL and MAE. NeRF can be extracted from videos and contains rich 3D information for robot learning, as exemplified by F3RM [41], and 3D-LLM [42]. The recent 3D Gaussian Splatting (3D-GS) [43], [44] surpasses NeRF in visual quality and rendering speed. In addition, many videos contain audio, which can provide important cues for robot policies [45].

4) Dynamics Learning: Dynamics learning encompasses objectives aimed at endowing the model $f(\cdot)$ with an understanding of forward or inverse dynamics. Forward dynamics involves predicting the subsequent state resulting from a given action, whereas inverse dynamics entails determining the action required to transition from a previous state to a known subsequent state:

$$\begin{aligned} \text{Forward dynamics: } \hat{s}_{t+1} &\leftarrow f_{\text{fwd}}(s_t, a_t) \\ \text{Inverse dynamics: } \hat{a}_t &\leftarrow f_{\text{inv}}(s_t, s_{t+1}). \end{aligned} \tag{1}$$

Some approaches also frame these objectives as reordering problems for shuffled state sequences. Some forward dynamics methods closely resemble the image or video prediction pretraining used in PVRs. We compare them in Table II.

Vi-PRoM [46] presents three distinct pretraining objectives. The first involves a contrastive self-supervised learning objective designed to distinguish between different videos. The remaining two objectives are centered around supervised learning tasks: temporal dynamics learning, aimed at recovering

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4

表一

PVRS。五：愿景。 L：语言。 CL：对比学习。 SIM / REAL：模拟/现实世界。 MANI / NAVI：操纵/导航。为了简单起见，我们仅显示目标方程的主要部分，省略了温度、辅助损耗等元素。 S (·) 表示相似性度量

模型	网络	类型	目标	符号	机器人任务
CLIP [23]	ViT-B	VL-CL
i=1 _−_log
exp(S(xi,yi))
N
j=1 exp(S(xi,yj))
(xi, yi)：图像文本对	（由 CLIPort [29]、EmbCLIP [30] 和 CoW [31] 使用）
R3M [24]	ResNet-50	Time-CL
_i,j,k __−_log
exp(S(xi,xj))
exp(S(xi,xj))+exp(S(xi,xk))|xi：时间 i 的视频帧； i < j < k|Sim-Mani：元世界、Franka Kitchen、Adroit|
|MVP [26]|ViT-B/L|MAE|
iεM MSE (_xi, f_mae(x/εM))
|xi：图像补丁； M：蒙面套装|真实玛尼 (xArm7)：拾取、到达方块、推动立方体、关闭冰箱|
|VIP [25]|ResNet-50|时间-CL|
_τ __−_log
exp(S(_x_0_,xT_))
i,j exp(S(xi,xT)−γS(xj,xT))|xi：时间 i 处的视频帧； 0_≤i < j < T_; γ：折扣因子； τ：轨迹|Real-Mani (Franka)：拾取放置、关闭抽屉、推瓶、折叠毛巾|
|VC-1 [32]|ViT-L|MAE、CL|（之前作品的组合，包括 CLIP、R3M、MVP 和 VIP。）||Sim-Mani：Meta-World、Adroit、DMC、TriFinger、Habitat 2.0； Sim-Navi：栖息地|
|战神金刚[33]|ViT-S/B|MAE，Lang-Gen|
iεM MSE (_xi, f_mae(x/εM, y));
N
i=1 _−_log_P_ (yi | x, y<i)|xi：图像补丁； M：面具套装； y：语言教学|Sim-Mani：弗兰卡厨房； Real-Mani（弗兰卡）：定制学习桌|
|RPT [28]|ViT|MAE|

i∈M MSE (_xi, f_mae(x/εM, y, z, . . .))|x, y, z：三种不同的模态； M：
蒙面套装|真实玛尼 (Franka)：堆叠、挑选、挑选
从垃圾箱|
|DINOv2
[34]|ViT|自
蒸馏|
x
_x_/′_=_x __H_(Pt(x), Ps(x′))|_x, x__′_：图像浏览次数； H()：交叉熵；
Pt，Ps：教师，学生|（由OpenVLA [35]，ReKep [36]使用）|
|I-JEPA [37]|ViT|JEPA|
i∈M MSE (f′(xi), g(f(x/εM)))|xi：图像块； M：目标设定； _f, f__′_:
编码器，EMA编码器； g：预测器||
|Theia [38]|ViT-T/S/B|蒸馏|（视觉蒸馏 基础模型（Foundation Models）：
ViT、CLIP、SAM、DINOv2、Depth-Anything。）||Sim-Mani&Navi：CortexBench (VC-1)；
Real-Mani：拾取、放置、打开门/抽屉|

MVP [26] 将计算机视觉中的屏蔽自动编码器 (MAE) 应用到机器人数据集。 MAE 涉及将 ViT 模型的部分输入补丁屏蔽掉，并对其进行训练以重建损坏的补丁。这种方法与 BERT [27] 中使用的掩码语言建模技术非常相似，属于自监督训练的范围

。 RPT [28] 经过预训练，不仅关注重建视觉输入，还关注机器人动作和本体感受状态。

Voltron [33] 通过将语言调节和语言生成纳入 MAE 目标中，引入了预训练目标。该模型采用编码器-解码器 Transformer 结构，在语言条件蒙版图像重建和蒙版图像语言生成之间交替。这增强了语言和视觉模式之间的一致性。

VC-1 [32] 对先前的 PVR 进行了深入检查，并通过系统地探索不同数据集的最佳 ViT 配置来引入增强的 PVR 模型。此外，他们还对各种操纵和导航数据集上的模型与之前的方法进行了全面的比较分析，揭示了有助于改进 PVR 的关键因素。另一项研究[39]也比较了之前在监督学习或自监督学习下获得的 PVR。

DINOv2 [34] 提出了一种新的 PVR 自监督训练范例，其性能超越了 MAE。它采用自蒸馏框架，其中教师和学生网络接收同一图像的不同视图并匹配其编码表示。学生网络通过梯度下降进行更新，而教师网络则保持为学生网络的指数移动平均值。

I-JEPA [37] 受到 LeCun [40] 提出的联合嵌入预测架构的启发。它通过比较补丁的嵌入来构造一个“原始”内部 世界模型（World Model）。与使用裁剪图像的 DINO 不同，I-JEPA

采用屏蔽补丁。此外，它与 MAE 不同，因为它是一种非生成方法。

Theia [38]建议将各种视觉 基础模型（Foundation Models） 提炼成单个模型。通过融合不同的视觉信息，例如分割、深度和语义，它优于以前的 PVR，同时需要更少的数据和更小的模型大小。

3) 视频表示： 视频是简单的图像序列，可以通过连接每帧的常用 PVR 来表示。然而，它们的多视图性质使得除上述之外的各种独特的表示技术成为可能，例如时间 CL 和 MAE。 NeRF 可以从视频中提取，包含丰富的用于机器人学习的 3D 信息，如 F3RM [41] 和 3DLLM [42] 所示。最近的 3D 高斯泼溅（3D-GS）[43]、[44] 在视觉质量和渲染速度上超越了 NeRF。此外，许多视频包含音频，可以为机器人策略提供重要线索[45]。

4) 动力学学习： 动力学学习包含旨在赋予模型 f (·) 理解正向或逆向动力学的目标。正向动力学涉及预测给定动作产生的后续状态，而逆向动力学则需要确定从先前状态转换到已知后续状态所需的动作：

正向动态： s ˆ t +1 ← f fwd(st, at)
(1)
逆向动态： a ˆ t ← f inv(st, st +1) .

一些方法还将这些目标构建为混洗状态序列的重新排序问题。一些前向动力学方法与 PVR 中使用的图像或视频预测预训练非常相似。我们在表二中对它们进行了比较。

Vi-PRoM [46] 提出了三个不同的预训练目标。第一个涉及对比自我监督学习目标，旨在区分不同的视频。剩下的两个目标围绕监督学习任务：时间动态学习，旨在恢复

Table II
DYNAMICS LEARNING METHODS FOR VLAS. $f(\cdot)$ IS THE DYNAMICS MODEL. FWD, INV: FORWARD & INVERSE DYNAMICS

Model	Vision Encoder	Type	Objective	Notations	Robotic Tasks
Vi-PRoM [46]	ResNet	Temporal dynamics	$\sum_{t=1}^{T} \mathrm{CE}(t, f(s'_t))$	s'_t: frame in shuffled seq.; $f(\cdot)$ predicts original frame index	**Sim-Mani**: Meta-World, Franka Kitchen; **Real-Mani**: open & close drawer/door
MaskDP [47]	ViT (MAE)	MAE (implicit fwd & inv dynamics)	$\sum_{t=1}^{T} \mathrm{MSE}\left(x_t, f_{\text{mask}}(\tau')\right)$	$x_t \in \{s_t, a_t\}$; τ': masked sequence	**Sim**: DeepMind Control Suite
MIDAS [48]	ViT, Mask R-CNN	Inverse dynamics	$\sum_{t=1}^{T-1} -\log P\left(a_t \mid f_{\text{inv}}(s_{1:T})\right)$	$s_{1:T}$: full state seq.	**Sim-Mani**: VIMA-Bench
SMART [49]	CNN	Forward & inverse dynamics	$\sum_{t=1}^{T-1} \big(\mathrm{MSE}\left(s_{t+1}, f_{\text{fwd}}(s_t, a_t)\right) + \mathrm{MSE}\left(a_t, f_{\text{inv}}(s_t, s_{t+1})\right)\big) + \sum \mathrm{MSE}\left(a_t, f_{\text{mask}}(\tau')\right)$	s_t, a_t: state, action; τ': masked sequence	**Sim**: DeepMind Control Suite
PACT [50]	ResNet-18, PointNet	Forward dynamics	$\sum_{t=1}^{T-1} \mathrm{MSE}\left(s_{t+1}, f_{\text{fwd}}(s_t, a_t)\right)$	s_t, a_t: state, action	**Sim-Navi**: Habitat; **Real-Navi** (MuSHR vehicle)
VPT [51]	ResNet	Inverse dynamics	$\sum_{t=1}^{T} -\log P\left(a_t \mid f_{\text{inv}}(s_{1:T})\right)$	s_t, a_t: state, action	**Sim**: Minecraft
GR-1 [52]	ViT (MAE)	Forward dynamics	$\sum_{t=1}^{T-1} \mathrm{MSE}\left(s_{t+1}, f_{\text{fwd}}(s_t, a_t)\right)$	s_t, a_t: state, action	**Sim-Mani**: CALVIN

shuffled video frames, and image classification employing pseudo labels. Through a comprehensive comparison with preceding pretraining methods, Vi-PRoM demonstrates its effectiveness for behavior cloning (BC) and PPO.

MIDAS [48] introduces an inverse dynamics prediction task as part of its pretraining. The objective is to train the model to predict actions from observations, formulated as a motion-following task. This approach enhances the model's understanding of the transition dynamics of the environment.

SMART [49] presents a pretraining scheme encompassing three distinct objectives: forward dynamics prediction, inverse dynamics prediction, and randomly masked hindsight control. The forward dynamics prediction task involves predicting the next latent state, while the inverse dynamics prediction task entails predicting the previous action. In the case of hindsight control, the entire control sequence is provided as input, with some actions masked, and the model is trained to recover these masked actions. The incorporation of the first two dynamics prediction tasks facilitates capturing local and short-term dynamics, while the third task is designed to capture global and long-term temporal dependencies.

MaskDP [47] features the masked decision prediction task, wherein both state and action tokens are masked for reconstruction. This masked modeling task is specifically crafted to equip the model with an understanding of both forward and inverse dynamics. Notably, in contrast to preceding masked modeling approaches like BERT or MAE, MaskDP can be applied directly to downstream tasks in a zero-shot manner.

PACT [50] introduces a pretraining objective aimed at modeling state-action transitions. It receives sequences of states and actions as input, and predicts each state and action token autoregressively. This pretrained model serves as a dynamics model, which can then be finetuned for various downstream tasks such as localization, mapping, and navigation.

VPT [51] proposes a video pretraining method that harnesses unlabeled internet data to pretrain a foundation model for the game of Minecraft. The approach begins by training an inverse dynamics model using a limited amount of labeled data, which is then utilized to label internet videos. Subsequently, this newly auto-labeled data is employed to train the VPT foundation model through BC. This methodology follows semi-supervised imitation learning. As a result of this process, the model demonstrates human-level performance across a multitude of tasks.

GR-1 [52] introduces video prediction pretraining tailored for a GPT-style model. The ability to anticipate future frames aligns with forward dynamics learning, contributing to more accurate action prediction.

5) World Models: A world model $P(\cdot)$ encodes commonsense knowledge about the world and predicts the future state for a given action [40]:

$$\hat{s}_{t+1} \sim P(s_{t+1} \mid s_t, a_t). \tag{2}$$

It enables model-based control and planning for embodied agents, as they can search for an optimal action sequence in imaginary space before executing any real actions. Although forward dynamics learning also attempts to predict the next state, it is usually treated as a pretraining task or an auxiliary loss to enhance the action decoder for the primary robotics tasks, rather than serving as a standalone module. In addition, new embodied experiences can be sampled from visual world models that explicitly generate images or videos of future states.

Dreamer [53] employs three primary modules to construct a latent dynamics model: a representation model, responsible for encoding images into latent states; a transition model, which captures transitions between latent states; and a reward model, predicting the reward associated with a given state. Under the actor-critic framework, Dreamer utilizes an action model and a value model to learn behavior through imagination by propagating analytic gradients through the learned dynamics. Building upon this foundation, DreamerV2 [54] introduces a discrete latent state space along with an improved objective. DreamerV3 [55] extends its focus to a broader range of domains with fixed hyperparameters. DayDreamer [56] applies this method to physical robots performing real-world tasks.

IRIS [57] employs a GPT-like autoregressive Transformer as the foundation of its world model, with a VQ-VAE serving as the vision encoder. Subsequently, a policy is trained using imagined trajectories, which are unrolled from a real observation by the world model. TWM [58] also investigates the application of Transformers in building world models.

6) LLM-Induced World Models: LLMs encompass a wealth of commonsense knowledge about the world, prompting many approaches to leverage that knowledge for improving VLAs.

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表二

VLAS 的动力学学习方法。 f (·) 是动力学模型。 FWD、INV：正向和反向动力学

模型	视觉编码器	类型	目标	符号	机器人任务
Vi-PRoM [46]	ResNet	时间动力学
t=1 CE(t, f(s′
t))
_s__′_
t：帧中打乱顺序
； f(·) 预测
原始帧索引	Sim-Mani：元世界、Franka
厨房； Real-Mani：打开和关闭抽屉/门
MaskDP [47]	ViT (MAE)	MAE（隐式 fwd
& inv 动态）
t=1 MSE (_xt, f_mask(τ ′))
xt ∈{st, at}; _τ __′_:
屏蔽序列	Sim：DeepMind Control Suite
MIDAS [48]	ViT，Mask R-CNN	逆动力学
t=1 _−_log_P_ (_at	f_inv(_s_1:T))
_s_1:T : full状态序列	Sim-Mani：VIMA-Bench
SMART [49]	CNN	正向和逆向
动力学
t=1
MSE (st+1_, f_fwd(st, at))
+ MSE (_at, f_inv(st, st+1))
+ MSE(_at, f_mask(_τ __′_))
|st, at: 状态，动作； _τ __′_:
屏蔽序列|Sim：DeepMind Control Suite|
|PACT [50]|ResNet-18，PointNet|前向动力学|
t=1 MSE (st+1_, f_fwd(st, at))|st, at：状态，动作|Sim-Navi：栖息地； Real-Navi
（MuSHR 车辆）|
|VPT [51]|ResNet|逆动力学|
t=1 _−_log_P_ (_at | f_inv(_s_1:T))
|st, at: 状态，动作|Sim:我的世界|
|GR-1 [52]|ViT (MAE)|前向动力学|

t=1 MSE (st+1_, f_fwd(st, at))|st, at：状态，动作|Sim-Mani：CALVIN|

打乱视频帧，以及使用伪标签进行图像分类。通过与之前的预训练方法的综合比较，Vi-PRoM 证明了其对于 行为克隆（Behavior Cloning）(BC) 和 PPO 的有效性。

MIDAS [48] 引入了逆动态预测任务作为其预训练的一部分。目标是训练模型根据观察预测动作，将其表述为运动跟踪任务。这种方法增强了模型对环境转变动态的理解。

SMART [49] 提出了一种预训练方案，包含三个不同的目标：正向动态预测、逆向动态预测和随机屏蔽后视控制。正向动态预测任务涉及预测下一个潜在状态，而逆向动态预测任务需要预测先前的动作。在事后控制的情况下，整个控制序列作为输入提供，其中一些动作被屏蔽，并且模型被训练以恢复这些被屏蔽的动作。前两个动态预测任务的结合有助于捕获局部和短期动态，而第三个任务旨在捕获全局和长期时间依赖性。

MaskDP [47] 具有屏蔽决策预测任务的特点，其中状态和动作标记都被屏蔽以进行重建

。这个掩码建模任务是专门设计的，旨在让模型了解正向和逆向动力学。值得注意的是，与之前的 BERT 或 MAE 等屏蔽建模方法相比，MaskDP 可以以零样本的方式直接应用于下游任务。

PACT [50] 引入了一个旨在建模状态-动作转换的预训练目标。它接收状态和动作序列作为输入，并自回归预测每个状态和动作标记。这个预训练模型充当动态模型，然后可以针对各种下游任务（例如定位、地图和导航）进行微调。

VPT[51]提出了一种视频预训练方法，即运行支撑框架（Harnesses）未标记的互联网数据来为Minecraft游戏预训练基础模型（Foundation Model）。该方法首先使用有限数量的标记数据训练逆动态模型，然后将其用于标记互联网视频。随后，使用这些新自动标记的数据通过 BC 来训练 VPT 基础模型（Foundation Model）。该方法遵循半监督 模仿学习（Imitation Learning）。这一过程的结果是，该模型在各个领域展示了人类水平的表现

大量的任务。

GR-1 [52] 引入了为 GPT 风格模型量身定制的视频预测预训练。预测未来帧的能力与前向动态学习相一致，有助于更准确的动作预测。

5) 世界模型（World Models）: A 世界模型（World Model） P（·）编码有关世界的常识知识并预测给定动作的未来状态 [40]：

s ^ t +1 ~ P (st +1 | st, at) 。 (2)

它为具身智能体提供基于模型的控制和规划，因为它们可以在执行任何实际动作之前在想象空间中搜索最佳动作序列。尽管前向动态学习也尝试预测下一个状态，但它通常被视为预训练任务或辅助损失，以增强主要机器人任务的动作解码器，而不是作为独立模块。此外，新的具体体验可以从视觉 世界模型（World Models） 中采样，明确生成未来状态的图像或视频。

Dreamer [53]采用三个主要模块来构建潜在动态模型：表示模型，负责将图像编码为潜在状态；转换模型，捕获潜在状态之间的转换；和奖励模型，预测与给定状态相关的奖励。在演员-评论家框架下，梦想家利用行动模型和价值模型，通过学习动态传播分析梯度，通过想象力来学习行为。在此基础上，DreamerV2 [54] 引入了离散潜在状态空间以及改进的目标。 DreamerV3 [55] 将其重点扩展到具有固定超参数的更广

泛的领域。 DayDreamer [56]将此方法应用于执行现实世界任务的物理机器人。

IRIS [57] 采用类似 GPT 的自回归 Transformer 作为其 世界模型（World Model） 的基础，并使用 VQ-VAE 作为视觉编码器。随后，使用想象的轨迹来训练策略，这些轨迹是根据 世界模型（World Model） 的真实观察展开的

。 TWM [58]还研究了 Transformers 在构建 世界模型（World Models） 中的应用。

6) LLM 引发的 世界模型（World Models）: LLM 包含丰富的关于世界的常识性知识，促使许多方法利用这些知识来改进 VLA。

DECKARD [59] prompts an LLM to generate abstract world models (AWMs) represented as directed acyclic graphs [60], [61], specifically tailored for the task of item crafting in Minecraft. DECKARD iterates between two phases: in the Dream phase, it samples a subgoal guided by AWM; in the Wake phase, DECKARD executes the subgoal and updates the AWM through interactions with the game. This guided approach enables DECKARD to achieve markedly faster item crafting compared with baselines lacking such guidance.

LLM-DM [62] uses an LLM to construct world models in planning domain definition language (PDDL)—a capability that LLM+P [63] did not achieve, as its PDDL world model was hand-crafted. The LLM also acts as an interface, mediating between the generated PDDL model and corrective feedback from syntactic validators and human experts. Finally, the PDDL world model serves as a symbolic simulator, enabling plan generation.

RAP [64] repurposes an LLM to act as both a policy that predicts actions and a world model that provides the state transition distribution. Unlike previous chain-of-thought (CoT) prompting methods, RAP incorporates Monte Carlo tree search (MCTS) to enable structured planning, allowing the LLM to build a reasoning tree incrementally. This reasoning strategy helps RAP find a high-reward path that balances exploration and exploitation. Tree-Planner [65] improves efficiency by prompting the LLM only once to generate diverse paths within an action tree.

LLM-MCTS [66] extends the problem setting to POMDPs. As a world model, the LLM generates the initial belief of the current state; as a policy, it serves as a heuristic to guide action selection. By leveraging its commonsense knowledge, the LLM reduces the search space of MCTS, thereby improving search efficiency.

7) Visual World Models: Unlike LLM-induced world models, which are in textual form, visual world models generate images, videos, or 3D scenes of future states—aligning more closely with the physical world. They can further be utilized to generate new trajectories. This direction has been gaining increasing attention since Sora demonstrated world-simulation capabilities [67], as investigated by a dedicated survey [68].

Genie [69] introduces a new class of generative models, termed generative interactive environments. It consists of three main components: a spatiotemporal video tokenizer, an autoregressive dynamics model, and a latent action model. After being trained on unlabeled videos in an unsupervised manner, Genie allows users to interact with the generative environment on a frame-by-frame basis. Consequently, Genie establishes a foundation world model.

3D-VLA [70] proposes a 3D world model capable of goal generation. It processes visual inputs, such as images, depth maps, and point clouds, and then generates a goal state—either as an image or a point cloud—using diffusion models in response to the user's query. The generated goal state can subsequently be used to guide robot control.

UniSim [71] builds a generative model based on real-world interaction videos. It is capable of simulating visual outcomes for both high-level and low-level actions, which can then be leveraged as new experiences for training embodied agents. E2WM [72] even treats existing simulators as world models to collect embodied experiences via MCTS.

8) Reasoning: Reasoning has become a key capability of LLMs, as demonstrated by chain-of-thought (CoT) methods [73], [74]. In embodied AI, researchers are exploring how to leverage CoT reasoning to refine the decision-making process.

Reasoning is naturally compatible with high-level task planning, as both often occur in the language domain. ThinkBot [75] applies CoT to recover missing action descriptions in sparse human instructions, thereby enhancing instruction coherence and boosting success rates in difficult tasks. ReAct [76] interleaves reasoning traces and actions, where CoT can help create action plans, inject commonsense knowledge, and handle exceptions, ultimately improving embodied decision-making. RAT [77] integrates CoT with retrieval-augmented generation (RAG) to mitigate hallucinations and thus improve embodied planning. Tree-Planner [65] employs a tree-of-thoughts approach for task planning.

Another paradigm involves equipping low-level control policies with reasoning capabilities, as pioneered by ECoT [78]. This method trains OpenVLA [35] to conduct embodied CoT reasoning about plans, subtasks, motions, and visual features, before predicting low-level actions. By relying on this multi-step reasoning rather than the "muscle memory" of VLAs, it improves success rates on challenging generalization tasks without requiring additional robot data. CoT-VLA [79] introduces visual CoT reasoning for VLAs.

9) Policy Steering: Policy steering can enhance VLA performance at test-time without the need for expensive retraining. V-GPS [80] re-ranks generated actions based on a learned value function, while RoboMonkey [81] employs a VLM-based verifier to select the optimal action from a sampled set.

10) Strengths and Limitations:

a) PVRs: Although time CL with videos and text-guided pretraining methods, such as CLIP, provide image-level information, they lack the pixel-level details offered by MAE-based self-supervised learning methods. Pixel-level information contains rich details—including segmentation masks, object positions, and depth estimation—that are generally more useful for robot manipulation tasks requiring high precision, as demonstrated by VC-1's comparison [32]. DINOv2 learns both pixel- and image-level features by combining masked image modeling with a momentum encoder and multicrop augmentation, with benefits on downstream robotic tasks evidenced by OpenVLA [35]. I-JEPA focuses on patches in the representation space and, as a result, captures low-level image features more effectively than view-invariance methods such as DINO. Theia distills various off-the-shelf vision foundation models into a single model that surpasses isolated individual models, as demonstrated by comprehensive evaluations against most existing PVRs in robot learning [38].

b) Forward & Inverse Dynamics: Forward dynamics learning is generally more challenging than inverse dynamics learning because predicting future states is more complex than predicting past actions. Consequently, the difficulty of forward dynamics often leads to greater performance improvements, as demonstrated in SMART. However, inverse dynamics models

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DECKARD [59] 提示法学硕士生成抽象 世界模型（World Models）(AWM)，表示为有向无环图 [60]、[61]，专门为 Minecraft 中的物品制作任务量身定制。 DECKARD 在两个阶段之间迭代：在 Dream 阶段，它对 AWM 引导的子目标进行采样；在Wake阶段，DECKARD执行子目标并通过与游戏的交互更新AWM。与缺乏此类指导的基线相比，这种引导方法使 DECKARD 能够显着加快物品制作速度。

LLM-DM [62] 使用 LLM 在规划域定义语言 (PDDL) 中构建 世界模型（World Models），这是 LLM+P [63] 未实现的功能，因为其 PDDL 世界模型（World Model） 是手工制作的。 LLM 还充当接口，在生成的 PDDL 模型与语法验证器和人类专家的纠正反馈之间进行调解。最后，PDDL 世界模型（World Model） 充当符号模拟器，支持计划生成。

RAP [64] 重新利用 LLM 来充当预测操作的策略和提供状态转换分布的 世界模型（World Model）。与之前的思想链（CoT）提示方法不同，RAP 结合了蒙特卡罗树搜索（MCTS）来实现结构化规划，允许法学硕士逐步构建推理树。这种推理策略帮助 RAP 找到一条平衡探索和利用的高回报路径。 Tree-Planner [65] 通过仅提示 LLM 一次在动作树中生成不同的路径来提高效率。

LLM-MCTS [66] 将问题设置扩展到 POMDP。作为世界模型（World Model），LLM生成当前状态的初始信念；作为一项策略，它可以作为指导行动选择的启发式方法。通过利用其常识性知识，LLM减少了MCTS的搜索空间，从而提高了搜索效率。

7) 视觉 世界模型（World Models）： 与 LLM 引发的文本形式的 世界模型（World Models） 不同，视觉世界模型（World Models） 生成未来状态的图像、视频或 3D 场景 - 与物理世界更加紧密地结合。它们可以进一步用于生成新的轨迹。自从 Sora 展示了世界模拟能力 [67] 以来，这一方向受到了越来越多的关注，正如一项专门调查 [68] 所调查的那样。

Genie [69] 引入了一类新的生成模型，称为生成交互环境。它由三个主要组件组成：时空视频分词器、自回归动力学模型和潜在动作模型。在以无人监督的方式对未标记视频进行训练后，Genie 允许用户逐帧与生成环境进行交互。因此，Genie 建立了一个基金会 世界模型（World Model）。

3D-VLA [70]提出了一种能够生成目标的 3D 世界模型（World Model）。它处理视觉输入，例如图像、深度图和点云，然后使用扩散模型响应用户的查询生成目标状态（作为图像或点云）。生成的目标状态随后可用于指导机器人控制。

UniSim [71]基于现实世界的交互视频构建生成模型

。它能够模拟高级和低级动作的视觉结果，然后可以将其用作训练具体智能体的新体验。

E2WM [72]甚至将现有模拟器视为 世界模型（World Models），以通过 MCTS 收集具体体验。

8) 推理： 正如思想链 (CoT) 方法 [73]、[74] 所证明的那样，推理已成为法学硕士的一项关键能力。在 具身智能（Embodied AI） 中，研究人员正在探索如何利用 CoT 推理来完善决策过程。

推理与高级任务规划自然兼容，因为两者经常发生在语言领域。 ThinkBot [75] 应用 CoT 来恢复稀疏人类指令中缺失的动作描述，从而增强指令的一致性并提高困难任务的成功率。 ReAct [76] 交织推理痕迹和行动，CoT 可以帮助创建行动计划、注入常识知识并处理异常，最终改善具体决策。 RAT [77] 将 CoT 与检

索增强生成 (RAG) 相结合，以减轻幻觉，从而改善具体规划。 Tree-Planner [65] 采用思想树方法进行任务规划。

另一种范式涉及为低级控制策略配备推理能力，如 ECoT [78] 所开创的。该方法训练 OpenVLA [35] 在预测低级动作之前对计划、子任务、运动和视觉特征进行具体的 CoT 推理。通过依靠这种多步骤推理而不是 VLA 的“肌肉记忆”，它可以提高具有挑战性的泛化任务的成功率，而无需额外的机器人数据。
CoT-VLA [79] 引入了 VLA 的视觉 CoT 推理。

9) 策略引导： 策略引导可以增强测试时的 VLA 性能，而不需要昂贵的重新训练。 V-GPS [80] 根据学习的价值函数对生成的动作重新排序，而 RoboMonkey [81] 采用基于 VLM 的验证器从采样集中选择最佳动作。 10) 优点和局限性：

_a) PVR：_尽管带有视频和文本引导预训练方法（例如 CLIP ）的时间 CL 提供了图像级信息，但它们缺乏基于 MAE 的自监督学习方法提供的像素级细节。像素级信息包含丰富的细节，包括分割掩模、物体位置和深度估计，这些细节通常对于需要高精度的机器人操作任务更有用，如 VC-1 的比较所示 [32]。
DINOv2 通过将蒙版图像建模与动量编码器和多作物增强相结合来学习像素级和图像级特征，OpenVLA [35] 证明了它对下游机器人任务的好处。 I-JEPA 专注于表示空间中的补丁，因此比 DINO 等视图不变方法更有效地捕获低级图像特征。 Theia 将各种现成的视觉 基础模型（Foundation Models） 提炼成一个单一模型，该模型超越了孤立的单个模型，正如对机器人学习中大多数现有 PVR 的综合评估所证明的那样 [38]。

b) 正向和逆向动态： 正向动态学习通常比逆向动态学习更具挑战性，因为预测未来状态比预测过去的行为更复杂

。因此，正向动态的难度通常会带来更大的性能改进，如 SMART 所示。然而，逆动力学模型

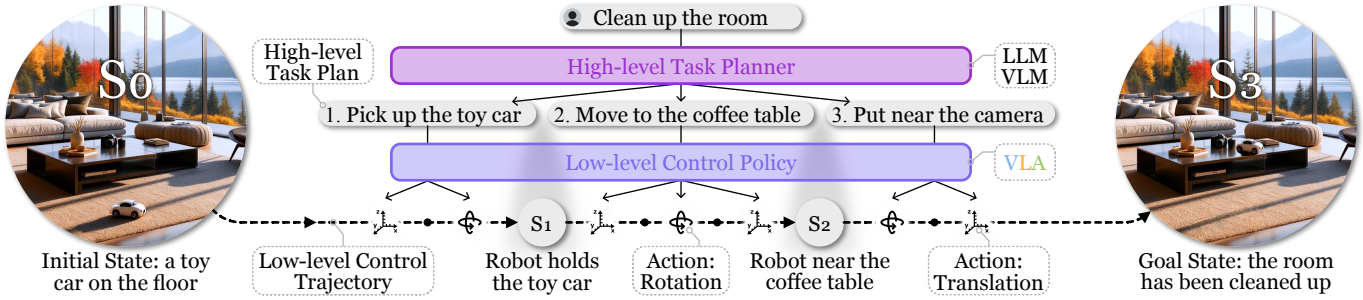


Fig. 4. Illustration of a hierarchical robot policy. The high-level task planner decomposes the user instruction into subtasks, which are then executed step by step by the low-level control policy.

can be used to generate action labels for datasets that contain only states, such as raw robot manipulation videos.

c) World Models & Reasoning: Although both world models and reasoning methods can be applied to low-level control policies and high-level task planners, current approaches remain distinct. World models are predominantly used to interact with control policies because they excel at generating the immediate next state given low-level actions. In contrast, CoT-based reasoning methods focus on task planning since they express thought chains in text, making them well-suited for refining text-based task plans.

B. Low-Level Control Policies

Through the integration of an action decoder with perception modules, such as vision encoders and language encoders, a VLA model π_θ with parameters θ is formed as a control policy to execute language instructions p:

$$\hat{a}_t \sim \pi_\theta(a_t \mid p, s_{\leq t}, a_{<t}). \tag{3}$$

It can also be referred to as a low-level policy, low-level controller, or action primitive. The diversity among VLAs arises from individual modules and overall architectures. The general architecture is shown in Fig. 1. This section explores various approaches to designing low-level control policies. Table III summarizes their technical details.

1) Non-Transformer Control Policies: Prior to the adoption of Transformer models, early control policies for language-conditioned robotic tasks varied significantly in architecture.

CLIPort [29] integrates CLIP with the Transporter network, creating a two-stream architecture. The CLIP vision encoder extracts "semantic" information from the RGB image, while the Transporter network extracts "spatial" information from the RGB-D image. The CLIP sentence encoder encodes the language instruction and guides the output $SE(2)$ action, consisting of paired pick and place end-effector poses. It represents an early demonstration of language-conditioned pick-and-place capabilities.

BC-Z [82] processes two types of task instructions: a language instruction or a human demonstration video. The environment is presented to the model in the form of an RGB image. Then the instruction embedding and the image embedding are combined through the FiLM layer, culminating in the generation of actions. This conditional policy is asserted to exhibit zero-shot task generalization to unseen tasks.

MCIL [83] represents a pioneering robot policy that integrates free-form natural language conditioning. This is in contrast to earlier approaches that typically rely on conditions in the form of task IDs or goal images. MCIL introduces the capability to leverage unlabeled and unstructured demonstration data. This is achieved by training the policy to follow either image or language goals, with a small fraction of the training dataset consisting of paired image and language goals.

HULC [84] introduces several techniques aimed at enhancing robot learning architectures. These include a hierarchical decomposition of robot learning, a multimodal Transformer, and discrete latent plans. The Transformer learns high-level behaviors, hierarchically dividing low-level local policies and the global plan. In addition, HULC incorporates a visuolingual semantic alignment loss based on CL to align VL modalities. HULC++ [85] further integrates a self-supervised affordance model. This model guides HULC to the actionable region specified by a language instruction, enabling it to fulfill tasks within this designated area.

UniPi [86] treats the decision-making problem as a text-conditioned video generation problem. To predict actions, UniPi generates a video based on a given text instruction, and actions are extracted from the frames of the video through inverse dynamics. This innovative policy-as-video formulation offers several advantages, including enhanced generalization across diverse robot tasks and the potential for knowledge transfer from internet videos to real robots.

2) Transformer-Based Control Policies: Since the introduction of Transformers, control policies have converged to similar Transformer-based architectures.

Interactive language [88] presents a robotic system wherein the low-level control policy can be guided in real-time by human instructions conveyed through language, enabling the completion of long-horizon rearrangement tasks. The efficacy of such language-based guidance is primarily attributed to the utilization of a meticulously collected dataset containing diverse language instructions, which surpasses previous datasets by an order of magnitude in scale.

Hiveformer [89] places significant emphasis on leveraging multiview scene observations and maintaining the full observation history for a language-conditioned policy. This approach represents an advancement over previous systems, such as CLIPort and BC-Z, that only use the current observation. Notably, Hiveformer stands out as one of the early adopters

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打扫房间
高级LLM
任务计划高级任务计划VLM
$S_0 S_3$
1。拿起玩具车 2. 移至咖啡桌 3. 放在相机附近
低级控制策略 VLA
S_1 S2
初始状态：玩具低级控制机器人握住动作：机器人靠近动作：目标状态：房间
汽车在地板上玩具车轨迹旋转咖啡桌翻译已清理

图 4. 分层机器人策略的图示。高层任务规划器将用户指令分解为子任务，然后由低层控制策略逐步执行。

可用于为仅包含状态的数据集生成动作标签，例如原始机器人操作视频。

c) 世界模型（World Models）和推理： 尽管 世界模型（World Models）和推理方法都可以应用于低级控制策略和高级任务规划器，但当前的方法仍然不同。 世界模型（World Models）主要用于与控制策略交互，因为它们擅长在给定低级操作的情况下生成立即的下一个状态。相比之下，基于 CoT 的推理方法专注于任务规划，因为它们用文本表达思维链，这使得它们非常适合完善基于文本的任务计划。
B。低层控制策略

通过动作解码器与感知模块（例如视觉编码器和语言编码器）的集成，形成带有参数_θ_的VLA模型_πθ_作为执行语言指令_p_的控制策略：

$$a \hat{} t \sim \pi\theta (at | p, s\leq t, a<t) 。 \quad (3)$$

它也可以称为低级策略、低级控制器或操作原语。 VLA 之间的多样性源于各个模块和整体架构。总体架构如图 1 所示。本节探讨设计低级控制策略的各种方法。表 III 总结了它们的技术细节。

1) 非 Transformer 控制策略： 在采用 Transformer 模型之前，语言条件机器人任务的早期控制策略在架构上存在显着差异。

CLIPort [29]将 CLIP 与 Transporter 网络集成，创建了双流架构。 CLIP视觉编码器从RGB图像中提取“语义”信息，而Transporter网络从RGB-D图像中提取“空间”信息。 CLIP 句子编码器对语言指令进行编码并指导输出 SE (2) 动作，包括成对的拾取和放置末端执行器姿势。它代表了语言条件拾取和放置功能的早期演示。

BC-Z [82] 处理两种类型的任务指令：语言指令或人类演示视频。环境以 RGB 图像的形式呈现给模型。然后指令嵌入和图像嵌入通过 FiLM 层组合，最终生成动作。据称，这种条件策略能够对未见过的任务表现出零样本任务泛化。

MCIL [83] 代表了一种集成了自由形式自然语言调节的开创性机器人策略

。这与早期的方法形成鲜明对比，早期的方法通常依赖于任务 ID 或目标图像形式的条件。 MCIL 引入了利用未标记和非结构化演示数据的功能。这是通过训练策略来遵循图像或语言目标来实现的，训练数据集的一小部分由成对的图像和语言目标组成。

HULC [84] 引入了几种旨在增强机器人学习架构的技术。其中包括机器人学习的层次分解、多模态 Transformer 和离散潜在计划。 Transformer 学习高层行为，分层划分低层本地策略和全局计划。此外，HULC 结合了基于 CL 的视觉语言语义对齐损失来对齐 VL 模态。 HULC++[85]进一步集成了自监督可供性（Affordance）模型。该模型将 HULC 引导至语言指令指定的可操作区域，使其能够完成该指定区域内的任务。

UniPi [86] 将决策问题视为文本条件视频生成问题。为了预测动作，UniPi 根据给定的文本指令生成视频，并通过逆动态从视频帧中提取动作。这种创新的策略视频表述具有多种优势，包括增强不同机器人任务的泛化能力以及从互联网视频到真实机器人的知识转移潜力。

2) 基于 Transformer 的控制策略： 自从 Transformer 引入以来，控制策略已经收敛到类似的基于 Transformer 的架构。

交互式语言[88]提出了一种机器人系统，其中低级控制策略可以由通过语言传达的人类指令实时引导，从而能够完成长时程（Long-Horizon）重排任务。这种基于语言的指导的有效性主要归因于使用了精心收集的包含多种语言指令的数据集，其规模超出了以前的数据集一个数量级。

Hiveformer [89] 非常重视利用多视图场景观察并维护语言条件策略的完整观察历史。这种方法比以前的系统（例如仅使用当前观测值的 CLIPort 和 BC-Z）有所进步。值得注意的是，Hiveformer 是早期采用者之一

Table III

VLAs THAT SERVE AS LOW-LEVEL CONTROL POLICIES. ◊ INDICATES LARGE VLAs. CLOSELY RELATED (NON-VLA MODELS) ARE INCLUDED IN BRACKETS. THE REMAINING ARE GENERALIZED VLAs. CONT/DISC: CONTINUOUS/DISCRETE ACTION. TFM: TRANSFORMER. XATTN: CROSS-ATTENTION. CONCAT: CONCATENATION. QUANT.: QUANTIZATION. p/s: PROMPT/STATE VISION ENCODER. [SC]: SELF-COLLECT DATA

Model	Vision Encoder	Language Encoder / VLM	Action Decoder/Head	Architecture	Training Objectives	Training Datasets	Environments, Embodiments, Tasks, and Skills
CLIPort [29]	CLIP-ResNet50, Transporter-ResNet	CLIP-GPT	LingUNet	Hadamard	BC ($SE(2)$)	[SC]	**Sim**: Ravens
MCIL [83]	Custom-CNN	USE	RNN	LMP	MCIL	Play-LMP, [SC]	**Sim**: 3D Playroom environment
HULC [84], [85]	MCIL CNN	Sentence-BERT	RNN	LMP	MCIL	CALVIN data	**Sim**: CALVIN
Language costs [87]	CLIP-ViT, UNet	CLIP-GPT	STORM	Concat	MLE (cost map)	[SC]	**Sim** & **Real** (Franka): pick, place
Interactive Language [88]	ResNet	CLIP-GPT	TFM	Xattn	BC (cont)	[SC: Language-Table]	**Sim** & **Real** (xArm6): rearrangement
Hiveformer [89]	UNet	CLIP-GPT	TFM, UNet	Concat	BC (cont)	RLBench data	**Sim**: RLBench
PerAct [90]	3D CNN	CLIP-GPT	PerceiverIO	Concat	3D affordance	RLBench data, [SC]	**Sim**: RLBench; **Real** (Franka): pick, place, stack, open drawer, sweep, insert peg, etc.
Act3D [91]	CLIP-ResNet50	CLIP-GPT	TFM	Xattn	3D affordance (voxel)	RLBench data, [SC]	**Sim**: RLBench; **Real** (Franka): reach target, wipe, stack, etc.
RVT [92], RVT-2 [93]	CLIP-ResNet50	CLIP-GPT	TFM	Concat	2D affordance (project to 3D)	RLBench data, [SC]	**Sim**: RLBench; **Real** (Franka): stack, press, place
RoboPoint [94]	ViT-L/14	Vicuna-V1.5 13B	—	Concat	2D affordance (project to 3D)	[SC]	**Real** (Franka): pick, place
Gato [18]	ViT	Sent.Piece	TFM	Concat	BC (cont & disc)	[SC]	**Sim** & **Real** (Sawyer): RGB-stacking
(RoboCat) [95]	VQ-GAN (p, s)	—	TFM	Quant.	BC, observation prediction	Self-improvement	**Sim** & **Real** (Sawyer, Franka, KUKA): stacking, building, lifting, insertion, removal
VIMA [96]	ViT, Mask R-CNN	T5	TFM	Xattn	BC ($SE(2)$)	[SC:VIMA-Data]	**Sim** (Ravens): VIMA-Bench
BC-Z [82]	ResNet18 (p, s)	USE	MLP	FiLM	BC (cont)	[SC]	**Real** (EDR): pick-place/wipe/drag, grasp, push
RT-1 [97]	EfficientNet	USE	TFM	FiLM	BC (disc)	[SC: Fractal]	**Real** (EDR): pick-place, move, knock
MOO [96]	OWL-ViT (p), EfficientNet (s)	USE	TFM	FiLM	BC (disc)	[SC]	**Real** (EDR): pick, move near, knock, place upright, place into
Q-Transformer [98]	EfficientNet	USE	TFM	FiLM	TD error	Fractal, Auto-collect	**Sim**: pick; **Real** (EDR): pick, place, open/close drawer, move near
(RT-Trajectory) [99]	EfficientNet	—	TFM	—	BC (disc)	[SC]	**Real** (EDR): pick, place, fold towel, swivel chair, etc.
(ACT) [100]	ResNet18	—	CVAE-TFM	—	BC (cont, action chunking)	[SC] with ALOHA	**Sim**: transfer cube, bimanual insertion; **Real** (ViperX, WidowX): slot battery, open cup, etc
RoboAgent / MT-ACT [101]	CNN	—	CVAE-TFM	FiLM	BC (cont, action chunking)	RoboSet	**Real** (Franka): pick, place, open/close drawers, pour, push, drag, etc.
RoboFlamingo [102]	CLIP-ViT-L/14	LLaMA, MPT, GPT-NeoX	LSTM, TFM	Xattn	BC (cont)	CALVIN data	**Sim**: CALVIN
RoboUniView [103]	UVFormer	(Model design follows RoboFlamingo)			BC (cont)	CALVIN data	**Sim**: CALVIN
DeeR-VLA [104]	(Model design follows RoboFlamingo++)				BC (cont)	CALVIN data	**Sim**: CALVIN
Instruct2Act [105]	CLIP, SAM	ChatGPT	Robot APIs	Tool-Use	—	—	**Sim** (Ravens): VIMA-Bench
VoxPoser [106]	ViLD, MDETR, OWL-ViT, SAM	GPT-4	MPC	Tool-Use	—	—	**Sim**: Sapien; **Real** (Franka): move&avoid, set table, close drawer, open bottle, sweep trash
UniPi [86]	Imagen video	T5-XXL	CNN	Xattn	Inverse dynamics	[SC:PF], Ravens, BridgeV1	**Sim**: Painting Factory (PF), Ravens
(Diffusion Policy) [107]	ResNet18	—	UNet, TFM	—	DDPM	[SC]	**Sim**: Robomimic, Franka Kitchen, etc; **Real** (UR5, Franka): push-T, flip mug, pour sauce
(DP3) [108]	DP3 Encoder	(Model design follows Diffusion Policy)			DDIM	[SC]	**Sim**: (72 tasks); **Real** (Franka, Allegro hand): roll-up, dumpling, drill, pour
SUDD [109]	ResNet18	CLIP-GPT	UNet, TFM	Concat	DDPM	Lang-guided data generation	**Sim**: MuJoCo; **Real** (UR5e): pick, place
Octo [110]	CNN	T5-base	TFM	Concat	DDPM	OXE	**Real**: BridgeV2, CMU Baking, Stanford Coffee, Berkeley Peg Insert, etc.
3D Diffuser Actor [111]	(Model design follows Act3D)				DDPM	RLBench, CALVIN, [SC]	**Sim**: RLBench, CALVIN; **Real** (Franka): put, open, close, stack, etc.
MDT [112]	CLIP-ViT-B/16 (p), Voltron/ResNet18 (s)	CLIP-GPT	DiT	Concat	DDIM	CALVIN, LIBERO, [SC]	**Sim**: CALVIN, LIBERO; **Real** (Franka): pick, push, open, close, etc.
RDT-1B [113]	SigLIP	T5-XXL	DiT	Xattn	DDPM	(Aggregated Datasets)	**Real** (ALOHA dual-arm): wash, pour, fold, etc.
RT-2 [1] ◊	ViT-4B, ViT-22B	PaLI-X, PaLM-E	Symbol-tuning	Concat	BC (disc), Co-fine-tuning	Fractal, VQA	**Sim**: Language-Table; **Real**: RT-1 evaluation tasks
RT-H [114] ◊	(Model design follows RT-2)				BC (disc)	Diverse+Kitchen	**Real**: Diverse+Kitchen eval tasks
RT-X [115] ◊	(Models from RT-1 and RT-2)				BC (disc)	[SC: OXE]	**Real**: BridgeV2, RT-1 evaluation tasks, etc.
OpenVLA [35] ◊	DINOv2, SigLIP	Prismatic-7B	Symbol-tuning	Concat	BC (disc)	OXE, DROID	**Real**: BridgeV2, RT-1 evaluation tasks, Franka-Tabletop, DROID, etc.
OpenVLA-OFT [116] ◊	(Improves OpenVLA with OFT recipe)				BC (cont, parallel decode w/ chunk.)	LIBERO, [SC]	**Sim**: LIBERO; **Real** (ALOHA setup): fold, scoop, put
TraceVLA [117] ◊	(Model design follows OpenVLA, adding visual trace prompting)				BC (disc)	BridgeV2, Fractal, [SC]	**Sim**: SimplerEnv; **Real** (WidowX): pick, push, fold, swipe
π_0 [118] ◊	SigLIP	PaliGemma	Action expert	Concat	Flow matching	OXE, [SC: π-cross-embod.]	**Real** (general robot control): sweep, open, pack, etc.
RoboMamba [119] ◊	CLIP-ViT	Mamba	MLP	Concat	BC (cont)	[SC]	**Sim**: Sapien; **Real** (Franka): open door/box, staple
SpatialVLA [120] ◊	SigLIP, Ego3D Position Encoding	PaliGemma 2	MLP	Concat	BC (Adaptive Action Grids)	OXE, BridgeV2, Fractal, RH20T	**Sim**: SimplerEnv, LIBERO; **Real** (WidowX): pick, place, close, push
TinyVLA [121] ◊	ViT	Pythia	Diffusion Policy	Concat	DDPM	OXE, [SC]	**Sim**: MetaWorld; **Real** (Franka, UR5): place, stack, flip mug, close drawer, open box
CogACT [122] ◊	DINOv2, SigLIP	LLaMA 2	DiT	Concat	DDIM	OXE, [SC]	**Sim**: SimplerEnv; **Real** (Realman, Franka): pick, place, stack, open/close oven
DexVLA [123] ◊	ViT, ResNet-50	Qwen2-VL 2B, DistilBERT,	ScaleDP	Concat	DDPM	[SC]	**Real** (Franka, UR5e, AgileX): pick, fold shirt, bus table, pour
HybridVLA [124] ◊	DINOv2, SigLIP, CLIP	LLaMA 2, Phi-2	MLP	Concat	DDIM	OXE, DROID	**Sim**: RLBench; **Real** (Franka, AgileX): pick-place, pour, open drawer, fold, etc.
LAPA [125] ◊	C-ViViT, VQ-GAN	LWM-Chat-1M	MLP	Concat	LAPA	OXE, BridgeV2, Something V2	**Sim**: Language-Table, SimplerEnv; **Real** (Franka): pick, cover with towel, knock over
WorldVLA [126] ◊	VQ-GAN	Chameleon	TFM	Quant.	BC (disc, chunk.), world model	LIBERO, OpenVLA data	**Sim**: LIBERO
UniVLA [127] ◊	MoVQ	Emu3	TFM	Quant.	BC (disc, chunk.), world model	(Sim benchmark data)	**Sim**: CALVIN, LIBERO, SimplerEnv

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8

表三

用作低级控制策略的 VLAS。 ♢ 表示大 VLAS。密切相关（非 VLA 型号）包含在括号中。其余的是通用 VLAS。 CONT/DISC：连续/离散动作。 TFM：变压器。 XATTN：交叉注意力。 CONCAT：串联。 QUANT.：量化。 p / s ：提示/状态视觉编码器。 [SC]：自收集数据

|模型|视觉
编码器|语言
编码器/VLM|动作解码器
编码器/头|架构|训练
目标|训练
数据集|环境、实施例、任务和
技能|
|---|---|---|---|---|---|---|---|
|CLIPort [29]|CLIP-ResNet50，
TtRNt|CLIP-GPT|LingUNet|Hadamard|BC (SE(2))|[SC]|模拟：乌鸦|
|MCIL [83]|ransporer-ese
Custom-CNN|USE|RNN|LMP|MCIL|Play-LMP，[SC]|Sim：3D 游戏室环境|
|HULC [84], [85]
Lanuaecosts[87]|MCIL CNN
CLIP-ViTUNet|Sentence-BERT
CLIP-GPT|RNN
STORM|LMP
Concat|MCIL
MLE(costma)|CALVIN 数据
[SC]|Sim: CALVIN
Sim&Real(Franka):icklace|
|gg
交互式语言
[88]|,
ResNet|CLIP-GPT|TFM|Xattn|p
BC（续）|[SC：语言-
表]|p、p
Sim & Real (xArm6)：重排|
|Hiveformer [89]
PerAct [90]|UNet
3D CNN|CLIP-GPT
CLIP-GPT|TFM、UNet
PerceiverIO|Concat

Concat|BC（续）
3D 可供性（Affordance）|RLBench 数据
RLBench 数据，
[SC]|Sim： RLBench
模拟：RLBench； Real (Franka)：拾取、放置、
stackopendrawersweepinsertpegetc。|
|Act3D [91]|CLIP-ResNet50|CLIP-GPT|TFM|Xattn|3D 可供性（Affordance）
(l)|RLBench 数据，
[SC]|, , , ,
Sim: RLBench; Real（弗兰卡）：达到目标，
itkt|
|RVT [92],
RVT2[93]|CLIP-ResNet50|CLIP-GPT|TFM|Concat|voxe
2D 可供性（Affordance）
(tt3D)|RLBench 数据，
[SC]|wpe、sac、ec。
Sim: RLBench; Real (Franka): 堆叠、按下、
l|
|-
RoboPoint [94]|ViT-L/14|Vicuna-V1.5
||Concat|项目
2D 可供性（Affordance）
|[SC]|pace
Real (Franka)：拾取、放置|
|Gato [18]|ViT|13B
Sent.Piece|TFM|Concat|（投影到 3D）
BC（连续和光盘）|[SC]|Sim 和 Real (Sawyer)：RGB 堆叠|
|(RoboCat) [95]|VQ-GAN (p, s)||TFM|定量|BC，观察
预测|自我
改进|Sim & Real (Sawyer, Franka, KUKA)：
stackingbuildingliftinginsertionremoval|
|VIMA [96]
|ViT，Mask R-CNN
|T5
|TFM
|Xattn
|BC (SE(2))
|[SC:VIMA-Data]
|, , , ,
Sim (乌鸦队): VIMA-Bench
|
|BC-Z [82]
RT-1 [97]|ResNet18 (p, s)

EfficientNet|USE
USE|MLP
TFM|FiLM
FiLM|BC（续）
BC（光盘）|[SC]
[SC：分形]|Real (EDR)：拾取/擦拭/拖动、抓取、推送
Real (EDR)：拾取、移动、敲击|
|MOO [96]|OWL-ViT (p),
EfficientNet(s)|USE|TFM|FiLM|BC (disc)|[SC]|Real (EDR)：拾取、靠近、敲击、放置
直立放置|
|Q-Transformer [98]|i
EfficientNet|USE|TFM|FiLM|TD 误差|分形，自动
ll|，
Sim：选择； Real (EDR)：拾取、放置、打开/关闭
d|
|(RT-Trajectory) [99]|EfficientNet||TFM||BC (disc)|coect
[SC]|rawer，靠近
Real (EDR)：拾取、放置、折叠毛巾、旋转
椅子等|
|(ACT) [100]|ResNet18||CVAE-TFM||BC（续，动作
|[SC] with
|, .
Sim：传输立方体，双手插入

;真实
|
|RoboAgent / MT-
ACT [101]|CNN||CVAE-TFM|FiLM|分块)
BC（续，动作
分块）|ALOHA
RoboSet|（ViperX、WidowX）：插槽电池、开杯等
真实（Franka）：拾取、放置、打开/关闭抽屉、
倒、推、拖动等|
|RoboFlamingo [102]|CLIP-ViT-L/14|LLaMA、MPT、
GPT-NeoX|LSTM、TFM|Xattn|BC（续）|CALVIN 数据|Sim：CALVIN|
|RbUniVi[103]|UVFrmr|(Mdldi|nfllRb|lmin)|BC(nt)|CALVINdt|SimCALVIN|
|ooew
DRVLA104|oe
M|oe es
ldiflIR|oows oo
Fli|ago|co
BC|aa
CALVINd|:

SiCALVIN|
|ee- []
Instruct2Act [105]|(o
CLIP, SAM|e esgn oows o
ChatGPT|oamngo++)
机器人 API|工具使用|（续）|ata|m:
Sim (Ravens): VIMA-Bench|
|VoxPoser [106]|ViLD、MDETR、|GPT-4|MPC|工具使用|||Sim：Sapien； Real (Franka): 移动&回避，设置|
|UniPi [86]|OWL-ViT、SAM
Imagen 视频|T5-XXL|CNN|Xattn|逆动力学|[SC:PF]、乌鸦队、
BrideV1|桌子、关闭抽屉、打开瓶子、扫垃圾
模拟：绘画工厂 (PF)、乌鸦队|
|(扩散策略（Diffusion Policy））
|ResNet18||UNet、TFM||DDPM|g
[SC]|Sim：Robomimic、Franka Kitchen 等；真实
||
|[107]
（DP3）[108]|DP3编码器|（模型设计|n遵循Diffusi|策略）|DDIM|[SC]|（UR5，Franka）：推T，翻转杯子，倒酱汁
Sim：（72个任务）； Real（Franka、Allegro 手）：
lldlidill|
|SUDD [109]|ResNet18|CLIP-GPT|UNet、TFM|Concat|DDPM|Lang-guided
数据生成|ro-up、umpng、r、pour
Sim：MuJoCo； Real (UR5e)：拾取、放置|
|Octo [110]|CNN|T5-base|TFM|Concat|DDPM|
OXE|真实：BridgeV2、CMU Baking、斯坦福
CoffeeBerkelePeInsertetc|
|3D 扩散器 Actor
||（模型设计遵循 Act3D）||DDPM|RLBench,
|, y g , .
Sim: RLBench, CALVIN; Real (Franka): put,
|
|[111]
MDT [112]|CLIP-ViT-B/16 (p),
|CLIP-GPT|DiT|Concat|DDIM|CALVIN, [SC]
CALVIN,
|打开、关闭、堆栈等。
模拟：CALVIN、LIBERO； Real（弗兰卡）：选择，
|
|RDT-1B [113]|Voltron/ResNet18 (s)
SigLIP|T5-XXL|DiT|Xattn|DDPM|LIBERO，[SC]
（聚合

数据集）|推、打开、关闭等
真实（ALOHA 双臂）：清洗、倾倒、折叠等|
|RT-2 [1] ♢|ViT-4B、ViT-22B|PaLI-X、PaLM-E|符号-
ti|Concat|BC（圆盘）、Co-
fiti|分形、VQA|。
Sim：语言表；真实：RT-1 评估
tk|
|RT-H [114] ♢
||（模型设计如下
|unng
ws RT-2）
||ine-unng
BC（光盘）
|多样化+厨房
|ass
真实：多样化+厨房评估任务
|
|RT-X [115] ♢
OpenVLA [35] ♢|DINOv2、SigLIP|（来自 RT-1 a
Prismatic-7B|nd RT-2 的型号）
符号-
调整|Concat|BC（光盘）
BC（光盘）|[SC: OXE]
OXE、DROID|真实：BridgeV2、 RT-1评估任务等
Real：BridgeV2、RT-1评估任务、
Franka-TabletopDROID等|
|OpenVLA-OFT
[116]♢|（使用 OFT 配方证明 OpenVLA）||BC（连续，并行
解码/块）|LIBERO，[SC]|，，
Sim：LIBERO；真实（ALOHA 设置）：折叠，
scoopput|
|
TraceVLA [117] ♢|（模型设计遵循 OpenVLA，添加视觉跟踪提示）|。
BC（光盘）|BridgeV2，
FtlSC|，
Sim：SimplerEnv； Real (WidowX)：拾取、推送、
fldi|
|_π_0 [118] ♢|SigLIP|PaliGemma|Action
|Concat|流匹配（Flow Matching）|raca, []
OXE, [SC: π-
|o, swpe
Real（通用机器人控制）：扫描、打开、
|

|RoboMamba [119] ♢|CLIP-ViT|Mamba|专家
MLP|Concat|BC（续）|交叉嵌入。]
[SC]|pack 等
Sim：Sapien

;真实（弗兰卡）：打开门/盒子，
|
|SpatialVLA [120] ♢|SigLIP、Ego3D
iidi|PaliGemma 2|MLP|Concat|BC（自适应
Aiid|OXE、BridgeV2、
l2|staple
Sim：SimplerEnv、LIBERO；真实 (WidowX)：
ikllh|
|TinyVLA [121] ♢|Poston Encong
ViT|Pythia|扩散
|Concat|cton Grs)
DDPM|Fracta、RH0T
OXE、[SC]|pc、pace、cose、pus
Sim：MetaWorld；真实（Franka，UR5）：地点，
l|
|CogACT [122] ♢|DINOv2、SigLIP|LLaMA 2|策略
DiT|Concat|DDIM|OXE、[SC]|堆叠、翻转杯子、关闭抽屉、打开盒子
Sim：SimplerEnv； Real（Realman，弗兰卡）：
|
|DexVLA [123] ♢|ViT、ResNet-50|Qwen2-VL 2B、
|ScaleDP|Concat|DDPM|[SC]|拾取、放置、堆叠、打开/关闭烤箱
真实（Franka、UR5e、AgileX）：拾取、折叠衬衫，
|
|HybridVLA [124] ♢|DINOv2、SigLIP、
CLIP|DistilBERT、
LLaMA 2、Phi-2|MLP|Concat|DDIM|OXE、DROID|总线表、倒
Sim：RLBench； Real（Franka、AgileX）：pick-
laceouroendrawerfoldetc|
|LAPA [125] ♢|C-ViViT、VQ-GAN|LWM-Chat-1M|MLP|Concat|LAPA|OXE、BridgeV2、
SomethinV2|p、p、p 、 、 。
Sim：语言表、SimplerEnv； Real
(Franka)ickcoerithtoelknockoer|
|WorldVLA [126] ♢|VQ-GAN|Chameleon|TFM|Quant.|BC（圆盘，块。），
|g
LIBERO，
|：p，v w w，v
Sim：LIBERO|
|UniVLA[127]♢|MoVQ|Emu3|TFM|Quant|世界模型（World Model）

BC(discchunk)|OpenVLA 数据
(Simbenchmark|Sim:CALVINLIBEROSimplerEnv|
|||||.|, .,
世界模型（World Model）|
数据)|, ,|

of Transformer architecture as its policy backbone.

Gato [18] proposes a model that can play Atari games, caption images, and stack blocks, all with a single set of model parameters. This achievement is facilitated by a unified tokenization scheme, harmonizing the input and output across diverse tasks and domains. Consequently, Gato enables the simultaneous training of different tasks. Astra [128] optimizes this architecture via trajectory attention.

RoboCat [95] proposes a self-improvement process designed to enable an agent to rapidly adapt to new tasks with as few as 100 demonstrations. This self-improvement process iteratively finetunes the model and self-generates new data with the finetuned model. Built upon the Gato model, RoboCat incorporates the VQ-GAN image encoder. During training, RoboCat predicts not only the next action but also future observations. The effectiveness of the self-improvement process is demonstrated through comprehensive experiments conducted in both simulated and real-world environments under multitask and multiembodiment settings.

RT-1 [97], developed by the same team as BC-Z, shares similarities with BC-Z but introduces some key distinctions. Notably, RT-1 employs a vision encoder based on the more efficient EfficientNet, departing from BC-Z's use of ResNet. However, RT-1 does not use video as a task instruction. In addition, RT-1 replaces the MLP action decoder in BC-Z with a Transformer decoder, producing discretized actions. This modification enables RT-1 to attend to past images, enhancing its performance compared with BC-Z.

Q-Transformer [98] extends RT-1 by introducing autoregressive Q-functions. In contrast to RT-1, which learns expert trajectories through imitation learning, Q-Transformer adopts Q-learning methods. Alongside the TD error objective of Q-learning, a conservative regularizer is incorporated to ensure that the maximum value action remains in-distribution. This approach allows Q-Transformer to leverage not only successful demonstrations but also failed trajectories for learning.

RT-Trajectory [99] adopts trajectory sketches as policy conditions instead of relying on language conditions or goal conditions. These trajectory sketches consist of curves that delineate the intended trajectory for the robot end-effector to follow. They can be manually specified through a graphical user interface, extracted from human demonstration videos, or generated by foundation models. RT-Trajectory's policy is built upon the backbone of RT-1 and trained to control the robot arm to accurately follow the trajectory sketches. This approach facilitates generalization to novel objects, tasks, and skills, as trajectories from various tasks are transferable.

ACT [100] builds a conditional VAE policy with action chunking, requiring the policy to predict a sequence of actions rather than a single one. During inference, the action sequence is averaged using a method called temporal ensembling. RoboAgent [101] extends ACT via MT-ACT, demonstrating that action chunking improves temporal consistency. It also introduces an inpainting-based semantic augmentation method.

RoboFlamingo [102] adapts the OpenFlamingo VLM to a robot policy by attaching an LSTM-based policy head. This demonstrates that pretrained VLMs can be effectively transferred to language-conditioned robotic manipulation tasks.

A recent trend in LLMs is equipping them with tool-use capabilities by generating code that calls tools via APIs [129]. Instruct2Act [105] follows this paradigm by integrating vision and action tools, enabling LLMs to perform robotic tasks.

3) Control Policies for Multimodal Instructions: Multimodal instruction enables new ways to specify tasks, such as through demonstrations, by naming novel objects, or by pointing with a finger or mouse click.

VIMA [130] places a significant emphasis on multimodal prompts and the generalization capabilities of models. By incorporating multimodal prompts, more specific and intricate tasks can be formulated compared with pure text prompts. VIMA introduces six main types of tasks: object manipulation, visual goal reaching, novel concept grounding, one-shot video imitation, visual constraint satisfaction, and visual reasoning. These tasks are often challenging or even infeasible to express using only language prompts. VIMA-Bench has been developed to evaluate across four generalizability levels: placement, novel combinatorial, novel object, and novel task.

MOO [96] extends RT-1 to handle multimodal prompts. Leveraging the backbone of RT-1, it incorporates OWL-ViT to localize target objects specified by the prompt. By expanding the RT-1 dataset with new objects and additional prompt images, MOO enhances the generalization capabilities of RT-1. This extension also facilitates new methods of specifying target objects, such as pointing with a finger and clicking on graphical user interfaces.

4) Control Policies with 3D Vision: In our 3D world, it is reasonable to assume that 3D vision provides richer information than 2D images.

Point clouds are a common representation for 3D visual inputs due to their straightforward derivation from RGB-D inputs, as demonstrated by both DP3 [108] and 3D Diffuser Actor [111]. Voxels have likewise been extensively studied. VER [131] proposes voxelizing multiview images into 3D cells in a coarse-to-fine manner, enhancing performance in vision-language navigation tasks. PerAct [90] facilitates efficient task learning with only a few demonstrations by leveraging voxel representations in both the observation and action spaces. The input comprises voxel maps reconstructed from multiview RGB-D images, while the output corresponds to the best voxel for guiding the gripper's movement. RoboUniView [103] improves performance by injecting 3D information from multi-perspective images through a novel UVFormer vision encoder, which is pretrained on a 3D occupancy task.

In contrast, Act3D [91] introduces a continuous resolution 3D feature field, with adaptive resolutions based on the current task, addressing the computational cost of voxelization. RVT, RVT-2 [92], [93] propose re-rendering images from virtual views of the scene point cloud and using these images as inputs, rather than directly relying on 3D inputs.

5) Diffusion-Based Control Policies: Diffusion-based action generation leverages the success of diffusion models in the field of CV.

Diffusion Policy [107] formulates a robot policy as a DDPM. This approach incorporates a variety of techniques, including receding horizon control, visual conditioning, and the time-series diffusion Transformer. The effectiveness of this

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Transformer 架构作为其策略支柱。

Gato [18] 提出了一个可以玩 Atari 游戏、字幕图像和堆栈块的模型，所有这些都具有一组模型参数。这一成就得益于统一的标记化方案，协调不同任务和领域的输入和输出。因此，Gato 能够同时训练不同的任务。 Astra [128] 通过轨迹注意力优化了该架构。

RoboCat [95] 提出了一种 自我改进（Self-Improvement） 流程，旨在使智能体能够通过少至 100 个演示快速适应新任务。此 自我改进（Self-Improvement） 过程迭代地微调模型，并使用微调后的模型自行生成新数据。 RoboCat 基于 Gato 模型构建，集成了 VQ-GAN 图像编码器。在训练过程中，RoboCat 不仅会预测下一步的行动，还会预测未来的观察结果。 自我改进（Self-Improvement） 过程的有效性通过在多任务和多实施设置下的模拟和现实环境中进行的综合实验得到证明。

RT-1 [97] 与 BC-Z 由同一团队开发，与 BC-Z 有相似之处，但引入了一些关键区别。值得注意的是，RT-1 采用了基于更高效的 EfficientNet 的视觉编码器，这与 BC-Z 使用的 ResNet 不同。然而，RT-1并不使用视频作为任务指令。此外，RT-1 用 Transformer 解码器替换了 BC-Z 中的 MLP 动作解码器，产生离散化动作。这一修改使 RT-1 能够关注过去的图像，与 BC-Z 相比增强了其性能。

Q-Transformer [98] 通过引入自回归 Q 函数来扩展 RT-1。与RT-1通过模仿学习（Imitation Learning）学习专家轨迹不同，Q-Transformer采用Q-learning方法。除了 Q 学习的 TD 误差目标之外，还结合了保守的正则化器以确保最大值动作保持分布。这种方法使 Q-Transformer 不仅可以利用成功的演示，还可以利用失败的轨迹进行学习。

RT-Trajectory[99]采用轨迹草图作为策略条件，而不是依赖于语言条件或目标条件。这些轨迹草图由描绘机器人末端执行器遵循的预期轨迹的曲线组成。它们可以通过图形用户界面手动指定、从人类演示视频中提取或由 基础模型（Foundation Models） 生成。 RT-Trajectory 的策略建立在 RT-1 的基础上，经过训练可以控制机器人手臂准确地遵循轨迹草图。这种方法有助于泛化新的对象、任务和技能，因为各种任务的轨迹是可转移的。

ACT [100] 使用 动作分块（Action Chunking） 构建了一个条件 VAE 策略，要求该策略预测一系列操作而不是单个操作。在推理过程中，使用称为时间集成的方法对动作序列进行平均。 RoboAgent [101] 通过 MT-ACT 扩展了 ACT，证明 动作分块（Action Chunking） 提高了时间一致性。它还引入了一种基于修复的语义增强方法。

RoboFlamingo [102] 通过附加基于 LSTM 的策略头，使 OpenFlamingo VLM 适应机器人策略。这表明预训练的 VLM 可以有效地转移到语言条件的机器人操作任务中。

LLM 的最新趋势是通过生成通过 API 调用工具的代码来为他们配备工具使用能力 [129]。 Instruct2Act [105] 通过集成视觉和行动工具遵循这一范式，使法学硕士能够执行机器人任务。

3) 多模式指令的控制策略： 多模式指令支持指定任务的新方法，例如通过演示、命名新对象或通过手指指向或鼠标点击。

VIMA [130] 非常重视多模态提示和模型的泛化能力。与纯文本提示相比，通过合并多模式提示，可以制定更具体、更复杂的任务。 VIMA 引入了六种主要任务类型：对象操纵、视觉目标达成、新颖概念 落地关联

（Grounding）、一次性视频模仿、视觉约束满足和视觉推理。这些任务通常具有挑战性，甚至无法仅使用语言提示来表达。 VIMA-Bench 的开发目的是评估四个普遍性级别：放置、新颖组合、新颖对象和新颖任务。

MOO [96] 扩展了 RT-1 以处理多模式提示。利用 RT-1 的主干，它结合了 OWL-ViT 来本地化提示指定的目标对象。通过使用新对象和附加提示图像扩展 RT-1 数据集，MOO 增强了 RT1 的泛化能力。此扩展还促进了指定目标对象的新方法，例如用手指指向和单击图形用户界面。

4) 3D 视觉控制策略： 在我们的 3D 世界中，可以合理地假设 3D 视觉提供比 2D 图像更丰富的信息。

点云是 3D 视觉输入的常见表示形式，因为它们直接源自 RGB-D 输入，如 DP3 [108] 和 3D Diffuser Actor [111] 所证明的那样。体素同样得到了广泛的研究。 VER [131] 提出以从粗到细的方式将多视图图像体素化为 3D 单元，从而增强视觉语言导航任务的性能。 PerAct [90] 通过利用观察和动作空间中的体素表示，只需少量演示即可促进高效的任务学习。输入包括从多视图 RGB-D 图像重建的体素图，而输出对应于引导夹具运动的最佳体素。 RoboUniView [103] 通过新颖的 UVFormer 视觉编码器注入来自多视角图像的 3D 信息来提高性能，该编码器在 3D 占用任务上进行了预训练。

相比之下，Act3D [91] 引入了连续分辨率 3D 特征场，具有基于当前任务的自适应分辨率，解决了体素化的计算成本。 RVT、RVT-2 [92]、[93] 提出从场景点云的虚拟视图重新渲染图像并使用这些图像作为输入，而不是直接依赖 3D 输入。

5) 基于扩散的控制策略： 基于扩散的动作生成利用了 CV 领域扩散模型的成功。

扩散策略（Diffusion Policy） [107]将机器人策略制定为DDPM。这种方法结合了多种技术，包括后退地平线控制、视觉调节和时间序列扩散变压器。此方法的有效性

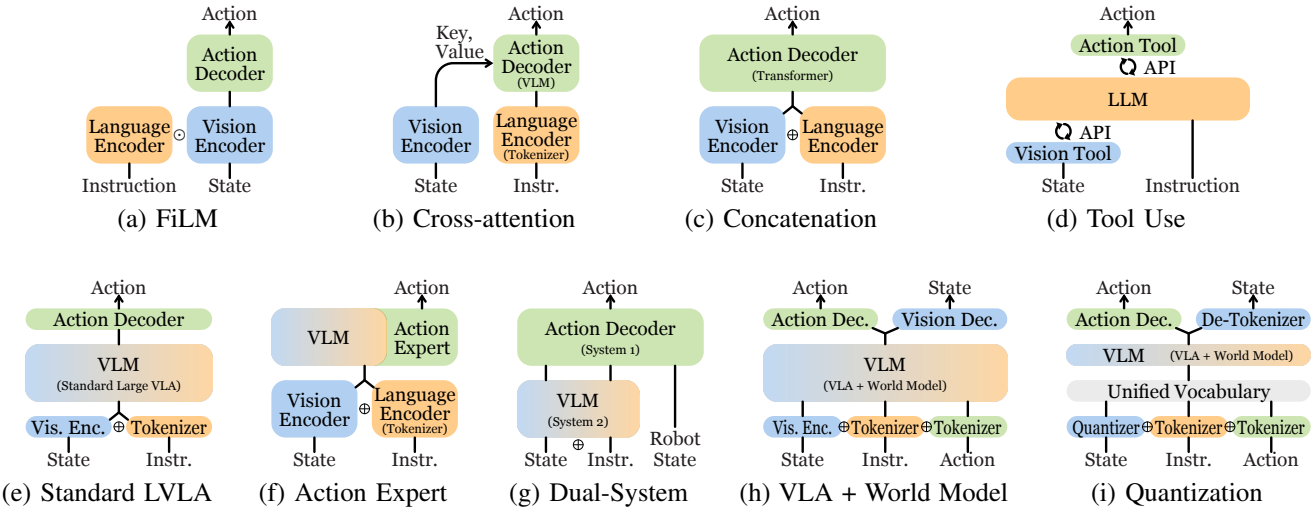


Fig. 5. Representative architectures of VLA models. Some correspond to the "Architecture" column in Table III. $\odot$: Hadamard product. $\oplus$: Concatenation.

diffusion-based visuomotor policy is underscored by its proficiency in multimodal action distributions, high-dimensional action spaces, and training stability.

SUDD [109] presents a framework where an LLM guides data generation, and subsequently, the filtered dataset is distilled into a visuo-linguo-motor policy. This framework achieves language-guided data generation by composing an LLM with a suite of primitive robot utilities, such as grasp samplers and motion planners. It then extends Diffusion Policy by incorporating language-based conditioning for multitask learning and facilitates the distillation of the filtered dataset.

Octo [110] introduces a Transformer-based diffusion policy characterized by a modular open-framework design, allowing for the flexible integration of different task definition encoders, observation encoders, and action decoders with the Octo Transformer. Being among the first to utilize the OXE dataset [115], Octo demonstrates positive transfer and generalizability across diverse robots and tasks.

MDT [112] builds a variant of the DiT model from CV for the action prediction head. Originally proposed as a Transformer-based diffusion model, DiT replaced the classical U-Net architecture in image generation. Coupled with two auxiliary objectives—masked generative foresight and contrastive latent alignment—MDT similarly demonstrates better performance than prior U-Net-based diffusion models.

RDT-1B [113] is a diffusion-based foundation model for bimanual manipulation, also built on DiT. It addresses data scarcity by introducing a physically interpretable unified action format across various robots, enabling pretraining on heterogeneous multirobot datasets with over 1M trajectories. This allows RDT to scale up to 1.2B parameters and demonstrate zero-shot generalization.

6) Diffusion-Based Control Policies with 3D Vision: Several works have proposed combining 3D vision with diffusion-based policies. DP3 [108] introduces 3D point cloud inputs to a diffusion policy. Similarly, 3D Diffuser Actor [111] shares the core idea of DP3 but differs in model architecture by combining Act3D with Diffusion Policy. 3D-MoE [132] explores an efficient mixture-of-experts architecture with DiT-based action diffusion using a rectified flow framework.

7) Control Policies for Motion Planning: Motion planning involves decomposing movement tasks into discrete waypoints while satisfying constraints such as obstacle avoidance and kinematic limits.

The *language costs* framework [87] presents a novel approach to robot correction using natural language for human-in-the-loop control. This method leverages predicted cost maps generated from human instructions, which are then utilized by the motion planner to compute the optimal path. This framework enables users to correct goals, specify preferences, or recover from errors through intuitive language commands.

VoxPoser [106] employs an LLM and a VLM to create 3D voxel maps that represent affordances and constraints. It leverages the programming capability of LLMs and the perception capability of VLMs. The LLM translates language instructions into executable code, invoking the VLM to obtain object coordinates. Based on the composed affordance and constraint maps, VoxPoser employs model predictive control to generate a feasible trajectory for the robot arm's end-effector. Notably, VoxPoser does not require any training, as it directly connects LLMs and VLMs for motion planning.

RoboTAP [133] breaks down demonstrations into stages marked by the opening and closing of the gripper. In each stage, RoboTAP uses the TAPIR algorithm to detect active points that track the relevant object from the source to the target pose. These tracked points are then used by a visual servoing controller to guide the robot. A motion plan is created by chaining these stages together, enabling few-shot visual imitation.

8) Control Policies with Point-Based Actions: Recent research has explored leveraging the capabilities of VLMs to select or predict point-based actions—a cost-effective alternative to building full VLAs.

PIVOT [134] casts robotics tasks as a VQA problem, leveraging VLMs to select the best robot action from a set of visual proposals. Visual proposals are annotated in the form

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Action Action Action Action
Key,
Action Value Action Action Decoder Action Tool
Decoder Decoder(VLM) (Transformer) A PI
LLM
LanguagEncoder e EncoderVision EVision ncoder LanguaEnco de gre EncodVision er Language Encoder API
(Tokenizer) Visi on Tool
指令状态 状态指令国家研究所状态指令
(a) FiLM (b) 交叉注意力 (c) 连接 (d) 工具使用
动作动作动作动作状态动作状态
动作解码器动作解码器视觉解码器动作解码器去标记器
动作动作解码器
VLM
专家（系统1）VLM（VLA + 世界模型（World Model））
VLM VLM
（标准大）VLA) 视觉语言 VL M (VLA + 世界模型（World Model））统一词汇
Vis。 恩c。 Tokenizer 编码器 E(Tokenizer) nc oder (系统 2) 机器人 Vis。恩科。分词器 分词器 量化器 分词器 分词器
状态指令。国家研究所国家研究所国家国家研究所。行动状态指令。 Action
(e) 标准 LVLA (f) Action Expert (g) 双系统 (h) V LA + 世界模型（World Model） (i) 量化
⊙

图 5. VLA 模型的代表性架构。有些对应于表 III 中的“架构”列。 ___ ：阿达玛积。 ⊕：串联。

基于扩散的视觉运动策略因其在多模式动作分布、高维动作空间和训练稳定性方面的熟练程度而得到强调。

SUDD [109] 提出了一个框架，其中 LLM 指导数据~~g~~生成，随后，过滤后的数据~~et~~被提炼成 <u>visuo-linguo-motor</u>策略。该框架通过<u>组合</u>法学硕士与一套原始机器人实用程序（例如抓取采样器和运动规划器）来实现语言引导的数据生成。然后，它通过合并用于多任务学习的基于语言的调节来扩展 扩散策略（Diffusion Policy），并促进过滤数据集的提炼。

Octo [110] 引入了一种基于 Transformer 的 扩散策略（Diffusion Policy），其特点是模块化开放框架设计，允许不同的任务定义编码器、观察编码器和动作解码器与 <u>Octo</u> Transformer 灵活集成。作为最早使用 OXE 数据集 [115] 的人之一，Octo 展示了跨不同机器人和任务的积极迁移和泛化性。

MDT [112] 为动作预测头构建了 CV 的 DiT 模型的变体。 DiT 最初是作为基于 Transformer 的扩散模型提出的，在图像生成中取代了经典的 U-Net 架构。与两个辅助目标（掩蔽生成前瞻和对比潜在对齐）相结合，MDT 同样表现出比之前基于 U-Net 的扩散模型更好的性能。

RDT-1B [113] 是一种基于扩散的 基础模型（Foundation Model），用于双手操作，也是基于 DiT 构建的。它通过在各种机器人之间引入物理上可解释的统一动作格式来解决数据稀缺问题，从而能够对具有超过 1M 轨迹的异构多机器人数据集进行预训练。这使得 RDT 能够扩展到 1.2B 参数并演示零样本泛化。

6) 具有 3D 视觉的基于扩散的控制策略： 一些工作提出了将 3D 视觉与基于扩散的策略相结合。 DP3 [108] 将 3D 点云输入引入到 扩散策略（Diffusion Policy）。类似地，3D Diffuser Actor [111] 共享 DP3 的核心思想，但模型架构有所不同，将 Act3D 与 扩散策略（Diffusion Policy） 结合起来。 3D-教育部 [132]

使用修正流框架探索一种高效的专家混合架构，该架构具有基于 DiT 的动作扩散。

7) 运动规划的控制策略： 运动规划涉及将运动任务分解为离散路径点，同时满足避障和运动学限制等约束。

_语言成本_框架[87]提出了一种使用自然语言进行人机交互控制的机器人校正的新方法。该方法利用从人类指令生成的预测成本图，然后由运动规划器利用这些图来计算最佳路径。该框架使用户能够通过直观的语言命令纠正目标、指定首选项或从错误中恢复。

VoxPoser [106] 使用 LLM 和 VLM 来创建表示 可供性（Affordances） 和约束的 3D 体素图。它利用了LLM的编程能力和VLM的感知能力。 LLM将语言指令翻译成可执行代码，调用VLM来获取对象坐标。基于组合的 可供性（Affordance） 和约束图，VoxPoser 采用模型预测控制为机械臂末端执行器生成可行的轨迹。值得注意的是，VoxPoser 不需要任何训练，因为它直接连接 LLM 和 VLM 进行运动规划。

RoboTAP [133] 将演示分为以夹具的打开和关闭为标志的阶段。在每个阶段，RoboTAP 使用 TAPIR 算法来检测跟踪相关对象从源姿势到目标姿势的活动点。然后，视觉伺服控制器使用这些跟踪点来引导机器人。通过将这些阶段链接在一起来创建运动计划，从而实现少量视觉模仿。

8) 具有基于点的操作的控制策略： 最近的研究探索了利用 VLM 的功能来选择或预测基于点的操作--这是构建完整 VLA 的一种经济高效的替代方案。

PIVOT [134] 将机器人任务视为 VQA 问题，利用 VLM 从一组视觉建议中选择最佳机器人动作。视觉提案以表格形式注释

of keypoints on images. The VLM is iteratively prompted to refine them until the best option is identified.

RoboPoint [94] finetunes a VLM on the task of spatial affordance prediction, in which the model outputs 2D points indicating where to act on the image. These affordance points are subsequently projected into 3D space using depth maps, forming the predicted robot action.

ReKep [36] introduces constraint functions that map 3D keypoints in the scene to a numerical cost. Robotics manipulation tasks can be represented as a sequence of these constraints, which are generated by large vision models and VLMs. Consequently, robot actions can be obtained by solving the resulting constrained optimization problem.

9) Large VLA: Large VLA is equivalent to the original VLA definition proposed by RT-2 [1], as illustrated in Fig. 2(a). This terminology is analogous to the distinction between LLMs and general language models, or between large VLMs and general VLMs.

RT-2 [1] leverages the capabilities of large multimodal models for robotics tasks, building upon architectures like PaLI-X and PaLM-E (see the architecture in Fig. 5(e)). The approach introduces a co-fine-tuning strategy, jointly training the model on Internet-scale VQA data and robot data. This training scheme enhances the model's generalizability and enables emergent capabilities.

RT-H [114] introduces an action hierarchy that includes an intermediate prediction layer of language motions, situated between language instructions and low-level actions (translation and rotation). This additional layer facilitates improved data sharing across different tasks. For example, both the language instructions "pick" and "pour" may involve the language motion "move the arm up". Moreover, this action hierarchy enables users to specify corrections to recover from failures, from which the model can subsequently learn.

RT-X [115] builds upon the previous RT-1 and RT-2 models. These models are retrained using a newly introduced open-source dataset named open X-embodiment (OXE), which is orders of magnitude larger than previous datasets. The resulting models, RT-1-X and RT-2-X, both outperform their original versions.

OpenVLA [35] was later developed as an open-source counterpart to RT-2-X. It additionally explored efficient finetuning methods, including LoRA and model quantization. OpenVLA-OFT [116] applies an Optimized Fine-Tuning (OFT) recipe for improved efficiency and performance. TraceVLA [117] finetunes OpenVLA to enable visual trace prompting, enhancing the model's spatial-temporal awareness.

π_0 [118] proposes a flow-matching architecture for transforming VLMs into VLAs. By incorporating an additional action expert based on the mixture-of-experts framework (see the architecture in Fig. 5(f)), it effectively inherits the Internet-scale knowledge of the base VLM while extending its capabilities to address robotic tasks.

RoboMamba [119] replaces the computationally expensive Transformer architecture with a Mamba state space model featuring linear inference complexity, thereby achieving efficient robotic reasoning and action capabilities.

SpatialVLA [120] introduces Ego3D position encoding for injecting 3D information and represents actions with adaptive action grids for enhanced generalizability and robustness.

LAPA [125] devises the first unsupervised pretraining method for VLAs based on latent actions [69]. This approach employs a three-stage process to learn from Internet-scale unlabeled videos. First, a VQ-VAE framework is pretrained to extract quantized latent actions between image frames. Next, a VLA model is pretrained to predict these latent actions. Finally, the model is finetuned using only a small robotic dataset to map the latent actions to actual robot actions.

The integration of LVLAs with diffusion has also been gaining popularity. TinyVLA [121] leverages Diffusion Policy, while CogACT [122] utilizes a DiT action diffusion module. DexVLA [123] proposes embodied curriculum learning to progressively train a diffusion action expert, incorporating substep reasoning for decomposing long-horizon tasks. HybridVLA [124] integrates diffusion with the autoregressive paradigm to fully leverage VLMs' reasoning capabilities.

GR00T N1 [135] introduces a dual-system architecture to build a robot foundation model for humanoid robots (see the architecture in Fig. 5(g)). Its VLM-based System 2 processes image observations and language instructions at 10 Hz, while System 1 is a diffusion module that generates closed-loop motor actions in real time (120 Hz).

NORA-1.5 [136] unifies a VLA with a world model through reward-guided post-training. Genie Envisioner [137] is a world foundation platform that integrates a world model and a VLA within a single video-generative framework. See the general architecture in Fig. 5(h).

Visual autoregressive (VAR) modeling [138] with quantized visual tokens demonstrates improved performance over diffusion models in image generation. This suggests that the three modalities of VLAs can be unified under the autoregressive paradigm [139]. WorldVLA [126] and UniVLA [127] advance this direction by integrating VLAs with world models (see the architecture in Fig. 5(i)). They quantize multimodal data into discrete tokens, forming a shared vocabulary of quantized multimodal tokens. Consequently, all modalities can be modeled autoregressively, enabling not only action and text generation but also image generation, thereby constituting a world model.

10) Strengths and Limitations:

a) Architectures: Representative VLA architectures are illustrated in Fig. 5. FiLM is used in RT-1, and thus its follow-up models inherit this mechanism. While cross-attention may offer superior performance with smaller model sizes, concatenation is simpler to implement and can achieve comparable results with larger models [130]. Quantization unifies multimodal tokens into a shared vocabulary, thus enabling integration with world models. The tool-use paradigm of LLMs can also be applied to robotic tasks.

b) Action Types and Their Training Objectives: Most low-level control policies predict actions for the end-effector pose while abstracting away the motion planning module that produces more fine-grained motions. While this abstraction facilitates better generalization to different embodiments, it also imposes limitations on dexterity.

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图像上的关键点。 VLM 会被反复提示以完善它们，直到确定最佳选项。

RoboPoint [94] 在空间 可供性（Affordance） 预测任务上对 VLM 进行微调，其中模型输出 2D 点，指示对图像的何处进行操作。随后使用深度图将这些 可供性（Affordance） 点投影到 3D 空间，形成预测的机器人动作。

ReKep [36] 引入了将场景中的 3D 关键点映射到数值成本的约束函数。机器人操作任务可以表示为这些约束的序列，这些约束是由大型视觉模型和 VLM 生成的。因此，可以通过解决由此产生的约束优化问题来获得机器人动作。

9) Large VLA: Large VLA 等价于 RT-2 [1] 提出的原始 VLA 定义，如图 2(a) 所示。该术语类似于 LLM 和通用语言模型之间的区别，或者大型 VLM 和通用 VLM 之间的区别。

RT-2 [1] 利用大型多模态模型的功能来执行机器人任务，建立在 PaLI-X 和 PaLM-E 等架构的基础上（参见图 5(e) 中的架构）。该方法引入了协同微调策略，在互联网规模的 VQA 数据和机器人数据上联合训练模型。这种训练计划增强了模型的通用性并实现了突发能力。

RT-H [114]引入了一个动作层次结构，其中包括位于语言指令和低级动作（平移和旋转）之间的语言运动的中间预测层。这个附加层有助于改进不同任务之间的数据共享。例如，语言指令“拾取”和“倾倒”都可能涉及语言动作“向上移动手臂”。此外，此操作层次结构使用户能够指定从故障中恢复的更正，模型随后可以从中学习。

RT-X [115] 建立在之前的 RT-1 和 RT-2 模型的基础上。这些模型使用新引入的名为 open X-embodiment (OXE) 的开源数据集进行重新训练，该数据集比以前的数据集大几个数量级。由此产生的模型 RT-1-X 和 RT-2-X 的性能均优于其原始版本。

OpenVLA [35] 后来被开发为 RT-2-X 的开源版本。它还探索了有效的微调方法，包括 LoRA 和模型量化。 OpenVLAOFT [116] 应用优化微调（OFT）方法来提高效率和性能。 TraceVLA [117]对OpenVLA进行了微调，以实现视觉轨迹提示，增强模型的时空感知。

π 0 [118]提出了一种将 VLM 转换为 VLA 的流匹配架构。通过结合基于专家混合框架的额外行动专家（参见图 5（f）中的架构），它有效地继承了基础 VLM 的互联网规模知识，同时扩展了其处理机器人任务的能力。

RoboMamba [119]用具有线性推理复杂性的 Mamba 状态空间模型取代了计算成本高昂的 Transformer 架构，从而实现了高效的机器人推理和动作能力。

SpatialVLA [120] 引入了用于注入 3D 信息的 Ego3D 位置编码，并使用自适应动作网格表示动作，以增强通用性和鲁棒性。

LAPA [125] 设计了第一个基于潜在动作的 VLA 无监督预训练方法 [69]。这种方法采用三阶段过程来从互联网规模的未标记视频中学习。首先，预训练 VQ-VAE 框架以提取图像帧之间的量化潜在动作。接下来，预训练 VLA 模型来预测这些潜在动作。最后，仅使用小型机器人数据集对模型进行微调，以将潜在动作映射到实际的机器人动作。

LVLA 与扩散的集成也越来越受欢迎。 TinyVLA [121] 利用 扩散策略（Diffusion Policy），而 CogACT [122] 利用 DiT 动作扩散模块。 DexVLA [123]提出了体现课程学习来逐步训练扩散行动专家，结合分解 长时程（Long-Horizon） 任务的子步骤推理。 HybridVLA [124]将扩散与自回归范式相结合，以充分利用 VLM 的推理能力。

GR00T N1 [135]引入了双系统架构来构建仿人机器人基础模型（Foundation Model）（见图5（g）中的架构）。其基于 VLM 的系统 2 以 10 Hz 的频率处理图像观察和语言指令，而系统 1 是一个扩散模块，可实时生成闭环运动动作 (120 Hz)。

NORA-1.5 [136]通过奖励引导的后训练将 VLA 与 世界模型（World Model） 统一起来。 Genie Envisioner [137] 是一个世界基础平台，它将 世界模型（World Model） 和 VLA 集成在单个视频生成框架内。总体架构见图5(h)。

具有量化视觉标记的视觉自回归（VAR）建模[138]证明了在图像生成方面比扩散模型具有更高的性能。这表明 VLA 的三种模式可以在自回归范式下统一[139]。 WorldVLA [126] 和 UniVLA [127] 通过将 VLA 与 世界模型（World Models） 集成来推进这一方向（参见图 5(i) 中的架构）。它们将多模态数据量化为离散标记，形成量化多模态标记的共享词汇表。因此，所有模态都可以进行自回归建模，不仅可以生成动作和文本，还可以生成图像，从而构成 世界模型（World Model）。 10) 优点和局限性：

a) 架构： 代表性的VLA架构如图5所示。RT-1中使用FiLM，因此其后续模型继承了这一机制。虽然交叉注意力可以通过较小的模型大小提供卓越的性能，但串联更容易实现，并且可以通过较大的模型获得可比的结果[130]。量化将多模态标记统一到共享词汇表中，从而实现与 世界模型（World Models） 的集成。法学硕士的工具使用范例也可以应用于机器人任务。

b) 动作类型及其训练目标： 大多数低级控制策略预测末端执行器姿势的动作，同时抽象出产生更细粒度运动的运动规划模块。虽然这种抽象有利于更好地推广到不同的实施例，但它也对灵活性造成了限制。

Table IV
HIGH-LEVEL TASK PLANNERS

Model	Vision Model	Language Model / VLM	Low-level Control Policy	Architecture	Require Training	Environments, Embodiments, Tasks, and Skills
PaLM-E [11]	ViT, OSRT	PaLM	Interactive Language, RT-1	Concat	✓	**Sim**: TAMP; **Sim&Real-Mani**: Language-Table; **Real**: SayCan setup
EmbodiedGPT [140]	EVA ViT-G/14	LLaMA-7B	MLP	Xattn	✓	**Sim-Mani**: Franka Kitchen, Meta-World
LEO [141]	ConvNeXt, PointNet++	Vicuna-7B	MLP	Concat	✓	**Sim-Mani**: CLIPort task subset; **Sim-Navi**: Habitat-web
3D-LLM [42]	CLIP, 3D-CLR, ConceptFusion	Flamingo, BLIP-2	DD-PPO	Concat	✓	**Sim-Navi**: Habitat
ShapeLLM [142]	ReCon++	LLaMA	–	Concat	✓	3D MM-Vet Benchmark
SayCan [10]	–	PaLM, FLAN	BC-Z, MT-Opt	–	✗	**Real-Navi&Mani** (EDR): office kitchen
Translated ⟨LM⟩ [143]	–	GPT-3, Codex	–	–	✗	**Sim**: VirtualHome
$(SL)^3$ [144]	–	T5-small	seq2seq	–	✓	**Sim-Navi**: ALFRED
Inner Monologue [9]	MDETR, CLIP	InstructGPT, PaLM	CLIPort, heuristics, SayCan policies	Language	✗	**Sim&Real-Mani**: Tabletop Rearrangement; **Real-Navi&Mani**: Kitchen Mobile Mani.
LLM-Planner [145]	HLSM object detector	GPT-3	HLSM	Language	✓	**Sim-Navi**: ALFRED
LID [146]	CNN	GPT-2	–	Lang., Embed	✓	**Sim**: VirtualHome
SMs [147]	CLIP, ViLD	GPT-3, RoBERTa	CLIPort	Lang., Code	✗	**Sim-Mani**: pick, place
ProgPrompt [148]	ViLD	GPT-3	API	Code	✗	**Sim**: VirtualHome
ChatGPT for Robotics [149]	YOLOv8	ChatGPT	API	Code	✗	**Real-Navi**: drone flight; **Sim-Navi**: AirSim, Habitat
CaP [150]	ViLD, MDETR	GPT-3, Codex	API	Code	✗	**Sim-Mani** & **Sim-Navi**: pick, place, etc.
DEPS [151]	CLIP	ChatGPT	MC-Controller, Steve-1	Code	✗	**Sim**: Minecraft
ConceptGraphs [152]	SAM, CLIP, LLaVA	GPT-4	API	Code (JSON)	✗	**Sim**: AI2-THOR; **Real** (Spot Arm): pick-place

The BC objective is used in imitation learning, with different variants for different action types. The BC objective for continuous actions can be written as

$$\mathcal{L}_{\text{Cont}} = \sum_t \text{MSE}\left(a_t, \hat{a}_t\right), \tag{4}$$

where $\text{MSE}(\cdot)$ stands for mean squared error, a_t is the action annotation from expert demonstrations, and $\hat{a}_t$ is the predicted action.

Discrete actions are achieved by dividing the action value range into a fixed number of bins. Their BC objective is

$$\mathcal{L}_{\text{Disc}} = \sum_t \text{CE}\left(a_t, \hat{a}_t\right), \tag{5}$$

where $\text{CE}(\cdot)$ stands for cross-entropy loss.

The BC objective for $SE(2)$ actions is

$$\mathcal{L}_{SE(2)} = \text{CE}(a_{\text{pick}}, \hat{a}_{\text{pick}}) + \text{CE}(a_{\text{place}}, \hat{a}_{\text{place}}). \tag{6}$$

The DDPM objective in diffusion-based control policies is

$$\mathcal{L}_{\text{DDPM}} = \text{MSE}\left(\varepsilon^k, \varepsilon_\theta(a_t^k, k)\right), \tag{7}$$

where a_t^k is the action a_t corrupted by random noise ε^k at diffusion iteration k, and ε_θ is the noise prediction network, i.e., the VLA model.

While discrete action has demonstrated superior performance in RT-1 [97], Octo [110] argues that it leads to early grasping issues. $SE(2)$ actions require the model to predict only the pick pose and the place pose of the end-effector, which are sufficient for many tabletop manipulation tasks. However, more complex tasks—such as "pouring water into a cup"—may require additional DoFs, thereby necessitating $SE(3)$ actions [36]. Although point-based actions can be coarse-grained, they are more easily obtained from VLMs in a zero-shot manner.

c) RT Series: RT-1 [97] inspired a series of "Robotic Transformer" models. The Transformer backbone surpasses previous RNN backbones by harnessing the higher capacity of Transformers to absorb larger robot datasets. Preceding RT-1 was BC-Z, which solely utilized MLP layers for action prediction. Subsequent to RT-1, several works emerged to introduce new capabilities. MOO adapted RT-1 to accommodate multimodal prompts. RT-Trajectory enabled RT-1 to process trajectory sketches as prompts. Q-Transformer utilized Q-learning to train RT-1. RT-2, based on ViT and LLM, introduced a completely different architecture from RT-1. RT-X retrained RT-1 and RT-2 with a significantly larger dataset, resulting in improved performance. Based on RT-2, RT-H [114] introduced action hierarchies for better data sharing.

d) LVLA vs. Generalized VLA: While LVLAs can greatly enhance instruction-following abilities because they can better parse user intentions, concerns arise regarding their training cost and deployment speed. Slow inference speed, in particular, can significantly impact performance in dynamic environments, as changes in the environment may occur during inference. Therefore, several methods have been proposed to improve efficiency. TinyVLA [121] focuses on inference speed and data efficiency through a smaller VLM and diffusion head for robot action. DeeR-VLA [104] proposes only partially activating the model via early-exit dynamic inference.

e) Scaling Laws: As in LLMs, scaling laws have also been observed in robotics [153], [154], revealing the importance of model size, dataset size, and the diversity of environments and objects. Further research in this area can guide the development of VLAs with robust in-the-wild generalization capabilities.

IV. TASK PLANNERS

A high-level task planner π_ϕ aims to decompose a complex task ℓ into a sequence of subtasks $\hat{\mathbf{p}} = [p_1, p_2, \ldots, p_N]$ (i.e., a

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表四

高级任务规划器

|模型|愿景
模型|语言模型 /
VLM|低级控制
策略|架构|要求
训练|环境、实施例、任务和
技能|
|---|---|---|---|---|---|---|
|PaLM-E [11]|ViT、OSRT|PaLM|交互式语言，
RT-1|Concat|✓|Sim：TAMP； Sim&Real-Mani：语言-
表；真实：SayCan 设置|
|EmbodiedGPT [140]|EVA ViT-G/14|LLaMA-7B|MLP|Xattn|✓|Sim-Mani：弗兰卡厨房，元世界|
|LEO [141]|ConvNeXt、PointNet++|Vicuna-7B|MLP|Concat|✓|Sim-Mani：CLIPort 任务子集； Sim-Navi
：
Habitat-web|
|3D-LLM [42]|CLIP、3D-CLR、
ConceptFusion|Flamingo、BLIP-2|DD-PPO|Concat|✓|Sim-Navi：栖息地|
|ShapeLLM [142]|ReCon++|LLaMA||Concat|✓|3D MM-Vet 基准|
|SayCan [10]||PaLM、FLAN|BC-Z、MT-Opt||✗|Real-Navi&Mani (EDR)：办公室厨房|
|翻译_□LM□_[143]
||GPT-3，Codex|||✗|模拟：VirtualHome|
|(SL)3 [144]||T5-small|seq2seq||✓|Sim-Navi：ALFRED|
|内心独白 [9]|MDETR、CLIP|InstructGPT、PaLM|CLIPort、启发式、
SayCan 策略|语言|✗|Sim&Real-Mani：桌面重新排列；
Real-Navi&Mani：厨房移动 Mani。
LLM-Planner [145]	HLSM 物体检测器	GPT-3	HLSM	语言	✓	Sim-Navi：ALFRED
LID [146]	CNN	GPT-2		Lang.，嵌入	✓	Sim：VirtualHome
SM [147]	CLIP、ViLD	GPT-3、RoBERTa	CLIPort	语言、代码	✗	Sim-Mani：拾取、放置
ProgPrompt [148]	ViLD	GPT-3	API	代码	✗	Sim：VirtualHome
ChatGPT 机器人技术
[149]	YOLOv8	ChatGPT	API	代码	✗	Real-Navi：无人机飞行； Sim-Navi：AirSim，
Habitat
CaP [150]	ViLD、MDETR	GPT-3、Codex	API	代码	✗	Sim-Mani 和 Sim-Navi：拾取、放置等
DEPS [151]	CLIP	ChatGPT	MC-控制器，Steve-1	代码	✗	模拟：我的世界
ConceptGraphs [152]	SAM、CLIP、LLaVA	GPT-4	API	代码 (JSON)	✗	Sim：AI2-THOR；真实（点臂）
：拾取位置

模仿学习（Imitation Learning）中使用BC目标，针对不同的动作类型有不同的变体。连续行动的 BC 目标可以写为

L Cont = ∑ MSE (at, ˆ at) , (4)
t

其中 MSE(·) 代表均方误差，at 是专家演示的动作注释，^ at 是预测动作。

离散动作是通过将动作值范围划分为固定数量的 bin 来实现的。他们的 BC 目标是

L Disc = ∑ CE (at, ˆ at) , (5)
t

其中CE(·)代表交叉熵损失。 SE (2) 操作的 BC 目标是

LSE (2) = CE(一个挑选，^ 一个挑选) + CE(一个地点，^ 一个地点) 。 (6)

基于扩散的控制策略中的 DDPM 目标是

L DDPM = MSE ∑ ε k , εθ (a k t , k) ∑ , (7)

其中_a__k_ _t_istheaction_at_corruptedbyrandomnoise_εk_at扩散迭代_k_，_εθ_是噪声预测网络，即VLA模型。

虽然离散动作在 RT-1 [97] 中表现出了卓越的性能，但 Octo [110] 认为它会导致早期抓取问题

。 SE (2) 动作要求模型仅预测末端执行器的拾取姿势和放置姿势，这足以满足许多桌面操作任务。然而，更复杂的任务--例如“将水倒入杯子”--可能需要额外的自由度，从而需要 SE (3) 动作 [36]。尽管基于点的动作可以是粗粒度的，但它们更容易以零样本方式从 VLM 中获得。

c) RT 系列： RT-1 [97] 激发了一系列“机器人变压器”模型。 Transformer 主干通过利用 Transformer 的更高容量来吸收更大的机器人数据集，从而超越了之前的 RNN 主干。之前的 RT1 是 BC-Z，它仅利用 MLP 层进行动作预测。 RT-1 之后，出现了一些引入新功能的作品。 MOO 改编了 RT-1 以适应多模式提示。 RT-Trajectory 使 RT-1 能够根据提示处理轨迹草图。 Q-Transformer 利用 Q-learning 来训练 RT-1。 RT-2基于ViT和LLM，引入了与RT-1完全不同的架构。 RT-X 使用更大的数据集重新训练 RT-1 和 RT-2，从而提高了性能。基于 RT2，RT-H [114] 引入了动作层次结构以实现更好的数据共享。

d) LVLA 与广义 VLA： 虽然 LVLA 可以极大地增强指令跟踪能力，因为它们可以更好地解析用户意图，但人们仍然担心它们的训练成本和部署速度。特别是，推理速度慢可能会显着影响动态环境中的性能，因为推理过程中可能会发生环境变化。因此，已经提出了多种方法来提高效率。 TinyVLA [121] 通过较小的 VLM 和用于机器人动作的扩散头专注于推理速度和数据效率。 DeeR-VLA [104]建议通过提前退出动态推理仅部分激活模型。

e) 缩放定律： 与法学硕士一样，机器人技术中也观察到了缩放定律 [153]、[154]，揭示了模型大小、数据集大小以及环境和对象多样性的重要性。该领域的进一步研究可以指导具有强大的野外泛化能力的 VLA 的开发。

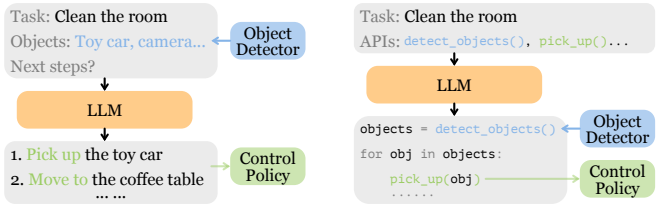
四．任务规划器

高级任务规划器 πphi 旨在将复杂任务 ℓ 分解为一系列子任务 p ˆ = [p 1 , p 2 , 。 。 。 , pN] (即

task plan), where each p_i serves as an instruction to low-level control policies π_θ:

$$\hat{\mathbf{p}} \sim \pi_\phi(\mathbf{p} \mid \ell, s_t). \tag{8}$$

This process is sometimes referred to as task or subgoal decomposition and is closely related to task and motion planning (TAMP) and embodied decision making. When equipped with task planners, VLAs can complete more complex, long-horizon tasks, as illustrated in Fig. 4. Details are summarized in Table IV. Ideally, task plans should also incorporate optimal scheduling for these subtasks.



(a) Language-based (b) Code-based

Fig. 6. Different approaches to connect LLM to multimodal modules in modular task planners.

A. Monolithic Task Planners

A single LLM or multimodal LLM (MLLM) can typically generate task plans by employing a tailored framework or through finetuning on embodied datasets. We refer to these as *monolithic* models.

1) End-to-End Task Planners: Similar to LVLAs, task planners can be implemented as end-to-end MLLMs, leveraging their Internet-scale knowledge for task planning.

PaLM-E [11] integrates ViT and PaLM to create an embodied MLLM capable of performing high-level embodied reasoning tasks. Based on perceived images and high-level language instructions, PaLM-E generates a text plan that serves as instructions for low-level robotic policies. In a mobile manipulation environment, the generated plan is mapped to executable low-level instructions using SayCan [10]. As the low-level policy executes actions, PaLM-E can also replan based on changes in the environment. With PaLM as its backbone, PaLM-E can handle standard VQA tasks along with additional embodied VQA tasks.

EmbodiedGPT [140] introduces the embodied-former, which outputs task-relevant instance-level features. This is achieved by incorporating vision encoder embeddings and embodied planning information provided by an LLM. These instance features serve to inform the low-level policy about the immediate next action to take.

2) End-to-End Task Planners with 3D Vision: Some task planners also explore the use of 3D vision. Because the majority of current MLLMs deal with images as visual inputs, they require architectural changes to incorporate 3D visual inputs. Consequently, they are usually end-to-end models.

LEO [141] uses a two-stage training strategy to integrate a point cloud encoder with an LLM: the first stage focuses on 3D vision-language alignment, followed by the second stage of 3D vision-language-action instruction tuning. LEO performs well not only in 3D question-answering tasks but also in manipulation, navigation, and task planning.

3D-LLM [42] injects 3D information into LLMs and empowers them to perform 3D tasks, such as 3D-assisted dialog and navigation, using features from point clouds, gradSLAM, and neural voxel fields. MultiPLY [155] is an object-centric embodied LLM that incorporates even more modalities, including audio, tactile, and thermal.

ShapeLLM [142] is built on the novel 3D vision encoder ReCon++, which distills knowledge from multiview image and text teacher models, along with a point cloud MAE. By integrating ReCon++ with LLaMA, ShapeLLM improves embodied interaction performance on their newly proposed 3D benchmark, 3D MM-Vet.

3) Grounded Task Planners: Grounded task planning involves generating high-level actions while considering whether they can be executed by low-level control policies.

SayCan [10] is a framework designed to integrate high-level LLM planners with low-level control policies. In this framework, the LLM planner accepts a user's high-level instruction and "says" what the most probable next low-level skill is, a concept referred to as *task-grounding*. The low-level policy provides the value function as the affordance function, determining the probability that the policy "can" complete the skill, known as *world-grounding*. By considering both the LLM's plan and the affordances, the framework selects the optimal skill for the current state.

Translated ⟨LM⟩ [143] employs a two-step process to translate high-level instructions into executable actions. Initially, a pretrained causal LLM is utilized for *plan generation*, breaking down the high-level instruction into the next action, expressed as a free-form language phrase. Then, as this phrase may not directly map to VirtualHome actions, a pretrained masked LLM is employed for *action translation*. This step involves calculating the similarity between the generated action phrase and the admissible VirtualHome actions. The translated action is appended to the plan, and the updated plan is fed back into the LLM to generate the next action phrase. This two-step process is repeated until a complete plan is formed. A "re-prompting" strategy [156] is further proposed to generate corrective actions when the agent encounters precondition errors.

$(SL)^3$ [144] is a learning algorithm that alternates between three steps: segmentation, labeling, and parameter update. In the segmentation step, high-level subtasks are aligned with low-level actions; subtask descriptions are then inferred in the labeling step; and finally, the network parameters are updated. This approach enables a hierarchical policy to discover reusable skills with sparse natural language annotations.

B. Modular Task Planners

Finetuning end-to-end models on embodied data can be expensive; therefore, some approaches adopt a modular design by assembling off-the-shelf LLMs and VLMs into task planners. These approaches can also be viewed as following the tool-use architecture [129].

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任务计划），其中每个 pi 作为低级控制策略 πθ 的指令：

p ˆ ~ πphi (p | ℓ, st) 。 (8)

这个过程有时被称为任务或子目标分解，与任务和运动规划（TAMP）以及具体决策密切相关。当配备任务规划器时，VLA 可以完成更复杂的长期任务，如图 4 所示。表 IV 总结了详细信息。理想情况下，任务计划还应包含这些子任务的最佳调度。

一个。整体任务规划器

单一法学硕士或多模式法学硕士（MLLM）通常可以通过采用定制框架或通过对具体数据集进行微调来生成任务计划。我们将这些称为“整体”模型。

1) 端到端任务规划器： 与 LVLA 类似，任务规划器可以作为端到端 MLLM 来实现，利用其互联网规模的知识进行任务规划。

PaLM-E [11] 集成了 ViT 和 PaLM，创建了一个能够执行高级具体推理任务的具体 MLLM。基于感知图像和高级语言指令，PaLM-E 生成一个文本计划，作为低级机器人策略的指令。在移动操作环境中，生成的计划使用 SayCan [10] 映射到可执行的低级指令。当低级策略执行操作时，PaLM-E还可以根据环境的变化重新规划。以 PaLM 作为骨干，PaLM-E 可以处理标准 VQA 任务以及附加的具体 VQA 任务。

EmbodiedGPT [140]引入了embodied-former，它输出与任务相关的实例级特征。这是通过结合视觉编码器嵌入和法学硕士提供的具体规划信息来实现的。这些实例功能用于通知低级策略立即采取的下一个操作。

2) 具有 3D 视觉的端到端任务规划器： 一些任务规划器还探索 3D 视觉的使用。由于当前大多数 MLLM 将图像作为视觉输入进行处理，因此它们需要更改架构以合并 3D 视觉输入。因此，它们通常是端到端模型。

LEO [141] 使用两阶段训练策略将点云编码器与 LLM 集成：第一阶段侧重于 3D 视觉语言对齐，其次是 3D 视觉语言动作（Vision-Language-Action） 指令调整的第二阶段。 LEO 不仅在 3D 问答任务中表现出色，而且在操纵、导航和任务规划方面也表现出色。

3D-LLM [42] 将 3D 信息注入 LLM，并使其能够使用点云、gradSLAM 和神经体素场的特征执行 3D 任务，例如 3D 辅助对话和导航。 MultiPLY [155] 是一种以对象为中心的体现法学硕士，它融合了更多模式，包括音频、触觉和热学。

ShapeLLM [142] 基于新颖的 3D 视觉编码器 ReCon++ 构建，它从多视图图像和文本教师模型以及点云 MAE 中提取知识。

任务：打扫房间 任务：打扫房间
对象：玩具车、相机..

。 DetectorObject API： detector_objects()、pick_up()...
后续步骤？
LLM
LLM
objects = detector_objects() DetectorObject
1.拾取对象中 obj 的玩具车 Control：
2。移至茶几... ... 策略 .....pick_up(obj) 控制策略
(a) 基于语言 (b) 基于代码

图 6. 将 LLM 连接到模块化任务规划器中的多模式模块的不同方法。

通过将 ReCon++ 与 LLaMA 集成，ShapeLLM 改进了其新提出的 3D 基准 3D MM-Vet 上的具体交互性能。

3) 接地 ~~ 固定任务~~ 规划器： 接地任务规划涉及生成高级操作，同时考虑它们是否可以通过低级控制策略执行。

SayCan [10] 是一个旨在将高级 LLM 规划者与低级控制策略集成的框架。在此框架中，LLM 规划器接受用户的高级指令并“说出”最有可能的下一个低级技能是什么，这个概念称为 task-落地关联（Grounding）。低级策略提供价值函数 可供性（Affordance） 函数，确定策略“可以”完成技能的概率，称为 world-落地关联（Grounding）。通过考虑 LLM 的计划和 可供性（Affordances），该框架为当前状态选择最佳技能。

Translated ☐ LM ☐ [143] 采用两步过程将高级指令转换为可执行动作。最初，预训练的因果 LLM 用于_计划生成_，将高级指令分解为下一个动作，并表示为自由格式语言短语。然后，由于该短语可能不会直接映射到 VirtualHome 操作，因此采用预训练的屏蔽 LLM 进行_操作翻译_。此步骤涉及计算生成的动作短语和可接受的 VirtualHome 动作之间的相似度。翻译后的行动将附加到计划中，更新后的计划将反馈到 LLM 以生成下一个行动短语。重复这个两步过程，直到形成完整的计划。进一步提出了“重新提示”策略 [156]，以在智能体遇到前提条件错误时生成纠正动作。

(SL)3 [144] 是一种在三个步骤之间交替的学习算法：分割、标记和参数更新。在分割步骤中，高级子任务与低级动作对齐；然后在标记步骤中推断子任务描述；最后，更新网络参数。这种方法使分层策略能够通过稀疏的自然语言注释发现可重用的技能。
B。模块化任务规划器

根据具体数据微调端到端模型的成本可能很高；因此，一些方法采用模块化设计，将现成的 LLM 和 VLM 组装到任务规划器中。这些方法也可以被视为遵循工具使用架构[129]。

1) Language-Based Task Planners: Language-based approaches use natural language descriptions as the medium for exchanging multimodal information, as shown in Fig. 6(a).

Inner monologue [9] sits between high-level commands and low-level policies to enable closed-loop control planning. It employs an LLM to generate language instructions for low-level policies and dynamically updates these instructions based on the received feedback. The feedback encompasses various sources: success feedback, object and scene feedback, and human feedback. As the feedback is communicated to the LLM in a textual format, no additional training is required for the LLM. A similar approach is used in ReAct [76].

LLM-Planner [145] introduces a novel approach to constructing a hierarchical policy comprising a high-level planner and a low-level planner. The high-level planner harnesses the capabilities of an LLM to generate natural language plans, while the low-level planner translates each subgoal within the plan into primitive actions. While sharing similarities with previous methods in its overall architecture, LLM-Planner distinguishes itself by incorporating a re-planning mechanism, enabling the robot to "get unstuck".

LID [146] introduces a novel data collection procedure termed active data gathering (ADG). A key aspect of ADG is hindsight relabeling, which reassigns labels to unsuccessful trajectories, effectively maximizing the utilization of data irrespective of trajectory success. By converting all environmental inputs into textual descriptions, the proposed language-model-based policy gains enhanced combinatorial generalization.

Socratic models (SMs) [147] present a unique framework wherein diverse pretrained models are effectively composed without the need for finetuning. The framework relies on a key mechanism called multimodal-informed prompting, which facilitates information exchange among models with varied multimodal capabilities. The idea is to utilize multimodal models to convert non-language inputs into language descriptions, effectively unifying different modalities within the language space. Beyond excelling in conventional multimodal tasks, SMs showcase their versatility in robot perception and planning. In addition to natural language plans, task plans can also be represented in the form of pseudocode.

2) Code-Based Task Planners: Code-based task planners leverage the coding ability of LLMs to generate task plans in the form of executable programs. Object detectors, VLMs, and control policies can be invoked via APIs, as shown in Fig. 6(b).

ProgPrompt [148] introduces a novel task-planning approach by prompting LLMs with program-like specifications detailing available actions and objects. This enables LLMs to generate high-level plans for household tasks in a few-shot manner. Environmental feedback can be incorporated through assertions within the program.

ChatGPT for robotics [149] takes advantage of the programming ability of ChatGPT to facilitate "user on the loop" control, a departure from the conventional "engineer in the loop" methodology. The procedure includes several steps. First, a list of APIs is defined, such as an object-detection API, a grasp API, and a move API. Second, a prompt is constructed for ChatGPT, specifying the environment, API functionality, and task goal. Third, ChatGPT is iteratively prompted to write code utilizing the defined APIs to execute the task, provided access to simulation and user feedback for evaluating code quality and safety. Finally, the ChatGPT-generated code is executed. In this procedure, ChatGPT serves as a high-level task planner, and actions are generated through function calls to corresponding low-level APIs.

Code as policies (CaP) [150] also leverages the code-writing capability of LLMs. It employs GPT-3 or Codex to generate policy code, which, in turn, invokes perception modules and control APIs. CaP exhibits proficiency in spatial-geometric reasoning, generalization to new instructions, and parameterization for low-level control primitives. By leveraging the multimodal capabilities of GPT-4V, COME-robot [157] eliminates the need for perception APIs required by CaP. This opens up possibilities for open-ended reasoning and adaptive planning within a closed-loop framework, enabling capabilities such as failure recovery and free-form instruction following.

DEPS [151] stands for "describe, explain, plan, and select." This approach employs an LLM to generate plans and explain failures based on feedback descriptions collected from the environment—a process referred to as "self-explanation," which aids in replanning. In addition, DEPS introduces a trainable goal selector to choose among parallel candidate subgoals based on how easily they can be achieved, a crucial aspect often overlooked by other high-level task planners.

ConceptGraphs [152] introduces a method to convert observation sequences into open-vocabulary 3D scene graphs. Objects are extracted from RGB images using 2D segmentation models, and VLMs are employed to caption objects and establish inter-object relations, resulting in the construction of a 3D scene graph. This graph can then be translated into a text description (JSON), providing LLMs with rich semantic and spatial relationships between entities for task planning.

Table V
EMBODIED DATASETS. PER RT-X, SKILLS CORRESPOND TO VERBS, AND TASKS ARE DIFFERENT COMBINATIONS OF VERBS AND OBJECTS. ∗ DATASET COLLECTED IN SIMULATION RATHER THAN THE REAL WORLD. SOME DATASETS ARE CONTINUALLY UPDATED, AND WE INCLUDE ONLY THE ORIGINAL VERSION. TABLE ADAPTED FROM [158], [159]

Dataset	Skills	Tasks	Scenes	Episodes	Collection	Obs.	Instruction	Robots
MIME [160]	12	20	1	8.3K	Human teleop.	RGBD	Demo	Baxter
∗RoboTurk [161]	2	3	1	2.1K	Human teleop.	RGB	–	Sawyer
RoboNet [162]	–	–	10	162K	Script	RGB	Goal image	(7 robots)
MT-Opt [163]	2	12	1	800K	Script, RL	RGB	Task ID	(7 robots)
BC-Z [82]	3	100	1	25.9K	Human teleop.	RGB	Lang, demo	EDR
Fractal [97]	12	700+	2	130K	Human teleop.	RGB	Lang	EDR
MOO [96]	5	–	–	59.1K	Human teleop.	RGB	Multimodal	EDR
∗VIMA [130]	17	–	1	650K	Script	RGB	Multimodal	UR5
RoboSet [164]	12	38	11	98.5K	Human, script	RGBD	Lang	Franka
BridgeV2 [165]	13	–	24	60.1K	Human, script	RGB(D)	Lang	WidowX
RH20T [158]	42	147	7	110K+	Human teleop.	RGBD	Lang	(4 robots)
DROID [159]	86	–	564	76K	Human teleop.	RGBD	Lang	Franka
OXE [115]	527	160K	311	1M+	Aggregate data	RGBD	Lang	(22 robots)

C. Strengths and Limitations

Monolithic task planners that utilize grounded task planning focus on generating executable plans. End-to-end models share an architecture similar to most LVLAs and can be finetuned on specialized embodied data to achieve better performance. However, the training costs of such large models are often substantial. In contrast, *modular* task planners are more readily

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IEEE 神经网络和学习系统交易

14

_1）基于语言的任务规划器：_基于语言的方法使用自然语言描述作为交换多模态信息的媒介，如图6（a）所示。

内部独白[9]位于高层命令和低层策略之间，以实现闭环控制规划。它采用法学硕士来生成低级策略的语言指令，并根据收到的反馈动态更新这些指令。反馈包括各种来源：成功反馈、对象和场景反馈以及人类反馈。由于反馈以文本格式传达给法学硕士，因此法学硕士不需要额外的训练。 ReAct [76] 使用了类似的方法。

LLM-Planner [145] 引入了一种新颖的方法来构建由高级规划器和低级规划器组成的分层策略。高级规划器 运行支撑框架（Harnesses） 具有 LLM 生成自然语言计划的能力，而低级规划器则将计划中的每个子目标转换为原始操作。虽然其整体架构与之前的方法有相似之处，但 LLM-Planner 的独特之处在于，它融入了重新规划机制，使机器人能够“摆脱困境”。

LID [146] 引入了一种新颖的数据收集程序，称为主动数据收集（ADG）。 ADG 的一个关键方面是事后重新标记，它将标签重新分配给不成功的轨迹，从而有效地最大化数据的利用率，而不管轨迹是否成功。通过将所有环境输入转换为文本描述，所提出的基于语言模型的策略获得了增强的组合泛化。

苏格拉底模型（SM）[147]提出了一个独特的框架，其中可以有效地组合不同的预训练模型，而无需进行微调。该框架依赖于一种称为多模式通知提示的关键机制，该机制促进具有不同多模式功能的模型之间的信息交换。这个想法是利用多模态模型将非语言输入转换为语言描述，有效地统一语言空间内的不同模态。除了在传统的多模式任务中表现出色之外，SM 还展示了其在机器人感知和规划方面的多功能性。除了自然语言计划之外，任务计划也可以以伪代码的形式表示。

2) 基于代码的任务规划器： 基于代码的任务规划器利用法学硕士的编码能力以可执行程序的形式生成任务计划。对象检测器、VLM 和控制策略可以通过 API 调用，如图 6（b）所示。

ProgPrompt [148] 引入了一种新颖的任务规划方法，通过类似程序的规范来提示法学硕士，详细说明可用的操作和对象。这使得法学硕士能够以几次方式生成家庭任务的高级计划。环境反馈可以通过程序中的断言纳入。

用于机器人技术的 ChatGPT [149] 利用 ChatGPT 的编程能力来促进“用户在环”控制，这与传统的“工程师在环”方法不同。该过程包括几个步骤

。首先，定义 API 列表，例如对象检测 API、抓取 API 和移动 API。其次，为ChatGPT构建一个提示，指定环境、API功能、

表五

体现的数据集。根据 RT-X，技能对应于动词，任务是动词和宾语的不同组合。 ** 数据集是在模拟而非现实世界中收集的。一些数据集不断更新，我们仅包含原始版本。表格改编自[158]、[159]

数据集	技能	任务	场景	剧集	集合	观测。	说明	机器人
MIME [160]	12	20	1	8.3K	人类遥控。	RGBD	演示	Baxter

__**_RoboTurk [161]	2	3	1	2.1K	人类遥控。	RGB		Sawyer
RoboNet [162]			10	162K	脚本	RGB	目标图像	(7 个机器人)
MT-Opt [163]	2	12	1	800K	脚本，RL	RGB	任务 ID	（7 个机器人）
BC-Z [82]	3	100	1	25.9K	人体遥控。	RGB	Lang、演示	EDR
分形 [97]	12	700+	2	130K	人类遥控。	RGB	Lang	EDR
MOO [96]	5			59.1K	人类遥控。	RGB	多模式	EDR
_**_VIMA [130]	17		1	650K	脚本	RGB	多模式	UR5
RoboSet [164]	12	38	11	98.5K	人类，脚本	RGBD	Lang	Franka
BridgeV2 [165]	13		24	60.1K	人类、脚本	RGB(D)	Lang	WidowX
RH20T [158]	42	147	7	110K+	人体遥控。	RGBD	Lang	(4 个机器人)
DROID [159]	86		564	76K	人类遥控。	RGBD	Lang	Franka
OXE [115]	527	160K	311	1M+	聚合数据	RGBD	Lang	(22个机器人)

和任务目标。第三，ChatGPT 被迭代提示使用定义的 API 编写代码来执行任务，并提供模拟和用户反馈的访问权限，以评估代码质量和安全性。最后，执行 ChatGPT 生成的代码。在此过程中，ChatGPT 充当高级任务规划器，并通过对相应低级 API 的函数调用来生成操作。

代码即策略（CaP）[150]也利用了法学硕士的代码编写能力。它采用 GPT-3 或 Codex 生成策略代码，进而调用感知模块和控制 API。 CaP 表现出在空间几何推理、新指令泛化以及低级控制原语参数化方面的熟练程度。通过利用 GPT-4V 的多模态功能，COME-robot [157] 消除了 CaP 所需的感知 API 的需要。这为闭环框架内的开放式推理和自适应规划提供了可能性，从而实现了故障恢复和自由格式指令跟踪等功能。

DEPS [151] 代表“描述、解释、计划和选择”。这种方法采用法学硕士根据从环境收集的反馈描述来生成计划并解释失败--这个过程被称为“自我解释”，有助于重新规划。此外，DEPS 引入了一个可训练的目标选择器，可以根据实现子目标的难易程度在并行候选子目标中进行选择，这是其他高级任务规划器经常忽视的一个关键方面。

ConceptGraphs [152] 引入了一种将观察序列转换为开放词汇 3D 场景图的方法。使用 2D 分割模型从 RGB 图像中提取对象，并使用 VLM 为对象添加标题并建立对象间关系，从而构建 3D 场景图

。然后可以将该图转换为文本描述 (JSON)，为法学硕士提供实体之间丰富的语义和空间关系以进行任务规划。

C。优点和局限性

利用扎根任务规划的单一任务规划器专注于生成可执行计划。端到端模型共享与大多数 LVLA 类似的架构，并且可以根据专门的具体数据进行微调，以实现更好的性能。然而，如此大的模型的训练成本通常是巨大的。相比之下，_模块化_任务规划器更容易

Table VI
SIMULATORS AND SIMULATED BENCHMARKS. CONTROL: CONTINUOUS CONTROL TASKS. D, S, A, N: DEPTH, SEGMENTATION, AUDIO, NORMAL. FORCE: SIMULATED CONTACT FORCE BETWEEN END-EFFECTOR AND ITEM. PD: PRE-DEFINED. TABLE ADAPTED FROM [166], [167]

Name	Scenes /Rooms	Objects /Cat	UI	Physics Engine	Task	Observation	Action	Agent	Description	Related
Gibson [168]	572/-	–	–	Pybullet	Navi	RGB, D, N, S	–	–	Navi only	–
iGibson [167], [169], [170]	15/108	152/5	Mouse, VR	Pybullet	Navi, Mani	RGB, D, S, N, Flow, LiDAR	Force	TurtleBot v2, LoCoBot, etc.	VR, Continuous Extended States. Versions: 0.5, 1.0, 2.0	Benchmarks: BEHAVIOR-100 [166], BEHAVIOR-1K [171]
SAPIEN [172]	–	2346/-	Code	PhysX	Navi, Mani	RGB, D, S	Force	Franka	Articulation, Ray Tracing	VoxPoser. Benchmark: SIMPLER [173]
AI2-THOR [174]	-/120	118/118	Mouse	Unity	Navi, Mani	RGB, D, S, A	Force, PD	ManipulaTHOR, LoCoBot, etc.	Object States, Task Planning. Versions: [175], [176]	Benchmarks: ALFRED, RoPOR [177]
VirtualHome [178]	7/-	-/509	Lang	Unity	Navi, Mani	RGB, D, S	Force, PD	Human	Object States, Task Planning	LID, Translated⟨LM⟩, ProgPrompt
TDW [179]	15/120	112/50	VR	Unity, Flex	Navi, Mani	RGB, D, S, A	Force	Fetch, Sawyer, Baxter	Audio, Fluids	–
RLBench [180]	1/-	28/28	Code	V-REP	Mani	RGB, D, S	Force	Franka	Tiered Task Difficulty	Hiveformer, PerAct
Meta-World [181]	1/-	80/7	Code	MuJoCo	Mani	Pose	Force	Sawyer	Meta-RL	R3M, VC-1, Vi-PRoM, EmbodiedGPT
CALVIN [182]	4/-	7/5	–	Pybullet	Mani	RGB, D	Force	Franka	Long-horizon Lang-cond tasks	GR-1, HULC, RoboFlamingo
Franka Kitchen [183]	1/-	10/6	VR	MuJoCo	Mani	Pose	Force	Franka	Extended by R3M with RGB	R3M, Voltron, Vi-PRoM, Diffusion Policy, EmbodiedGPT
Habitat [184], [185]	(Matterport + Gibson)		Mouse	Bullet	Navi	RGB, D, S, A	Force	Fetch, Franka, AlienGO	Fast, Navi only. Versions: 1.0, 2.0, Rearrangement [186]	VC-1, PACT; Benchmark: OVMM [187]
ALFRED [188]	-/120	84/84	–	Unity	Navi, Mani	RGB, D, S	PD	Human	Diverse long-horizon tasks	$(SL)^3$, LLM-Planner
DMC [189]	1/-	4/4	Code	MuJoCo	Control	RGB, D	Force	–	Continuous RL	VC-1, SMART
OpenAI Gym [190]	1/-	4/4	Code	MuJoCo	Control	RGB	Force	–	Single agent RL environments	–
Genesis [191]	(Rigid, deformable, liquid, etc.)		Code	(Custom)	Navi, Mani	RGB, D, S, N	Force	Franka, Unitree, etc.	High-speed comprehensive physics simulation	–

deployable because they leverage off-the-shelf LLMs and VLMs. Language-based task planners offer the advantage of seamless integration of LLMs and VLMs, as they are designed to operate in the natural language space. However, they often require extra steps to align the generated task plans with language instructions that are admissible to low-level control policies. Conversely, while code-based task planners may require manually wrapping VLMs and control policies within APIs and preparing clear documentation in advance, they enable code debugging and provide greater controllability. Ultimately, their performance remains constrained by the programming capabilities of the underlying models.

V. DATASETS AND BENCHMARKS

A. Real-World Robot Datasets & Benchmarks

Embodied AI faces significant data scarcity issues because real-world robot data are not as readily available as language data. Collecting real-world robot datasets poses multiple challenges. First, the process is impeded by the cost and time required to procure robotic equipment, set up environments, and gather expert data through dedicated policies or human teleoperation. Second, the diverse types and configurations of robots introduce inconsistencies in factors such as sensory data, control modes, and gripper types. Finally, accurately capturing 6D object poses and reproducing setups remain elusive. We summarize recent embodied datasets in Table V. In addition, real-world benchmarks are further complicated by the need for human evaluation.

B. Simulators, Simulated Robot Datasets & Benchmarks

Many researchers resort to simulated environments to circumvent real-world obstacles and scale the data collection process. We compare simulators and simulated benchmarks in Table VI. Nevertheless, this strategy presents its own challenges, chief among which is the sim-to-real gap. This discrepancy arises when models trained on simulated data exhibit poor performance during real-world deployment. The causes of this gap are multifaceted, encompassing unrealistic rendering quality, inaccuracies in physics simulations, and domain shifts characterized by object properties and robot motion planners. For instance, simulating non-rigid entities such as deformable objects or liquids presents significant difficulties. Moreover, importing new objects into simulators requires considerable effort, often involving techniques such as 3D scanning and mesh editing. Despite these hurdles, simulated environments provide automated evaluation metrics that aid researchers in consistently evaluating robotic models. Many benchmarks are based on simulators because they can precisely reproduce the experimental setup and yield fair comparisons of different models. Another technique, known as real-to-sim, can improve simulation fidelity, recreate failure cases, or facilitate digital twins.

C. Automated Dataset Collection

Several approaches advocate for automated dataset collection. RoboGen [192] employs a generative simulation paradigm that proposes interesting skills, simulates corresponding environments, and selects optimal learning approaches to train policies for acquiring those skills. AutoRT [193] functions as a robot orchestrator driven by LLMs, generating tasks, filtering them by affordance, and utilizing either autonomous policies or human teleoperators to collect and evaluate data. DIAL [194] focuses on augmenting language instructions in existing datasets using VLMs. RoboPoint [94] generates scenes procedurally with randomized 3D layouts, objects, and camera viewpoints.

D. Human Datasets

An alternative strategy to address data scarcity in real-world settings is to leverage human data. Human behavior offers plentiful guidance for robot policies due to its dexterity and diversity [195]. However, this strategy also comes with inherent limitations. Capturing and transferring human hand and body motions to robot embodiments is difficult. Moreover, the inconsistency in human data poses a hurdle, as some

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表六

模拟器和模拟基准。控制：连续的控制任务。 D、S、A、N：深度、分段、音频、正常。力：末端执行器和物品之间的模拟接触力。 PD：预先定义。表格改编自[166]、[167]

|名称|场景
/房间|对象
/Cat|UI|物理
引擎|任务|观察|操作|智能体|描述|相关|
|---|---|---|---|---|---|---|---|---|---|---|---|
|Gibson [168]|572/-|||Pybullet|Navi|RGB、D、N、S|||仅限 Navi||
|iGibson [167]、[169]、
[170]|15/108|152/5|鼠标、
VR|Pybullet|Navi、Mani|RGB、D、S、N、
Flow、LiDAR|Force|TurtleBot v2、
LoCoBot 等|VR、连续扩展状态。
版本：0.5、 1.0、2.0|基准：BEHAVIOR-100
[166]、BEHAVIOR-1K [171]|
|SAPIEN [172]||2346/-|代码|PhysX|Navi、Mani|RGB、D、S|Force|Franka|关节、光线追踪|VoxPoser。
基准：更简单
[173]|
|AI2-THOR [174]|-/120|118/118|鼠标|Unity|Navi、Mani|RGB、D、S、A|Force、PD|ManipulaTHOR、
LoCoBot 等|对象状态、任务规划。
版本：[175]、[176]|基准：ALFRED、RoPOR
[177]|
|VirtualHome [178]|7/-|-/509|Lang|Unity|Navi、Mani|RGB、D、S|Force、PD|人类|对象状态、任务规划
|LID、Translated<LM>、ProgPrompt|
|TDW [179]|15/120|112/50|VR|Unity、Flex|Navi、Mani|RGB、D、S、A|Force|Fetch、Sawyer、
Baxter|音频、流体||
|RLBench [180]|1/-|28/28|代码|V-REP|Mani|RGB、D、S|Force|Franka|分层任务难度|Hiveformer、
PerAct|
|元世界 [181]|1/-|80/7|代码|MuJoCo|Mani|姿势|Force|Sawyer|Meta-RL|R3M、VC-1、Vi-PRoM、
EmbodiedGPT|
|CALVIN [182]|4/-|7/5||Pybullet|Mani|RGB、D|Force|Franka|长时程（Long-Horizon） Lang-cond 任务
|GR-1、HULC、RoboFlamingo|
|Franka 厨房 [183]|1/-|10/6|VR|MuJoCo|Mani|Pose|Force|Franka|R3M 扩展，带 RGB|R3M、Voltron、
Vi-PRoM、
扩散策略（Diffusion Policy）、EmbodiedGPT|
|Habitat [184]、[185]|(Matterp|ort + Gibson)|鼠标|Bullet|Navi|RGB、D、S、A|Force|Fetch、Franka、

AlienGO|快速，仅限 Navi。版本：1.0、
2.0、重排[186]|VC-1、PACT；基准：
OVMM [187]|
|ALFRED [188]|-/120|84/84||Unity|Navi、Mani|RGB、D、S|PD|人类|多种 长时程（Long-Horizon） 任务
|(SL)3、LLM-Planner|
|DMC [189]|1/-|4/4|代码|MuJoCo|控制|RGB、D|强制||连续 RL|VC-1、SMART|
|OpenAI Gym [190]|1/-|4/4|代码|MuJoCo|控制|RGB|Force||单智能体 RL 环境||
|Genesis [191]|(Rigid、d
liquid、et|eformable、
c.)|Code|(自定义)|Navi、Mani|RGB、D、S、N|Force|Franka、Unitree等|高速综合
物理模拟||

可部署，因为它们利用现成的 LLM 和 VLM。基于语言的任务规划器具有 LLM 和 VLM 无缝集成的优势，因为它们被设计为在自然语言空间中运行。然而，它们通常需要额外的步骤来使生成的任务计划与低级控制策略可接受的语言指令保持一致。相反，虽然基于代码的任务规划器可能需要在 API 中手动包装 VLM 和控制策略并提前准备清晰的文档，但它们支持代码调试并提供更大的可控性。最终，它们的性能仍然受到底层模型的编程能力的限制。

五、数据集和基准

A。真实世界的机器人数据集和基准

具身智能（Embodied AI） 面临着严重的数据稀缺问题，因为现实世界的机器人数据不像语言数据那么容易获得。收集真实世界的机器人数据集面临着多重挑战。首先，这一过程受到采购机器人设备、设置环境以及通过专门策略或人工远程操作收集专家数据所需的成本和时间的阻碍。其次，机器人的类型和配置多种多样，导致传感数据、控制模式、夹具类型等因素不一致。最后，准确捕捉 6D 物体姿势并再现设置仍然难以实现。我们在表五中总结了最近的具体数据集。此外，由于人类评估的需要，现实世界的基准变得更加复杂。

造成这种差距的原因是多方面的，包括不切实际的渲染质量、物理模拟的不准确以及以对象属性和机器人运动规划器为特征的域转移。例如，模拟非刚性实体（例如可变形物体或液体）存在很大的困难。此外，将新对象导入模拟器需要付出相当大的努力，通常涉及 3D 扫描和网格编辑等技术。尽管存在这些障碍，模拟环境提供了自动化评估指标，帮助研究人员持续评估机器人模型。许多基准测试都基于模拟器，因为它们可以精确地重现实验设置并产生不同模型的公平比较。另一种技术称为“真实到模拟”，可以提高仿真保真度、重现故障案例或促进数字孪生。

C。自动化数据集收集

有几种方法提倡自动化数据集收集。 RoboGen [192] 采用生成模拟范式，提出有趣的技能，模拟相应的环境，并选择最佳的学习方法来训练获取这些技能的策略。 AutoRT [193] 充当由 LLM 驱动的机器人协调器，生成任务，通过 可供性（Affordance） 过滤任务，并利用自主策略或人类远程操作员来收集和评估数据。 DIAL [194] 专注于使用 VLM 增强现有数据集中的语言指令。 RoboPoint [94] 使用随机 3D 布局、对象和摄像机视点按程序生成场景。

B。模拟器、模拟机器人数据集和基准

许多研究人员求助于模拟环境来规避现实世界的障碍并扩展数据收集过程。我们在表六中比较了模拟器和模拟基准。然而，该策略也存在其自身的挑战，其中最主要的是 仿真到现实（Sim-to-Real） 差距。当基于模拟数据训练的模型在实际部署过程中表现不佳时，就会出现这种差异。的

D。人类数据集

解决现实世界中数据稀缺问题的另一种策略是利用人类数据。由于其灵活性和多样性，人类行为为机器人策略提供了大量指导[195]。然而，这种策略也有其固有的局限性。捕捉人手和身体的运动并将其转移到机器人实施例上是很困难的。此外，人类数据的不一致构成了障碍，因为一些人

Table VII
EMBODIED QUESTION ANSWERING BENCHMARKS. EXPLORE: ACTIVE EXPLORATION

Benchmark	QA Pairs	Scenes	Source	Answer Type	Collection	Explore	Metrics
EQA [203]	5K	750 envs	House3D simulator	Answer set (172 answers)	Template	✓	Accuracy
IQUAD [204]	75K	30 rooms	AI2-THOR	Multiple choice	Template	✓	Accuracy
MT-EQA [205]	19.3K	588 envs	House3D simulator	Binary answer	Template	✓	Accuracy
MP3D-EQA [206]	1,136	83 envs	MatterPort3D	Answer set (53 answers)	Template	✓	Accuracy
EgoVQA [207]	600	16 videos	IU Multi-view	Multiple choice (1 out of 5)	Human annotators	✗	Accuracy
EgoTaskQA [208]	40K	2K videos	LEMMA dataset	Open answer, binary verification	Human annotators & template	✗	Accuracy
EQA-MX [209]	8.2M	750K images	CAESAR simulator	Answer set	Question templates, answer set	✗	Accuracy
OpenEQA [210]	557 + 1079	180 envs	HM3D + ScanNet	Open answer	Human annotators	✗	LLM Score

data are egocentric while others are captured from third-person perspectives. In addition, filtering human data to extract useful information can be labor-intensive. These obstacles underscore the complexities involved in incorporating human data into robot learning processes. UMI [196] proposes a method to mitigate these issues using hand-held grippers. For a more comprehensive comparison of human datasets, we refer interested readers to [197].

E. Task Planning Benchmarks

EgoPlan-Bench [198] focuses on benchmarking real-world task planning with human annotations. PlanBench [199], [200] comprehensively assesses various aspects of task planning ability, such as cost optimality, plan verification, and replanning. LoTa-Bench [201] directly evaluates task planning by executing the generated plans in simulators and calculating success rates. Embodied agent interface (EAI) [202] argues that this approach fails to pinpoint issues in LLMs. By formalizing the input-output of LLM-based modules for decision-making tasks, EAI enables more fine-grained metrics beyond success rates.

F. Embodied Question Answering (EQA) Benchmarks

EQA benchmarks, as summarized in Table VII, do not directly evaluate robotic tasks like manipulation and navigation, but instead target other relevant abilities for embodied AI, such as spatial reasoning, physics understanding, and world knowledge. EQA is akin to traditional VQA benchmarks, but differs in that the agent can actively explore the environment before providing an answer. EmbodiedQA [203] and IQUAD [204] were among the first works to introduce this type of benchmark. MT-EQA [205] focuses on complex questions involving multiple targets. MP3D-EQA [206] converts previous RGB inputs to point clouds, testing 3D perception capabilities.

Active exploration requires access to a simulator, thereby restricting the ability to utilize real-world data. EgoVQA [207] shifts the focus of VQA to egocentric videos. EgoTaskQA [208] emphasizes spatial, temporal, and causal relationship reasoning. EQA-MX [209] investigates multimodal expressions (MXs), including regular verbal utterances and nonverbal gestures like eye gaze and pointing. OpenEQA [210] evaluates seven main categories, including functional reasoning and world knowledge.

VI. CHALLENGES AND FUTURE DIRECTIONS

a) Safety First: Safety is paramount in robotics, as robots interact directly with the physical world. Ensuring the safety of robotic systems requires the integration of real-world commonsense. This involves the incorporation of robust safety guardrails, risk assessment frameworks, and human-robot interaction protocols. RLHF and "evaluation without execution" can also significantly lower safety risks [20]. Interpretability and explainability of VLA decision-making processes are also crucial for enhancing robot safety through error diagnosis and troubleshooting.

b) Datasets & Benchmarks: In addition to the issues discussed in Section V, comprehensive benchmarks that cover a wide range of skills, objects, embodiments, and environments remain to be developed. Moreover, metrics beyond the success rate are needed for a fine-grained diagnosis of issues in VLA models, as highlighted by EAI [202] for LLMs.

c) Foundation Models & Generalization: VLA foundation models or robotic foundation models (RFMs) for embodied AI remain an open research topic primarily due to the diversity in embodiments, environments, and tasks. Many have made significant progress [113], [118], but still lack generalization capability on par with LLMs in NLP. Attaining such a level of generalization is very challenging, as it requires the development of many core AGI capabilities.

d) Multimodality: VLAs inherit many challenges associated with multimodal models, such as obtaining useful embeddings and aligning different modalities. Current approaches, like ImageBind [211] and LanguageBind [212], align different modalities to the image or language embedding space, respectively. Within a unified embedding space, MLLMs can accommodate diverse modalities and become more general. However, whether focusing on embeddings alone is sufficient remains under debate. Although beyond the scope of VLAs, other modalities—such as audio [45], haptics [213], and gaze [214]—have proven useful for certain embodied AI applications. For instance, modeling human gaze data using an additional gaze network [215] or incorporating it via an auxiliary loss [216] has been shown to enhance the performance of the primary policy network. These approaches demonstrate that RL policies can benefit from human-like visual attention, as human gaze often reveals the most salient locations in the environment [217]. While incorporating additional modalities is often advantageous, it inevitably introduces extra complexity into the model design.

e) Framework for Long-Horizon Tasks: The hierarchical framework is currently the most practical approach for long-horizon tasks. However, it increases system complexity and potential points of failure. Frequent task execution failures can trigger replanning, which can cause significant latency.

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表七

体现问答基准。探索：积极探索

基准	QA 对	场景	来源	答案类型	集合	探索	指标
EQA [203]	5K	750 envs	House3D模拟器	答案集（172个答案）	模板	✓	准确性
IQUAD [204]	75K	30个房间	AI2-THOR	多选	模板	✓	准确度
MT-EQA [205]	19.3K	588 envs	House3D 模拟器	二进制答案	模板	✓	准确性
MP3D-EQA [206]	1,136	83 envs	MatterPort3D	答案集（53 个答案）	模板	✓	准确性
EgoVQA [207]	600	16 个视频	IU 多视图	多项选择（5 项中的 1 项）	人工注释者	✗	准确性
EgoTaskQA [208]	40K	2K 视频	LEMMA 数据集	开放答案，二进制验证	人工注释器和模板	✗	准确性
EQA-MX [209]	8.2M	750K 图像	CAESAR 模拟器	答案集	问题模板、答案集	✗	准确性
OpenEQA [210]	557 + 1079	180 envs	HM3D + ScanNet	开放答案	人类注释者	✗	法学硕士分数

数据是以自我为中心的，而其他数据则是从第三人的角度捕获的。此外，过滤人类数据以提取有用信息可能会耗费大量人力。这些障碍凸显了将人类数据纳入机器人学习过程的复杂性。 UMI [196]提出了一种使用手持式夹具来缓解这些问题的方法。为了对人类数据集进行更全面的比较，我们建议感兴趣的读者参考[197]。

E。任务规划基准

EgoPlan-Bench [198] 专注于通过人工注释对现实世界的任务规划进行基准测试。 PlanBench[199]、[200]综合评估任务规划能力的各个方面，例如成本最优性、计划验证和重新规划。 LoTa-Bench [201] 通过在模拟器中执行生成的计划并计算成功率来直接评估任务计划。体现智能体接口（EAI）[202]认为这种方法无法查明法学硕士中的问题。通过形式化基于 LLM 的决策任务模块的输入输出，EAI 可以实现成功率以外的更细粒度的指标。

F。具身问答 (EQA) 基准

如表七所示，EQA 基准并不直接评估操纵和导航等机器人任务，而是针对 具身智能（Embodied AI） 的其他相关能力，例如空间推理、物理理解和世界知识。 EQA 类似于传统的 VQA 基准，但不同之处在于智能体可以在提供答案之前主动探索环境。 EmbodiedQA [203] 和 IQUAD [204] 是最早引入此类基准的作品之一。 MT-EQA [205] 专注于涉及多个目标的复杂问题。 MP3D-EQA [206] 将以前的 RGB 输入转换为点云，测试 3D 感知能力。主动探索需要访问模拟器，从而限制了利用现实世界数据的能力。 EgoVQA [207] 将 VQA 的焦点转移到以自我为中心的视频。 EgoTaskQA [208]强调空间、时间和因果关系推理。 EQA-MX [209] 研究多模态表达（MX），包括常规言语表达和非言语手势（例如眼神和指向）。 OpenEQA [210]评估七个主要类别，包括函数推理和世界知识。

六、

。挑战和未来方向

a) 安全第一： 安全在机器人技术中至关重要，因为机器人直接与物理世界交互。确保

机器人系统的安全需要整合现实世界的常识。这涉及到强大的安全护栏、风险评估框架和人机交互协议的结合。 RLHF和“评估而不执行”也可以显着降低安全风险[20]。 VLA 决策过程的可解释性和可解释性对于通过错误诊断和故障排除来增强机器人安全性也至关重要。

b) 数据集和基准： 除了第五节中讨论的问题之外，涵盖广泛技能、对象、实施例和环境的综合基准仍有待开发。此外，正如法学硕士的 EAI [202] 所强调的那样，对 VLA 模型中的问题进行细粒度诊断还需要成功率之外的指标。

c) 基础模型（Foundation Models） 和概括： 具身智能（Embodied AI） 的 VLA 基础模型（Foundation Models） 或机器人 基础模型（Foundation Models）(RFM) 仍然是一个开放的研究主题，主要是由于实施例、环境和任务的多样性。许多人已经取得了重大进展[113]、[118]，但仍然缺乏与 NLP 法学硕士同等的泛化能力。达到这样的通用化水平非常具有挑战性，因为它需要开发许多核心 AGI 功能。

d) 多模态： VLA 继承了与多模态模型相关的许多挑战，例如获得有用的嵌入和对齐不同的模态。当前的方法，如 ImageBind [211] 和 LanguageBind [212]，分别将不同的模式与图像或语言嵌入空间对齐。在统一的嵌入空间中，MLLM 可以适应不同的模式并变得更加通用。然而，仅关注嵌入是否足够仍然存在争议。尽管超出了 VLA 的范围，但其他模式（例如音频 [45]、触觉 [213] 和凝视 [214]）已被证明对某些 具身智能（Embodied AI） 应用有用。例如，使用额外的注视网络[215]对人类注视数据进行建模或通过辅助损失[216]将其合并已被证明可以增强主要策略网络的性能。这些方法表明，强化学习策略可以受益于类人的视觉注意力，因为人类的凝视通常会揭示环境中最显着的位置[217]。虽然合并额外的模式通常是有利的，但它不可避免地会给模型设计带来额外的复杂性。

e) 长时程（Long-Horizon） 任务框架： 分层框架是目前最实用的长期任务方法。然而，它增加了系统复杂性和潜在的故障点。频繁的任务执行失败可能会触发重新计划，从而导致显着的延迟。

Moreover, monolithic task planners share similar architectures with LVLAs, so employing two large models can be redundant and may hinder scalability. In addition, although modular task planners typically do not require training, they are not plug-and-play: language-based models may generate subtasks that control policies cannot execute, whereas code-based models require modules to be prewrapped in APIs manually. Therefore, developing a unified framework that directly translates long-horizon tasks into low-level control signals in an end-to-end fashion is worth exploring.

f) Real-Time Responsiveness: Unlike conversational AI, many robotic applications require real-time decision-making to respond to dynamic environments. If inference time cannot keep pace with environmental changes, the model may generate obsolete actions repeatedly. However, current VLA models—especially LVLAs and task planners—face a tradeoff between speed and capacity. Novel mechanisms are thus needed to strike an optimal balance.

g) Multiagent Systems: Cooperative multiagent systems offer benefits such as distributed perception and collaborative fault recovery. However, they also face challenges, including effective communication, coordinated dispatching, and fleet heterogeneity. In certain scenarios, individual agents may have conflicting goals, which further increases complexity.

h) Ethical and Societal Implications: Robotics has always raised various ethical, societal, and legal concerns. These include risks related to privacy, job displacement, decision-making bias, and the impact on social norms and human relationships.

i) Applications: Most current VLAs focus on household or industrial settings, but a wider range of applications is possible, such as virtual assistants, autonomous vehicles, and agricultural robots. Various embodiments may also call for specialized VLAs, including dexterous hands, drones, quadruped robots, and humanoid robots. One particularly important field is healthcare, encompassing surgical robots [218] and care robots [219]. Healthcare demands higher safety and privacy standards and may necessitate novel techniques such as human-in-the-loop (HITL) control and federated learning. Moreover, due to the significant domain gap, specialized vision models might be needed for medical images.

VII. Conclusion

VLA models hold immense promise for enabling embodied agents to interact with the physical world and fulfill user instructions. This article is the first survey to review LVLAs alongside generalized VLAs. Our taxonomy provides a high-level overview of three main lines of research: key components, control policies, and task planners. We meticulously analyze and compare their technical details, including model architectures, training strategies, and individual modules. In addition, we highlight essential resources for training and evaluating VLAs, such as datasets, simulators, and benchmarks. We hope this survey captures the rapidly evolving landscape of embodied AI and inspires future research.

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References

[1] A. Brohan, N. Brown, J. Carbajal, Y. Chebotar, X. Chen, K. Choromanski, T. Ding, D. Driess, A. Dubey, C. Finn, P. Florence, C. Fu, M. G. Arenas, K. Gopalakrishnan, K. Han, K. Hausman, A. Herzog, J. Hsu, B. Ichter, A. Irpan, N. J. Joshi, R. Julian, D. Kalashnikov, Y. Kuang, I. Leal, L. Lee, T. E. Lee, S. Levine, Y. Lu, H. Michalewski, I. Mordatch, K. Pertsch, K. Rao, K. Reymann, M. S. Ryoo, G. Salazar, P. Sanketi, P. Sermanet, J. Singh, A. Singh, R. Soricut, H. T. Tran, V. Vanhoucke, Q. Vuong, A. Wahid, S. Welker, P. Wohlhart, J. Wu, F. Xia, T. Xiao, P. Xu, S. Xu, T. Yu, and B. Zitkovich, "RT-2: vision-language-action models transfer web knowledge to robotic control," *CoRR*, vol. abs/2307.15818, 2023.

[2] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "Imagenet classification with deep convolutional neural networks," in *NIPS*, 2012, pp. 1106–1114.

[3] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, "Attention is all you need," in *NIPS*, 2017, pp. 5998–6008.

[4] V. Mnih, K. Kavukcuoglu, D. Silver, A. A. Rusu, J. Veness, M. G. Bellemare, A. Graves, M. A. Riedmiller, A. Fidjeland, G. Ostrovski, S. Petersen, C. Beattie, A. Sadik, I. Antonoglou, H. King, D. Kumaran, D. Wierstra, S. Legg, and D. Hassabis, "Human-level control through deep reinforcement learning," *Nat.*, vol. 518, no. 7540, pp. 529–533, 2015.

[5] S. Levine, P. Pastor, A. Krizhevsky, and D. Quillen, "Learning hand-eye coordination for robotic grasping with large-scale data collection," in *ISER*, ser. Springer Proceedings in Advanced Robotics, vol. 1. Springer, 2016, pp. 173–184.

[6] J. Alayrac, J. Donahue, P. Luc, A. Miech, I. Barr, Y. Hasson, K. Lenc, A. Mensch, K. Millican, M. Reynolds, R. Ring, E. Rutherford, S. Cabi, T. Han, Z. Gong, S. Samangooei, M. Monteiro, J. L. Menick, S. Borgeaud, A. Brock, A. Nematzadeh, S. Sharifzadeh, M. Binkowski, R. Barreira, O. Vinyals, A. Zisserman, and K. Simonyan, "Flamingo: a visual language model for few-shot learning," in *NeurIPS*, 2022.

[7] J. Li, D. Li, S. Savarese, and S. C. H. Hoi, "BLIP-2: bootstrapping language-image pre-training with frozen image encoders and large language models," in *ICML*, vol. 202. PMLR, 2023, pp. 19 730–19 742.

[8] H. Liu, C. Li, Q. Wu, and Y. J. Lee, "Visual instruction tuning," *CoRR*, vol. abs/2304.08485, 2023.

[9] W. Huang, F. Xia, T. Xiao, H. Chan, J. Liang, P. Florence, A. Zeng, J. Tompson, I. Mordatch, Y. Chebotar, P. Sermanet, T. Jackson, N. Brown, L. Luu, S. Levine, K. Hausman, and B. Ichter, "Inner monologue: Embodied reasoning through planning with language models," in *CoRL*, vol. 205. PMLR, 2022, pp. 1769–1782.

[10] B. Ichter, A. Brohan, Y. Chebotar, C. Finn, K. Hausman, A. Herzog, D. Ho, J. Ibarz, A. Irpan, E. Jang, R. Julian, D. Kalashnikov, S. Levine, Y. Lu, C. Parada, K. Rao, P. Sermanet, A. Toshev, V. Vanhoucke, F. Xia, T. Xiao, P. Xu, M. Yan, N. Brown, M. Ahn, O. Cortes, N. Sievers, C. Tan, S. Xu, D. Reyes, J. Rettinghouse, J. Quiambao, P. Pastor, L. Luu, K. Lee, Y. Kuang, S. Jesmonth, N. J. Joshi, K. Jeffrey, R. J. Ruano, J. Hsu, K. Gopalakrishnan, B. David, A. Zeng, and C. K. Fu, "Do as I can, not as I say: Grounding language in robotic affordances," in *CoRL*, vol. 205. PMLR, 2022, pp. 287–318.

[11] D. Driess, F. Xia, M. S. M. Sajjadi, C. Lynch, A. Chowdhery, B. Ichter, A. Wahid, J. Tompson, Q. Vuong, T. Yu, W. Huang, Y. Chebotar, P. Sermanet, D. Duckworth, S. Levine, V. Vanhoucke, K. Hausman, M. Toussaint, K. Greff, A. Zeng, I. Mordatch, and P. Florence, "Palm-e: An embodied multimodal language model," in *ICML*, vol. 202. PMLR, 2023, pp. 8469–8488.

[12] R. Firoozi, J. Tucker, S. Tian, A. Majumdar, J. Sun, W. Liu, Y. Zhu, S. Song, A. Kapoor, K. Hausman, B. Ichter, D. Driess, J. Wu, C. Lu, and M. Schwager, "Foundation models in robotics: Applications, challenges, and the future," *CoRR*, vol. abs/2312.07843, 2023.

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此外，整体任务规划器与 LVLA 共享相似的架构，因此采用两个大型模型可能是多余的，并且可能会阻碍可扩展性。此外，虽然模块化任务规划器通常不需要训练，但它们不是即插即用的：基于语言的模型可能会生成控制策略无法执行的子任务，而基于代码的模型则需要手动将模块预先包装在 API 中。因此，开发一个统一的框架，以端到端的方式直接将 长时程（Long-Horizon） 任务转换为低级控制信号是值得探索的。

f) 实时响应： 与对话式人工智能不同，许多机器人应用程序需要实时决策来响应动态环境。如果推理时间跟不上环境变化，模型可能会重复生成过时的动作。然而，当前的 VLA 模型（尤其是 LVLA 和任务规划器）面临着速度和容量之间的权衡。因此需要新的机制来达到最佳平衡。

g) 多智能体系统： 协作多智能体系统提供分布式感知和协作故障恢复等优点。但他们也面临着有效沟通、协调调度、机队异构等挑战。在某些情况下，各个智能体可能有相互冲突的目标，这进一步增加了复杂性。

h) 伦理和社会影响： 机器人技术总是引起各种伦理、社会和法律问题。其中包括与隐私、工作替代、决策偏见以及对社会规范和人际关系的影响相关的风险。

i) 应用： 当前大多数 VLA 侧重于家庭或工业环境，但更广泛的应用也是可能的，例如虚拟助手、自动驾驶车辆和农业机器人。各种实施例还可能需要专门的VLA，包括灵巧的手、无人机、四足机器人和人形机器人。一个特别重要的领域是医疗保健，包括手术机器人[218]和护理机器人[219]。医疗保健需要更高的安全和隐私标准，并且可能需要人机交互 (HITL) 控制和联合学习等新技术。此外，由于显着的域差距，医学图像可能需要专门的视觉模型。

七．结论

VLA 模型在使具身智能体能够与物理世界交互并执行用户指令方面具有巨大的前景。本文是第一篇回顾 LVLA 和广义 VLA 的调查。我们的分类提供了三个主要研究方向的高级概述：关键组成部分、控制策略和任务规划器。我们仔细分析和比较他们的技术细节，包括模型架构、训练策略和各个模块。此外，我们还重点介绍了训练和评估 VLA 的重要资源，例如数据集、模拟器和基准测试。我们希望这项调查能够捕捉到 具身智能（Embodied AI） 快速发展的情况并激发未来的研究。

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参考文献

- [1] A. Brohan、N. Brown、J. Carbajal、Y. Chebotar、X. Chen、K. Choromanski、T. Ding、D. Driess、A. Dubey、C. Finn、P. Florence、C. Fu、M. G. Arenas、K. Gopalakrishnan、K. Han、K. Hausman、A. Herzog、J. Hsu、B. Ichter、A. Irpan、 N. J. Joshi、R. Julian、D. Kalashnikov、Y. Kuang、I. Leal、L. Lee、T. E. Lee、S. Levine、Y. Lu、H. Michalewski、I. Mordatch、K. Pertsch、K. Rao、K. Reymann、M. S. Ryoo、G. Salazar、P. Sanketi、P. Sermanet、J. Singh、A. Singh、R. Soricut、H. T. Tran、V. Vanhoucke、Q. Vuong、A. Wahid、S. Welker、P. Wohlhart、J. Wu、F. Xia、T. Xiao、P. Xu、S. Xu、T. Yu 和 B. Zitkovich，“RT-2：视觉语言动作模型将网络知识转移到机器人控制”，CoRR，卷。绝对 /2307.15818，2023。

- [2] A. Krizhevsky、I. Sutskever 和 G. E. Hinton，“使用深度卷积神经网络进行 Imagenet 分类”，载于 NIPS，2012 年，第 1106-1114 页。

- [3] A. Vaswani、N. Shazeer、N. Parmar、J. Uszkoreit、L. Jones、A. N. Gomez、L. Kaiser 和 I. Polosukhin，“注意力就是你所需要的”，载于 NIPS，2017 年，第 5998-6008 页。

- [4] V. Mnih、K. Kavukcuoglu、D. Silver、A. A. Rusu、J. Veness、M. G. Bellemare、A. Graves、M. A. Riedmiller、A. Fidjeland、G. Ostrovski、S. Petersen、C. Beattie、A. Sadik、I. Antonoglou、H. King、D. Kumaran、D. Wierstra、S. Legg 和D. Hassabis，“通过深层 强化学习（Reinforcement Learning） 进行人性化控制”，Nat.，卷。 518，没有。 7540，第 529-533 页 ，2015 年。

- [5] S. Levine、P. Pastor、A. Krizhevsky 和 D. Quillen，“通过大规模数据收集学习机器人抓取的手眼协调”，《ISER》，系列。施普林格高级机器人学报，卷。 1. 施普林格，2016 年，第 173-184 页。

- [6] J. Alayrac、J. Donahue、P. Luc、A. Miech、I. Barr、Y. Hasson、K. Lenc、A. Mensch、K. Millican、M. Reynolds、R. Ring、E. Rutherford、S. Cabi、T. Han、Z. Kong、S. Samangooei、M. Monteiro、J. L. Menick、S. Borgeaud、A. Brock、A. Nematzadeh、S. Sharifzadeh、M. Binkowski、R. Barreira、O. Vinyals、A. Zisserman 和 K. Simonyan，“Flamingo：用于小样本学习的视觉语言模型”，《NeurIPS》，2022 年。

- [7] J. Li、D. Li、S. Savarese 和 S. C. H. Hoi，“BLIP-2：使用冻结图像编码器和大型语言模型进行语言图像预训练”，载于 ICML，第 1 卷。 202.PMLR，2023 年，第 19 730-19 742 页。

- [8] H. Liu、C. Li、Q. Wu 和 Y. J. Lee，“视觉指令调整”，CoRR，卷。绝对/2304.08485，2023。

- [9] W. Huang、F. Xia、T. Xiao、H. Chan、J. Liang、P. Florence、A. Zeng、J. Tompson、I. Mordatch、Y. Chebotar、P. Sermanet、T. Jackson、N. Brown、L. Luu、S. Levine、K. Hausman 和 B. Ichter，“内心独白：通过语言模型规划进行具体推理”，载于_CoRL_，卷。 205.PMLR，2022 年，第 1769-1782 页。

- [10] B. Ichter、A. Brohan、Y. Chebotar、C. Finn、K. Hausman、A. Herzog、D. Ho、J. Ibarz、A. Irpan、E. Jang、R. Julian、D. Kalashnikov、S. Levine、Y. Lu、C. Parada、K. Rao、P. Sermanet、A. Toshev、V. Vanhoucke、F. Xia、T.肖，P. Xu，M. Yan，N. Brown，M. Ahn，O. Cortes，N. Sievers，C. Tan，S. Xu，D. Reyes，J. Rettinghouse，J. Quiambao，P. Pastor，L. Luu，K. Lee，Y. Kuang，S. Jesmonth，N. J. Joshi，K

。 Jeffrey、R. J. Ruano、J. Hsu、K. Gopalakrishnan、B. David、A. Zeng 和 C. K. Fu，“尽我所能，而不是照我说的：机器人 可供性（Affordances） 中的 落地关联（Grounding） 语言”，载于_CoRL_，卷。205.PMLR，2022 年，第 287-318 页。

- [11] D. Driess、F. Xia、M. S. M. Sajjadi、C. Lynch、A. Chowdhery、B. Ichter、A. Wahid、J. Tompson、Q. Vuong、T. Yu、W. Huang、Y. Chebotar、P. Sermanet、D. Duckworth、S. Levine、V. Vanhoucke、K. Hausman、M. Toussaint、K. Greff、A. Zeng、I. Mordatch 和 P. Florence，“Palm-e：一种体现的多模态语言模型”，载于 ICML，第 1 卷。 202.PMLR，2023 年，第 8469-8488 页。

- [12] R. Firoozi、J. Tucker、S. Tian、A. Majumdar、J. Sun、W. Liu、Y. Zhu、S. Song、A. Kapoor、K. Hausman、B. Ichter、D. Driess、J. Wu、C. Lu 和 M. Schwager，“机器人技术中的 基础模型（Foundation Models）：应用、挑战和未来”CoRR，卷。绝对/2312.07843，2023。

[13] J. Wang, Z. Wu, Y. Li, H. Jiang, P. Shu, E. Shi, H. Hu, C. Ma, Y. Liu, X. Wang, Y. Yao, X. Liu, H. Zhao, Z. Liu, H. Dai, L. Zhao, B. Ge, X. Li, T. Liu, and S. Zhang, "Large language models for robotics: Opportunities, challenges, and perspectives," *CoRR*, vol. abs/2401.04334, 2024.

[14] Y. Hu, Q. Xie, V. Jain, J. Francis, J. Patrikar, N. V. Keetha, S. Kim, Y. Xie, T. Zhang, S. Zhao, Y. Q. Chong, C. Wang, K. P. Sycara, M. Johnson-Roberson, D. Batra, X. Wang, S. A. Scherer, Z. Kira, F. Xia, and Y. Bisk, "Toward general-purpose robots via foundation models: A survey and meta-analysis," *CoRR*, vol. abs/2312.08782, 2023.

[15] K. Kawaharazuka, T. Matsushima, A. Gambardella, J. Guo, C. Paxton, and A. Zeng, "Real-world robot applications of foundation models: a review," *Adv. Robotics*, vol. 38, no. 18, pp. 1232–1254, 2024.

[16] L. Chen, K. Lu, A. Rajeswaran, K. Lee, A. Grover, M. Laskin, P. Abbeel, A. Srinivas, and I. Mordatch, "Decision transformer: Reinforcement learning via sequence modeling," in *NeurIPS*, 2021, pp. 15 084–15 097.

[17] M. Janner, Q. Li, and S. Levine, "Offline reinforcement learning as one big sequence modeling problem," in *NeurIPS*, 2021, pp. 1273–1286.

[18] S. E. Reed, K. Zolna, E. Parisotto, S. G. Colmenarejo, A. Novikov, G. Barth-Maron, M. Gimenez, Y. Sulsky, J. Kay, J. T. Springenberg, T. Eccles, J. Bruce, A. Razavi, A. Edwards, N. Heess, Y. Chen, R. Hadsell, O. Vinyals, M. Bordbar, and N. de Freitas, "A generalist agent," *Trans. Mach. Learn. Res.*, vol. 2022, 2022.

[19] A. A. Physical Intelligence and, R. Aniceto, A. Balakrishna, K. Black, K. Conley, G. Connors, J. Darpinian, K. Dhabalia, J. DiCarlo, D. Driess, M. Equi, A. Esmail, Y. Fang, C. Finn, C. Glossop, T. Godden, I. Goryachev, L. Groom, H. Hancock, K. Hausman, G. Hussein, B. Ichter, S. Jakubczak, R. Jen, T. Jones, B. Katz, L. Ke, C. Kuchi, M. Lamb, D. LeBlanc, S. Levine, A. Li-Bell, Y. Lu, V. Mano, M. Mothukuri, S. Nair, K. Pertsch, A. Z. Ren, C. Sharma, L. X. Shi, L. Smith, J. T. Springenberg, K. Stachowicz, W. Stoeckle, A. Swerdlow, J. Tanner, M. Torne, Q. Vuong, A. Walling, H. Wang, B. Williams, S. Yoo, L. Yu, U. Zhilinsky, and Z. Zhou, "$\pi^*_{0.6}$: a VLA that learns, from experience," *CoRR*, vol. abs/2511.14759, 2025.

[20] A. Hiranaka, M. Hwang, S. Lee, C. Wang, L. Fei-Fei, J. Wu, and R. Zhang, "Primitive skill-based robot learning from human evaluative feedback," in *IROS*, 2023, pp. 7817–7824.

[21] N. Shinn, F. Cassano, A. Gopinath, K. Narasimhan, and S. Yao, "Reflexion: language agents with verbal reinforcement learning," in *NeurIPS*, 2023.

[22] Y. J. Ma, W. Liang, G. Wang, D. Huang, O. Bastani, D. Jayaraman, Y. Zhu, L. Fan, and A. Anandkumar, "Eureka: Human-level reward design via coding large language models," in *ICLR*, 2024.

[23] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever, "Learning transferable visual models from natural language supervision," in *ICML*, vol. 139. PMLR, 2021, pp. 8748–8763.

[24] S. Nair, A. Rajeswaran, V. Kumar, C. Finn, and A. Gupta, "R3M: A universal visual representation for robot manipulation," in *CoRL*, vol. 205. PMLR, 2022, pp. 892–909.

[25] Y. J. Ma, S. Sodhani, D. Jayaraman, O. Bastani, V. Kumar, and A. Zhang, "VIP: towards universal visual reward and representation via value-implicit pre-training," in *ICLR*, 2023.

[26] I. Radosavovic, T. Xiao, S. James, P. Abbeel, J. Malik, and T. Darrell, "Real-world robot learning with masked visual pre-training," in *CoRL*, vol. 205. PMLR, 2022, pp. 416–426.

[27] J. Devlin, M. Chang, K. Lee, and K. Toutanova, "BERT: pre-training of deep bidirectional transformers for language understanding," in *NAACL-HLT (1)*. Association for Computational Linguistics, 2019, pp. 4171–4186.

[28] I. Radosavovic, B. Shi, L. Fu, K. Goldberg, T. Darrell, and J. Malik, "Robot learning with sensorimotor pre-training," in *CoRL*, vol. 229. PMLR, 2023, pp. 683–693.

[29] M. Shridhar, L. Manuelli, and D. Fox, "Cliport: What and where pathways for robotic manipulation," in *CoRL*, vol. 164. PMLR, 2021, pp. 894–906.

[30] A. Khandelwal, L. Weihs, R. Mottaghi, and A. Kembhavi, "Simple but effective: CLIP embeddings for embodied AI," in *CVPR*. IEEE, 2022, pp. 14 809–14 818.

[31] S. Y. Gadre, M. Wortsman, G. Ilharco, L. Schmidt, and S. Song, "Cows on pasture: Baselines and benchmarks for language-driven zero-shot object navigation," in *CVPR*. IEEE, 2023, pp. 23 171–23 181.

[32] A. Majumdar, K. Yadav, S. Arnaud, Y. J. Ma, C. Chen, S. Silwal, A. Jain, V. Berges, T. Wu, J. Vakil, P. Abbeel, J. Malik, D. Batra, Y. Lin, O. Maksymets, A. Rajeswaran, and F. Meier, "Where are we in the search for an artificial visual cortex for embodied intelligence?" in *NeurIPS*, 2023.

[33] S. Karamcheti, S. Nair, A. S. Chen, T. Kollar, C. Finn, D. Sadigh, and P. Liang, "Language-driven representation learning for robotics," in *RSS*, 2023.

[34] M. Oquab, T. Darcet, T. Moutakanni, H. V. Vo, M. Szafraniec, V. Khalidov, P. Fernandez, D. Haziza, F. Massa, A. El-Nouby, M. Assran, N. Ballas, W. Galuba, R. Howes, P. Huang, S. Li, I. Misra, M. Rabbat, V. Sharma, G. Synnaeve, H. Xu, H. Jégou, J. Mairal, P. Labatut, A. Joulin, and P. Bojanowski, "Dinov2: Learning robust visual features without supervision," *Trans. Mach. Learn. Res.*, vol. 2024, 2024.

[35] M. J. Kim, K. Pertsch, S. Karamcheti, T. Xiao, A. Balakrishna, S. Nair, R. Rafailov, E. P. Foster, G. Lam, P. Sanketi, Q. Vuong, T. Kollar, B. Burchfiel, R. Tedrake, D. Sadigh, S. Levine, P. Liang, and C. Finn, "Openvla: An open-source vision-language-action model," *CoRR*, vol. abs/2406.09246, 2024.

[36] W. Huang, C. Wang, Y. Li, R. Zhang, and L. Fei-Fei, "Rekep: Spatio-temporal reasoning of relational keypoint constraints for robotic manipulation," *CoRR*, vol. abs/2409.01652, 2024.

[37] M. Assran, Q. Duval, I. Misra, P. Bojanowski, P. Vincent, M. G. Rabbat, Y. LeCun, and N. Ballas, "Self-supervised learning from images with a joint-embedding predictive architecture," in *CVPR*. IEEE, 2023, pp. 15 619–15 629.

[38] J. Shang, K. Schmeckpeper, B. B. May, M. V. Minniti, T. Kelestemur, D. Watkins, and L. Herlant, "Theia: Distilling diverse vision foundation models for robot learning," in *CoRL*, ser. Proceedings of Machine Learning Research, vol. 270. PMLR, 2024, pp. 724–748.

[39] S. Parisi, A. Rajeswaran, S. Purushwalkam, and A. Gupta, "The unsurprising effectiveness of pre-trained vision models for control," in *ICML*, vol. 162. PMLR, 2022, pp. 17 359–17 371.

[40] Y. LeCun, "A path towards autonomous machine intelligence," 2022. [Online]. Available: https://openreview.net/pdf?id=BZ5a1r-kVsf

[41] W. Shen, G. Yang, A. Yu, J. Wong, L. P. Kaelbling, and P. Isola, "Distilled feature fields enable few-shot language-guided manipulation," in *CoRL*, vol. 229. PMLR, 2023, pp. 405–424.

[42] Y. Hong, H. Zhen, P. Chen, S. Zheng, Y. Du, Z. Chen, and C. Gan, "3d-llm: Injecting the 3d world into large language models," in *NeurIPS*, 2023.

[43] B. Kerbl, G. Kopanas, T. Leimkühler, and G. Drettakis, "3d gaussian splatting for real-time radiance field rendering," *ACM Trans. Graph.*, vol. 42, no. 4, pp. 139:1–139:14, 2023.

[44] M. Qin, W. Li, J. Zhou, H. Wang, and H. Pfister, "Langsplat: 3d language gaussian splatting," in *CVPR*. IEEE, 2024, pp. 20 051–20 060.

[45] A. Thankaraj and L. Pinto, "That sounds right: Auditory self-supervision for dynamic robot manipulation," in *CoRL*, vol. 229. PMLR, 2023, pp. 1036–1049.

[46] Y. Jing, X. Zhu, X. Liu, Q. Sima, T. Yang, Y. Feng, and T. Kong, "Exploring visual pre-training for robot manipulation: Datasets, models and methods," in *IROS*, 2023, pp. 11 390–11 395.

[47] F. Liu, H. Liu, A. Grover, and P. Abbeel, "Masked autoencoding for scalable and generalizable decision making," in *NeurIPS*, 2022.

[48] J. Li, Q. Gao, M. Johnston, X. Gao, X. He, H. Shi, S. Shakiah, R. Ghanadan, and W. Y. Wang, "Mastering robot manipulation with multimodal prompts through pretraining and multi-task fine-tuning," in *ICML*, 2024.

[49] Y. Sun, S. Ma, R. Madaan, R. Bonatti, F. Huang, and A. Kapoor, "SMART: self-supervised multi-task pretraining with control transformers," in *ICLR*, 2023.

[50] R. Bonatti, S. Vemprala, S. Ma, F. Frujeri, S. Chen, and A. Kapoor, "PACT: perception-action causal transformer for autoregressive robotics pre-training," in *IROS*, 2023, pp. 3621–3627.

[51] B. Baker, I. Akkaya, P. Zhokhov, J. Huizinga, J. Tang, A. Ecoffet, B. Houghton, R. Sampedro, and J. Clune, "Video pretraining (VPT): learning to act by watching unlabeled online videos," in *NeurIPS*, 2022.

[52] H. Wu, Y. Jing, C. Cheang, G. Chen, J. Xu, X. Li, M. Liu, H. Li, and T. Kong, "Unleashing large-scale video generative pre-training for visual robot manipulation," in *ICLR*, 2024.

[53] D. Hafner, T. P. Lillicrap, J. Ba, and M. Norouzi, "Dream to control: Learning behaviors by latent imagination," in *ICLR*, 2020.

[54] D. Hafner, T. P. Lillicrap, M. Norouzi, and J. Ba, "Mastering atari with discrete world models," in *ICLR*, 2021.

[55] D. Hafner, J. Pasukonis, J. Ba, and T. P. Lillicrap, "Mastering diverse domains through world models," *CoRR*, vol. abs/2301.04104, 2023.

原文第 18 页 | 中文翻译初稿

- [13] J. Wang、Z. Wu、Y. Li、H. Jiang、P. Shu、E. Shi、H. Hu、C. Ma、Y. Liu、X. Wang、Y. Yao、X. Liu、H. Zhu、Z. Liu、H. Dai、L. Zhao、B. Ge、X. Li、T. Liu 和 S. Zhang，“机器人技术的大型语言模型：机遇、挑战和观点”CoRR，卷。绝对/2401.04334，2024。

- [14] Y. Hu、Q. Xie、V. Jain、J. Francis、J. Patrikar、N. V. Keetha、S. Kim、Y. Xie、T. Zhang、S. Zhu、Y. Q. Chong、C. Wang、K. P. Sycara、M. Johnson-Roberson、D. Batra、X. Wang、S. A. Scherer、Z. Kira、F. Xia 和 Y. Bisk，“走向通用（General-Purpose）机器人通过 基础模型（Foundation Models）：一项调查和荟萃分析，”CoRR，卷。绝对/2312.08782，2023。

- [15] K. Kawaharazuka、T. Matsushima、A. Gambardella、J.Guo、C. Paxton 和 A. Zeng，“基础模型（Foundation Models）的真实机器人应用：综述”，Adv。机器人学，卷。 38，没有。 18，第 1232-1254 页，2024 年。

- [16] L. Chen、K. Lu、A. Rajeswaran、K. Lee、A. Grover、M. Laskin、P. Abbeel、A. Srinivas 和 I. Mordatch，“决策转换器：通过序列建模的 强化学习（Reinforcement Learning）”，载于 NeurIPS，2021 年，第 15 084-15 097 页。

- [17] M. Janner、Q. Li 和 S. Levine，“离线 强化学习（Reinforcement Learning）作为一个大序列建模问题”，载于 NeurIPS，2021 年，第 1273-1286 页。

- [18] S. E. Reed、K. Zolna、E. Parisotto、S. G. Colmenarejo、A. Novikov、G. Barth-Maron、M. Gimenez、Y. Sulsky、J. Kay、J. T. Springenberg、T. Eccles、J. Bruce、A. Razavi、A. Edwards、N. Heess、Y. Chen、R. Hadsell、O. Vinyals、M. Bordbar 和 N. de Freitas，“多才多艺的特工”，Trans。马赫。学习。 Res.，卷。 2022 年，2022 年。

- [19] A. A. 身体智力和 R. Aniceto、A. Balakrishna、K. Black、K. Conley、G. Connors、J. Darpinian、K. Dhabalia、J. DiCarlo、D. Driess、M. Equi、A. Esmail、Y. Fang、C. Finn、C. Glossop、T. Godden、I. Goryachev、L. Groom、H. Hancock、K. Hausman、G. Hussein、B. Ichter、S. Jakubczak、R. Jen、T. Jones、B. Katz、L. Ke、C. Kuchi、M. Lamb、D. LeBlanc、S. Levine、A. Li-Bell、Y. Lu、V. Mano、M. Mothukuri、S. Nair、K. Pertsch、A. Z. Ren、C. Sharma、L. X. Shi、L. Smith、 J. T. Springenberg、K. Stachowicz、W. Stoeckle、A. Swerdlow、J. Tanner、M. Torne、Q. Vuong、A. Walling、H. Wang、B. Williams、S. Yoo、L. Yu、U.zhilinsky 和 Z. Zhou，“π* 0.6：从经验中学习的 VLA”，CoRR，卷。绝对/2511.14759，2025。

- [20] A. Hiranaka、M. Hwang、S. Lee、C. Wang、L. Fei-Fei、J. Wu 和 R. Zhu，“基于人类评价反馈的基于原始技能的机器人学习”，载于 IROS，2023 年，第 7817-7824 页。

- [21] N. Shinn、F. Cassano、A. Gopinath、K. Narasimhan 和 S. Yao，“反射：具有言语 强化学习（Reinforcement Learning）的语言智能体”，载于 NeurIPS，2023 年。

- [22] Y. J. Ma、W. Liang、G. Wang、D. Huang、O. Bastani、D. Jayaraman、Y. Zhu、L. Fan 和 A. Anandkumar，“Eureka：通过编码大型语言模型进行人类级奖励设计”，载于 ICLR，2024 年。

- [23] A. Radford、J. W. Kim、C. Hallacy、A. Ramesh、G. Goh、S. Agarwal、G. Sastry、A. Askell、P. Mishkin、J. Clark、G. Krueger 和 I. Sutskever，“从自然语言监督中学习可迁移视觉模型”，载于 ICML ，第 1 卷。 139.PMLR，2021 年，第 8748-8763 页。

- [24] S. Nair、A. Rajeswaran、V. Kumar、C. Finn 和 A

。 Gupta，“R3M：机器人操作的通用视觉表示”，《CoRL》，卷。 205.PMLR，2022 年，第 892-909 页。

- [25] Y. J. Ma、S. Sodhani、D. Jayaraman、O. Bastani、V. Kumar 和 A. Zhu，“VIP：通过价值隐式预训练实现通用视觉奖励和表示”，载于 ICLR，2023 年。

- [26] I.Radosavovic、T.Xiao、S.James、P.Abbeel、J.Malik 和 T.Darrell，“使用蒙版视觉预训练进行现实世界机器人学习”，载于 CoRL，第 1 卷。 205.PMLR，2022 年，第 416-426 页。

- [27] J. Devlin、M. Chang、K. Lee 和 K. Toutanova，“BERT：用于语言理解的深度双向转换器的预训练”，载于_NAACL-HLT (1)_。计算语言学协会，2019 年，第 4171-4186 页。

- [28] I. Radosavovic、B. Shi、L. Fu、K. Goldberg、T. Darrell 和 J. Malik，“通过感觉运动预训练进行机器人学习”，载于 CoRL，第 1 卷。 229.PMLR，2023 年，第 683-693 页。

- [29] M. Shridhar、L. Manuelli 和 D. Fox，“Cliport：机器人操作的路径和位置”，载于_CoRL_，卷。164.PMLR，2021 年，第 894-906 页。

- [30] A. Khandelwal、L. Weihs、R. Mottaghi 和 A. Kembhavi，“简单但有效：具身智能（Embodied AI）的 CLIP 嵌入”，参见 CVPR。 IEEE，2022 年，第 14 809-14 818 页。

- [31] S. Y. Gadre、M. Wortsman、G. Ilharco、L. Schmidt 和 S. Song，“牧场上的奶牛：语言驱动的零样本对象导航的基线和基准”，载于_CVPR_。 IEEE，2023 年，第 23 171-23 181 页。

- [32] A. Majumdar、K. Yadav、S. Arnaud、Y. J. Ma、C. Chen、S. Silwal、A. Jain、V. Berges、T. Wu、J. Vakil、P. Abbeel、J. Malik、D. Batra、

Y. Lin、O. Maksymets、A. Rajeswaran 和 F. Meier，“我们在寻找 具身智能（Embodied Intelligence） 的人工视觉皮层时处于什么阶段？”于_NeurIPS_，2023 年。

- [33] S. Karamcheti、S. Nair、A. S. Chen、T. Kollar、C. Finn、D. Sadigh 和 P. Liang，“机器人技术的语言驱动表示学习”，载于 RSS，2023 年。

- [34] M. Oquab、T. Darcet、T. Moutakanni、H. V. Vo、M. Szafraniec、V. Khalidov、P. Fernandez、D. Haziza、F. Massa、A. El-Nouby、M. Assran、N. Ballas、W. Galuba、R. Howes、P. Huang、S. Li、I. Misra、M. Rabbat、V. Sharma， G. Synnaeve、H. Xu、H. J´egou、J. Mairal、P. Labatut、A. Joulin 和 P. Bojanowski，“Dinov2：在没有监督的情况下学习鲁棒的视觉特征”，Trans。马赫。学习。 Res.，卷。 2024 年，2024 年。

- [35] M. J. Kim、K. Pertsch、S. Karamcheti、T. Xiao、A. Balakrishna、S. Nair、R. Rafailov、E. P. Foster、G. Lam、P. Sanketi、Q. Vuong、T. Kollar、B. Burchfiel、R. Tedrake、D. Sadigh、S. Levine、P. Liang 和 C. Finn，“Openvla：开源 视觉语言动作（Vision-Language-Action） 模型，”CoRR，卷。绝对/2406.09246，2024。

- [36] W. Huang、C. Wang、Y. Li、R. Zhang 和 L. Fei-Fei，“Rekep：机器人操作关系关键点约束的时空推理”，CoRR，卷。绝对/2409.01652，2024。

- [37] M. Assran、Q. Duval、I. Misra、P. Bojanowski、P. Vincent、M. G. Rabbat、Y. LeCun 和 N. Ballas，“使用联合嵌入预测架构从图像中进行自我监督学习”，载于_CVPR_。 IEEE，2023 年，第 15 619-15 629 页。

- [38] J. Shang、K. Schmeckpeper、B. B. May、M. V. Minniti、T. Kelestemur、D. Watkins 和 L. Herlant，“Theia：为机器人学习提炼多样化的视觉 基础模型（Foundation Models）”，载于_CoRL_，系列。机器学习研究论文集，卷。 270

。 PMLR，2024 年，第 724-748 页。

- [39] S. Parisi、A. Rajeswaran、S. Purushwalkam 和 A. Gupta，“用于控制的预训练视觉模型的有效性不足为奇”，载于 ICML，第 1 卷。 162.PMLR，2022 年，第 17 359-17 371 页。

- [40] Y. LeCun，“通往自主机器智能的道路”，2022 年。[在线]。可用：https://openreview.net/pdf?id=BZ5a1r-kVsf

- [41] W. Shen、G. Yang、A. Yu、J. Wong、L. P. Kaelbling 和 P. Isola，“蒸馏特征字段可实现少量语言引导操作”，载于 CoRL，第 1 卷。 229.PMLR，2023 年，第 405-424 页。

- [42] Y. Hong、H.zhen、P.Chen、S.Zheng、Y.Du、Z.Chen 和 C.Gan，“3dllm：将 3d 世界注入大型语言模型”，《NeurIPS》，2023 年。

- [43] B. Kerbl、G. Kopanas、T. Leimkühler 和 G. Drettakis，“用于实时辐射场渲染的 3d 高斯泼溅”，ACM Trans。图.，卷。 42、没有。 4，第 139:1-139:14，2023 年。

- [44] M.qin、W.Li、J.Zhou、H.Wang 和 H.Pfister，“Langsplat：3d 语言高斯分布”，载于 CVPR。 IEEE，2024 年，第 20 051-20 060 页。

- [45] A.Thankaraj 和 L.Pinto，“听起来不错：动态机器人操作的听觉自我监督”，载于_CoRL_，卷。229.PMLR，2023 年，第 1036-1049 页。

- [46] Y. Jing、X. Zhu、X. Liu、Q. Sima、T. Yang、Y. Feng 和 T. Kong，“探索机器人操作的视觉预训练：数据集、模型和方法”，载于 IROS，2023 年，第 11 390-11 395 页。

- [47] F. Liu、H. Liu、A. Grover 和 P. Abbeel，“用于可扩展和泛化决策的屏蔽自动编码”，载于 NeurIPS，2022 年。

- [48] J. Li、Q. Gau、M. Johnston、X. Gau、X. He、H. Shi、S. Shakiah、R. Ghanadan 和 W. Y. Wang，“通过预训练和多任务微调掌握多模式提示的机器人操作”，载于 ICML，2024 年。

- [49] Y. Sun、S. Ma、R. Madaan、R. Bonatti、F. Huang 和 A. Kapoor，“SMART：使用控制变压器进行自监督多任务预训练”，载于 ICLR，2023 年。

- [50] R. Bonatti、S. Vemprala、S. Ma、F. Frujeri、S. Chen 和 A. Kapoor，“PACT：自回归机器人预训练的感知-动作因果变换器”，载于 IROS，2023 年，第 3621-3627 页。

- [51] B. Baker、I. Akkaya、P. Zhokhov、J. Huizinga、J. Tang、A. Ecoffet、B. Houghton、R. Sampedro 和 J. Clune，“视频预训练 (VPT)：通过观看未标记的在线视频学习行动”，《NeurIPS》，2022 年。

- [52] H. Wu、Y. Jing、C. Cheang、G. Chen、J. Xu、X. Li、M. Liu、H. Li 和 T. Kong，“为视觉机器人操作释放大规模视频生成预训练”，载于 ICLR，2024 年。

- [53] D. Hafner、T. P. Lillicrap、J. Ba 和 M. Norouzi，“控制梦想：通过潜在想象力学习行为”，《ICLR》，2020 年。

- [54] D. Hafner、T. P. Lillicrap、M. Norouzi 和 J. Ba，“使用离散 世界模型（World Models） 掌握 atari”，载于 ICLR，2021 年。

- [55] D. Hafner、J. Pasukonis、J. Ba 和 T. P. Lillicrap，“通过 世界模型（World Models） 掌握不同领域”，CoRR，卷。绝对/2301.04104，2023。

[56] P. Wu, A. Escontrela, D. Hafner, P. Abbeel, and K. Goldberg, "Daydreamer: World models for physical robot learning," in *CoRL*, vol. 205. PMLR, 2022, pp. 2226–2240.

[57] V. Micheli, E. Alonso, and F. Fleuret, "Transformers are sample-efficient world models," in *ICLR*, 2023.

[58] J. Robine, M. Höftmann, T. Uelwer, and S. Harmeling, "Transformer-based world models are happy with 100k interactions," in *ICLR*, 2023.

[59] K. Nottingham, P. Ammanabrolu, A. Suhr, Y. Choi, H. Hajishirzi, S. Singh, and R. Fox, "Do embodied agents dream of pixelated sheep: Embodied decision making using language guided world modelling," in *ICML*, vol. 202. PMLR, 2023, pp. 26 311–26 325.

[60] Z. Song, Y. Zhang, and I. King, "No change, no gain: Empowering graph neural networks with expected model change maximization for active learning," in *NeurIPS*, 2023.

[61] Y. Ma, Z. Song, X. Hu, J. Li, Y. Zhang, and I. King, "Graph component contrastive learning for concept relatedness estimation," in *AAAI*. AAAI Press, 2023, pp. 13 362–13 370.

[62] L. Guan, K. Valmeekam, S. Sreedharan, and S. Kambhampati, "Leveraging pre-trained large language models to construct and utilize world models for model-based task planning," in *NeurIPS*, 2023.

[63] B. Liu, Y. Jiang, X. Zhang, Q. Liu, S. Zhang, J. Biswas, and P. Stone, "LLM+P: empowering large language models with optimal planning proficiency," *CoRR*, vol. abs/2304.11477, 2023.

[64] S. Hao, Y. Gu, H. Ma, J. J. Hong, Z. Wang, D. Z. Wang, and Z. Hu, "Reasoning with language model is planning with world model," in *EMNLP*, 2023, pp. 8154–8173.

[65] M. Hu, Y. Mu, X. Yu, M. Ding, S. Wu, W. Shao, Q. Chen, B. Wang, Y. Qiao, and P. Luo, "Tree-planner: Efficient close-loop task planning with large language models," in *ICLR*, 2024.

[66] Z. Zhao, W. S. Lee, and D. Hsu, "Large language models as commonsense knowledge for large-scale task planning," in *NeurIPS*, 2023.

[67] OpenAI. (2024) Video generation models as world simulators. [Online]. Available: https://openai.com/index/video-generation-models-as-world-simulators/

[68] Z. Zhu, X. Wang, W. Zhao, C. Min, N. Deng, M. Dou, Y. Wang, B. Shi, K. Wang, C. Zhang, Y. You, Z. Zhang, D. Zhao, L. Xiao, J. Zhao, J. Lu, and G. Huang, "Is sora a world simulator? A comprehensive survey on general world models and beyond," *CoRR*, vol. abs/2405.03520, 2024.

[69] J. Bruce, M. D. Dennis, A. Edwards, J. Parker-Holder, Y. Shi, E. Hughes, M. Lai, A. Mavalankar, R. Steigerwald, C. Apps, Y. Aytar, S. Bechtle, F. M. P. Behbahani, S. C. Y. Chan, N. Heess, L. Gonzalez, S. Osindero, S. Ozair, S. E. Reed, J. Zhang, K. Zolna, J. Clune, N. de Freitas, S. Singh, and T. Rocktäschel, "Genie: Generative interactive environments," in *ICML*, 2024.

[70] H. Zhen, X. Qiu, P. Chen, J. Yang, X. Yan, Y. Du, Y. Hong, and C. Gan, "3d-vla: A 3d vision-language-action generative world model," in *ICML*, 2024.

[71] S. Yang, Y. Du, S. K. S. Ghasemipour, J. Tompson, L. P. Kaelbling, D. Schuurmans, and P. Abbeel, "Learning interactive real-world simulators," in *ICLR*, 2024.

[72] J. Xiang, T. Tao, Y. Gu, T. Shu, Z. Wang, Z. Yang, and Z. Hu, "Language models meet world models: Embodied experiences enhance language models," in *NeurIPS*, 2023.

[73] T. Kojima, S. S. Gu, M. Reid, Y. Matsuo, and Y. Iwasawa, "Large language models are zero-shot reasoners," in *NeurIPS*, 2022.

[74] J. Wei, X. Wang, D. Schuurmans, M. Bosma, B. Ichter, F. Xia, E. H. Chi, Q. V. Le, and D. Zhou, "Chain-of-thought prompting elicits reasoning in large language models," in *NeurIPS*, 2022.

[75] G. Lu, Z. Wang, C. Liu, J. Lu, and Y. Tang, "Thinkbot: Embodied instruction following with thought chain reasoning," *CoRR*, vol. abs/2312.07062, 2023.

[76] S. Yao, J. Zhao, D. Yu, N. Du, I. Shafran, K. R. Narasimhan, and Y. Cao, "React: Synergizing reasoning and acting in language models," in *ICLR*, 2023.

[77] Z. Wang, A. Liu, H. Lin, J. Li, X. Ma, and Y. Liang, "RAT: retrieval augmented thoughts elicit context-aware reasoning in long-horizon generation," *CoRR*, vol. abs/2403.05313, 2024.

[78] M. Zawalski, W. Chen, K. Pertsch, O. Mees, C. Finn, and S. Levine, "Robotic control via embodied chain-of-thought reasoning," *CoRR*, vol. abs/2407.08693, 2024.

[79] Q. Zhao, Y. Lu, M. J. Kim, Z. Fu, Z. Zhang, Y. Wu, Z. Li, Q. Ma, S. Han, C. Finn, A. Handa, T. Lin, G. Wetzstein, M. Liu, and D. Xiang, "Cot-vla: Visual chain-of-thought reasoning for vision-language-action models," in *CVPR*, 2025, pp. 1702–1713.

[80] M. Nakamoto, O. Mees, A. Kumar, and S. Levine, "Steering your generalists: Improving robotic foundation models via value guidance," in *CoRL*, ser. Proceedings of Machine Learning Research, vol. 270. PMLR, 2024, pp. 4996–5013.

[81] J. Kwok, C. Agia, R. Sinha, M. Foutter, S. Li, I. Stoica, A. Mirhoseini, and M. Pavone, "Robomonkey: Scaling test-time sampling and verification for vision-language-action models," *CoRR*, vol. abs/2506.17811, 2025.

[82] E. Jang, A. Irpan, M. Khansari, D. Kappler, F. Ebert, C. Lynch, S. Levine, and C. Finn, "BC-Z: zero-shot task generalization with robotic imitation learning," in *CoRL*, vol. 164. PMLR, 2021, pp. 991–1002.

[83] C. Lynch and P. Sermanet, "Language conditioned imitation learning over unstructured data," in *RSS*, 2021.

[84] O. Mees, L. Hermann, and W. Burgard, "What matters in language conditioned robotic imitation learning over unstructured data," *IEEE Robotics Autom. Lett.*, vol. 7, no. 4, pp. 11 205–11 212, 2022.

[85] O. Mees, J. Borja-Diaz, and W. Burgard, "Grounding language with visual affordances over unstructured data," in *ICRA*. IEEE, 2023, pp. 11 576–11 582.

[86] Y. Du, S. Yang, B. Dai, H. Dai, O. Nachum, J. Tenenbaum, D. Schuurmans, and P. Abbeel, "Learning universal policies via text-guided video generation," in *NeurIPS*, 2023.

[87] P. Sharma, B. Sundaralingam, V. Blukis, C. Paxton, T. Hermans, A. Torralba, J. Andreas, and D. Fox, "Correcting robot plans with natural language feedback," in *RSS*, 2022.

[88] C. Lynch, A. Wahid, J. Tompson, T. Ding, J. Betker, R. Baruch, T. Armstrong, and P. Florence, "Interactive language: Talking to robots in real time," *CoRR*, vol. abs/2210.06407, 2022.

[89] P. Guhur, S. Chen, R. G. Pinel, M. Tapaswi, I. Laptev, and C. Schmid, "Instruction-driven history-aware policies for robotic manipulations," in *CoRL*, vol. 205. PMLR, 2022, pp. 175–187.

[90] M. Shridhar, L. Manuelli, and D. Fox, "Perceiver-actor: A multi-task transformer for robotic manipulation," in *CoRL*, vol. 205. PMLR, 2022, pp. 785–799.

[91] T. Gervet, Z. Xian, N. Gkanatsios, and K. Fragkiadaki, "Act3d: 3d feature field transformers for multi-task robotic manipulation," in *CoRL*, vol. 229. PMLR, 2023, pp. 3949–3965.

[92] A. Goyal, J. Xu, Y. Guo, V. Blukis, Y. Chao, and D. Fox, "RVT: robotic view transformer for 3d object manipulation," in *CoRL*, vol. 229. PMLR, 2023, pp. 694–710.

[93] A. Goyal, V. Blukis, J. Xu, Y. Guo, Y. Chao, and D. Fox, "RVT-2: learning precise manipulation from few demonstrations," *CoRR*, vol. abs/2406.08545, 2024.

[94] W. Yuan, J. Duan, V. Blukis, W. Pumacay, R. Krishna, A. Murali, A. Mousavian, and D. Fox, "Robopoint: A vision-language model for spatial affordance prediction for robotics," *CoRR*, vol. abs/2406.10721, 2024.

[95] K. Bousmalis, G. Vezzani, D. Rao, C. Devin, A. X. Lee, M. Bauzá, T. Davchev, Y. Zhou, A. Gupta, A. Raju, A. Laurens, C. Fantacci, V. Dalibard, M. Zambelli, M. F. Martins, R. Pevceviciute, M. Blokzijl, M. Denil, N. Batchelor, T. Lampe, E. Parisotto, K. Zolna, S. E. Reed, S. G. Colmenarejo, J. Scholz, A. Abdolmaleki, O. Groth, J. Regli, O. Sushkov, T. Rothörl, J. E. Chen, Y. Aytar, D. Barker, J. Ortiz, M. A. Riedmiller, J. T. Springenberg, R. Hadsell, F. Nori, and N. Heess, "Robocat: A self-improving foundation agent for robotic manipulation," *CoRR*, vol. abs/2306.11706, 2023.

[96] A. Stone, T. Xiao, Y. Lu, K. Gopalakrishnan, K. Lee, Q. Vuong, P. Wohlhart, B. Zitkovich, F. Xia, C. Finn, and K. Hausman, "Open-world object manipulation using pre-trained vision-language models," *CoRR*, vol. abs/2303.00905, 2023.

[97] A. Brohan, N. Brown, J. Carbajal, Y. Chebotar, J. Dabis, C. Finn, K. Gopalakrishnan, K. Hausman, A. Herzog, J. Hsu, J. Ibarz, B. Ichter, A. Irpan, T. Jackson, S. Jesmonth, N. J. Joshi, R. Julian, D. Kalashnikov, Y. Kuang, I. Leal, K. Lee, S. Levine, Y. Lu, U. Malla, D. Manjunath, I. Mordatch, O. Nachum, C. Parada, J. Peralta, E. Perez, K. Pertsch, J. Quiambao, K. Rao, M. S. Ryoo, G. Salazar, P. R. Sanketi, K. Sayed, J. Singh, S. Sontakke, A. Stone, C. Tan, H. T. Tran, V. Vanhoucke, S. Vega, Q. Vuong, F. Xia, T. Xiao, P. Xu, S. Xu, T. Yu, and B. Zitkovich, "RT-1: robotics transformer for real-world control at scale," in *RSS*, 2023.

[98] Y. Chebotar, Q. Vuong, A. Irpan, K. Hausman, F. Xia, Y. Lu, A. Kumar, T. Yu, A. Herzog, K. Pertsch, K. Gopalakrishnan, J. Ibarz, O. Nachum, S. Sontakke, G. Salazar, H. T. Tran, J. Peralta, C. Tan, D. Manjunath, J. Singh, B. Zitkovich, T. Jackson, K. Rao, C. Finn, and S. Levine, "Q-transformer: Scalable offline reinforcement learning via autoregressive q-functions," *CoRR*, vol. abs/2309.10150, 2023.

[99] J. Gu, S. Kirmani, P. Wohlhart, Y. Lu, M. G. Arenas, K. Rao, W. Yu, C. Fu, K. Gopalakrishnan, Z. Xu, P. Sundaresan, P. Xu,

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- [56] P. Wu、A. Escontrela、D. Hafner、P. Abbeel 和 K. Goldberg，“Daydreamer：用于物理机器人学习的 世界模型（World Models）”，载于_CoRL_，卷。 205.PMLR，2022 年，第 2226-2240 页。

- [57] V. Micheli、E. Alonso 和 F. Fleuret，“Transformers 具有样本效率 世界模型（World Models）”，载于 ICLR，2023 年。

- [58] J. Robine、M. Hěftmann、T. Uelwer 和 S. Harmeling，“基于 Transformer 的 世界模型（World Models）对 10 万次交互感到满意”，载于 ICLR，2023 年。

- [59] K. Nottingham、P. Ammanabrolu、A. Suhr、Y. Choi、H. Hajishirzi、S. Singh 和 R. Fox，“具身智能体会梦想像素化的羊吗：使用语言引导的世界建模进行具身决策”，载于 ICML，第 1 卷。 202.PMLR，2023 年，第 26 311-26 325 页。

- [60] Z. Song、Y. Zhang 和 I. King，“没有变化，就没有收获：通过预期模型变化最大化来增强图神经网络的主动学习能力”，载于 NeurIPS，2023 年。

- [61] Y. Ma、Z. Song、X. Hu、J. Li、Y. Zhang 和 I. King，“用于概念相关性估计的图组件对比学习”，载于_AAAI_。 AAAI 出版社，2023 年，第 13 362-13 370 页。

- [62] L.guan、K.Valmeekam、S.Sreedharan 和 S.Kambhampati，“利用预训练的大型语言模型构建和利用 世界模型（World Models）进行基于模型的任务规划”，《NeurIPS》，2023 年。

- [63] B. Liu、Y. Jiang、X. Zhang、Q. Liu、S. Zhang、J. Biswas 和 P. Stone，“LLM+P：以最佳规划能力增强大型语言模型”，CoRR，卷。 ABS/2304.11477，2023。

- [64] S.Hao、Y.Gu、H.Ma、J.J.Hong、Z.Wang、D.Z.Wang、Z.Hu，“语言模型推理正在用 世界模型（World Model）进行规划”，载于 EMNLP，2023 年，第 8154-8173 页。

- [65] M. Hu、Y. Mu、X. Yu、M. Ding、S. Wu、W. Shao、Q. Chen、B. Wang、Y. Qiao 和 P. Luo，“Tree-planner：使用大型语言模型进行高效闭环任务规划”，载于 ICLR，2024 年。

- [66] Z.Zhao、W.S.Lee 和 D.Hsu，“大型语言模型作为大规模任务规划的常识知识”，载于 NeurIPS，2023 年。

- [67] 开放人工智能。 (2024) 视频生成模型作为世界模拟器。 [在线的]。可用：https://openai.com/index/video- Generation-models-as-world-simulators/

- [68] Z. Zhu, X. Wang, W. Zhao, C. Min, N. Deng, M. Dou, Y. Wang, B. Shi, K. Wang, C. Zhang, Y. You, Z. Zhu, D. Zhao, L. Xiao, J. Zhao, J. Lu, and G. Huang, “sora是一个世界模拟器吗？对一般世界模型（World Models）及其他的综合调查”，CoRR，卷。绝对/2405.03520，2024。

- [69] J. Bruce、M. D. Dennis、A. Edwards、J. Parker-Holder、Y. Shi、E. Hughes、M. Lai、A. Mavalankar、R. Steigerwald、C. Apps、Y. Aytar、S. Bechtle、F. M. P. Behbahani、S. C. Y. Chan、N. Heess、L. Gonzalez、S. Osindero、S. Ozair、S.E.Reed、J.Zhang、K.Zolna、J.Clune、N.de Freitas、S.Singh 和 T.Rocktäschel，“精灵：生成交互环境”，载于 ICML，2024 年。

- [70] H.Zhen、X.Qiu、P.Chen、J.Yang、X.Yan、Y.Du、Y.Hong 和 C.Gan，“3d-vla：3d 视觉语言动作（Vision-Language-Action） 生成式 世界模型（World Model）”，载于 ICML，2024 年。

- [71] S. Yang、Y. Du、S. K. S. Ghasemipour、J. Tompson、L. P. Kaelbling、D. Schuurmans 和 P. Abbeel，“学习交互式现实世界模拟器”，载于 ICLR，2024 年。

- [72] J.向、T.陶、Y.顾、T.舒、Z.王、Z.杨和Z

。 Hu，“语言模型满足 世界模型（World Models）：具体体验增强语言模型”，《NeurIPS》，2023 年。

- [73] T. Kojima、S. S. Gu、M. Reid、Y. Matsuo 和 Y. Iwasawa，“大型语言模型是零样本推理器”，载于 NeurIPS，2022 年。

- [74] J. Wei、X. Wang、D. Schuurmans、M. Bosma、B. Ichter、F. Xia、E. H. Chi、Q. V. Le 和 D. Zhou，“思想链提示在大型语言模型中引发推理”，《NeurIPS》，2022 年。

- [75] G. Lu、Z. Wang、C. Liu、J. Lu 和 Y. Tang，“Thinkbot：遵循思想链推理的具体指令”，CoRR，卷。 ABS/2312.07062，2023。

- [76] S. Yao、J. Zhu、D. Yu、N. Du、I. Shafran、K. R. Narasimhan 和 Y. Cao，“React：在语言模型中协同推理和行动”，载于 ICLR，2023 年。

- [77] Z. Wang、A. Liu、H. Lin、J. Li、X. Ma 和 Y. Liang，“RAT：检索增强思维在 长时程（Long-Horizon） 一代中引发上下文感知推理”，CoRR，卷。绝对/2403.05313，2024。

- [78] M. Zawalski、W. Chen、K. Pertsch、O. Mees、C. Finn 和 S. Levine，“通过具体思想链推理进行机器人控制”，CoRR，卷。绝对/2407.08693，2024。

- [79] Q.Zhao、Y.Lu、M.J.Kim、Z.Fu、Z.Zhang、Y.Wu、Z.Li、Q.Ma、S.Han、C.Finn、A.Handa、T.Lin、G.Wetzstein、M.Liu 和 D.Xiang，“Cot-vla：视觉语言动作（Vision-Language-Action） 模型的视觉链式思维推理”，载于_CVPR_， 2025 年，第 1702-1713 页。

- [80] M. Nakamoto、O. Mees、A. Kumar 和 S. Levine，“引导多面手：通过价值指导改进机器人 基础模型（Foundation Models）”，

在_CoRL_，爵士。机器学习研究论文集，卷。 270.PMLR，2024 年，第 4996-5013 页。

- [81] J. Kwok、C. Agia、R. Sinha、M. Foutter、S. Li、I. Stoica、A. Mirhoseini 和 M. Pavone，“Robomonkey：视觉语言动作（Vision-Language-Action） 模型的缩放测试时间采样和验证”，CoRR，卷。绝对/2506.17811，2025。

- [82] E. Jang、A. Irpan、M. Khansari、D. Kappler、F. Ebert、C. Lynch、S. Levine 和 C. Finn，“BC-Z：使用机器人 模仿学习（Imitation Learning） 进行零样本任务泛化”，载于_CoRL_，卷。 164.PMLR，2021 年，第 991-1002 页。

- [83] C. Lynch 和 P. Sermanet，“非结构化数据上的语言条件 模仿学习（Imitation Learning）”，载于 RSS，2021 年。

- [84] O. Mees、L. Hermann 和 W. Burgard，“语言调节机器人 模仿学习（Imitation Learning） 对非结构化数据的影响”，IEEE Robotics Autom。 Lett.，卷。 7、没有。 4，第 11 205-11 212 页，2022 年。

- [85] O. Mees、J. Borja-Diaz 和 W. Burgard，“在非结构化数据上使用可视化 可供性（Affordances） 的落地关联（Grounding） 语言”，载于_ICRA_。 IEEE，2023 年，第 11 576-11 582 页。

- [86] Y. Du、S. Yang、B. Dai、H. Dai、O. Nachum、J. Tenenbaum、D. Schuurmans 和 P. Abbeel，“通过文本引导视频生成学习通用策略”，《NeurIPS》，2023 年。

- [87] P. Sharma、B. Sundaralingam、V. Blukis、C. Paxton、T. Hermans、A. Torralba、J. Andreas 和 D. Fox，“利用自然语言反馈纠正机器人计划”，载于 RSS，2022 年。

- [88] C. Lynch、A. Wahid、J. Tompson、T. Ding、J. Betker、R. Baruch、T. Armstrong 和 P. Florence，“交互式语言：与机器人实时对话”，CoRR，卷。绝对/2210.06407，2022。

- [89] P. Guhur、S. Chen、R. G. Pinel、M. Tapaswi、I. Laptev 和 C. Schmid，“用于机器人操作的指令驱动的历史感知策略”，载于_CoRL_，卷。 205

。 PMLR，2022 年，第 175-187 页。

- [90] M. Shridhar、L. Manuelli 和 D. Fox，“感知器演员：用于机器人操作的多任务变压器”，载于_CoRL_，卷。 205.PMLR，2022 年，第 785-799 页。

- [91] T. Gervet、Z. Xian、N. Gkanatsios 和 K. Fragkiadaki，“Act3d：用于多任务机器人操作的 3d 特征场转换器”，载于_CoRL_，卷。 229.PMLR，2023 年，第 3949-3965 页。

- [92] A. Goyal、J. Xu、Y.Guo、V. Blukis、Y. Chao 和 D. Fox，“RVT：用于 3d 对象操作的机器人视图转换器”，载于_CoRL_，卷。 229.PMLR，2023 年，第 694-710 页。

- [93] A. Goyal、V. Blukis、J. Xu、Y.Guo、Y.Chao 和 D.Fox，“RVT-2：从少量演示中学习精确操作”，CoRR，卷。绝对/2406.08545，2024。

- [94] W. Yuan、J. Duan、V. Blukis、W. Pumacay、R. Krishna、A. Murali、A. Mousavian 和 D. Fox，“Robopoint：机器人空间 可供性（Affordance） 预测的视觉语言模型”，CoRR，卷。绝对/2406.10721，2024。

- [95] K. Bousmalis、G. Vezzani、D. Rao、C. Devin、A. X. Lee、M. Bauz´a、T. Davchev、Y. Zhou、A. Gupta、A. Raju、A. Laurens、C. Fantacci、V. Dalibard、M. Zambelli、M. F. Martins、R. Pevceviciute、M. Blokzijl、M. Denil、 N. Batchelor、T. Lampe、E. Parisotto、K. Zolna、S. E. Reed、S. G. Colmenarejo、J. Scholz、A. Abdolmaleki、O. Groth、J. Regli、O. Sushkov、T. Rothorl、J. E. Chen、Y. Aytar、D. Barker、J. Ortiz、M. A. Riedmiller、J. T. Springenberg、R. Hadsell、F. Nori 和 N. Heess，“Robocat：用于机器人操作的 自我改进（Self-Improving） 基础智能体”，CoRR，卷。ABS/2306.11706，2023。

- [96] A. Stone、T. Xiao、Y. Lu、K. Gopalakrishnan、K. Lee、Q. Vuong、P. Wohlhart、B. Zitkovich、F. Xia、C. Finn 和 K. Hausman，“使用预先训练的视觉语言模型进行开放世界对象操作”，CoRR，卷。绝对/2303.00905，2023。

- [97] A. Brohan、N. Brown、J. Carbajal、Y. Chebotar、J. Dabis、C. Finn、K. Gopalakrishnan、K. Hausman、A. Herzog、J. Hsu、J. Ibarz、B. Ichter、A. Irpan、T. Jackson、S. Jesmonth、N. J. Joshi、R. Julian、D. Kalashnikov、Y. Kuang、 I. Leal、K. Lee、S. Levine、Y. Lu、U. Malla、D. Manjunath、I. Mordatch、O. Nachum、C. Parada、J. Peralta、E. Perez、K. Pertsch、J. Quiambao、K. Rao、M. S. Ryoo、G. Salazar、P. R. Sanketi、K. Sayed、J. Singh、S. Sontakke、A. Stone、 C. Tan、H. T. Tran、V. Vanhoucke、S. Vega、Q. Vuong、F. Xia、T. Xiao、P. Xu、S. Xu、T. Yu 和 B. Zitkovich，“RT-1：用于大规模现实世界控制的机器人变压器”，载于 RSS，2023 年。

- [98] Y. Chebotar、Q. Vuong、A. Irpan、K. Hausman、F. Xia、Y. Lu、A. Kumar、T. Yu、A. Herzog、K. Pertsch、K. Gopalakrishnan、J. Ibarz、O. Nachum、S. Sontakke、G. Salazar、H. T. Tran、J. Peralta、C. Tan、D. Manjunath、J. Singh、B. Zitkovich、T. Jackson、K. Rao、C. Finn 和 S. Levine，“Qtransformer：通过自回归 q 函数可扩展离线 强化学习（Reinforcement Learning）”，CoRR，卷。绝对/2309.10150，2023。

- [99] J. Gu、S. Kirmani、P. Wohlhart、Y. Lu、M. G. Arenas、K. Rao、W. Yu、C. Fu、K. Gopalakrishnan、Z. Xu、P. Sundaresan、P. Xu、

H. Su, K. Hausman, C. Finn, Q. Vuong, and T. Xiao, "Rt-trajectory: Robotic task generalization via hindsight trajectory sketches," *CoRR*, vol. abs/2311.01977, 2023.

[100] T. Z. Zhao, V. Kumar, S. Levine, and C. Finn, "Learning fine-grained bimanual manipulation with low-cost hardware," in *RSS*, 2023.

[101] H. Bharadhwaj, J. Vakil, M. Sharma, A. Gupta, S. Tulsiani, and V. Kumar, "Roboagent: Generalization and efficiency in robot manipulation via semantic augmentations and action chunking," in *ICRA*. IEEE, 2024, pp. 4788–4795.

[102] X. Li, M. Liu, H. Zhang, C. Yu, J. Xu, H. Wu, C. Cheang, Y. Jing, W. Zhang, H. Liu, H. Li, and T. Kong, "Vision-language foundation models as effective robot imitators," *CoRR*, vol. abs/2311.01378, 2023.

[103] F. Liu, F. Yan, L. Zheng, C. Feng, Y. Huang, and L. Ma, "Robouniview: Visual-language model with unified view representation for robotic manipulaiton," *CoRR*, vol. abs/2406.18977, 2024.

[104] Y. Yue, Y. Wang, B. Kang, Y. Han, S. Wang, S. Song, J. Feng, and G. Huang, "Deer-vla: Dynamic inference of multimodal large language models for efficient robot execution," in *NeurIPS*, 2024.

[105] S. Huang, Z. Jiang, H. Dong, Y. Qiao, P. Gao, and H. Li, "Instruct2act: Mapping multi-modality instructions to robotic actions with large language model," *CoRR*, vol. abs/2305.11176, 2023.

[106] W. Huang, C. Wang, R. Zhang, Y. Li, J. Wu, and L. Fei-Fei, "Voxposer: Composable 3d value maps for robotic manipulation with language models," *CoRR*, vol. abs/2307.05973, 2023.

[107] C. Chi, S. Feng, Y. Du, Z. Xu, E. Cousineau, B. Burchfiel, and S. Song, "Diffusion policy: Visuomotor policy learning via action diffusion," in *RSS*, 2023.

[108] Y. Ze, G. Zhang, K. Zhang, C. Hu, M. Wang, and H. Xu, "3d diffusion policy: Generalizable visuomotor policy learning via simple 3d representations," in *RSS*, 2024.

[109] H. Ha, P. Florence, and S. Song, "Scaling up and distilling down: Language-guided robot skill acquisition," in *CoRL*, vol. 229. PMLR, 2023, pp. 3766–3777.

[110] D. Ghosh, H. R. Walke, K. Pertsch, K. Black, O. Mees, S. Dasari, J. Hejna, T. Kreiman, C. Xu, J. Luo, Y. L. Tan, L. Y. Chen, Q. Vuong, T. Xiao, P. R. Sanketi, D. Sadigh, C. Finn, and S. Levine, "Octo: An open-source generalist robot policy," in *RSS*, 2024.

[111] T. Ke, N. Gkanatsios, and K. Fragkiadaki, "3d diffuser actor: Policy diffusion with 3d scene representations," *CoRR*, vol. abs/2402.10885, 2024.

[112] M. Reuss, Ö. E. Yagmurlu, F. Wenzel, and R. Lioutikov, "Multimodal diffusion transformer: Learning versatile behavior from multimodal goals," *CoRR*, vol. abs/2407.05996, 2024.

[113] S. Liu, L. Wu, B. Li, H. Tan, H. Chen, Z. Wang, K. Xu, H. Su, and J. Zhu, "Rdt-1b: a diffusion foundation model for bimanual manipulation," *arXiv preprint arXiv:2410.07864*, 2024.

[114] S. Belkhale, T. Ding, T. Xiao, P. Sermanet, Q. Vuong, J. Tompson, Y. Chebotar, D. Dwibedi, and D. Sadigh, "RT-H: action hierarchies using language," *CoRR*, vol. abs/2403.01823, 2024.

[115] O. X. Collaboration, A. Padalkar, A. Pooley, A. Jain, A. Bewley, A. Herzog, A. Irpan, A. Khazatsky, A. Raj, A. Singh, A. Brohan, A. Raffin, A. Wahid, B. Burgess-Limerick, B. Kim, B. Schölkopf, B. Ichter, C. Lu, C. Xu, C. Finn, C. Xu, C. Chi, C. Huang, C. Chan, C. Pan, C. Fu, C. Devin, D. Driess, D. Pathak, D. Shah, D. Büchler, D. Kalashnikov, D. Sadigh, E. Johns, F. Ceola, F. Xia, F. Stulp, G. Zhou, G. S. Sukhatme, G. Salhotra, G. Yan, G. Schiavi, G. Kahn, H. Su, H. Fang, H. Shi, H. B. Amor, H. I. Christensen, H. Furuta, H. Walke, H. Fang, I. Mordatch, I. Radosavovic, and et al., "Open x-embodiment: Robotic learning datasets and RT-X models," *CoRR*, vol. abs/2310.08864, 2023.

[116] M. J. Kim, C. Finn, and P. Liang, "Fine-tuning vision-language-action models: Optimizing speed and success," *CoRR*, vol. abs/2502.19645, 2025.

[117] R. Zheng, Y. Liang, S. Huang, J. Gao, H. D. III, A. Kolobov, F. Huang, and J. Yang, "Tracevla: Visual trace prompting enhances spatial-temporal awareness for generalist robotic policies," in *ICLR*, 2025.

[118] K. Black, N. Brown, D. Driess, A. Esmail, M. Equi, C. Finn, N. Fusai, L. Groom, K. Hausman, B. Ichter, S. Jakubczak, T. Jones, L. Ke, S. Levine, A. Li-Bell, M. Mothukuri, S. Nair, K. Pertsch, L. X. Shi, J. Tanner, Q. Vuong, A. Walling, H. Wang, and U. Zhilinsky, "π_0: A vision-language-action flow model for general robot control," *CoRR*, vol. abs/2410.24164, 2024.

[119] J. Liu, M. Liu, Z. Wang, P. An, X. Li, K. Zhou, S. Yang, R. Zhang, Y. Guo, and S. Zhang, "Robomamba: Efficient vision-language-action model for robotic reasoning and manipulation," in *NeurIPS*, 2024.

[120] D. Qu, H. Song, Q. Chen, Y. Yao, X. Ye, Y. Ding, Z. Wang, J. Gu, B. Zhao, D. Wang, and X. Li, "Spatialvla: Exploring spatial representations for visual-language-action model," *CoRR*, vol. abs/2501.15830, 2025.

[121] J. Wen, Y. Zhu, J. Li, M. Zhu, K. Wu, Z. Xu, N. Liu, R. Cheng, C. Shen, Y. Peng, F. Feng, and J. Tang, "Tinyvla: Towards fast, data-efficient vision-language-action models for robotic manipulation," *CoRR*, vol. abs/2409.12514, 2024.

[122] Q. Li, Y. Liang, Z. Wang, L. Luo, X. Chen, M. Liao, F. Wei, Y. Deng, S. Xu, Y. Zhang, X. Wang, B. Liu, J. Fu, J. Bao, D. Chen, Y. Shi, J. Yang, and B. Guo, "Cogact: A foundational vision-language-action model for synergizing cognition and action in robotic manipulation," *CoRR*, vol. abs/2411.19650, 2024.

[123] J. Wen, Y. Zhu, J. Li, Z. Tang, C. Shen, and F. Feng, "Dexvla: Vision-language model with plug-in diffusion expert for general robot control," *CoRR*, vol. abs/2502.05855, 2025.

[124] J. Liu, H. Chen, P. An, Z. Liu, R. Zhang, C. Gu, X. Li, Z. Guo, S. Chen, M. Liu, C. Hou, M. Zhao, K. alex Zhou, P. Heng, and S. Zhang, "Hybridvla: Collaborative diffusion and autoregression in a unified vision-language-action model," *CoRR*, vol. abs/2503.10631, 2025.

[125] S. Ye, J. Jang, B. Jeon, S. J. Joo, J. Yang, B. Peng, A. Mandlekar, R. Tan, Y. Chao, B. Y. Lin, L. Liden, K. Lee, J. Gao, L. Zettlemoyer, D. Fox, and M. Seo, "Latent action pretraining from videos," in *ICLR*, 2025.

[126] J. Cen, C. Yu, H. Yuan, Y. Jiang, S. Huang, J. Guo, X. Li, Y. Song, H. Luo, F. Wang, D. Zhao, and H. Chen, "Worldvla: Towards autoregressive action world model," *CoRR*, vol. abs/2506.21539, 2025.

[127] Y. Wang, X. Li, W. Wang, J. Zhang, Y. Li, Y. Chen, X. Wang, and Z. Zhang, "Unified vision-language-action model," *CoRR*, vol. abs/2506.19850, 2025.

[128] Y. Ma, D. Chi, S. Wu, Y. Liu, Y. Zhuang, and I. King, "Astra: Efficient transformer architecture and contrastive dynamics learning for embodied instruction following," in *EMNLP*, 2025.

[129] C. Qu, S. Dai, X. Wei, H. Cai, S. Wang, D. Yin, J. Xu, and J. Wen, "Tool learning with large language models: a survey," *Frontiers Comput. Sci.*, vol. 19, no. 8, p. 198343, 2025.

[130] Y. Jiang, A. Gupta, Z. Zhang, G. Wang, Y. Dou, Y. Chen, L. Fei-Fei, A. Anandkumar, Y. Zhu, and L. Fan, "VIMA: general robot manipulation with multimodal prompts," *CoRR*, vol. abs/2210.03094, 2022.

[131] R. Liu, W. Wang, and Y. Yang, "Volumetric environment representation for vision-language navigation," in *CVPR*. IEEE, 2024, pp. 16 317–16 328.

[132] Y. Ma and I. King, "3d-moe: A mixture-of-experts multi-modal LLM for 3d vision and pose diffusion via rectified flow," *CoRR*, vol. abs/2501.16698, 2025.

[133] M. Vecerík, C. Doersch, Y. Yang, T. Davchev, Y. Aytar, G. Zhou, R. Hadsell, L. Agapito, and J. Scholz, "Robotap: Tracking arbitrary points for few-shot visual imitation," in *ICRA*. IEEE, 2024, pp. 5397–5403.

[134] S. Nasiriany, F. Xia, W. Yu, T. Xiao, J. Liang, I. Dasgupta, A. Xie, D. Driess, A. Wahid, Z. Xu, Q. Vuong, T. Zhang, T. E. Lee, K. Lee, P. Xu, S. Kirmani, Y. Zhu, A. Zeng, K. Hausman, N. Heess, C. Finn, S. Levine, and B. Ichter, "PIVOT: iterative visual prompting elicits actionable knowledge for vlms," in *ICML*, 2024.

[135] J. Bjorck, F. Castañeda, N. Cherniadev, X. Da, R. Ding, Linxi, Y. Fang, D. Fox, F. Hu, S. Huang, J. Jang, Z. Jiang, J. Kautz, K. Kundalia, L. Lao, Z. Li, Z. Lin, K. Lin, G. Liu, E. LLontop, L. Magne, A. Mandlekar, A. Narayan, S. Nasiriany, S. Reed, Y. L. Tan, G. Wang, Z. Wang, J. Wang, Q. Wang, J. Xiang, Y. Xie, Y. Xu, Z. Xu, S. Ye, Z. Yu, A. Zhang, H. Zhang, Y. Zhao, R. Zheng, and Y. Zhu, "GR00T N1: an open foundation model for generalist humanoid robots," *CoRR*, vol. abs/2503.14734, 2025.

[136] N. M. Chia-Yu Hung and, H. Deng, L. Renhang, Y. Ang, A. Zadeh, C. Li, D. Herremans, Z. Wang, and S. Poria, "NORA-1.5: A vision-language-action model trained using world model-, and action-based preference rewards," *CoRR*, vol. abs/2511.14659, 2025.

[137] P. Z. Yue Liao and, S. Huang, D. Yang, S. Chen, Y. Jiang, H. Yue, J. Cai, S. Liu, J. Luo, L. Chen, S. Yan, M. Yao, and G. Ren, "Genie envisioner: A unified world foundation platform for robotic, manipulation," *CoRR*, vol. abs/2508.05635, 2025.

[138] K. Tian, Y. Jiang, Z. Yuan, B. Peng, and L. Wang, "Visual autoregressive modeling: Scalable image generation via next-scale prediction," in *NeurIPS*, 2024.

[139] X. Zhuang, Y. Xie, Y. Deng, L. Liang, J. Ru, Y. Yin, and Y. Zou, "VARGPT: unified understanding and generation in a visual autoregres-

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H. Su、K. Hausman、C. Finn、Q. Vuong 和 T. Xiao，“Rt 轨迹：通过事后轨迹草图概括机器人任务”，CoRR，卷。绝对/2311.01977，2023。

- [100] T.Z.Zhao、V.Kumar、S.Levine 和 C.Finn，“使用低成本硬件学习细粒度双手操作”，载于 RSS，2023 年。

- [101] H. Bharadhwaj、J. Vakil、M. Sharma、A. Gupta、S. Tulsiani 和 V. Kumar，“Roboagent：通过语义增强和 动作分块（Action Chunking）实现机器人操作的泛化和效率”，载于_ICRA_。 IEEE，2024 年，第 4788-4795 页。

- [102] X. Li、M. Liu、H. Zhang、C. Yu、J. Xu、H. Wu、C. Cheang、Y. Jing、W. Zhang、H. Liu、H. Li 和 T. Kong，“视觉语言 基础模型（Foundation Models）作为有效的机器人模仿者”，CoRR，卷。绝对/2311.01378，2023。

- [103] F. Liu、F. Yan、L. Cheng、C. Feng、Y. Huang 和 L. Ma，“Robouniview：用于机器人操作的统一视图表示的视觉语言模型”，CoRR，卷。绝对/2406.18977，2024。

- [104] Y. Yue、Y. Wang、B. Kang、Y. Han、S. Wang、S. Song、J. Feng 和 G. Huang，“Deer-vla：用于高效机器人执行的多模态大语言模型的动态推理”，《NeurIPS》，2024 年。

- [105] S. Huang、Z. Jiang、H. Dong、Y. Qiao、P. Gau 和 H. Li，“Instruct2act：使用大语言模型将多模态指令映射到机器人动作”，CoRR，卷。 ABS/2305.11176，2023。

- [106] W. Huang、C. Wang、R. Zhu、Y. Li、J. Wu 和 L. Fei-Fei，“Voxposer：使用语言模型进行机器人操作的可组合 3D 值图”，CoRR，卷。绝对/2307.05973，2023。

- [107] C. Chi、S. Feng、Y. Du、Z. Xu、E. Cousineau、B. Burchfiel 和 S. Song，“扩散策略（Diffusion Policy）：通过动作扩散进行视觉运动策略学习”，载于_RSS_，2023 年。

- [108] Y. Ze、G.Zhang、K.Zhang、C.Hu、M.Wang 和 H.Xu，“3d 扩散策略（Diffusion Policy）：通过简单 3D 表示进行泛化视觉运动策略学习”，载于 RSS，2024 年。

- [109] H. Ha、P. Florence 和 S. Song，“扩大规模和缩小规模：语言引导的机器人技能获取”，载于 CoRL，第 1 卷。 229.PMLR，2023 年，第 3766-3777 页。

- [110] D. Ghosh、H. R. Walke、K. Pertsch、K. Black、O. Mees、S. Dasari、J. Hejna、T. Kreiman、C. Xu、J. Luo、Y. L. Tan、L. Y. Chen、Q. Vuong、T. Xiao、P. R. Sanketi、D. Sadigh、C. Finn 和 S. Levine，“Octo：开源多面手机器人策略”，载于 RSS，2024 年。

- [111] T. Ke、N. Gkanatsios 和 K. Fragkiadaki，“3d 扩散器演员：具有 3d 场景表示的策略扩散”，CoRR，卷。绝对/2402.10885，2024。

- [112] M. Reuss、O.¡ E. Yagmurlu、F. Wenzel 和 R. Lioutikov，“多模态扩散转换器：从多模态目标中学习多功能行为”，CoRR，卷。绝对/2407.05996，2024。

- [113] S. Liu、L. Wu、B. Li、H. Tan、H. Chen、Z. Wang、K. Xu、H. Su 和 J. Zhu，“Rdt-1b：用于双手操作的扩散 基础模型（Foundation Model）”，arXiv 预印本 arXiv：2410.07864，2024 年。

- [114] S. Belkhale、T. Ding、T. Xiao、P. Sermanet、Q. Vuong、J. Tompson、Y. Chebotar、D. Dwibedi 和 D. Sadigh，“RT-H：使用语言的动作层次结构”，CoRR，卷。绝对/2403.01823，2024。

- [115] O. X. 协作，A. Padalkar、A. Pooley、A. Jain、A. Bewley、A. Herzog、A. Irpan、A. Khazatsky、A. Raj、A. Singh、A. Brohan、A. Raffin、A

。 Wahid、B. Burgess-Limerick、B. Kim、B. Schöolkopf、B. Ichter、C. Lu、C. Xu、C. Finn、C. Xu、C. Chi、C. Huang、C. Chan、C. Pan、C. Fu、C. Devin、D. Driess、D. Pathak、D. Shah、D. Buchler、D. Kalashnikov、D. Sadigh、E. Johns、F. Ceola、F. Xia、F. Stulp、G. Zhou、G. S. Sukhatme、G. Salhotra、G. Yan、G. Schiavi、G. Kahn、H. Su、H. Fang、H. Shi、H. B. Amor、H. I. Christensen、H. Furuta、H. Walke、H. Fang、I. Mordatch、I. Radosavovic 等人，“开放 x-体现：机器人学习数据集和 RT-X 模型”，CoRR，卷。绝对/2310.08864，2023。

- [116] M. J. Kim、C. Finn 和 P. Liang，“微调 视觉语言动作（Vision-Language-Action） 模型：优化速度和成功”，CoRR，卷。绝对/2502.19645，2025。

- [117] R. Cheng、Y. Liang、S. Huang、J. Gau、H. D. III、A. Kolobov、F. Huang 和 J. Yang，“Tracevla：视觉跟踪提示增强通用机器人策略的时空意识”，载于 ICLR，2025 年。

- [118] K. Black、N. Brown、D. Driess、A. Esmail、M. Equi、C. Finn、N. Fusai、L. Groom、K. Hausman、B. Ichter、S. Jakubczak、T. Jones、L. Ke、S. Levine、A. Li-Bell、M. Mothukuri、S. Nair、K. Pertsch、L. X. Shi、J. Tanner、Q. Vuong、A. Walling、H. Wang 和 U.zhilinsky，“π 0：用于通用机器人控制的 视觉语言动作（Vision-Language-Action） 流模型”，CoRR，卷。绝对/2410.24164，2024。

- [119] J. Liu、M. Liu、Z. Wang、P. An、X. Li、K. Zhou、S. Yang、R. Zhu、Y.Guo 和 S.Zhang，“Robomamba：用于机器人推理和操作的高效 视觉语言动作（Vision-Language-Action） 模型”，《NeurIPS》，2024 年。

- [120] D. Qu、H. Song、Q. Chen、Y. Yao、X. Ye、Y. Ding、Z. Wang、J. Gu、B. Zhu、D. Wang 和 X. Li，“Spatialvla：探索视觉语言动作模型的空间表示”，CoRR，卷。绝对/2501.15830，2025。

- [121] J. Wen、Y. Zhu、J. Li、M. Zhu、K. Wu、Z. Xu、N. Liu、R. Cheng、C. Shen、Y. Peng、F. Feng 和 J. Tang，“Tinyvla：面向快速、数据高效的机器人操作 视觉语言动作（Vision-Language-Action） 模型”，CoRR，卷。绝对/2409.12514，2024。

- [122] Q. Li、Y. Liang、Z. Wang、L. Luo、X. Chen、M. Liao、F. Wei、Y. Deng、S. Xu、Y. Zhuang、X. Wang、B. Liu、J. Fu、J. Bao、D. Chen、Y. Shi、J. Yang 和 B.Guo，“Cogact：用于在机器人操作中协同认知和动作的基础 视觉语言动作（Vision-Language-Action） 模型”， CoRR，卷。绝对/2411.19650，2024。

- [123] J. Wen，Y. Zhu，J. Li，Z. Tang，C. Shen，and F. Feng，“Dexvla：用于通用机器人控制的具有插件扩散专家的 Visionlanguage 模型”，CoRR，卷。绝对/2502.05855，2025。

- [124] J. Liu, H. Chen, P. An, Z. Liu, R. Zhu, C. Gu, X. Li, Z.Guo, S. Chen, M. Liu, C. Hou, M. Zhu, K. alex Zhou, P. Heng, and S. Zhang, “Hybridvla: 统一 视觉语言动作（Vision-Language-Action） 模型中的协作扩散和自回归”，CoRR，卷。绝对/2503.10631，2025。

- [125] S. Ye、J. Jang、B. Jeon、S. J. Joo、J. Yang、B. Peng、A. Mandlekar、R. Tan、Y. Chao、B. Y. Lin、L. Liden、K. Lee、J. Gau、L. Zettlemoyer、D. Fox 和 M. Seo，“来自视频的潜在动作预训练”，载于 ICLR，2025 年。

- [126] J. Cen、C. Yu、H. Yuan、Y. Jiang、S. Huang、J.Guo、X. Li、Y. Song、H. Luo、F. Wang、D. Zhu 和 H. Chen，“Worldvla：迈向自回归作用 世界模型（World Model）”，CoRR，卷。绝对/2506.21539，2025。

- [127] Y.王、X.李、W.王、J.张、Y.李、Y.陈、X.王和Z

。张，“统一 视觉语言动作（Vision-Language-Action） 模型”，CoRR，卷。绝对/2506.19850，2025。

- [128] Y. Ma、D. Chi、S. Wu、Y. Liu、Y. Zhuang 和 I. King，“Astra：用于具体指令跟踪的高效 Transformer 架构和对比动态学习”，载于 EMNLP，2025 年。

- [129] C. Qu、S. Dai、X. Wei、H. Cai、S. Wang、D. Yin、J. Xu 和 J. Wen，“大型语言模型的工具学习：一项调查”，Frontiers Comput。科学，卷。 19、没有。 8，p。 198343, 2025。

- [130] Y. Jiang，A. Gupta，Z. Zhang，G. Wang，Y. Dou，Y. Chen，L. FeiFei，A. Anandkumar，Y. Zhu，and L. Fan，“VIMA：具有多模式提示的通用机器人操作”，CoRR，卷。绝对/2210.03094，2022。

- [131] R. Liu、W. Wang 和 Y. Yang，“视觉语言导航的体积环境表示”，载于_CVPR_。 IEEE，2024 年，第 16 317-16 328 页。

- [132] Y. Ma 和 I. King，“3d-moe：通过校正流实现 3d 视觉和姿势扩散的专家混合多模态法学硕士”，CoRR，卷。绝对/2501.16698，2025。

- [133] M. Vecer´ık、C. Doersch、Y. Yang、T. Davchev、Y. Aytar、G. Zhou、R. Hadsell、L. Agapito 和 J. Scholz，“Robotap：跟踪少量视觉模仿的任意点”，载于_ICRA_。 IEEE，2024 年，第 5397-5403 页。

- [134] S. Nasiriany、F. Xia、W. Yu、T. Xiao、J. Liang、I. Dasgupta、A. Xie、D. Driess、A. Wahid、Z. Xu、Q. Vuong、T. Zhang、T. E. Lee、K. Lee、P. Xu、S. Kirmani、Y. Zhu、A. Zeng、K. Hausman、N. Heess、C. Finn、S. Levine 和 B. Ichter，“PIVOT：迭代视觉提示为 vlms 引出可操作的知识”，载于 ICML，2024 年。

- [135] J. Bjorck、F. Casta&#xfffd;neda、N. Cherniadev、X. Da、R. Ding、Linxi、Y. Fang、D. Fox、F. Hu、S. Huang、J. Jang、Z. Jiang、J. Kautz、K. Kundalia、L. Lao、Z. Li、Z. Lin、K. Lin、G. Liu、E. LLontop、L. Magne、A. Mandlekar、A. Narayan、S. Nasiriany、S. Reed、Y. L. Tan、G. Wang、Z. Wang、J. Wang、Q. Wang、J. Shang、Y. Xie、Y. Xu、Z. Xu、S. Ye、Z. Yu、A. Zhang、H. Zhu、Y. Zhu、R. Cheng 和 Y. Zhu，“GR00T N1：用于通用人形机器人的开放式 基础模型（Foundation Model）”，CoRR，卷。绝对/2503.14734，2025。

- [136] N. M. Chia-Yu Hung 和 H. Deng、L. Renhang、Y. Ang、A. Zadeh、C. Li、D. Herremans、Z. Wang 和 S. Poria，“NORA-1.5：使用 世界模型（World Model） 和基于动作的偏好奖励训练的视觉语言动作模型”，CoRR，卷。绝对/2511.14659，2025。

- [137] P. Z. Yue Liao 和 S. Huang、D. Yang、S. Chen、Y. Jiang、H. Yue、J. Cai、S. Liu、J. Luo、L. Chen、S. Yan、M. Yao 和 G. Ren，“Genie envisioner：机器人操作的统一世界基础平台”，CoRR，卷。绝对/2508.05635，2025。

- [138] K. Tian、Y. Jiang、Z. Yuan、B. Peng 和 L. Wang，“视觉自回归建模：通过下一尺度预测生成可扩展图像”，载于 NeurIPS，2024 年。

- [139] X. Zhuang，Y. Xie，Y. Deng，L. Liang，J. Ru，Y. Yin，Y. Zou，“VARGPT：视觉自回归中的统一理解和生成-

sive multimodal large language model," *CoRR*, vol. abs/2501.12327, 2025.

[140] Y. Mu, Q. Zhang, M. Hu, W. Wang, M. Ding, J. Jin, B. Wang, J. Dai, Y. Qiao, and P. Luo, "Embodiedgpt: Vision-language pre-training via embodied chain of thought," *CoRR*, vol. abs/2305.15021, 2023.

[141] J. Huang, S. Yong, X. Ma, X. Linghu, P. Li, Y. Wang, Q. Li, S. Zhu, B. Jia, and S. Huang, "An embodied generalist agent in 3d world," in *ICML*, 2024.

[142] Z. Qi, R. Dong, S. Zhang, H. Geng, C. Han, Z. Ge, L. Yi, and K. Ma, "Shapellm: Universal 3d object understanding for embodied interaction," in *ECCV (43)*, ser. Lecture Notes in Computer Science, vol. 15101. Springer, 2024, pp. 214–238.

[143] W. Huang, P. Abbeel, D. Pathak, and I. Mordatch, "Language models as zero-shot planners: Extracting actionable knowledge for embodied agents," in *ICML*, vol. 162. PMLR, 2022, pp. 9118–9147.

[144] P. Sharma, A. Torralba, and J. Andreas, "Skill induction and planning with latent language," in *ACL (1)*, 2022, pp. 1713–1726.

[145] C. H. Song, B. M. Sadler, J. Wu, W. Chao, C. Washington, and Y. Su, "Llm-planner: Few-shot grounded planning for embodied agents with large language models," in *ICCV*. IEEE, 2023, pp. 2986–2997.

[146] S. Li, X. Puig, C. Paxton, Y. Du, C. Wang, L. Fan, T. Chen, D. Huang, E. Akyürek, A. Anandkumar, J. Andreas, I. Mordatch, A. Torralba, and Y. Zhu, "Pre-trained language models for interactive decision-making," in *NeurIPS*, 2022.

[147] A. Zeng, M. Attarian, B. Ichter, K. M. Choromanski, A. Wong, S. Welker, F. Tombari, A. Purohit, M. S. Ryoo, V. Sindhwani, J. Lee, V. Vanhoucke, and P. Florence, "Socratic models: Composing zero-shot multimodal reasoning with language," in *ICLR*, 2023.

[148] I. Singh, V. Blukis, A. Mousavian, A. Goyal, D. Xu, J. Tremblay, D. Fox, J. Thomason, and A. Garg, "Progprompt: Generating situated robot task plans using large language models," in *ICRA*. IEEE, 2023, pp. 11 523–11 530.

[149] S. Vemprala, R. Bonatti, A. Bucker, and A. Kapoor, "Chatgpt for robotics: Design principles and model abilities," *CoRR*, vol. abs/2306.17582, 2023.

[150] J. Liang, W. Huang, F. Xia, P. Xu, K. Hausman, B. Ichter, P. Florence, and A. Zeng, "Code as policies: Language model programs for embodied control," in *ICRA*. IEEE, 2023, pp. 9493–9500.

[151] Z. Wang, S. Cai, A. Liu, X. Ma, and Y. Liang, "Describe, explain, plan and select: Interactive planning with large language models enables open-world multi-task agents," *CoRR*, vol. abs/2302.01560, 2023.

[152] Q. Gu, A. Kuwajerwala, S. Morin, K. M. Jatavallabhula, B. Sen, A. Agarwal, C. Rivera, W. Paul, K. Ellis, R. Chellappa, C. Gan, C. M. de Melo, J. B. Tenenbaum, A. Torralba, F. Shkurti, and L. Paull, "Conceptgraphs: Open-vocabulary 3d scene graphs for perception and planning," *CoRR*, vol. abs/2309.16650, 2023.

[153] F. Lin, Y. Hu, P. Sheng, C. Wen, J. You, and Y. Gao, "Data scaling laws in imitation learning for robotic manipulation," in *ICLR*, 2025.

[154] T. Pearce, T. Rashid, D. Bignell, R. Georgescu, S. Devlin, and K. Hofmann, "Scaling laws for pre-training agents and world models," in *ICML*, 2025.

[155] Y. Hong, Z. Zheng, P. Chen, Y. Wang, J. Li, and C. Gan, "Multiply: A multisensory object-centric embodied large language model in 3d world," in *CVPR*. IEEE, 2024, pp. 26 396–26 406.

[156] S. S. Raman, V. Cohen, E. Rosen, I. Idrees, D. Paulius, and S. Tellex, "Planning with large language models via corrective re-prompting," *CoRR*, vol. abs/2211.09935, 2022.

[157] P. Zhi, Z. Zhang, M. Han, Z. Zhang, Z. Li, Z. Jiao, B. Jia, and S. Huang, "Closed-loop open-vocabulary mobile manipulation with GPT-4V," *CoRR*, vol. abs/2404.10220, 2024.

[158] H. Fang, H. Fang, Z. Tang, J. Liu, C. Wang, J. Wang, H. Zhu, and C. Lu, "RH20T: A comprehensive robotic dataset for learning diverse skills in one-shot," in *ICRA*. IEEE, 2024, pp. 653–660.

[159] A. Khazatsky, K. Pertsch, S. Nair, A. Balakrishna, S. Dasari, S. Karamcheti, S. Nasiriany, M. K. Srirama, L. Y. Chen, K. Ellis, P. D. Fagan, J. Hejna, M. Itkina, M. Lepert, Y. J. Ma, P. T. Miller, J. Wu, S. Belkhale, S. Dass, H. Ha, A. Jain, A. Lee, Y. Lee, M. Memmel, S. Park, I. Radosavovic, K. Wang, A. Zhan, K. Black, C. Chi, K. B. Hatch, S. Lin, J. Lu, J. Mercat, A. Rehman, P. R. Sanketi, A. Sharma, C. Simpson, Q. Vuong, H. R. Walke, B. Wulfe, T. Xiao, J. H. Yang, A. Yavary, T. Z. Zhao, C. Agia, R. Baijal, M. G. Castro, D. Chen, Q. Chen, T. Chung, J. Drake, E. P. Foster, and et al., "DROID: A large-scale in-the-wild robot manipulation dataset," *CoRR*, vol. abs/2403.12945, 2024.

[160] P. Sharma, L. Mohan, L. Pinto, and A. Gupta, "Multiple interactions made easy (MIME): large scale demonstrations data for imitation," in *CoRL*, vol. 87. PMLR, 2018, pp. 906–915.

[161] A. Mandlekar, Y. Zhu, A. Garg, J. Booher, M. Spero, A. Tung, J. Gao, J. Emmons, A. Gupta, E. Orbay, S. Savarese, and L. Fei-Fei, "ROBOTURK: A crowdsourcing platform for robotic skill learning through imitation," in *CoRL*, vol. 87. PMLR, 2018, pp. 879–893.

[162] S. Dasari, F. Ebert, S. Tian, S. Nair, B. Bucher, K. Schmeckpeper, S. Singh, S. Levine, and C. Finn, "Robonet: Large-scale multi-robot learning," in *CoRL*, vol. 100. PMLR, 2019, pp. 885–897.

[163] D. Kalashnikov, J. Varley, Y. Chebotar, B. Swanson, R. Jonschkowski, C. Finn, S. Levine, and K. Hausman, "Mt-opt: Continuous multi-task robotic reinforcement learning at scale," *CoRR*, vol. abs/2104.08212, 2021.

[164] V. Kumar, R. M. Shah, G. Zhou, V. Moens, V. Caggiano, A. Gupta, and A. Rajeswaran, "Robohive: A unified framework for robot learning," in *NeurIPS*, 2023.

[165] F. Ebert, Y. Yang, K. Schmeckpeper, B. Bucher, G. Georgakis, K. Daniilidis, C. Finn, and S. Levine, "Bridge data: Boosting generalization of robotic skills with cross-domain datasets," in *RSS*, 2022.

[166] S. Srivastava, C. Li, M. Lingelbach, R. Martín-Martín, F. Xia, K. E. Vainio, Z. Lian, C. Gokmen, S. Buch, C. K. Liu, S. Savarese, H. Gweon, J. Wu, and L. Fei-Fei, "BEHAVIOR: benchmark for everyday household activities in virtual, interactive, and ecological environments," in *CoRL*, vol. 164. PMLR, 2021, pp. 477–490.

[167] C. Li, F. Xia, R. Martín-Martín, M. Lingelbach, S. Srivastava, B. Shen, K. E. Vainio, C. Gokmen, G. Dharan, T. Jain, A. Kurenkov, C. K. Liu, H. Gweon, J. Wu, L. Fei-Fei, and S. Savarese, "igibson 2.0: Object-centric simulation for robot learning of everyday household tasks," in *CoRL*, vol. 164. PMLR, 2021, pp. 455–465.

[168] F. Xia, A. R. Zamir, Z. He, A. Sax, J. Malik, and S. Savarese, "Gibson env: Real-world perception for embodied agents," in *CVPR*. Computer Vision Foundation / IEEE Computer Society, 2018, pp. 9068–9079.

[169] F. Xia, W. B. Shen, C. Li, P. Kasimbeg, M. Tchapmi, A. Toshev, R. Martín-Martín, and S. Savarese, "Interactive gibson benchmark: A benchmark for interactive navigation in cluttered environments," *IEEE Robotics Autom. Lett.*, vol. 5, no. 2, pp. 713–720, 2020.

[170] B. Shen, F. Xia, C. Li, R. Martín-Martín, L. Fan, G. Wang, C. Pérez-D'Arpino, S. Buch, S. Srivastava, L. Tchapmi, M. Tchapmi, K. Vainio, J. Wong, L. Fei-Fei, and S. Savarese, "igibson 1.0: A simulation environment for interactive tasks in large realistic scenes," in *IROS*. IEEE, 2021, pp. 7520–7527.

[171] C. Li, R. Zhang, J. Wong, C. Gokmen, S. Srivastava, R. Martín-Martín, C. Wang, G. Levine, W. Ai, B. Martinez, H. Yin, M. Lingelbach, M. Hwang, A. Hiranaka, S. Garlanka, A. Aydin, S. Lee, J. Sun, M. Anvari, M. Sharma, D. Bansal, S. Hunter, K. Kim, A. Lou, C. R. Matthews, I. Villa-Renteria, J. H. Tang, C. Tang, F. Xia, Y. Li, S. Savarese, H. Gweon, C. K. Liu, J. Wu, and L. Fei-Fei, "BEHAVIOR-1K: A human-centered, embodied AI benchmark with 1, 000 everyday activities and realistic simulation," *CoRR*, vol. abs/2403.09227, 2024.

[172] F. Xiang, Y. Qin, K. Mo, Y. Xia, H. Zhu, F. Liu, M. Liu, H. Jiang, Y. Yuan, H. Wang, L. Yi, A. X. Chang, L. J. Guibas, and H. Su, "SAPIEN: A simulated part-based interactive environment," in *CVPR*. Computer Vision Foundation / IEEE, 2020, pp. 11 094–11 104.

[173] X. Li, K. Hsu, J. Gu, K. Pertsch, O. Mees, H. R. Walke, C. Fu, I. Lunawat, I. Sieh, S. Kirmani, S. Levine, J. Wu, C. Finn, H. Su, Q. Vuong, and T. Xiao, "Evaluating real-world robot manipulation policies in simulation," *CoRR*, vol. abs/2405.05941, 2024.

[174] E. Kolve, R. Mottaghi, D. Gordon, Y. Zhu, A. Gupta, and A. Farhadi, "AI2-THOR: an interactive 3d environment for visual AI," *CoRR*, vol. abs/1712.05474, 2017.

[175] K. Ehsani, W. Han, A. Herrasti, E. VanderBilt, L. Weihs, E. Kolve, A. Kembhavi, and R. Mottaghi, "Manipulathor: A framework for visual object manipulation," in *CVPR*. Computer Vision Foundation / IEEE, 2021, pp. 4497–4506.

[176] L. Weihs, M. Deitke, A. Kembhavi, and R. Mottaghi, "Visual room rearrangement," in *CVPR*. Computer Vision Foundation / IEEE, 2021, pp. 5922–5931.

[177] K. Mirakhor, S. Ghosh, D. Das, and B. Bhowmick, "Task planning for visual room rearrangement under partial observability," in *ICLR*, 2024.

[178] X. Puig, K. Ra, M. Boben, J. Li, T. Wang, S. Fidler, and A. Torralba, "Virtualhome: Simulating household activities via programs," in *2018 IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2018, Salt Lake City, UT, USA, June 18-22, 2018*. Computer Vision Foundation / IEEE Computer Society, 2018, pp. 8494–8502.

[179] C. Gan, J. Schwartz, S. Alter, D. Mrowca, M. Schrimpf, J. Traer, J. D. Freitas, J. Kubilius, A. Bhandwaldar, N. Haber, M. Sano, K. Kim, E. Wang, M. Lingelbach, A. Curtis, K. T. Feigelis, D. Bear, D. Gutfreund, D. D. Cox, A. Torralba, J. J. DiCarlo, J. Tenenbaum, J. H. McDermott, and D. Yamins, "Threedworld: A platform for

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sive 多模态大语言模型，”CoRR，vol.abs/2501.12327，2025。

- [140] Y. Mu，Q. 张，M. Hu，W. Wang，M. Ding，J. Jin，B. Wang，J. Dai，Y. Qiao，and P. Luo，“Embodiedgpt：通过体现思想链进行视觉语言预训练”，CoRR，卷。 ABS/2305.15021，2023。

- [141] J. Huang、S. Yong、X. Ma、X. Linghu、P. Li、Y. Wang、Q. Li、S. Zhu、B. Jia 和 S. Huang，“3d 世界中的具体多面手智能体”，载于 ICML，2024 年。

- [142] Z. Qi、R. Dong、S. Zhang、H. Geng、C. Han、Z. Ge、L. Yi 和 K. Ma，“Shapellm：体现交互的通用 3d 对象理解”，载于_ECCV (43)_，系列。计算机科学讲义，卷。 15101。施普林格，2024 年，第 214-238 页。

- [143] W. Huang、P. Abbeel、D. Pathak 和 I. Mordatch，“语言模型作为零样本规划器：为具体智能体提取可操作的知识”，载于_ICML_，卷。 162.PMLR，2022 年，第 9118-9147 页。

- [144] P. Sharma、A. Torralba 和 J. Andreas，“潜在语言的技能归纳和规划”，载于 ACL (1)，2022 年，第 1713-1726 页。

- [145] C. H. Song、B. M. Sadler、J. Wu、W. Chao、C. Washington 和 Y. Su，“Llm-planner：针对具有大型语言模型的具体智能体的少镜头扎根规划”，载于 ICCV。 IEEE，2023 年，第 2986-2997 页。

- [146] S. Li、X. Puig、C. Paxton、Y. Du、C. Wang、L. Fan、T. Chen、D. Huang、E. Akyùurek、A. Anandkumar、J. Andreas、I. Mordatch、A. Torralba 和 Y. Zhu，“用于交互式决策的预训练语言模型”，载于 NeurIPS，2022 年。

- [147] A. Zeng、M. Attarian、B. Ichter、K. M. Choromanski、A. Wong、S. Welker、F. Tombari、A. Purohit、M. S. Ryoo、V. Sindhwani、J. Lee、V. Vanhoucke 和 P. Florence，“苏格拉底模型：用语言构建零样本多模态推理”，载于_ICLR_， 2023 年。

- [148] I. Singh、V. Blukis、A. Mousavian、A. Goyal、D. Xu、J. Tremblay、D. Fox、J. Thomason 和 A. Garg，“Progprompt：使用大型语言模型生成情境机器人任务计划”，载于_ICRA_。 IEEE，2023 年，第 11 523-11 530 页。

- [149] S. Vemprala、R. Bonatti、A. Bucker 和 A. Kapoor，“机器人技术的 Chatgpt：设计原理和模型能力”，CoRR，卷。绝对/2306.17582，2023。

- [150] J. Liang、W. Huang、F. Xia、P. Xu、K. Hausman、B. Ichter、P. Florence 和 A. Zeng，“代码作为策略：用于体现控制的语言模型程序”，载于 ICRA。 IEEE，2023 年，第 9493-9500 页。

- [151] Z. Wang、S. Cai、A. Liu、X. Ma 和 Y. Liang，“描述、解释、规划和选择：使用大型语言模型进行交互式规划可实现开放世界的多任务智能体”，CoRR，第 1 卷。绝对/2302.01560，2023。

- [152] Q. Gu、A. Kuwajerwala、S. Morin、K. M. Jatavallabhula、B. Sen、A. Agarwal、C. Rivera、W. Paul、K. Ellis、R. Chellappa、C. Gan、C. M. de Melo、J. B. Tenenbaum、A. Torralba、F. Shkurti 和 L. Paull，“概念图：用于感知和规划的开放词汇 3D 场景图”，CoRR，卷。绝对/2309.16650，2023。

- [153] F. Lin、Y. Hu、P. Shen、C. Wen、J. You 和 Y. Gau，“用于机器人操作的 模仿学习（Imitation Learning） 中的数据缩放定律”，载于 ICLR，2025 年。

- [154] T. Pearce、T. Rashid、D. Bignell、R. Georgescu、S. Devlin 和 K. Hofmann，“预训练智能体和世界模型（World Models） 的缩放法则”，载于 ICML，2025 年。

- [155] Y.洪，Z.郑，P.陈，Y

。 Wang、J. Li 和 C. Gan，“乘法：3D 世界中以多感官对象为中心的大型语言模型”，载于_CVPR_。IEEE，2024 年，第 26 396-26 406 页。

- [156] S. S. Raman、V. Cohen、E. Rosen、I. Idrees、D. Paulius 和 S. Tellex，“通过纠正性重新提示规划大型语言模型”，CoRR，卷。绝对/2211.09935，2022。

- [157] P.zhi、Z.Zhang、M.Han、Z.Zhang、Z.Li、Z.Jiao、B.Jia 和 S.Huang，“GPT-4V 的闭环开放词汇移动操作”，CoRR，卷。 ABS/2404.10220，2024。

- [158] H. Fang、H. Fang、Z. Tang、J. Liu、C. Wang、J. Wang、H. Zhu 和 C. Lu，“RH20T：一次性学习多种技能的综合机器人数据集”，载于_ICRA_。 IEEE，2024 年，第 653-660 页。

- [159] A. Khazatsky、K. Pertsch、S. Nair、A. Balakrishna、S. Dasari、S. Karamcheti、S. Nasiriany、M. K. Srirama、L. Y. Chen、K. Ellis、P. D. Fagan、J. Hejna、M. Itkina、M. Lepert、Y. J. Ma、P. T. Miller、J. Wu、S. Belkhale、S. Dass、H. Ha、A. Jain、A. Lee、Y. Lee、M. Memmel、S. Park、I. Radosavovic、K. Wang、A. Zhan、K. Black、C. Chi、K. B. Hatch、S. Lin、J. Lu、J. Mercat、A. Rehman、P. R. Sanketi、A. Sharma、C. Simpson、Q. Vuong、H. R. Walke、B. Wulfe、T.Xiao、J.H.Yang、A.Yavary、T.Z.Zhao、C.Agia、R.Baijal、M.G.Castro、D.Chen、Q.Chen、T.Chung、J.Drake、E.P.Foster 等人，“DROID：大规模野外机器人操作数据集”，CoRR，卷。绝对/2403.12945，2024。

- [160] P. Sharma、L. Mohan、L. Pinto 和 A. Gupta，“多重交互变得简单 (MIME)：用于模仿的大规模演示数据”，载于 CoRL，第 1 卷。 87.PMLR，2018 年，第 906-915 页。

- [161] A. Mandlekar、Y. Zhu、A. Garg、J. Booher、M. Spero、A. Tung、J. Gau、J. Emmons、A. Gupta、E. Orbay、S. Savarese 和 L. Fei-Fei，“ROBOTURK：通过模仿学习机器人技能的众包平台”，载于 CoRL，第 1 卷。 87.PMLR，2018 年，第 879-893 页。

- [162] S. Dasari、F. Ebert、S. Tian、S. Nair、B. Bucher、K. Schmeckpeper、S. Singh、S. Levine 和 C. Finn，“Robonet：大规模多机器人学习”，载于_CoRL_，卷。 100.PMLR，2019 年，第 885-897 页。

- [163] D. Kalashnikov、J. Varley、Y. Chebotar、B. Swanson、R. Jonschkowski、C. Finn、S. Levine 和 K. Hausman，“Mt-opt：大规模连续多任务机器人 强化学习（Reinforcement Learning）”，CoRR，卷。绝对/2104.08212，2021。

- [164] V. Kumar、R. M. Shah、G. Zhou、V. Moens、V. Caggiano、A. Gupta 和 A. Rajeswaran，“Robohive：机器人学习的统一框架”，载于 NeurIPS，2023 年。

- [165] F. Ebert、Y. Yang、K. Schmeckpeper、B. Bucher、G. Georgakis、K. Daniilidis、C. Finn 和 S. Levine，“桥梁数据：利用跨域数据集促进机器人技能的泛化”，载于 RSS，2022 年。

- [166] S. Srivastava、C. Li、M. Lingelbach、R. Mart´ın-Mart´ın、F. Xia、K. E. Vainio、Z. Lian、C. Gokmen、S. Buch、C. K. Liu、S. Savarese、H. Gweon、J. Wu 和 L. Fei-Fei，“行为：虚拟、互动和生

态环境中日常家庭活动的基准”，载于_CoRL_，卷。 164.PMLR，2021 年，第 477-490 页。

- [167] C. Li、F. Xia、R. Mart´ın-Mart´ın、M. Lingelbach、S. Srivastava、B. Shen、K. E. Vainio、C. Gokmen、G. Dharan、T. Jain、A. Kurenkov、C. K. Liu、H. Gweon、J. Wu、L. Fei-Fei 和 S

。 Savarese，“igibson 2.0：机器人学习日常家务任务的以对象为中心的模拟”，《CoRL_》，卷。 164.PMLR，2021 年，第 455-465 页。

- [168] F. Xia、A. R. Zamir、Z. He、A. Sax、J. Malik 和 S. Savarese，“Gibson env：体现智能体的真实世界感知”，载于_CVPR_。计算机视觉基金会/IEEE 计算机协会，2018 年，第 9068-9079 页。

- [169] F. Xia、W. B. Shen、C. Li、P. Kasimbeg、M. Tchapmi、A. Toshev、R. Mart´ın-Mart´ın 和 S. Savarese，“交互式 gibson 基准：杂乱环境中交互式导航的基准”，IEEE Robotics Autom。 Lett.，卷。 5、没有。 2，第 713-720 页，2020 年。

- [170] B. Shen、F. Xia、C. Li、R. Mart´ın-Mart´ın、L. Fan、G. Wang、C. P´erezD'Arpino、S. Buch、S. Srivastava、L. Tchapmi、M. Tchapmi、K. Vainio、J. Wong、L. Fei-Fei 和 S. Savarese，“igibson 1.0：交互式任务的模拟环境在大型现实场景中，”在 IROS 中。 IEEE，2021 年，第 7520-7527 页。

- [171] C. Li、R. 张、J. Wong、C. Gokmen、S. Srivastava、R. Mart´ın-Mart´ın、C. Wang、G. Levine、W. Ai、B. Martinez、H. Yin、M. Lingelbach、M. Hwang、A. Hiranaka、S. Garlanka、A. Aydin、S. Lee、J. Sun、M. Anvari、M. Sharma、D. Bansal、S. Hunter、K. Kim、A. Lou、C. R. Matthews、I. Villa-Renteria、J. H. Tang、C. Tang、F. Xia、Y. Li、S. Savarese、H. Gweon、C. K. Liu、J. Wu 和 L. Fei-Fei，“BEHAVIOR1K：以人为本的 具身智能（Embodied AI） 基准，包含 1, 000 项日常活动和现实模拟，”CoRR，卷。绝对/2403.09227，2024。

- [172] F.向，Y.Qin，K.Mo，Y.Xia，H.Zhu，F.Liu，M.Liu，H.Jiang，Y.Yuan，H.Wang，L.Yi，A.X.Chang，L.J.Guibas，和H.Su，“SAPIEN：基于部件的模拟交互环境”，载于_CVPR_。计算机视觉基金会/IEEE，2020，第 11 094-11 104 页。

- [173] X. Li、K. Hsu、J. Gu、K. Pertsch、O. Mees、H. R. Walke、C. Fu、I. Lunawat、I. Sieh、S. Kirmani、S. Levine、J. Wu、C. Finn、H. Su、Q. Vuong 和 T. Xiao，“在模拟中评估现实世界的机器人操纵策略”，CoRR，卷。绝对/2405.05941，2024。

- [174] E. Kolve、R. Mottaghi、D. Gordon、Y. Zhu、A. Gupta 和 A. Farhadi，“AI2-THOR：视觉 AI 的交互式 3D 环境”，CoRR，卷。绝对/1712.05474，2017。

- [175] K. Ehsani、W. Han、A. Herrasti、E. VanderBilt、L. Weihs、E. Kolve、A. Kembhavi 和 R. Mottaghi，“Manipulathor：视觉对象操作框架”，载于_CVPR_。计算机视觉基金会/IEEE，2021 年，第 4497-4506 页。

- [176] L. Weihs、M. Deitke、A. Kembhavi 和 R. Mottaghi，“视觉房间重新布置”，载于 CVPR。计算机视觉基金会/IEEE，2021 年，第 5922-5931 页。

- [177] K. Mirakhor、S. Ghosh、D. Das 和 B. Bhowmick，“部分可观察性下视觉房间重排的任务规划”，载于 ICLR，2024 年。

- [178] X. Puig、K. Ra、M. Boben、J. Li、T. Wang、S. Fidler 和 A. Torralba，“Virtualhome：通过程序模拟家庭活动”，2018 年 IEEE 计算机视觉和模式识别会议，CVPR 2018，美国犹他州盐湖城，2018 年 6 月 18-22 日_。计算机视觉基金会/IEEE 计算机协会，2018 年，第 8494-8502 页。

- [179] C. Gan、J. Schwartz、S. Alter、D. Mrowca、M. Schrimpf、J. Traer、J. D. Freitas、J. Kubilius、A. Bhandwaldar、N. Haber、M. Sano、K. Kim、E. Wang、M. Lingelbach、A. Curtis、K. T. Feigelis、D. Bear、D

。 Gutfreund、D. D. Cox、A. Torralba、J. J. DiCarlo、J. Tenenbaum、J. H. McDermott 和 D. Yamins，“Threedworld：一个平台

interactive multi-modal physical simulation," in *NeurIPS Datasets and Benchmarks*, 2021.

[180] S. James, Z. Ma, D. R. Arrojo, and A. J. Davison, "Rlbench: The robot learning benchmark & learning environment," *IEEE Robotics Autom. Lett.*, vol. 5, no. 2, pp. 3019–3026, 2020.

[181] T. Yu, D. Quillen, Z. He, R. Julian, K. Hausman, C. Finn, and S. Levine, "Meta-world: A benchmark and evaluation for multi-task and meta reinforcement learning," in *CoRL*, vol. 100. PMLR, 2019, pp. 1094–1100.

[182] O. Mees, L. Hermann, E. Rosete-Beas, and W. Burgard, "CALVIN: A benchmark for language-conditioned policy learning for long-horizon robot manipulation tasks," *IEEE Robotics Autom. Lett.*, vol. 7, no. 3, pp. 7327–7334, 2022.

[183] A. Gupta, V. Kumar, C. Lynch, S. Levine, and K. Hausman, "Relay policy learning: Solving long-horizon tasks via imitation and reinforcement learning," in *CoRL*, vol. 100. PMLR, 2019, pp. 1025–1037.

[184] M. Savva, J. Malik, D. Parikh, D. Batra, A. Kadian, O. Maksymets, Y. Zhao, E. Wijmans, B. Jain, J. Straub, J. Liu, and V. Koltun, "Habitat: A platform for embodied AI research," in *ICCV*. IEEE, 2019, pp. 9338–9346.

[185] A. Szot, A. Clegg, E. Undersander, E. Wijmans, Y. Zhao, J. M. Turner, N. Maestre, M. Mukadam, D. S. Chaplot, O. Maksymets, A. Gokaslan, V. Vondrus, S. Dharur, F. Meier, W. Galuba, A. X. Chang, Z. Kira, V. Koltun, J. Malik, M. Savva, and D. Batra, "Habitat 2.0: Training home assistants to rearrange their habitat," in *NeurIPS*, 2021, pp. 251–266.

[186] D. Batra, A. X. Chang, S. Chernova, A. J. Davison, J. Deng, V. Koltun, S. Levine, J. Malik, I. Mordatch, R. Mottaghi, M. Savva, and H. Su, "Rearrangement: A challenge for embodied AI," *CoRR*, vol. abs/2011.01975, 2020.

[187] S. Yenamandra, A. Ramachandran, K. Yadav, A. S. Wang, M. Khanna, T. Gervet, T. Yang, V. Jain, A. Clegg, J. M. Turner, Z. Kira, M. Savva, A. X. Chang, D. S. Chaplot, D. Batra, R. Mottaghi, Y. Bisk, and C. Paxton, "Homerobot: Open-vocabulary mobile manipulation," in *CoRL*, vol. 229. PMLR, 2023, pp. 1975–2011.

[188] M. Shridhar, J. Thomason, D. Gordon, Y. Bisk, W. Han, R. Mottaghi, L. Zettlemoyer, and D. Fox, "ALFRED: A benchmark for interpreting grounded instructions for everyday tasks," in *CVPR*. Computer Vision Foundation / IEEE, 2020, pp. 10 737–10 746.

[189] Y. Tassa, Y. Doron, A. Muldal, T. Erez, Y. Li, D. de Las Casas, D. Budden, A. Abdolmaleki, J. Merel, A. Lefrancq, T. P. Lillicrap, and M. A. Riedmiller, "Deepmind control suite," *CoRR*, vol. abs/1801.00690, 2018.

[190] G. Brockman, V. Cheung, L. Pettersson, J. Schneider, J. Schulman, J. Tang, and W. Zaremba, "Openai gym," *CoRR*, vol. abs/1606.01540, 2016.

[191] Genesis-Team, "Genesis: A universal and generative physics engine for robotics and beyond," December 2024. [Online]. Available: https://github.com/Genesis-Embodied-AI/Genesis

[192] Y. Wang, Z. Xian, F. Chen, T. Wang, Y. Wang, K. Fragkiadaki, Z. Erickson, D. Held, and C. Gan, "Robogen: Towards unleashing infinite data for automated robot learning via generative simulation," in *ICML*, 2024.

[193] M. Ahn, D. Dwibedi, C. Finn, M. G. Arenas, K. Gopalakrishnan, K. Hausman, B. Ichter, A. Irpan, N. J. Joshi, R. Julian, S. Kirmani, I. Leal, T. E. Lee, S. Levine, Y. Lu, S. Maddineni, K. Rao, D. Sadigh, P. Sanketi, P. Sermanet, Q. Vuong, S. Welker, F. Xia, T. Xiao, P. Xu, S. Xu, and Z. Xu, "Autort: Embodied foundation models for large scale orchestration of robotic agents," *CoRR*, vol. abs/2401.12963, 2024.

[194] T. Xiao, H. Chan, P. Sermanet, A. Wahid, A. Brohan, K. Hausman, S. Levine, and J. Tompson, "Robotic skill acquisition via instruction augmentation with vision-language models," in *RSS*, 2023.

[195] Y. Ma, D. Chi, J. Li, K. Song, Y. Zhuang, and I. King, "VOLTA: improving generative diversity by variational mutual information maximizing autoencoder," in *NAACL-HLT (Findings)*, 2024.

[196] C. Chi, Z. Xu, C. Pan, E. Cousineau, B. Burchfiel, S. Feng, R. Tedrake, and S. Song, "Universal manipulation interface: In-the-wild robot teaching without in-the-wild robots," in *RSS*, 2024.

[197] G. Moon, S. Saito, W. Xu, R. Joshi, J. Buffalini, H. Bellan, N. Rosen, J. Richardson, M. Mize, P. de Bree, T. Simon, B. Peng, S. Garg, K. McPhail, and T. Shiratori, "A dataset of relighted 3d interacting hands," in *NeurIPS*, 2023.

[198] Y. Chen, Y. Ge, Y. Ge, M. Ding, B. Li, R. Wang, R. Xu, Y. Shan, and X. Liu, "Egoplan-bench: Benchmarking egocentric embodied planning with multimodal large language models," *CoRR*, vol. abs/2312.06722, 2023.

[199] K. Valmeekam, M. Marquez, A. O. Hernandez, S. Sreedharan, and S. Kambhampati, "Planbench: An extensible benchmark for evaluating large language models on planning and reasoning about change," in *NeurIPS*, 2023.

[200] K. Valmeekam, M. Marquez, S. Sreedharan, and S. Kambhampati, "On the planning abilities of large language models - A critical investigation," in *NeurIPS*, 2023.

[201] J. Choi, Y. Yoon, H. Ong, J. Kim, and M. Jang, "Lota-bench: Benchmarking language-oriented task planners for embodied agents," *CoRR*, vol. abs/2402.08178, 2024.

[202] M. Li, S. Zhao, Q. Wang, K. Wang, Y. Zhou, S. Srivastava, C. Gokmen, T. Lee, L. E. Li, R. Zhang, W. Liu, P. Liang, L. Fei-Fei, J. Mao, and J. Wu, "Embodied agent interface: Benchmarking llms for embodied decision making," in *NeurIPS*, 2024.

[203] A. Das, S. Datta, G. Gkioxari, S. Lee, D. Parikh, and D. Batra, "Embodied question answering," in *CVPR*. Computer Vision Foundation / IEEE Computer Society, 2018, pp. 1–10.

[204] D. Gordon, A. Kembhavi, M. Rastegari, J. Redmon, D. Fox, and A. Farhadi, "IQA: visual question answering in interactive environments," in *CVPR*. Computer Vision Foundation / IEEE Computer Society, 2018, pp. 4089–4098.

[205] L. Yu, X. Chen, G. Gkioxari, M. Bansal, T. L. Berg, and D. Batra, "Multi-target embodied question answering," in *CVPR*. Computer Vision Foundation / IEEE, 2019, pp. 6309–6318.

[206] E. Wijmans, S. Datta, O. Maksymets, A. Das, G. Gkioxari, S. Lee, I. Essa, D. Parikh, and D. Batra, "Embodied question answering in photorealistic environments with point cloud perception," in *CVPR*. Computer Vision Foundation / IEEE, 2019, pp. 6659–6668.

[207] C. Fan, "Egovqa - an egocentric video question answering benchmark dataset," in *ICCV Workshops*. IEEE, 2019, pp. 4359–4366.

[208] B. Jia, T. Lei, S. Zhu, and S. Huang, "Egotaskqa: Understanding human tasks in egocentric videos," in *NeurIPS*, 2022.

[209] M. M. Islam, A. Gladstone, R. Islam, and T. Iqbal, "EQA-MX: embodied question answering using multimodal expression," in *ICLR*, 2024.

[210] A. Majumdar, A. Ajay, X. Zhang, P. Putta, S. Yenamandra, M. Henaff, S. Silwal, P. Mcvay, O. Maksymets, S. Arnaud, K. Yadav, Q. Li, B. Newman, M. Sharma, V. Berges, S. Zhang, P. Agrawal, Y. Bisk, D. Batra, M. Kalakrishnan, F. Meier, C. Paxton, S. Sax, and A. Rajeswaran, "OpenEQA: Embodied Question Answering in the Era of Foundation Models," in *CVPR*, 2024.

[211] R. Girdhar, A. El-Nouby, Z. Liu, M. Singh, K. V. Alwala, A. Joulin, and I. Misra, "Imagebind one embedding space to bind them all," in *CVPR*. IEEE, 2023, pp. 15 180–15 190.

[212] B. Zhu, B. Lin, M. Ning, Y. Yan, J. Cui, H. Wang, Y. Pang, W. Jiang, J. Zhang, Z. Li, C. Zhang, Z. Li, W. Liu, and L. Yuan, "Languagebind: Extending video-language pretraining to n-modality by language-based semantic alignment," in *ICLR*, 2024.

[213] I. Güzey, B. Evans, S. Chintala, and L. Pinto, "Dexterity from touch: Self-supervised pre-training of tactile representations with robotic play," in *CoRL*, vol. 229. PMLR, 2023, pp. 3142–3166.

[214] R. Zhang, A. Saran, B. Liu, Y. Zhu, S. Guo, S. Niekum, D. Ballard, and M. Hayhoe, "Human gaze assisted artificial intelligence: A review," in *IJCAI-20*, 7 2020, pp. 4951–4958, survey track.

[215] R. Zhang, Z. Liu, L. Zhang, J. A. Whritner, K. S. Muller, M. M. Hayhoe, and D. H. Ballard, "AGIL: learning attention from human for visuomotor tasks," in *ECCV (11)*, vol. 11215. Springer, 2018, pp. 692–707.

[216] A. Saran, R. Zhang, E. S. Short, and S. Niekum, "Efficiently guiding imitation learning agents with human gaze," in *AAMAS*. ACM, 2021, pp. 1109–1117.

[217] S. Guo, R. Zhang, B. Liu, Y. Zhu, D. H. Ballard, M. M. Hayhoe, and P. Stone, "Machine versus human attention in deep reinforcement learning tasks," in *NeurIPS*, 2021, pp. 25 370–25 385.

[218] H. M. Le, T. N. Do, and S. J. Phee, "A survey on actuators-driven surgical robots," *Sensors and Actuators A-physical*, vol. 247, pp. 323–354, 2016.

[219] M. S. Laursen, J. S. Pedersen, S. A. Just, T. R. Savarimuthu, B. Blomholt, J. K. H. Andersen, and P. J. Vinholt, "Factors facilitating the acceptance of diagnostic robots in healthcare: A survey," in *ICHI*. IEEE, 2022, pp. 442–448.

[220] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *CVPR*. IEEE Computer Society, 2016, pp. 770–778.

[221] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby, "An image is worth 16x16 words: Transformers for image recognition at scale," in *ICLR*, 2021.

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交互式多模态物理模拟”，《NeurIPS 数据集和基准》，2021 年。

- [180] S. James、Z. Ma、D. R. Arrojo 和 A. J. Davison，“Rlbench：机器人学习基准和学习环境”，IEEE Robotics Autom。 Lett.，卷。 5、没有。 2，第 3019-3026 页，2020 年。

- [181] T. Yu、D. Quillen、Z. He、R. Julian、K. Hausman、C. Finn 和 S. Levine，“元世界：多任务和元强化学习（Reinforcement Learning）的基准和评估”，载于_CoRL_，卷。 100.PMLR，2019 年，第 1094-1100 页。

- [182] O. Mees、L. Hermann、E. Rosete-Beas 和 W. Burgard，“CALVIN：长时程（Long-Horizon）机器人操作任务的语言条件策略学习基准”，IEEE Robotics Autom。 Lett.，卷。 7、没有。 3，第 7327-7334 页，2022 年。

- [183] A. Gupta、V. Kumar、C. Lynch、S. Levine 和 K. Hausman，“中继策略学习：通过模仿和 强化学习（Reinforcement Learning）解决 长时程（Long-Horizon）任务”，载于_CoRL_，卷。 100.PMLR，2019 年，第 1025-1037 页。

- [184] M. Savva、J. Malik、D. Parikh、D. Batra、A. Kadian、O. Maksymets、Y. Zhao、E. Wijmans、B. Jain、J. Straub、J. Liu 和 V. Koltun，“栖息地：具身智能（Embodied AI）研究平台”，载于_ICCV_。IEEE，2019 年，第 9338-9346 页。

- [185] A. Szot、A. Clegg、E. Undersander、E. Wijmans、Y. Zhao、J. M. Turner、N. Maestre、M. Mukadam、D. S. Chaplot、O. Maksymets、A. Gokaslan、V. Vondrus、S. Dharur、F. Meier、W. Galuba、A. X. Chang、Z. Kira、V. Koltun、J. Malik、M. Savva 和 D. Batra，“栖息地 2.0：训练家庭助理重新安排栖息地”，载于 NeurIPS，2021 年，第 251-266 页。

- [186] D. Batra、A. X. Chang、S. Chernova、A. J. Davison、J. Deng、V. Koltun、S. Levine、J. Malik、I. Mordatch、R. Mottaghi、M. Savva 和 H. Su，“重排：具身智能（Embodied AI）的挑战”，CoRR，卷。绝对/2011.01975，2020。

- [187] S. Yenamandra、A. Ramachandran、K. Yadav、A. S. Wang、M. Khanna、T. Gervet、T. Yang、V. Jain、A. Clegg、J. M. Turner、Z. Kira、M. Savva、A. X. Chang、D. S. Chaplot、D. Batra、R. Mottaghi、Y. Bisk 和 C. Paxton，“Homerobot：开放词汇移动操作，”，in CoRL，卷。 229.PMLR，2023 年，第 1975-2011 页。

- [188] M. Shridhar、J. Thomason、D. Gordon、Y. Bisk、W. Han、R. Mottaghi、L. Zettlemoyer 和 D. Fox，“ALFRED：解释日常任务的接地指令的基准”，载于_CVPR_。计算机视觉基金会/IEEE，2020，第 10 737-10 746 页。

- [189] Y. Tassa、Y. Doron、A. Muldal、T. Erez、Y. Li、D. de Las Casas、D. Budden、A. Abdolmaleki、J. Merel、A. Lefrancq、T. P. Lillicrap 和 M. A. Riedmiller，“Deepmind 控制套件”，CoRR，卷。绝对/1801.00690，2018。

- [190] G. Brockman、V. Cheung、L. Pettersson、J. Schneider、J. Schulman、J. Tang 和 W. Zaremba，“Openai 健身房”，CoRR，卷。绝对/1606.01540，2016。

- [191] Genesis-Team，“Genesis：用于机器人及其他领域的通用生成物理引擎”，2024 年 12 月。[在线]。可用：https://github.com/Genesis-Embodied-AI/Genesis

- [192] Y. Wang、Z. Xian、F. Chen、T. Wang、Y. Wang、K. Fragkiadaki、Z. Erickson、D. Held 和 C. Gan，“Robogen：通过生成模拟为自动机器人学习释放无限数据”，载于 ICML，2024 年。

- [193] M. Ahn、D. Dwibedi、C. Finn、M. G. Arenas、K. Gopalakrishnan、K. Hausman、B. Ichter、A

。 Irpan、N. J. Joshi、R. Julian、S. Kirmani、I. Leal、T. E. Lee、S. Levine、Y. Lu、S. Maddineni、K. Rao、D. Sadigh、P. Sanketi、P. Sermanet、Q. Vuong、S. Welker、F. Xia、T. Xiao、P. Xu、S. Xu 和 Z. Xu，《Autort：体现》 基础模型（Foundation Models） 用于大规模机器人智能体编排”，CoRR，卷。绝对/2401.12963，2024。

- [194] T.Xiao、H.Chan、P.Sermanet、A.Wahid、A.Brohan、K.Hausman、S.Levine 和 J.Tompson，“通过视觉语言模型的指令增强获得机器人技能”，载于 RSS，2023 年。

- [195] Y. Ma、D. Chi、J. Li、K. Song、Y. Zhuang 和 I. King，“VOLTA：通过变分互信息最大化自动编码器提高生成多样性”，载于_NAACL-HLT（调查结果）_，2024 年。

- [196] C. Chi、Z. Xu、C. Pan、E. Cousineau、B. Burchfiel、S. Feng、R. Tedrake 和 S. Song，“通用操作界面：无需野外机器人的野外机器人教学”，载于_RSS_，2024 年。

- [197] G. Moon、S. Saito、W. Xu、R. Joshi、J. Buffalini、H. Bellan、N. Rosen、J. Richardson、M. Mize、P. de Bree、T. Simon、B. Peng、S. Garg、K. McPhail 和 T. Shiratori，“重新点亮的 3d 交互手数据集”，《NeurIPS》，2023 年。

- [198] Y. Chen、Y. Ge、Y. Ge、M. Ding、B. Li、R. Wang、R. Xu、Y. Shan 和 X. Liu，“Egoplan-bench：使用多模态大语言模型对自我中心体现规划进行基准测试”，CoRR，卷。绝对/2312.06722，2023。

- [199] K. Valmeekam、M. Marquez、A. O. Hernandez、S. Sreedharan 和 S. Kambhampati，“Planbench：用于评估规划和推理变化的大型语言模型的可扩展基准”，载于 NeurIPS，2023 年。

- [200] K. Valmeekam、M. Marquez、S. Sreedharan 和 S. Kambhampati，“关于大型语言模型的规划能力 - 一项批判性调查”，《NeurIPS》，2023 年。

- [201] J. Choi、Y. Yoon、H. Ong、J. Kim 和 M. Jang，“Lota-bench：针对具体智能体的面向语言的任务规划器的基准测试”，CoRR，卷。绝对/2402.08178，2024。

- [202] M. Li、S. Zhu、Q. Wang、K. Wang、Y. Zhou、S. Srivastava、C. Gokmen、T. Lee、L. E. Li、R. Zhu、W. Liu、P. Liang、L. Fei-Fei、J. Mao 和 J. Wu，“体现智能体界面：体现决策的基准化 llms”，载于 NeurIPS，2024 年。

- [203] A. Das、S. Datta、G. Gkioxari、S. Lee、D. Parikh 和 D. Batra，“体现问答”，载于 CVPR。计算机视觉基金会/IEEE 计算机协会，2018 年，第 1-10 页。

- [204] D. Gordon、A. Kembhavi、M. Rastegari、J. Redmon、D. Fox 和 A. Farhadi，“IQA：交互式环境中的视觉问答”，载于_CVPR_。计算机视觉基金会/IEEE 计算机协会，2018 年，第 4089-4098 页。

- [205] L. Yu、X. Chen、G. Gkioxari、M. Bansal、T. L. Berg 和 D. Batra，“多目标具体问答”，载于 _CVPR_。计算机视觉基金会/IEEE，2019 年，第 6309-6318 页。

- [206] E. Wijmans、S. Datta、O. Maksymets、A. Das、G. Gkioxari、S. Lee、I. Essa、D. Parikh 和 D. Batra，“在具有点云感知的真实感环境中实现问答”，载于_CVPR_。计算机视觉基金会/IEEE，2019 年，第 6659-6668 页。

- [207] C. Fan，“Egovqa - 一个以自我为中心的视频问答基准数据集”，载于 ICCV Workshops。 IEEE，2019 年，第 4359-4366 页。

- [208] B. Jia、T. Lei、S. Zhu 和 S

。 Huang，“Egotaskqa：理解以自我为中心的视频中的人类任务”，《NeurIPS》，2022 年。

- [209] M. M. Islam、A. Gladstone、R. Islam 和 T. Iqbal，“EQA-MX：使用多模态表达的具体问答”，载于 ICLR，2024 年。

- [210] A. Majumdar、A. Ajay、X. 张、P. Putta、S. Yenamandra、M. Henaff、S. Silwal、P. Mcvay、O. Maksymets、S. Arnaud、K. Yadav、Q. Li、B. Newman、M. Sharma、V. Berges、S. 张、P. Agrawal、Y. Bisk、D. Batra、M. Kalakrishnan、F. Meier、C. Paxton、S. Sax 和 A. Rajeswaran，“OpenEQA：基础模型（Foundation Models）时代的嵌入式问答”，载于 CVPR，2024 年。

- [211] R. Girdhar、A. El-Nouby、Z. Liu、M. Singh、K. V. Alwala、A. Joulin 和 I. Misra，“图像绑定一个嵌入空间以将它们全部绑定”，载于_CVPR_。 IEEE，2023 年，第 15 180-15 190 页。

- [212] B. Zhu、B. Lin、M. Ning、Y. Yan、J. Cui、H. Wang、Y. Pang、W. Jiang、J. Zhang、Z. Li、C. Zhu、Z. Li、W. Liu 和 L. Yuan，“Languagebind：通过基于语言的语义对齐将视频语言预训练扩展到 n 模态”，载于 ICLR，2024 年。

- [213] I. Gúuzey、B. Evans、S. Chintala 和 L. Pinto，“触摸的敏捷性：通过机器人游戏进行触觉表征的自我监督预训练”，载于_CoRL_，卷。 229.PMLR，2023 年，第 3142-3166 页。

- [214] R.Zhang、A.Saran、B.Liu、Y.Zhu、S.Guo、S.Niekum、D.Ballard 和 M.Hayhoe，“人类凝视辅助人工智能：综述”，载于 IJCAI-20，2020 年 7 月，第 4951-4958 页，调查轨迹。

- [215] R.Zhang、Z.Liu、L.Zhang、J.A.Whritner、K.S.Muller、M.M.Hayhoe 和 D.H.Ballard，“AGIL：学习人类视觉运动任务的注意力”，载于 ECCV (11)，第 1 卷。 11215。施普林格，2018，第 692-707 页。

- [216] A. Saran、R. Zhu、E. S. Short 和 S. Niekum，“用人类凝视有效引导 模仿学习（Imitation Learning）智能体”，载于_AAMAS_。 ACM，2021 年，第 1109-1117 页。

- [217] S.Guo、R.Zhang、B.Liu、Y.Zhu、D.H.Ballard、M.M.Hayhoe 和 P.Stone，“深度 强化学习（Reinforcement Learning）任务中的机器与人类注意力”，载于 NeurIPS，2021 年，第 25 370-25 385 页。

- [218] H. M. Le、T. N. Do 和 S. J. Phee，“关于执行器驱动的手术机器人的调查”，传感器和执行器 A-physical，卷。 247，第 323-354 页，2016 年。

- [219] M. S. Laursen、J. S. Pedersen、S. A. Just、T. R. Savarimuthu、B. Blomholt、J. K. H. Andersen 和 P. J. Vinholt，“促进医疗保健领域接受诊断机器人的因素：一项调查”，载于_ICHI_。 IEEE，2022 年，第 442-448 页。

- [220] K. He、X. Zhang、S. Ren 和 J. Sun，“用于图像识别的深度残差学习”，载于_CVPR_。 IEEE 计算机协会，2016 年，第 770-778 页。

- [221] A. Dosovitskiy、L. Beyer、A. Kolesnikov、D. Weissenborn、X. Zhai、T. Unterthiner、M. Dehghani、M. Minderer、G. Heigold、S. Gelly、J. Uszkoreit 和 N. Houlsby，“一张图像相当于 16x16 个单词：用于大规模图像识别的 Transformers”，载于_ICLR_， 2021 年。

[222] A. Kirillov, E. Mintun, N. Ravi, H. Mao, C. Rolland, L. Gustafson, T. Xiao, S. Whitehead, A. C. Berg, W. Lo, P. Dollár, and R. B. Girshick, "Segment anything," *CoRR*, vol. abs/2304.02643, 2023.

[223] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Comput.*, vol. 9, no. 8, pp. 1735–1780, 1997.

[224] K. Cho, B. van Merrienboer, Ç. Gülçehre, D. Bahdanau, F. Bougares, H. Schwenk, and Y. Bengio, "Learning phrase representations using RNN encoder-decoder for statistical machine translation," in *EMNLP*. ACL, 2014, pp. 1724–1734.

[225] OpenAI. (2023) Introducing chatgpt. [Online]. Available: https://openai.com/blog/chatgpt

[226] D. Silver, A. Huang, C. J. Maddison, A. Guez, L. Sifre, G. van den Driessche, J. Schrittwieser, I. Antonoglou, V. Panneershelvam, M. Lanctot, S. Dieleman, D. Grewe, J. Nham, N. Kalchbrenner, I. Sutskever, T. P. Lillicrap, M. Leach, K. Kavukcuoglu, T. Graepel, and D. Hassabis, "Mastering the game of go with deep neural networks and tree search," *Nat.*, vol. 529, no. 7587, pp. 484–489, 2016.

[227] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," *CoRR*, vol. abs/1707.06347, 2017.

[228] OpenAI, M. Andrychowicz, B. Baker, M. Chociej, R. Józefowicz, B. McGrew, J. Pachocki, A. Petron, M. Plappert, G. Powell, A. Ray, J. Schneider, S. Sidor, J. Tobin, P. Welinder, L. Weng, and W. Zaremba, "Learning dexterous in-hand manipulation," *CoRR*, vol. abs/1808.00177, 2018.

[229] R. Rafailov, A. Sharma, E. Mitchell, C. D. Manning, S. Ermon, and C. Finn, "Direct preference optimization: Your language model is secretly a reward model," in *NeurIPS*, 2023.

[230] X. Chen, H. Fang, T. Lin, R. Vedantam, S. Gupta, P. Dollár, and C. L. Zitnick, "Microsoft COCO captions: Data collection and evaluation server," *CoRR*, vol. abs/1504.00325, 2015.

[231] S. Antol, A. Agrawal, J. Lu, M. Mitchell, D. Batra, C. L. Zitnick, and D. Parikh, "VQA: visual question answering," in *ICCV*. IEEE Computer Society, 2015, pp. 2425–2433.

[232] L. Yu, P. Poirson, S. Yang, A. C. Berg, and T. L. Berg, "Modeling context in referring expressions," in *ECCV (2)*, ser. Lecture Notes in Computer Science, vol. 9906. Springer, 2016, pp. 69–85.

[233] O. Vinyals, A. Toshev, S. Bengio, and D. Erhan, "Show and tell: A neural image caption generator," in *CVPR*. IEEE Computer Society, 2015, pp. 3156–3164.

[234] A. Radford, K. Narasimhan, T. Salimans, and I. Sutskever, "Improving language understanding by generative pre-training," *OpenAI blog*, 2018.

[235] J. Lu, D. Batra, D. Parikh, and S. Lee, "Vilbert: Pretraining task-agnostic visiolinguistic representations for vision-and-language tasks," in *NeurIPS*, 2019, pp. 13–23.

[236] A. Khan, A. Sohail, U. Zahoora, and A. S. Qureshi, "A survey of the recent architectures of deep convolutional neural networks," *Artif. Intell. Rev.*, vol. 53, no. 8, pp. 5455–5516, 2020.

[237] S. H. Khan, M. Naseer, M. Hayat, S. W. Zamir, F. S. Khan, and M. Shah, "Transformers in vision: A survey," *ACM Comput. Surv.*, vol. 54, no. 10s, pp. 200:1–200:41, 2022.

[238] Y. LeCun, B. E. Boser, J. S. Denker, D. Henderson, R. E. Howard, W. E. Hubbard, and L. D. Jackel, "Backpropagation applied to handwritten zip code recognition," *Neural Comput.*, vol. 1, no. 4, pp. 541–551, 1989.

[239] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," in *ICLR*, 2015.

[240] C. Szegedy, W. Liu, Y. Jia, P. Sermanet, S. E. Reed, D. Anguelov, D. Erhan, V. Vanhoucke, and A. Rabinovich, "Going deeper with convolutions," in *CVPR*. IEEE Computer Society, 2015, pp. 1–9.

[241] C. Szegedy, S. Ioffe, V. Vanhoucke, and A. A. Alemi, "Inception-v4, inception-resnet and the impact of residual connections on learning," in *AAAI*. AAAI Press, 2017, pp. 4278–4284.

[242] S. Xie, R. B. Girshick, P. Dollár, Z. Tu, and K. He, "Aggregated residual transformations for deep neural networks," in *CVPR*. IEEE Computer Society, 2017, pp. 5987–5995.

[243] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *CVPR*. Computer Vision Foundation / IEEE Computer Society, 2018, pp. 7132–7141.

[244] M. Tan and Q. V. Le, "Efficientnet: Rethinking model scaling for convolutional neural networks," in *ICML*, vol. 97. PMLR, 2019, pp. 6105–6114.

[245] R. B. Girshick, J. Donahue, T. Darrell, and J. Malik, "Rich feature hierarchies for accurate object detection and semantic segmentation," in *CVPR*. IEEE Computer Society, 2014, pp. 580–587.

[246] R. B. Girshick, "Fast R-CNN," in *ICCV*. IEEE Computer Society, 2015, pp. 1440–1448.

[247] S. Ren, K. He, R. B. Girshick, and J. Sun, "Faster R-CNN: towards real-time object detection with region proposal networks," in *NIPS*, 2015, pp. 91–99.

[248] K. He, G. Gkioxari, P. Dollár, and R. B. Girshick, "Mask R-CNN," in *ICCV*. IEEE Computer Society, 2017, pp. 2980–2988.

[249] J. Redmon, S. K. Divvala, R. B. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *CVPR*. IEEE Computer Society, 2016, pp. 779–788.

[250] T. Lin, P. Dollár, R. B. Girshick, K. He, B. Hariharan, and S. J. Belongie, "Feature pyramid networks for object detection," in *CVPR*. IEEE Computer Society, 2017, pp. 936–944.

[251] T. Lin, P. Goyal, R. B. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *ICCV*. IEEE Computer Society, 2017, pp. 2999–3007.

[252] P. Anderson, X. He, C. Buehler, D. Teney, M. Johnson, S. Gould, and L. Zhang, "Bottom-up and top-down attention for image captioning and visual question answering," in *CVPR*. Computer Vision Foundation / IEEE Computer Society, 2018, pp. 6077–6086.

[253] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *CVPR*. IEEE Computer Society, 2015, pp. 3431–3440.

[254] V. Badrinarayanan, A. Kendall, and R. Cipolla, "Segnet: A deep convolutional encoder-decoder architecture for image segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 12, pp. 2481–2495, 2017.

[255] O. Ronneberger, P. Fischer, and T. Brox, "U-net: Convolutional networks for biomedical image segmentation," in *MICCAI (3)*, ser. Lecture Notes in Computer Science, vol. 9351. Springer, 2015, pp. 234–241.

[256] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in *ECCV (1)*, ser. Lecture Notes in Computer Science, vol. 12346. Springer, 2020, pp. 213–229.

[257] R. Strudel, R. G. Pinel, I. Laptev, and C. Schmid, "Segmenter: Transformer for semantic segmentation," in *ICCV*. IEEE, 2021, pp. 7242–7252.

[258] A. Ioannidou, E. Chatzilari, S. Nikolopoulos, and I. Kompatsiaris, "Deep learning advances in computer vision with 3d data: A survey," *ACM Comput. Surv.*, vol. 50, no. 2, pp. 20:1–20:38, 2017.

[259] E. Ahmed, A. Saint, A. E. R. Shabayek, K. Cherenkova, R. Das, G. Gusev, D. Aouada, and B. E. Ottersten, "Deep learning advances on different 3d data representations: A survey," *CoRR*, vol. abs/1808.01462, 2018.

[260] Y. Guo, H. Wang, Q. Hu, H. Liu, L. Liu, and M. Bennamoun, "Deep learning for 3d point clouds: A survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 12, pp. 4338–4364, 2021.

[261] C. R. Qi, H. Su, M. Nießner, A. Dai, M. Yan, and L. J. Guibas, "Volumetric and multi-view cnns for object classification on 3d data," in *CVPR*. IEEE Computer Society, 2016, pp. 5648–5656.

[262] Y. Feng, Y. Feng, H. You, X. Zhao, and Y. Gao, "Meshnet: Mesh neural network for 3d shape representation," in *AAAI*. AAAI Press, 2019, pp. 8279–8286.

[263] D. W. Otter, J. R. Medina, and J. K. Kalita, "A survey of the usages of deep learning for natural language processing," *IEEE Trans. Neural Networks Learn. Syst.*, vol. 32, no. 2, pp. 604–624, 2021.

[264] P. Liu, W. Yuan, J. Fu, Z. Jiang, H. Hayashi, and G. Neubig, "Pre-train, prompt, and predict: A systematic survey of prompting methods in natural language processing," *ACM Comput. Surv.*, vol. 55, no. 9, pp. 195:1–195:35, 2023.

[265] Y. Bengio, R. Ducharme, and P. Vincent, "A neural probabilistic language model," in *NIPS*. MIT Press, 2000, pp. 932–938.

[266] T. Mikolov, I. Sutskever, K. Chen, G. S. Corrado, and J. Dean, "Distributed representations of words and phrases and their compositionality," in *NIPS*, 2013, pp. 3111–3119.

[267] T. Mikolov, K. Chen, G. Corrado, and J. Dean, "Efficient estimation of word representations in vector space," in *ICLR (Workshop Poster)*, 2013.

[268] J. Pennington, R. Socher, and C. D. Manning, "Glove: Global vectors for word representation," in *EMNLP*. ACL, 2014, pp. 1532–1543.

[269] J. L. Elman, "Finding structure in time," *Cogn. Sci.*, vol. 14, no. 2, pp. 179–211, 1990.

[270] D. Bahdanau, K. Cho, and Y. Bengio, "Neural machine translation by jointly learning to align and translate," in *ICLR*, 2015.

[271] Z. Huang, W. Xu, and K. Yu, "Bidirectional LSTM-CRF models for sequence tagging," *CoRR*, vol. abs/1508.01991, 2015.

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- [222] A. Kirillov、E. Mintun、N. Ravi、H. Mao、C. Rolland、L. Gustafson、T. Xiao、S. Whitehead、A. C. Berg、W. Lo、P. Doll´ar 和 R. B. Girshick，“分割任何东西”，CoRR，卷。绝对/2304.02643，2023。

- [223] S. Hochreiter 和 J. Schmidhuber，“长短期记忆”，Neural Comput.，卷。 9、不。 8，第 1735-1780 页，1997 年。

- [224] K. Cho, B. van Merrienboer, C¸. Gülc¸ehre、D. Bahdanau、F. Bougares、H. Schwenk 和 Y. Bengio，“使用 RNN 编码器-解码器进行统计机器翻译来学习短语表示”，载于_EMNLP_。 ACL，2014 年，第 1724-1734 页。

- [225] 开放人工智能。 (2023) 介绍chatgpt。 [在线的]。可用：https://openai.com/blog/chatgpt

- [226] D. Silver、A. Huang、C. J. Maddison、A. Guez、L. Sifre、G. van den Driessche、J. Schrittwieser、I. Antonoglou、V. Panneershelvam、M. Lanctot、S. Dieleman、D. Grewe、J. Nham、N. Kalchbrenner、I. Sutskever、T. P. Lillicrap、M. Leach， K. Kavukcuoglu、T. Graepel 和 D. Hassabis，“通过深度神经网络和树搜索掌握围棋游戏”，Nat.，卷。 529，没有。 7587，第 484-489 页，2016 年。

- [227] J. Schulman、F. Wolski、P. Dhariwal、A. Radford 和 O. Klimov，“近端策略优化算法”，CoRR，卷。绝对/1707.06347，2017。

- [228] OpenAI、M. Andrychowicz、B. Baker、M. Chociej、R. J´ozefowicz、B. McGrew、J. Pachocki、A. Petron、M. Plappert、G. Powell、A. Ray、J. Schneider、S. Sidor、J. Tobin、P. Welinder、L. Weng 和 W. Zaremba，“学习灵巧手动操作，”CoRR，卷。绝对/1808.00177，2018。

- [229] R. Rafailov、A. Sharma、E. Mitchell、C. D. Manning、S. Ermon 和 C. Finn，“直接偏好优化：你的语言模型实际上是一个奖励模型”，载于 NeurIPS，2023 年。

- [230] X. Chen、H. Fang、T. Lin、R. Vedantam、S. Gupta、P. Doll´ar 和 C. L. Zitnick，“Microsoft COCO 说明：数据收集和评估服务器”，CoRR，卷。绝对/1504.00325，2015。

- [231] S. Antol、A. Agrawal、J. Lu、M. Mitchell、D. Batra、C. L. Zitnick 和 D. Parikh，“VQA：视觉问答”，载于_ICCV_。 IEEE 计算机协会，2015 年，第 2425-2433 页。

- [232] L. Yu、P. Poirson、S. Yang、A. C. Berg 和 T. L. Berg，“在指称表达式中建模上下文”，载于_ECCV (2)_，序列号。计算机科学讲义，卷。 9906。施普林格，2016 年，第 69-85 页。

- [233] O. Vinyals、A. Toshev、S. Bengio 和 D. Erhan，“展示并讲述：神经图像字幕生成器”，载于 CVPR。 IEEE 计算机协会，2015 年，第 3156-3164 页。

- [234] A. Radford、K. Narasimhan、T. Salimans 和 I. Sutskever，“通过生成预训练提高语言理解”，OpenAI 博客，2018 年。

- [235] J. Lu、D. Batra、D. Parikh 和 S. Lee，“Vilbert：针对视觉和语言任务预训练任务无关的视觉语言表征”，载于 NeurIPS，2019 年，第 13-23 页。

- [236] A. Khan、A. Sohail、U. Zahoora 和 A. S. Qureshi，“深度卷积神经网络最新架构的调查”，Artif。英特尔。 Rev.，卷。 53，没有。 8，第 5455-5516 页，2020 年。

- [237] S. H. Khan、M. Naseer、M. Hayat、S. W. Zamir、F. S. Khan 和 M. Shah，“愿景中的变形金刚：一项调查”，ACM Comput。 Surv.，卷。 54，没有。 10 秒，第 200:1-200:41，2022 年。

- [238] Y. LeCun、B. E. Boser、J. S. Denker、D. Henderson、R

。 E. Howard、W. E. Hubbard 和 L. D. Jackel，“反向传播应用于手写邮政编码识别”，Neural Comput.，卷。 1、没有。 4，第 541-551 页，1989 年。

- [239] K. Simonyan 和 A. Zisserman，“用于大规模图像识别的非常深的卷积网络”，载于 ICLR，2015 年。

- [240] C. Szegedy、W. Liu、Y. Jia、P. Sermanet、S. E. Reed、D. Anguelov、D. Erhan、V. Vanhoucke 和 A. Rabinovich，“深入卷积”，载于_CVPR_。 IEEE 计算机协会，2015 年，第 1-9 页。

- [241] C. Szegedy、S. Ioffe、V. Vanhoucke 和 A. A. Alemi，“Inception-v4、inception-resnet 以及残差连接对学习的影响”，载于_AAAI_。 AAAI 出版社，2017 年，第 4278-4284 页。

- [242] S. Xie、R. B. Girshick、P. Doll´ar、Z. Tu 和 K. He，“深度神经网络的聚合残差变换”，载于_CVPR_。 IEEE 计算机协会，2017 年，第 5987-5995 页。

- [243] J. Hu、L. Shen 和 G. Sun，“挤压和激励网络”，载于 CVPR。计算机视觉基金会/IEEE 计算机协会，2018 年，第 7132-7141 页。

- [244] M. Tan 和 Q. V. Le，“Efficientnet：重新思考卷积神经网络的模型扩展”，载于_ICML_，卷。97.PMLR，2019 年，第 6105-6114 页。

- [245] R. B. Girshick、J. Donahue、T. Darrell 和 J. Malik，“用于精确对象检测和语义分割的丰富特征层次结构”，载于 CVPR。 IEEE 计算机协会，2014 年，第 580-587 页。

- [246] R. B. Girshick，“Fast R-CNN”，载于_ICCV_。 IEEE 计算机协会，2015 年，第 1440-1448 页。

- [247] S. Ren、K. He、R. B. Girshick 和 J. Sun，“Faster R-CNN：通过区域提议网络实现实时对象检测”，载于 NIPS，2015 年，第 91-99 页。

- [248] K. He、G. Gkioxari、P. Doll´ar 和 R. B. Girshick，“Mask R-CNN”，载于_ICCV_。 IEEE 计算机协会，2017 年，第 2980-2988 页。

- [249] J. Redmon、S. K. Divvala、R. B. Girshick 和 A. Farhadi，“你只看一次：统一的实时对象检测”，载于 CVPR。 IEEE 计算机协会，2016 年，第 779-788 页。

- [250] T. Lin、P. Doll´ar、R. B. Girshick、K. He、B. Hariharan 和 S. J. Belongie，“用于对象检测的特征金字塔网络”，载于_CVPR_。 IEEE 计算机协会，2017 年，第 936-944 页。

- [251] T. Lin、P. Goyal、R. B. Girshick、K. He 和 P. Doll´ar，“密集物体检测的焦点损失”，载于_ICCV_。 IEEE 计算机协会，2017 年，第 2999-3007 页。

- [252] P. Anderson、X. He、C. Buehler、D. Teney、M. Johnson、S. Gould 和 L. Zhang，“图像字幕和视觉问答的自下而上和自上而下的关注”，载于_CVPR_。计算机视觉基金会/IEEE 计算机协会，2018 年，第 6077-6086 页。

- [253] J. Long、E. Shelhamer 和 T. Darrell，“用于语义分割的全卷积网络”，载于_CVPR_。 IEEE 计算机协会，2015 年，第 3431-3440 页。

- [254] V. Badrinarayanan、A. Kendall 和 R. Cipolla，“Segnet：用于图像分割的深度卷积编码器-解码器架构”，IEEE Trans。模式肛门。马赫。情报.，卷。 39，没有。 12，第 2481-2495 页，2017 年。

- [255] O. Ronneberger、P. Fischer 和 T. Brox，“U-net：用于生物医学图像分割的卷积网络”，载于 MICCAI (3)，系列。计算机科学讲义，卷。 9351.施普林格，2015 年，第 234-241 页。

- [256] N. Carion、F. Massa、G. Synnaeve、N. Usunier、A. Kirillov 和 S

。 Zagoruyko，“使用 Transformer 进行端到端对象检测”，载于 ECCV (1)，ser。计算机科学讲义，卷。 12346。施普林格，2020 年，第 213-229 页。

- [257] R. Strudel、R. G. Pinel、I. Laptev 和 C. Schmid，“Segmenter：用于语义分割的 Transformer”，载于_ICCV_。 IEEE，2021 年，第 7242-7252 页。

- [258] A. Ioannidou、E. Chatzilari、S. Nikolopoulos 和 I. Kompatsiaris，“基于 3D 数据的计算机视觉深度学习进展：一项调查”，ACM Comput。 Surv.，卷。 50，没有。 2，第 20:1-20:38，2017 年。

- [259] E. Ahmed、A. Saint、A. E. R. Shabayek、K. Cherenkova、R. Das、G. Gusev、D. Aouada 和 B. E. Ottersten，“不同 3D 数据表示的深度学习进展：一项调查”，CoRR，卷。绝对/1808.01462，2018。

- [260] Y.Guo、H.Wang、Q.Hu、H.Liu、L.Liu 和 M.Bennamoun，“3d 点云深度学习：一项调查”，IEEE Trans。模式肛门。马赫。情报.，卷。 43，没有。 12，第 4338-4364 页，2021 年。

- [261] C. R. Qi、H. Su、M. Nießner、A. Dai、M. Yan 和 L. J. Guibas，“用于 3d 数据对象分类的体积和多视图 CNN”，参见 CVPR。 IEEE 计算机协会，2016 年，第 5648-5656 页。

- [262] Y. Feng、Y. Feng、H. You、X. Zhao 和 Y. Gau，“Meshnet：用于 3d 形状表示的网格神经网络”，载于_AAAI_。 AAAI 出版社，2019 年，第 8279-8286 页。

- [263] D. W. Otter、J. R. Medina 和 J. K. Kalita，“深度学习在自然语言处理中的应用调查”，IEEE Trans。神经网络学习。系统.，卷。 32、没有。 2，第 604-624 页，2021 年。

- [264] P. Liu、W. Yuan、J. Fu、Z. Jiang、H. Hayashi 和 G. Neubig，“预训练、提示和预测：自然语言处理中提示方法的系统调查”，ACM Comput。 Surv.，卷。 55，没有。 9，第 195:1-195:35，2023 年。

- [265] Y. Bengio、R. Ducharme 和 P. Vincent，“神经概率语言模型”，载于_NIPS_。麻省理工学院出版社，2000 年，第 932-938 页。

- [266] T. Mikolov、I. Sutskever、K. Chen、G. S. Corrado 和 J. Dean，“单词和短语的分布式表示及其组合性”，载于 NIPS，2013 年，第 3111-3119 页。

- [267] T. Mikolov、K. Chen、G. Corrado 和 J. Dean，“向量空间中单词表示的有效估计”，载于 ICLR（研讨会海报），2013 年。

- [268] J. Pennington、R. Socher 和 C. D. Manning，“Glove：单词表示的全局向量”，载于_EMNLP_。 ACL，2014 年，第 1532-1543 页。

- [269] J. L. Elman，“及时发现结构”，Cogn。科学，卷。 14、没有。 2，第 179-211 页，1990 年。

- [270] D. Bahdanau、K. Cho 和 Y. Bengio，“通过联合学习对齐和翻译进行神经机器翻译”，载于 ICLR，2015 年。

- [271] Z. Huang、W. Xu 和 K. Yu，“用于序列标记的双向 LSTM-CRF 模型”，CoRR，卷。绝对/1508.01991，2015。

[272] M. E. Peters, M. Neumann, M. Iyyer, M. Gardner, C. Clark, K. Lee, and L. Zettlemoyer, "Deep contextualized word representations," in *NAACL-HLT*. Association for Computational Linguistics, 2018, pp. 2227–2237.

[273] J. Howard and S. Ruder, "Universal language model fine-tuning for text classification," in *ACL (1)*. Association for Computational Linguistics, 2018, pp. 328–339.

[274] Y. Kim, "Convolutional neural networks for sentence classification," in *EMNLP*. ACL, 2014, pp. 1746–1751.

[275] X. Zhang, J. J. Zhao, and Y. LeCun, "Character-level convolutional networks for text classification," in *NIPS*, 2015, pp. 649–657.

[276] X. Ma and E. H. Hovy, "End-to-end sequence labeling via bi-directional lstm-cnns-crf," in *ACL (1)*. The Association for Computer Linguistics, 2016.

[277] A. Radford, J. Wu, R. Child, D. Luan, D. Amodei, I. Sutskever *et al.*, "Language models are unsupervised multitask learners," *OpenAI blog*, vol. 1, no. 8, p. 9, 2019.

[278] T. B. Brown, B. Mann, N. Ryder, M. Subbiah, J. Kaplan, P. Dhariwal, A. Neelakantan, P. Shyam, G. Sastry, A. Askell *et al.*, "Language models are few-shot learners," *arXiv preprint arXiv:2005.14165*, 2020.

[279] OpenAI, "GPT-4 technical report," *CoRR*, vol. abs/2303.08774, 2023.

[280] Y. Liu, M. Ott, N. Goyal, J. Du, M. Joshi, D. Chen, O. Levy, M. Lewis, L. Zettlemoyer, and V. Stoyanov, "Roberta: A robustly optimized BERT pretraining approach," *CoRR*, vol. abs/1907.11692, 2019.

[281] Z. Lan, M. Chen, S. Goodman, K. Gimpel, P. Sharma, and R. Soricut, "ALBERT: A lite BERT for self-supervised learning of language representations," in *ICLR*, 2020.

[282] K. Clark, M. Luong, Q. V. Le, and C. D. Manning, "ELECTRA: pre-training text encoders as discriminators rather than generators," in *ICLR*, 2020.

[283] Z. Yang, Z. Dai, Y. Yang, J. G. Carbonell, R. Salakhutdinov, and Q. V. Le, "Xlnet: Generalized autoregressive pretraining for language understanding," in *NeurIPS*, 2019, pp. 5754–5764.

[284] S. Zhang, S. Roller, N. Goyal, M. Artetxe, M. Chen, S. Chen, C. Dewan, M. T. Diab, X. Li, X. V. Lin, T. Mihaylov, M. Ott, S. Shleifer, K. Shuster, D. Simig, P. S. Koura, A. Sridhar, T. Wang, and L. Zettlemoyer, "OPT: open pre-trained transformer language models," *CoRR*, vol. abs/2205.01068, 2022.

[285] M. Lewis, Y. Liu, N. Goyal, M. Ghazvininejad, A. Mohamed, O. Levy, V. Stoyanov, and L. Zettlemoyer, "BART: denoising sequence-to-sequence pre-training for natural language generation, translation, and comprehension," in *ACL*. Association for Computational Linguistics, 2020, pp. 7871–7880.

[286] C. Raffel, N. Shazeer, A. Roberts, K. Lee, S. Narang, M. Matena, Y. Zhou, W. Li, and P. J. Liu, "Exploring the limits of transfer learning with a unified text-to-text transformer," *J. Mach. Learn. Res.*, vol. 21, pp. 140:1–140:67, 2020.

[287] A. Chowdhery, S. Narang, J. Devlin, M. Bosma, G. Mishra, A. Roberts, P. Barham, H. W. Chung, C. Sutton, S. Gehrmann, P. Schuh, K. Shi, S. Tsvyashchenko, J. Maynez, A. Rao, P. Barnes, Y. Tay, N. Shazeer, V. Prabhakaran, E. Reif, N. Du, B. Hutchinson, R. Pope, J. Bradbury, J. Austin, M. Isard, G. Gur-Ari, P. Yin, T. Duke, A. Levskaya, S. Ghemawat, S. Dev, H. Michalewski, X. Garcia, V. Misra, K. Robinson, L. Fedus, D. Zhou, D. Ippolito, D. Luan, H. Lim, B. Zoph, A. Spiridonov, R. Sepassi, D. Dohan, S. Agrawal, M. Omernick, A. M. Dai, T. S. Pillai, M. Pellat, A. Lewkowycz, E. Moreira, R. Child, O. Polozov, K. Lee, Z. Zhou, X. Wang, B. Saeta, M. Diaz, O. Firat, M. Catasta, J. Wei, K. Meier-Hellstern, D. Eck, J. Dean, S. Petrov, and N. Fiedel, "Palm: Scaling language modeling with pathways," *J. Mach. Learn. Res.*, vol. 24, pp. 240:1–240:113, 2023.

[288] R. Anil, A. M. Dai, O. Firat, M. Johnson, D. Lepikhin, A. Passos, S. Shakeri, E. Taropa, P. Bailey, Z. Chen, E. Chu, J. H. Clark, L. E. Shafey, Y. Huang, K. Meier-Hellstern, G. Mishra, E. Moreira, M. Omernick, K. Robinson, S. Ruder, Y. Tay, K. Xiao, Y. Xu, Y. Zhang, G. H. Ábrego, J. Ahn, J. Austin, P. Barham, J. A. Botha, J. Bradbury, S. Brahma, K. Brooks, M. Catasta, Y. Cheng, C. Cherry, C. A. Choquette-Choo, A. Chowdhery, C. Crepy, S. Dave, M. Dehghani, S. Dev, J. Devlin, M. Díaz, N. Du, E. Dyer, V. Feinberg, F. Feng, V. Fienber, M. Freitag, X. Garcia, S. Gehrmann, L. Gonzalez, and et al., "Palm 2 technical report," *CoRR*, vol. abs/2305.10403, 2023.

[289] H. Touvron, T. Lavril, G. Izacard, X. Martinet, M. Lachaux, T. Lacroix, B. Rozière, N. Goyal, E. Hambro, F. Azhar, A. Rodriguez, A. Joulin, E. Grave, and G. Lample, "Llama: Open and efficient foundation language models," *CoRR*, vol. abs/2302.13971, 2023.

[290] H. Touvron, L. Martin, K. Stone, P. Albert, A. Almahairi, Y. Babaei, N. Bashlykov, S. Batra, P. Bhargava, S. Bhosale, D. Bikel, L. Blecher, C. Canton-Ferrer, M. Chen, G. Cucurull, D. Esiobu, J. Fernandes, J. Fu, W. Fu, B. Fuller, C. Gao, V. Goswami, N. Goyal, A. Hartshorn, S. Hosseini, R. Hou, H. Inan, M. Kardas, V. Kerkez, M. Khabsa, I. Kloumann, A. Korenev, P. S. Koura, M. Lachaux, T. Lavril, J. Lee, D. Liskovich, Y. Lu, Y. Mao, X. Martinet, T. Mihaylov, P. Mishra, I. Molybog, Y. Nie, A. Poulton, J. Reizenstein, R. Rungta, K. Saladi, A. Schelten, R. Silva, E. M. Smith, R. Subramanian, X. E. Tan, B. Tang, R. Taylor, A. Williams, J. X. Kuan, P. Xu, Z. Yan, I. Zarov, Y. Zhang, A. Fan, M. Kambadur, S. Narang, A. Rodriguez, R. Stojnic, S. Edunov, and T. Scialom, "Llama 2: Open foundation and fine-tuned chat models," *CoRR*, vol. abs/2307.09288, 2023.

[291] Baidu. (2023) Introducing ernie 3.5: Baidu's knowledge-enhanced foundation model takes a giant leap forward. [Online]. Available: http://research.baidu.com/Blog/index-view?id=185

[292] L. Ouyang, J. Wu, X. Jiang, D. Almeida, C. L. Wainwright, P. Mishkin, C. Zhang, S. Agarwal, K. Slama, A. Ray, J. Schulman, J. Hilton, F. Kelton, L. Miller, M. Simens, A. Askell, P. Welinder, P. F. Christiano, J. Leike, and R. Lowe, "Training language models to follow instructions with human feedback," in *NeurIPS*, 2022.

[293] J. Wei, M. Bosma, V. Y. Zhao, K. Guu, A. W. Yu, B. Lester, N. Du, A. M. Dai, and Q. V. Le, "Finetuned language models are zero-shot learners," in *ICLR*, 2022.

[294] H. W. Chung, L. Hou, S. Longpre, B. Zoph, Y. Tay, W. Fedus, E. Li, X. Wang, M. Dehghani, S. Brahma, A. Webson, S. S. Gu, Z. Dai, M. Suzgun, X. Chen, A. Chowdhery, S. Narang, G. Mishra, A. Yu, V. Y. Zhao, Y. Huang, A. M. Dai, H. Yu, S. Petrov, E. H. Chi, J. Dean, J. Devlin, A. Roberts, D. Zhou, Q. V. Le, and J. Wei, "Scaling instruction-finetuned language models," *CoRR*, vol. abs/2210.11416, 2022.

[295] R. Taori, I. Gulrajani, T. Zhang, Y. Dubois, X. Li, C. Guestrin, P. Liang, and T. B. Hashimoto, "Stanford alpaca: An instruction-following llama model," https://github.com/tatsu-lab/stanford_alpaca, 2023.

[296] W.-L. Chiang, Z. Li, Z. Lin, Y. Sheng, Z. Wu, H. Zhang, L. Zheng, S. Zhuang, Y. Zhuang, J. E. Gonzalez, I. Stoica, and E. P. Xing, "Vicuna: An open-source chatbot impressing gpt-4 with 90%* chatgpt quality," March 2023. [Online]. Available: https://lmsys.org/blog/2023-03-30-vicuna/

[297] Y. Li, "Deep reinforcement learning: An overview," *CoRR*, vol. abs/1701.07274, 2017.

[298] K. Arulkumaran, M. P. Deisenroth, M. Brundage, and A. A. Bharath, "A brief survey of deep reinforcement learning," *CoRR*, vol. abs/1708.05866, 2017.

[299] W. Li, H. Luo, Z. Lin, C. Zhang, Z. Lu, and D. Ye, "A survey on transformers in reinforcement learning," *CoRR*, vol. abs/2301.03044, 2023.

[300] H. van Hasselt, A. Guez, and D. Silver, "Deep reinforcement learning with double q-learning," in *AAAI*. AAAI Press, 2016, pp. 2094–2100.

[301] M. Andrychowicz, D. Crow, A. Ray, J. Schneider, R. Fong, P. Welinder, B. McGrew, J. Tobin, P. Abbeel, and W. Zaremba, "Hindsight experience replay," in *NIPS*, 2017, pp. 5048–5058.

[302] S. Fujimoto, D. Meger, and D. Precup, "Off-policy deep reinforcement learning without exploration," in *ICML*, vol. 97. PMLR, 2019, pp. 2052–2062.

[303] A. Kumar, J. Fu, M. Soh, G. Tucker, and S. Levine, "Stabilizing off-policy q-learning via bootstrapping error reduction," in *NeurIPS*, 2019, pp. 11 761–11 771.

[304] A. Kumar, A. Zhou, G. Tucker, and S. Levine, "Conservative q-learning for offline reinforcement learning," in *NeurIPS*, 2020.

[305] S. Levine and V. Koltun, "Guided policy search," in *ICML (3)*, ser. JMLR Workshop and Conference Proceedings, vol. 28. JMLR.org, 2013, pp. 1–9.

[306] D. Silver, G. Lever, N. Heess, T. Degris, D. Wierstra, and M. A. Riedmiller, "Deterministic policy gradient algorithms," in *ICML*, ser. JMLR Workshop and Conference Proceedings, vol. 32. JMLR.org, 2014, pp. 387–395.

[307] T. P. Lillicrap, J. J. Hunt, A. Pritzel, N. Heess, T. Erez, Y. Tassa, D. Silver, and D. Wierstra, "Continuous control with deep reinforcement learning," in *ICLR (Poster)*, 2016.

[308] V. Mnih, A. P. Badia, M. Mirza, A. Graves, T. P. Lillicrap, T. Harley, D. Silver, and K. Kavukcuoglu, "Asynchronous methods for deep reinforcement learning," in *ICML*, ser. JMLR Workshop and Conference Proceedings, vol. 48. JMLR.org, 2016, pp. 1928–1937.

[309] S. Gu, T. P. Lillicrap, I. Sutskever, and S. Levine, "Continuous deep q-learning with model-based acceleration," in *ICML*, ser. JMLR Workshop and Conference Proceedings, vol. 48. JMLR.org, 2016, pp. 2829–2838.

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- [272] M. E. Peters、M. Neumann、M. Iyyer、M. Gardner、C. Clark、K. Lee 和 L. Zettlemoyer，“深度语境化单词表征”，载于_NAACL-HLT_。计算语言学协会，2018 年，第 2227-2237 页。

- [273] J. Howard 和 S. Ruder，“文本分类的通用语言模型微调”，参见 ACL (1)。计算语言学协会，2018 年，第 328-339 页。

- [274] Y. Kim，“用于句子分类的卷积神经网络”，载于_EMNLP_。 ACL，2014 年，第 1746-1751 页。

- [275] X.Zhang、J.J.Zhao 和 Y.LeCun，“用于文本分类的字符级卷积网络”，载于 NIPS，2015 年，第 649-657 页。

- [276] X. Ma 和 E. H. Hovy，“通过双向 lstm-cnns-crf 进行端到端序列标记”，位于 ACL (1) 中。计算机语言学协会，2016。

- [277] A. Radford、J. Wu、R. Child、D. Luan、D. Amodei、I. Sutskever 等人，“语言模型是无监督的多任务学习者”，OpenAI 博客，卷。 1、没有。 8，p。 2019 年 9 月。

- [278] T. B. Brown、B. Mann、N. Ryder、M. Subbiah、J. Kaplan、P. Dhariwal、A. Neelakantan、P. Shyam、G. Sastry、A. Askell 等人，“语言模型是小样本学习者”，arXiv 预印本 arXiv：2005.14165，2020 年。

- [279] OpenAI，“GPT-4 技术报告”，CoRR，卷。绝对/2303.08774，2023。

- [280] Y. Liu、M. Ott、N. Goyal、J. Du、M. Joshi、D. Chen、O. Levy、M. Lewis、L. Zettlemoyer 和 V. Stoyanov，“Roberta：一种稳健优化的 BERT 预训练方法”，CoRR，卷。 ABS/1907.11692，2019。

- [281] Z. Lan、M. Chen、S. Goodman、K. Gimpel、P. Sharma 和 R. Soricut，“ALBERT：用于语言表示自我监督学习的 lite BERT”，载于 ICLR，2020 年。

- [282] K. Clark、M. Luong、Q. V. Le 和 C. D. Manning，“ELECTRA：将文本编码器预训练为判别器而不是生成器”，载于 ICLR，2020 年。

- [283] Z. Yang、Z. Dai、Y. Yang、J. G. Carbonell、R. Salakhutdinov 和 Q. V. Le，“Xlnet：语言理解的广义自回归预训练”，载于 NeurIPS，2019 年，第 5754-5764 页。

- [284] S. 张、S. Roller、N. Goyal、M. Artetxe、M. Chen、S. Chen、C. Dewan、M. T. Diab、X. Li、X. V. Lin、T. Mihaylov、M. Ott、S. Shleifer、K. Shuster、D. Simig、P. S. Koura、A. Sridhar、T. Wang 和 L. Zettlemoyer，“OPT：开放预训练的 Transformer 语言模型，”CoRR，卷。绝对/2205.01068，2022。

- [285] M. Lewis、Y. Liu、N. Goyal、M. Ghazvininejad、A. Mohamed、O. Levy、V. Stoyanov 和 L. Zettlemoyer，“BART：用于自然语言生成、翻译和理解的去噪序列到序列预训练”，载于_ACL_。计算语言学协会，2020 年，第 7871-7880 页。

- [286] C. Raffel、N. Shazeer、A. Roberts、K. Lee、S. Narang、M. Matena、Y. Zhou、W. Li 和 P. J. Liu，“利用统一的文本到文本转换器探索迁移学习的局限性”，J.马赫。学习。 Res.，卷。 21，第 140:1-140:67，2020 年。

- [287] A. Chowdhery、S. Narang、J. Devlin、M. Bosma、G. Mishra、A. Roberts、P. Barham、H. W. Chung、C. Sutton、S. Gehrmann、P. Schuh、K. Shi、S. Tsvyashchenko、J. Maynez、A. Rao、P. Barnes、Y. Tay、N. Shazeer、V. Prabhakaran、E. Reif、N. Du、B. Hutchinson、R. Pope、J. Bradbury、J. Austin、M. Isard、G. Gur-Ari、P. Yin、T. Duke、A

。 Levskaya、S. Ghemawat、S. Dev、H. Michalewski、X. Garcia、V. Misra、K. Robinson、L. Fedus、D. Zhou、D. Ippolito、D. Luan、H. Lim、B. Zoph、A. Spiridonov、R. Sepassi、D. Dohan、S. Agrawal、M. Omernick、A. M. Dai、T. S. Pillai、M. Pellat、A. Lewkowycz、E. Moreira、R. Child、O. Polozov、K. Lee、Z. Zhou、X. Wang、B. Saeta、M. Diaz、O. Firat、M. Catasta、J. Wei、K. Meier-Hellstern、D. Eck、J. Dean、S. Petrov 和 N. Fiedel，“Palm：通过路径扩展语言建模”，J。马赫。学习。 Res.，卷。 24，第 240:1-240:113，2023 年。

- [288] R. Anil、A. M. Dai、O. Firat、M. Johnson、D. Lepikhin、A. Passos、S. Shakeri、E. Taropa、P. Bailey、Z. Chen、E. Chu、J. H. Clark、L. E. Shafey、Y. Huang、K. Meier-Hellstern、G. Mishra、E. Moreira、M. Omernick、K. Robinson、S. Ruder、Y. Tay、K. 肖、Y. Xu、Y. 张、G. H. Abrego、´ J. Ahn、J. Austin、P. Barham、J. A. Botha、J. Bradbury、S. Brahma、K. Brooks、M. Catasta、Y. Cheng、C. Cherry、C. A. Choquette-Choo、A. Chowdhery、C. Crepy、S. Dave、 M. Dehghani、S. Dev、J. Devlin、M. D´ıaz、N. Du、E. Dyer、V. Feinberg、F. Feng、V. Fienber、M. Freitag、X. Garcia、S. Gehrmann、L. Gonzalez 等人，“Palm 2 技术报告”，CoRR，卷。绝对/2305.10403，2023。

- [289] H. Touvron、T. Lavril、G. Izacard、X. Martinet、M. Lachaux、T. Lacroix、B. Roziere、N. Goyal、E. Hambro、F. Azhar、A. Rodriguez、A. Joulin、E. Grave 和 G. Lample，“Llama：开放且高效的基础语言模型”，CoRR，卷。 ABS/2302.13971，2023。

- [290] H. Touvron、L. Martin、K. Stone、P. Albert、A. Almahairi、Y. Babaei、N. Bashlykov、S. Batra、P. Bhargava、S. Bhosale、D. Bikel、L. Blecher、C. Canton-Ferrer、M. Chen、G. Cucurull、D. Esiobu、J. Fernandes，

- J. Fu、W. Fu、B. Fuller、C. Gau、V. Goswami、N. Goyal、A. Hartshorn、S. Hosseini、R. Hou、H. Inan、M. Kardas、V. Kerkez、M. Khabsa、I. Kloumann、A. Korenev、P. S. Koura、M. Lachaux、T. Lavril、J. Lee、D. Liskovich、Y. Lu、Y. Mao、X. Martinet、T. Mihaylov、P. Mishra、I. Molybog、Y. Nie、A. Poulton、J. Reizenstein、R. Rungta、K. Saladi、A. Schelten、R. Silva、E. M. Smith、R. Subramanian、X. E. Tan、B. Tang、R. Taylor、A. Williams、J. X. Kuan、P. Xu、Z. Yan、 I. Zarov、Y. Zhang、A. Fan、M. Kambadur、S. Narang、A. Rodriguez、R. Stojnic、S. Edunov 和 T. Scialom，“Llama 2：开放基础和微调聊天模型”，CoRR，卷。绝对/2307.09288，2023。

- [291] 百度. (2023) ernie 3.5隆重登场：百度知识增强基础模型（Foundation Model）实现巨大飞跃。 [在线的]。可用：http://research.baidu.com/Blog/index-view?id=185

- [292] L. Ouyang、J. Wu、X. Jiang、D. Almeida、C. L. Wainwright、P. Mishkin、C. Zhang、S. Agarwal、K. Slama、A. Ray、J. Schulman、J. Hilton、F. Kelton、L. Miller、M. Simens、A. Askel、P. Welinder、P. F. Christiano、J. Leike 和 R. Lowe，“训练语言模型遵循人类反馈的指令”，《NeurIPS》，2022 年。

- [293] J. Wei、M. Bosma、V. Y. Zhu、K. Guu、A. W. Yu、B. Lester、N. Du、A. M. Dai 和 Q. V. Le，“微调语言模型是零样本学习者”，载于 ICLR，2022 年。

- [294] H. W. Chung、L. Hou、S. Longpre、B. Zoph、Y. Tay、W. Fedus、E. Li、X. Wang、M. Dehghani、S. Brahma、A. Webson、S. S. Gu、Z. Dai、M. Suzgun、X. Chen、A. Chowdhery、S. Narang、G. Mishra、A. Yu、V. Y. Zhu、Y. Huang、A.戴明华

。 Yu、S. Petrov、E. H. Chi、J. Dean、J. Devlin、A. Roberts、D. Zhou、Q. V. Le 和 J. Wei，“扩展指令微调语言模型”，CoRR，卷。绝对/2210.11416，2022。

- [295] R. Taori、I. Gulrajani、T. Zhang、Y. Dubois、X. Li、C. Guestrin、P. Liang 和 T. B. Hashimoto，“斯坦福羊驼：遵循指令的美洲驼模型”，https://github.com/tatsu-lab/stanford <u>alpaca，</u> 2023。

- [296] W.-L。 Jiang、Z. Li、Z. Lin、Y. Shen、Z. Wu、H. Zhang、L. Cheng、S. Zhuang、Y. Zhuang、J. E. Gonzalez、I. Stoica 和 E. P. Xing，“Vicuna：一款以 90%* chatgpt 质量打动 gpt-4 的开源聊天机器人”，2023 年 3 月。[在线]。可用：https://lmsys.org/blog/2023-03-30-vicuna/

- [297] Y. Li，“Deep 强化学习（Reinforcement Learning）：概述”，CoRR，卷。绝对/1701.07274，2017。

- [298] K. Arulkumaran、M. P. Deisenroth、M. Brundage 和 A. A. Bharath，“深部 强化学习（Reinforcement Learning）的简要调查”，CoRR，卷。绝对/1708.05866，2017。

- [299] W. Li、H. Luo、Z. Lin、C. Zhu、Z. Lu 和 D. Ye，“强化学习（Reinforcement Learning） 中变压器的调查”，CoRR，卷。绝对/2301.03044，2023。

- [300] H. van Hasselt、A. Guez 和 D. Silver，“采用双 q 学习的深度 强化学习（Reinforcement Learning）”，载于_AAAI_。 AAAI 出版社，2016 年，第 2094-2100 页。

- [301] M. Andrychowicz、D. Crow、A. Ray、J. Schneider、R. Fong、P. Welinder、B. McGrew、J. Tobin、P. Abbeel 和 W. Zaremba，“事后经验重播”，载于 NIPS，2017 年，第 5048-5058 页。

- [302] S. Fujimoto、D. Meger 和 D. Precup，“无需探索的离策略深度 强化学习（Reinforcement Learning）”，载于_ICML_，卷。 97.PMLR，2019 年，第 2052-2062 页。

- [303] A. Kumar、J. Fu、M. Soh、G. Tucker 和 S. Levine，“通过引导误差减少稳定offpolicy q-learning”，载于 NeurIPS，2019 年，第 11 761-11 771 页。

- [304] A. Kumar、A. Zhou、G. Tucker 和 S. Levine，“离线 强化学习（Reinforcement Learning） 的保守 q 学习”，《NeurIPS》，2020 年。

- [305] S. Levine 和 V. Koltun，“指导性策略搜索”，载于 ICML (3)，系列。 JMLR 研讨会和会议论文集，卷。 28. JMLR.org，2013 年，第 1-9 页。

- [306] D. Silver、G. Lever、N. Heess、T. Degris、D. Wierstra 和 M. A. Riedmiller，“确定性策略梯度算法”，载于 ICML，系列。 JMLR 研讨会和会议论文集，卷。 32. JMLR.org，2014 年，第 387-395 页。

- [307] T. P. Lillicrap、J. J. Hunt、A. Pritzel、N. Heess、T. Erez、Y. Tassa、D. Silver 和 D. Wierstra，“深度 强化学习（Reinforcement Learning） 的连续控制”，载于_ICLR（海报）_，2016 年。

- [308] V. Mnih、A. P. Badia、M. Mirza、A. Graves、T. P. Lillicrap、T. Harley、D. Silver 和 K. Kavukcuoglu，“深度 强化学习（Reinforcement Learning） 的异步方法”，载于_ICML_，系列。 JMLR 研讨会和会议论文集，卷。 48. JMLR.org，2016 年，第 1928-1937 页。

- [309] S. Gu、T. P. Lillicrap、I. Sutskever 和 S. Levine，“基于模型加速的连续深度 q 学习”，载于 ICML，ser。 JMLR 研讨会和会议论文集，卷。 48. JMLR.org，2016 年，第 2829-2838 页。

[310] T. Haarnoja, A. Zhou, P. Abbeel, and S. Levine, "Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor," in *ICML*, vol. 80. PMLR, 2018, pp. 1856–1865.

[311] J. Schulman, S. Levine, P. Abbeel, M. I. Jordan, and P. Moritz, "Trust region policy optimization," in *ICML*, ser. JMLR Workshop and Conference Proceedings, vol. 37. JMLR.org, 2015, pp. 1889–1897.

[312] J. Schulman, P. Moritz, S. Levine, M. I. Jordan, and P. Abbeel, "High-dimensional continuous control using generalized advantage estimation," in *ICLR (Poster)*, 2016.

[313] J. Ho and S. Ermon, "Generative adversarial imitation learning," in *NIPS*, 2016, pp. 4565–4573.

[314] L. Pinto, J. Davidson, R. Sukthankar, and A. Gupta, "Robust adversarial reinforcement learning," in *ICML*, vol. 70. PMLR, 2017, pp. 2817–2826.

[315] P. F. Christiano, J. Leike, T. B. Brown, M. Martic, S. Legg, and D. Amodei, "Deep reinforcement learning from human preferences," in *NIPS*, 2017, pp. 4299–4307.

[316] A. S. Vezhnevets, S. Osindero, T. Schaul, N. Heess, M. Jaderberg, D. Silver, and K. Kavukcuoglu, "Feudal networks for hierarchical reinforcement learning," in *ICML*, vol. 70. PMLR, 2017, pp. 3540–3549.

[317] S. Levine, C. Finn, T. Darrell, and P. Abbeel, "End-to-end training of deep visuomotor policies," *J. Mach. Learn. Res.*, vol. 17, pp. 39:1–39:40, 2016.

[318] D. Kalashnikov, A. Irpan, P. Pastor, J. Ibarz, A. Herzog, E. Jang, D. Quillen, E. Holly, M. Kalakrishnan, V. Vanhoucke, and S. Levine, "Qt-opt: Scalable deep reinforcement learning for vision-based robotic manipulation," *CoRR*, vol. abs/1806.10293, 2018.

[319] OpenAI, I. Akkaya, M. Andrychowicz, M. Chociej, M. Litwin, B. McGrew, A. Petron, A. Paino, M. Plappert, G. Powell, R. Ribas, J. Schneider, N. Tezak, J. Tworek, P. Welinder, L. Weng, Q. Yuan, W. Zaremba, and L. Zhang, "Solving rubik's cube with a robot hand," *CoRR*, vol. abs/1910.07113, 2019.

[320] F. Scarselli, M. Gori, A. C. Tsoi, M. Hagenbuchner, and G. Monfardini, "The graph neural network model," *IEEE Trans. Neural Networks*, vol. 20, no. 1, pp. 61–80, 2009.

[321] Z. Wu, S. Pan, F. Chen, G. Long, C. Zhang, and P. S. Yu, "A comprehensive survey on graph neural networks," *IEEE Trans. Neural Networks Learn. Syst.*, vol. 32, no. 1, pp. 4–24, 2021.

[322] J. Bruna, W. Zaremba, A. Szlam, and Y. LeCun, "Spectral networks and locally connected networks on graphs," in *ICLR*, 2014.

[323] M. Defferrard, X. Bresson, and P. Vandergheynst, "Convolutional neural networks on graphs with fast localized spectral filtering," in *NIPS*, 2016, pp. 3837–3845.

[324] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," in *ICLR (Poster)*, 2017.

[325] A. Micheli, "Neural network for graphs: A contextual constructive approach," *IEEE Trans. Neural Networks*, vol. 20, no. 3, pp. 498–511, 2009.

[326] J. Gilmer, S. S. Schoenholz, P. F. Riley, O. Vinyals, and G. E. Dahl, "Neural message passing for quantum chemistry," in *ICML*, vol. 70. PMLR, 2017, pp. 1263–1272.

[327] K. Xu, W. Hu, J. Leskovec, and S. Jegelka, "How powerful are graph neural networks?" in *ICLR*, 2019.

[328] W. L. Hamilton, Z. Ying, and J. Leskovec, "Inductive representation learning on large graphs," in *NIPS*, 2017, pp. 1024–1034.

[329] P. Velickovic, G. Cucurull, A. Casanova, A. Romero, P. Liò, and Y. Bengio, "Graph attention networks," in *ICLR (Poster)*, 2018.

[330] S. Cao, W. Lu, and Q. Xu, "Deep neural networks for learning graph representations," in *AAAI*. AAAI Press, 2016, pp. 1145–1152.

[331] T. N. Kipf and M. Welling, "Variational graph auto-encoders," *CoRR*, vol. abs/1611.07308, 2016.

[332] Y. Seo, M. Defferrard, P. Vandergheynst, and X. Bresson, "Structured sequence modeling with graph convolutional recurrent networks," in *ICONIP (1)*, ser. Lecture Notes in Computer Science, vol. 11301. Springer, 2018, pp. 362–373.

[333] W. Du, H. Zhang, Y. Du, Q. Meng, W. Chen, N. Zheng, B. Shao, and T. Liu, "SE(3) equivariant graph neural networks with complete local frames," in *ICML*, vol. 162. PMLR, 2022, pp. 5583–5608.

[334] V. G. Satorras, E. Hoogeboom, and M. Welling, "E(n) equivariant graph neural networks," in *ICML*, vol. 139. PMLR, 2021, pp. 9323–9332.

[335] Y. Rong, Y. Bian, T. Xu, W. Xie, Y. Wei, W. Huang, and J. Huang, "Self-supervised graph transformer on large-scale molecular data," in *NeurIPS*, 2020.

[336] F. Fuchs, D. E. Worrall, V. Fischer, and M. Welling, "Se(3)-transformers: 3d roto-translation equivariant attention networks," in *NeurIPS*, 2020.

[337] R. Krishna, Y. Zhu, O. Groth, J. Johnson, K. Hata, J. Kravitz, S. Chen, Y. Kalantidis, L. Li, D. A. Shamma, M. S. Bernstein, and L. Fei-Fei, "Visual genome: Connecting language and vision using crowdsourced dense image annotations," *Int. J. Comput. Vis.*, vol. 123, no. 1, pp. 32–73, 2017.

[338] L. Wu, Y. Chen, K. Shen, X. Guo, H. Gao, S. Li, J. Pei, and B. Long, "Graph neural networks for natural language processing: A survey," *Found. Trends Mach. Learn.*, vol. 16, no. 2, pp. 119–328, 2023.

[339] D. Ghosal, N. Majumder, S. Poria, N. Chhaya, and A. F. Gelbukh, "Dialoguegcn: A graph convolutional neural network for emotion recognition in conversation," in *EMNLP/IJCNLP (1)*. Association for Computational Linguistics, 2019, pp. 154–164.

[340] W. Zhong, J. Xu, D. Tang, Z. Xu, N. Duan, M. Zhou, J. Wang, and J. Yin, "Reasoning over semantic-level graph for fact checking," in *ACL*. Association for Computational Linguistics, 2020, pp. 6170–6180.

[341] L. Wang, Y. Li, Ö. Aslan, and O. Vinyals, "Wikigraphs: A wikipedia text - knowledge graph paired dataset," *CoRR*, vol. abs/2107.09556, 2021.

[342] J. Tang, J. Zhang, L. Yao, J. Li, L. Zhang, and Z. Su, "Arnetminer: extraction and mining of academic social networks," in *KDD*. ACM, 2008, pp. 990–998.

[343] H. Tan and M. Bansal, "LXMERT: learning cross-modality encoder representations from transformers," in *EMNLP/IJCNLP (1)*. Association for Computational Linguistics, 2019, pp. 5099–5110.

[344] L. H. Li, M. Yatskar, D. Yin, C. Hsieh, and K. Chang, "Visualbert: A simple and performant baseline for vision and language," *CoRR*, vol. abs/1908.03557, 2019.

[345] W. Su, X. Zhu, Y. Cao, B. Li, L. Lu, F. Wei, and J. Dai, "VL-BERT: pre-training of generic visual-linguistic representations," in *ICLR*, 2020.

[346] Y. Chen, L. Li, L. Yu, A. E. Kholy, F. Ahmed, Z. Gan, Y. Cheng, and J. Liu, "UNITER: universal image-text representation learning," in *ECCV (30)*, ser. Lecture Notes in Computer Science, vol. 12375. Springer, 2020, pp. 104–120.

[347] W. Kim, B. Son, and I. Kim, "Vilt: Vision-and-language transformer without convolution or region supervision," in *ICML*, vol. 139. PMLR, 2021, pp. 5583–5594.

[348] Z. Wang, J. Yu, A. W. Yu, Z. Dai, Y. Tsvetkov, and Y. Cao, "Simvlm: Simple visual language model pretraining with weak supervision," in *ICLR*, 2022.

[349] Z. Dai, H. Liu, Q. V. Le, and M. Tan, "Coatnet: Marrying convolution and attention for all data sizes," in *NeurIPS*, 2021, pp. 3965–3977.

[350] J. Wang, Z. Yang, X. Hu, L. Li, K. Lin, Z. Gan, Z. Liu, C. Liu, and L. Wang, "GIT: A generative image-to-text transformer for vision and language," *Trans. Mach. Learn. Res.*, vol. 2022, 2022.

[351] L. Yuan, D. Chen, Y. Chen, N. Codella, X. Dai, J. Gao, H. Hu, X. Huang, B. Li, C. Li, C. Liu, M. Liu, Z. Liu, Y. Lu, Y. Shi, L. Wang, J. Wang, B. Xiao, Z. Xiao, J. Yang, M. Zeng, L. Zhou, and P. Zhang, "Florence: A new foundation model for computer vision," *CoRR*, vol. abs/2111.11432, 2021.

[352] W. Wang, H. Bao, L. Dong, J. Bjorck, Z. Peng, Q. Liu, K. Aggarwal, O. K. Mohammed, S. Singhal, S. Som, and F. Wei, "Image as a foreign language: BEIT pretraining for vision and vision-language tasks," in *CVPR*. IEEE, 2023, pp. 19 175–19 186.

[353] L. Yao, R. Huang, L. Hou, G. Lu, M. Niu, H. Xu, X. Liang, Z. Li, X. Jiang, and C. Xu, "FILIP: fine-grained interactive language-image pre-training," in *ICLR*, 2022.

[354] C. Jia, Y. Yang, Y. Xia, Y. Chen, Z. Parekh, H. Pham, Q. V. Le, Y. Sung, Z. Li, and T. Duerig, "Scaling up visual and vision-language representation learning with noisy text supervision," in *ICML*, vol. 139. PMLR, 2021, pp. 4904–4916.

[355] Q. Xie, M. Luong, E. H. Hovy, and Q. V. Le, "Self-training with noisy student improves imagenet classification," in *CVPR*. Computer Vision Foundation / IEEE, 2020, pp. 10 684–10 695.

[356] J. Li, R. R. Selvaraju, A. Gotmare, S. R. Joty, C. Xiong, and S. C. Hoi, "Align before fuse: Vision and language representation learning with momentum distillation," in *NeurIPS*, 2021, pp. 9694–9705.

[357] A. Singh, R. Hu, V. Goswami, G. Couairon, W. Galuba, M. Rohrbach, and D. Kiela, "FLAVA: A foundational language and vision alignment model," in *CVPR*. IEEE, 2022, pp. 15 617–15 629.

[358] Z. Liu, Y. Lin, Y. Cao, H. Hu, Y. Wei, Z. Zhang, S. Lin, and B. Guo, "Swin transformer: Hierarchical vision transformer using shifted windows," in *ICCV*. IEEE, 2021, pp. 9992–10 002.

[359] H. Wu, B. Xiao, N. Codella, M. Liu, X. Dai, L. Yuan, and L. Zhang, "Cvt: Introducing convolutions to vision transformers," in *ICCV*. IEEE, 2021, pp. 22–31.

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- [310] T. Haarnoja、A. Zhou、P. Abbeel 和 S. Levine，“软演员批评家：带有随机演员的离策略最大熵深度 强化学习（Reinforcement Learning）”，载于_ICML_，卷。 80.PMLR，2018 年，第 1856-1865 页。

- [311] J. Schulman、S. Levine、P. Abbeel、M. I. Jordan 和 P. Moritz，“信任区域策略优化”，载于_ICML_，系列。 JMLR 研讨会和会议论文集，卷。 37. JMLR.org，2015 年，第 1889-1897 页。

- [312] J. Schulman、P. Moritz、S. Levine、M. I. Jordan 和 P. Abbeel，“使用广义优势估计的高维连续控制”，载于_ICLR（海报）_，2016 年。

- [313] J. Ho 和 S. Ermon，“生成对抗 模仿学习（Imitation Learning）”，载于 NIPS，2016 年，第 4565-4573 页。

- [314] L. Pinto、J. Davidson、R. Sukthankar 和 A. Gupta，“鲁棒对抗性 强化学习（Reinforcement Learning）”，载于_ICML_，卷。 70.PMLR，2017 年，第 2817-2826 页。

- [315] P. F. Christiano、J. Leike、T. B. Brown、M. Martic、S. Legg 和 D. Amodei，“来自人类偏好的 Deep 强化学习（Reinforcement Learning）”，载于 NIPS，2017 年，第 4299-4307 页。

- [316] A. S. Vezhnevets、S. Osindero、T. Schaul、N. Heess、M. Jaderberg、D. Silver 和 K. Kavukcuoglu，“分层 强化学习（Reinforcement Learning） 的封建网络”，载于_ICML_，卷。 70.PMLR，2017 年，第 3540-3549 页。

- [317] S. Levine、C. Finn、T. Darrell 和 P. Abbeel，“深度视觉运动策略的端到端训练”，J。马赫。学习。 Res.，卷。 17，第 39:1-39:40，2016 年。

- [318] D. Kalashnikov、A. Irpan、P. Pastor、J. Ibarz、A. Herzog、E. Jang、D. Quillen、E. Holly、M. Kalakrishnan、V. Vanhoucke 和 S. Levine，“Qt-opt：用于基于视觉的机器人操作的可扩展深度 强化学习（Reinforcement Learning）”，CoRR，卷。绝对/1806.10293，2018。

- [319] OpenAI、I. Akkaya、M. Andrychowicz、M. Chociej、M. Litwin、B. McGrew、A. Petron、A. Paino、M. Plapert、G. Powell、R. Ribas、J. Schneider、N. Tezak、J. Tworek、P. Welinder、L. Weng、Q. Yuan、W. Zaremba 和 L. Zhang，“求解带有机器人手的魔方”，CoRR，卷。绝对/1910.07113，2019。

- [320] F. Scarselli、M. Gori、A. C. Tsoi、M. Hagenbuchner 和 G. Monfardini，“图神经网络模型”，IEEE Trans。神经网络，卷。 20、没有。 1，第 61-80 页，2009 年。

- [321] Z. Wu、S. Pan、F. Chen、G. Long、C. Zhang 和 P. S. Yu，“图神经网络的综合综述”，IEEE Trans。神经网络学习。系统.，卷。 32、没有。 1，第 4-24 页，2021 年。

- [322] J. Bruna、W. Zaremba、A. Szlam 和 Y. LeCun，“图上的谱网络和局部连接网络”，载于 ICLR，2014 年。

- [323] M. Deferrad、X. Bresson 和 P. Vandergheynst，“具有快速局部光谱过滤功能的图上的卷积神经网络”，载于 NIPS，2016 年，第 3837-3845 页。

- [324] T. N. Kipf 和 M. Welling，“图卷积网络的半监督分类”，载于 ICLR（海报），2017 年。

- [325] A. Micheli，“图神经网络：上下文构建方法”，IEEE Trans。神经网络，卷。 20、没有。 3，第 498-511 页，2009 年。

- [326] J. Gilmer、S. S. Schoenholz、P. F. Riley、O. Vinyals 和 G. E. Dahl，“量子化学的神经消息传递”，载于_ICML_，卷。 70.PMLR，2017 年，第 1263-1272 页。

- [327] K. Xu、W. Hu、J. Leskovec 和 S

。 Jegelka，“图神经网络有多强大？”在 ICLR 中，2019 年。

- [328] W. L. Hamilton、Z. Ying 和 J. Leskovec，“大图上的归纳表示学习”，载于 NIPS，2017 年，第 1024-1034 页。

- [329] P. Velickovic、G. Cucurull、A. Casanova、A. Romero、P. Lio 和 Y. Bengio，“图注意力网络”，载于 ICLR（海报），2018 年。

- [330] S. Cao、W. Lu 和 Q. Xu，“用于学习图表示的深度神经网络”，载于_AAAI_。 AAAI 出版社，2016 年，第 1145-1152 页。

- [331] T. N. Kipf 和 M. Welling，“变分图自动编码器”，CoRR，卷。绝对/1611.07308，2016。

- [332] Y. Seo、M. Defferrard、P. Vandergheynst 和 X. Bresson，“使用图卷积循环网络进行结构化序列建模”，载于 ICONIP (1)，序列号。计算机科学讲义，卷。 11301。施普林格，2018，第 362-373 页。

- [333] W. Du、H. 张、Y. Du、Q. Meng、W. Chen、N. Cheng、B. Shao 和 T. Liu，“具有完整局部框架的 SE(3) 等变图神经网络”，载于 ICML，第 1 卷。 162.PMLR，2022 年，第 5583-5608 页。

- [334] V. G. Satorras、E. Hoogeboom 和 M. Welling，“E(n) 等变图神经网络”，载于_ICML_，卷。 139.PMLR，2021 年，第 9323-9332 页。

- [335] Y. Rong、Y. Bian、T. Xu、W. Xie、Y. Wei、W. Huang 和 J. Huang，“大规模分子数据上的自监督图形转换器”，《NeurIPS》，2020 年。

- [336] F. Fuchs、D. E. Worrall、V. Fischer 和 M. Welling，“Se(3)transformers：3d 旋转翻译等变注意网络”，载于 NeurIPS，2020 年。

- [337] R. Krishna、Y. Zhu、O. Groth、J. Johnson、K. Hata、J. Kravitz、S. Chen、Y. Kalantidis、L. Li、D. A. Shamma、M. S. Bernstein 和 L. Fei-Fei，“视觉基因组：使用众包密集图像注释连接语言和视觉”，Int。 J. 计算机。视觉，卷。 123，没有。 1，第 32-73 页，2017 年。

- [338] L.Wu，Y.Chen，K.Shen，X.Guo，H.Gao，S.Li，J.Pei，B.Long，“用于自然语言处理的图神经网络：一项调查”，Found。趋势马赫。学习.，卷。 16、没有。 2，第 119-328 页，2023 年。

- [339] D. Ghosal、N. Majumder、S. Poria、N. Chhaya 和 A. F. Gelbukh，“Dialoguegcn：用于对话中情感识别的图卷积神经网络”，载于_EMNLP/IJCNLP (1)_。计算语言学协会，2019 年，第 154-164 页。

- [340] W. Zhu、J. Xu、D. Tang、Z. Xu、N. Duan、M. Zhou、J. Wang 和 J. Yin，“用于事实检查的语义级图推理”，载于 ACL。计算语言学协会，2020 年，第 6170-6180 页。

- [341] L. Wang、Y. Li、O.¡ Aslan 和 O. Vinyals，“Wikigraphs：维基百科文本 - 知识图配对数据集”，CoRR，卷。绝对/2107.09556，2021。

- [342] J. Tang、J. 张、L. Yao、J. Li、L. 张和 Z. Su，“Arnetminer：学术社交网络的提取和挖掘”，载于 KDD。 ACM，2008 年，第 990-998 页。

- [343] H. Tan 和 M. Bansal，“LXMERT：从 Transformer 中学习跨模态编码器表示”，参见 EMNLP/IJCNLP (1)。计算语言学协会，2019 年，第 5099-5110 页。

- [344] L. H. Li、M. Yatskar、D. Yin、C. Hsieh 和 K. Chang，“Visualbert：视觉和语言的简单且高性能的基线”，CoRR，卷。绝对/1908.03557，2019。

- [345] W. Su、X. Zhu、Y. Cao、B. Li、L. Lu、F. Wei 和 J

。 Dai，“VL-BERT：通用视觉语言表示的预训练”，载于 ICLR，2020 年。

- [346] Y. Chen、L. Li、L. Yu、A. E. Kholy、F. Ahmed、Z. Gan、Y. Cheng 和 J. Liu，“UNITER：通用图像文本表示学习”，载于_ECCV (30)_，序列号。计算机科学讲义，卷。 12375。施普林格，2020 年，第 104-120 页。

- [347] W. Kim、B. Son 和 I. Kim，“Vilt：没有卷积或区域监督的视觉和语言转换器”，载于_ICML_，卷。139.PMLR，2021 年，第 5583-5594 页。

- [348] Z. Wang、J. Yu、A. W. Yu、Z. Dai、Y. Tsvetkov 和 Y. Cao，“Simvlm：弱监督下的简单视觉语言模型预训练”，载于 ICLR，2022 年。

- [349] Z. Dai、H. Liu、Q. V. Le 和 M. Tan，“Coatnet：将卷积和注意力机制结合到所有数据大小”，载于 NeurIPS，2021 年，第 3965-3977 页。

- [350] J. Wang、Z. Yang、X. Hu、L. Li、K. Lin、Z. Gan、Z. Liu、C. Liu 和 L. Wang，“GIT：用于视觉和语言的生成式图像到文本转换器”，Trans。马赫。学习。 Res.，卷。 2022 年，2022 年。

- [351] L. Yuan、D. Chen、Y. Chen、N. Codella、X. Dai、J. Gau、H. Hu、X. Huang、B. Li、C. Li、C. Liu、M. Liu、Z. Liu、Y. Lu、Y. Shi、L. Wang、J. Wang、B. Shaw、Z. Shaw、J. Yang、M. Zeng、L. Zhou 和 P. Zhang，《佛罗伦萨：新的佛罗伦萨》用于计算机视觉的 基础模型（Foundation Model）”，CoRR，卷。 ABS/2111.11432，2021。

- [352] W. Wang、H. Bao、L. Dong、J. Bjorck、Z. Peng、Q. Liu、K. Aggarwal、O. K. Mohammed、S. Singhal、S. Som 和 F. Wei，“图像作为外语：视觉和视觉语言任务的 BEIT 预训练”，载于_CVPR_。IEEE，2023 年，第 19 175-19 186 页。

- [353] L. Yao、R. Huang、L. Hou、G. Lu、M. Niu、H. Xu、X. Liang、Z. Li、X. Jiang 和 C. Xu，“FILIP：细粒度交互式语言图像预训练”，载于 ICLR，2022 年。

- [354] C. Jia、Y. Yang、Y. Xia、Y. Chen、Z. Parekh、H. Pham、Q. V. Le、Y. Sung、Z. Li 和 T. Duerig，“利用噪声文本监督扩展视觉和视觉语言表示学习”，载于_ICML_，第 1 卷。 139.PMLR，2021 年，第 4904-4916 页。

- [355] Q. Xie、M. Luong、E. H. Hovy 和 Q. V. Le，“对嘈杂学生进行自我训练可改善图像网络分类”，载于 CVPR。计算机视觉基金会/IEEE，2020 年，第 10 684-10 695 页。

- [356] J. Li、R. R. Selvaraju、A. Gotmare、S. R. Joty、C. Xiong 和 S. C. Hoi，“融合前对齐：利用动量蒸馏进行视觉和语言表示学习”，载于 NeurIPS，2021 年，第 9694-9705 页。

- [357] A. Singh、R. Hu、V. Goswami、G. Couairon、W. Galuba、M. Rohrbach 和 D. Kiela，“FLAVA：基础语言和视觉对齐模型”，载于_CVPR_。 IEEE，2022 年，第 15 617-15 629 页。

- [358] Z. Liu、Y. Lin、Y. Cao、H. Hu、Y. Wei、Z.Zhang、S. Lin 和 B.Guo，“Swin 变压器：使用移位窗口的分层视觉变压器”，载于 ICCV。 IEEE，2021 年，第 9992-10 002 页。

- [359] H.Wu、B.Xiao、N.Codella、M.Liu、X.Dai、L.Yuan 和 L.Zhang，“Cvt：将卷积引入视觉变换器”，载于_ICCV_。 IEEE，2021 年，第 22-31 页。

[360] A. Brock, S. De, S. L. Smith, and K. Simonyan, "High-performance large-scale image recognition without normalization," in *ICML*, vol. 139. PMLR, 2021, pp. 1059–1071.

[361] J. Hoffmann, S. Borgeaud, A. Mensch, E. Buchatskaya, T. Cai, E. Rutherford, D. de Las Casas, L. A. Hendricks, J. Welbl, A. Clark, T. Hennigan, E. Noland, K. Millican, G. van den Driessche, B. Damoc, A. Guy, S. Osindero, K. Simonyan, E. Elsen, J. W. Rae, O. Vinyals, and L. Sifre, "Training compute-optimal large language models," *CoRR*, vol. abs/2203.15556, 2022.

[362] Y. Fang, W. Wang, B. Xie, Q. Sun, L. Wu, X. Wang, T. Huang, X. Wang, and Y. Cao, "EVA: exploring the limits of masked visual representation learning at scale," in *CVPR*. IEEE, 2023, pp. 19 358–19 369.

[363] X. Chen, X. Wang, S. Changpinyo, A. J. Piergiovanni, P. Padlewski, D. Salz, S. Goodman, A. Grycner, B. Mustafa, L. Beyer, A. Kolesnikov, J. Puigcerver, N. Ding, K. Rong, H. Akbari, G. Mishra, L. Xue, A. V. Thapliyal, J. Bradbury, and W. Kuo, "Pali: A jointly-scaled multilingual language-image model," in *ICLR*, 2023.

[364] L. Xue, N. Constant, A. Roberts, M. Kale, R. Al-Rfou, A. Siddhant, A. Barua, and C. Raffel, "mt5: A massively multilingual pre-trained text-to-text transformer," in *NAACL-HLT*. Association for Computational Linguistics, 2021, pp. 483–498.

[365] X. Chen, J. Djolonga, P. Padlewski, B. Mustafa, S. Changpinyo, J. Wu, C. R. Ruiz, S. Goodman, X. Wang, Y. Tay, S. Shakeri, M. Dehghani, D. Salz, M. Lucic, M. Tschannen, A. Nagrani, H. Hu, M. Joshi, B. Pang, C. Montgomery, P. Pietrzyk, M. Ritter, A. J. Piergiovanni, M. Minderer, F. Pavetic, A. Waters, G. Li, I. Alabdulmohsin, L. Beyer, J. Amelot, K. Lee, A. P. Steiner, Y. Li, D. Keysers, A. Arnab, Y. Xu, K. Rong, A. Kolesnikov, M. Seyedhosseini, A. Angelova, X. Zhai, N. Houlsby, and R. Soricut, "Pali-x: On scaling up a multilingual vision and language model," *CoRR*, vol. abs/2305.18565, 2023.

[366] M. Dehghani, J. Djolonga, B. Mustafa, P. Padlewski, J. Heek, J. Gilmer, A. Steiner, M. Caron, R. Geirhos, I. Alabdulmohsin, R. Jenatton, L. Beyer, M. Tschannen, A. Arnab, X. Wang, C. Riquelme, M. Minderer, J. Puigcerver, U. Evci, M. Kumar, S. van Steenkiste, G. F. Elsayed, A. Mahendran, F. Yu, A. Oliver, F. Huot, J. Bastings, M. P. Collier, A. A. Gritsenko, V. Birodkar, C. Vasconcelos, Y. Tay, T. Mensink, A. Kolesnikov, F. Pavetic, D. Tran, T. Kipf, M. Lucic, X. Zhai, D. Keysers, J. Harmsen, and N. Houlsby, "Scaling vision transformers to 22 billion parameters," *CoRR*, vol. abs/2302.05442, 2023.

[367] Y. Tay, M. Dehghani, V. Q. Tran, X. Garcia, J. Wei, X. Wang, H. W. Chung, D. Bahri, T. Schuster, H. S. Zheng, D. Zhou, N. Houlsby, and D. Metzler, "UL2: unifying language learning paradigms," in *ICLR*, 2023.

[368] R. Zhang, J. Han, A. Zhou, X. Hu, S. Yan, P. Lu, H. Li, P. Gao, and Y. Qiao, "Llama-adapter: Efficient fine-tuning of language models with zero-init attention," *CoRR*, vol. abs/2303.16199, 2023.

[369] P. Gao, J. Han, R. Zhang, Z. Lin, S. Geng, A. Zhou, W. Zhang, P. Lu, C. He, X. Yue, H. Li, and Y. Qiao, "Llama-adapter V2: parameter-efficient visual instruction model," *CoRR*, vol. abs/2304.15010, 2023.

[370] S. Huang, L. Dong, W. Wang, Y. Hao, S. Singhal, S. Ma, T. Lv, L. Cui, O. K. Mohammed, B. Patra, Q. Liu, K. Aggarwal, Z. Chi, J. Bjorck, V. Chaudhary, S. Som, X. Song, and F. Wei, "Language is not all you need: Aligning perception with language models," *CoRR*, vol. abs/2302.14045, 2023.

[371] Z. Peng, W. Wang, L. Dong, Y. Hao, S. Huang, S. Ma, and F. Wei, "Kosmos-2: Grounding multimodal large language models to the world," *CoRR*, vol. abs/2306.14824, 2023.

[372] H. Wang, S. Ma, S. Huang, L. Dong, W. Wang, Z. Peng, Y. Wu, P. Bajaj, S. Singhal, A. Benhaim, B. Patra, Z. Liu, V. Chaudhary, X. Song, and F. Wei, "Foundation transformers," *CoRR*, vol. abs/2210.06423, 2022.

[373] W. Dai, J. Li, D. Li, A. M. H. Tiong, J. Zhao, W. Wang, B. Li, P. Fung, and S. C. H. Hoi, "Instructblip: Towards general-purpose vision-language models with instruction tuning," *CoRR*, vol. abs/2305.06500, 2023.

[374] D. Zhu, J. Chen, X. Shen, X. Li, and M. Elhoseiny, "Minigpt-4: Enhancing vision-language understanding with advanced large language models," *CoRR*, vol. abs/2304.10592, 2023.

[375] H. Zhang, X. Li, and L. Bing, "Video-llama: An instruction-tuned audio-visual language model for video understanding," *CoRR*, vol. abs/2306.02858, 2023.

[376] Y. Su, T. Lan, H. Li, J. Xu, Y. Wang, and D. Cai, "Pandagpt: One model to instruction-follow them all," *CoRR*, vol. abs/2305.16355, 2023.

[377] K. Li, Y. He, Y. Wang, Y. Li, W. Wang, P. Luo, Y. Wang, L. Wang, and Y. Qiao, "Videochat: Chat-centric video understanding," *CoRR*, vol. abs/2305.06355, 2023.

[378] S. contributors. (2023) Stablelm: Stability ai language models. [Online]. Available: https://github.com/stability-AI/stableLM

[379] L. Zhao, E. Yu, Z. Ge, J. Yang, H. Wei, H. Zhou, J. Sun, Y. Peng, R. Dong, C. Han, and X. Zhang, "Chatspot: Bootstrapping multimodal llms via precise referring instruction tuning," *CoRR*, vol. abs/2307.09474, 2023.

[380] Q. Ye, H. Xu, G. Xu, J. Ye, M. Yan, Y. Zhou, J. Wang, A. Hu, P. Shi, Y. Shi, C. Li, Y. Xu, H. Chen, J. Tian, Q. Qi, J. Zhang, and F. Huang, "mplug-owl: Modularization empowers large language models with multimodality," *CoRR*, vol. abs/2304.14178, 2023.

[381] Q. Ye, H. Xu, J. Ye, M. Yan, A. Hu, H. Liu, Q. Qian, J. Zhang, F. Huang, and J. Zhou, "mplug-owl2: Revolutionizing multi-modal large language model with modality collaboration," *CoRR*, vol. abs/2311.04257, 2023.

[382] C. Xu, D. Guo, N. Duan, and J. J. McAuley, "Baize: An open-source chat model with parameter-efficient tuning on self-chat data," in *EMNLP*. Association for Computational Linguistics, 2023, pp. 6268–6278.

[383] C. Wu, S. Yin, W. Qi, X. Wang, Z. Tang, and N. Duan, "Visual chatgpt: Talking, drawing and editing with visual foundation models," *CoRR*, vol. abs/2303.04671, 2023.

[384] F. Chen, M. Han, H. Zhao, Q. Zhang, J. Shi, S. Xu, and B. Xu, "X-LLM: bootstrapping advanced large language models by treating multi-modalities as foreign languages," *CoRR*, vol. abs/2305.04160, 2023.

[385] X. Zhai, A. Kolesnikov, N. Houlsby, and L. Beyer, "Scaling vision transformers," in *CVPR*. IEEE, 2022, pp. 1204–1213.

[386] Z. Du, Y. Qian, X. Liu, M. Ding, J. Qiu, Z. Yang, and J. Tang, "GLM: general language model pretraining with autoregressive blank infilling," in *ACL (1)*. Association for Computational Linguistics, 2022, pp. 320–335.

[387] F. Chen, D. Zhang, M. Han, X. Chen, J. Shi, S. Xu, and B. Xu, "VLP: A survey on vision-language pre-training," *Int. J. Autom. Comput.*, vol. 20, no. 1, pp. 38–56, 2023.

[388] P. Xu, X. Zhu, and D. A. Clifton, "Multimodal learning with transformers: A survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 10, pp. 12 113–12 132, 2023.

[389] X. Wang, G. Chen, G. Qian, P. Gao, X. Wei, Y. Wang, Y. Tian, and W. Gao, "Large-scale multi-modal pre-trained models: A comprehensive survey," *Mach. Intell. Res.*, vol. 20, no. 4, pp. 447–482, 2023.

[390] J. Zhang, J. Huang, S. Jin, and S. Lu, "Vision-language models for vision tasks: A survey," *CoRR*, vol. abs/2304.00685, 2023.

[391] C. Sun, A. Myers, C. Vondrick, K. Murphy, and C. Schmid, "Videobert: A joint model for video and language representation learning," in *ICCV*. IEEE, 2019, pp. 7463–7472.

[392] H. Bao, W. Wang, L. Dong, Q. Liu, O. K. Mohammed, K. Aggarwal, S. Som, S. Piao, and F. Wei, "Vlmo: Unified vision-language pre-training with mixture-of-modality-experts," in *NeurIPS*, 2022.

[393] X. Zhai, X. Wang, B. Mustafa, A. Steiner, D. Keysers, A. Kolesnikov, and L. Beyer, "Lit: Zero-shot transfer with locked-image text tuning," in *CVPR*. IEEE, 2022, pp. 18 102–18 112.

[394] M. Tsimpoukelli, J. Menick, S. Cabi, S. M. A. Eslami, O. Vinyals, and F. Hill, "Multimodal few-shot learning with frozen language models," in *NeurIPS*, 2021, pp. 200–212.

[395] J. Yu, Z. Wang, V. Vasudevan, L. Yeung, M. Seyedhosseini, and Y. Wu, "Coca: Contrastive captioners are image-text foundation models," *Trans. Mach. Learn. Res.*, vol. 2022, 2022.

[396] P. Wang, A. Yang, R. Men, J. Lin, S. Bai, Z. Li, J. Ma, C. Zhou, J. Zhou, and H. Yang, "OFA: unifying architectures, tasks, and modalities through a simple sequence-to-sequence learning framework," in *ICML*, vol. 162. PMLR, 2022, pp. 23 318–23 340.

[397] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, "Lora: Low-rank adaptation of large language models," in *ICLR*, 2022.

[398] P. Esser, R. Rombach, and B. Ommer, "Taming transformers for high-resolution image synthesis," in *CVPR*. Computer Vision Foundation / IEEE, 2021, pp. 12 873–12 883.

[399] C. Zheng, T. Vuong, J. Cai, and D. Phung, "Movq: Modulating quantized vectors for high-fidelity image generation," in *NeurIPS*, 2022.

[400] X. Zhai, B. Mustafa, A. Kolesnikov, and L. Beyer, "Sigmoid loss for language image pre-training," in *ICCV*. IEEE, 2023, pp. 11 941–11 952.

[401] X. Gu, T. Lin, W. Kuo, and Y. Cui, "Open-vocabulary object detection via vision and language knowledge distillation," in *ICLR*, 2022.

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- [360] A. Brock、S. De、S. L. Smith 和 K. Simonyan，“无需标准化的高性能大规模图像识别”，载于 ICML，第 1 卷。 139.PMLR，2021 年，第 1059-1071 页。

- [361] J. Hoffmann、S. Borgeaud、A. Mensch、E. Buchatskaya、T. Cai、E. Rutherford、D. de Las Casas、L. A. Hendricks、J. Welbl、A. Clark、T. Hennigan、E. Noland、K. Millican、G. van den Driessche、B. Damoc、A. Guy、S. Osindero、K. Simonyan、E. Elsen、J. W. Rae、O. Vinyals 和 L. Sifre，“训练计算最优大型语言模型”，CoRR，卷。 ABS/2203.15556，2022。

- [362] Y. Fang、W. Wang、B. Xie、Q. Sun、L. Wu、X. Wang、T. Huang、X. Wang 和 Y. Cao，“EVA：探索大规模蒙版视觉表示学习的局限性”，载于_CVPR_。 IEEE，2023 年，第 19 358-19 369 页。

- [363] X. Chen、X. Wang、S. Changpinyo、A. J. Piergiovanni、P. Padlewski、D. Salz、S. Goodman、A. Grycner、B. Mustafa、L. Beyer、A. Kolesnikov、J. Puigcerver、N. Ding、K. Rong、H. Akbari、G. Mishra、L. Xu、A. V. Thapliyal、J. Bradbury 和 W. Kuo，“巴利语：联合缩放的多语言语言图像模型”，载于 ICLR，2023 年。

- [364] L. Xu、N. Constant、A. Roberts、M. Kale、R. Al-Rfou、A. Siddhant、A. Barua 和 C. Raffel，“mt5：大规模多语言预训练文本到文本转换器”，载于_NAACL-HLT_。计算语言学协会，2021 年，第 483-498 页。

- [365] X. Chen、J. Djolonga、P. Padlewski、B. Mustafa、S. Changpinyo、J. Wu、C. R. Ruiz、S. Goodman、X. Wang、Y. Tay、S. Shakeri、M. Dehghani、D. Salz、M. Lucic、M. Tschannen、A. Nagrani、H. Hu、M. Joshi、B. Pang、C.蒙哥马利、P. Pietrzyk、M. Ritter、A. J. Piergiovanni、M. Minderer、F. Pavetic、A. Waters、G. Li、I. Alabdulmohsin、L. Beyer、J. Amelot、K. Lee、A. P. Steiner、Y. Li、D. Keysers、A. Arnab、Y. Xu、K. Rong、A. Kolesnikov、M. Seyedhosseini、A. Angelova、X. Zhai、N. Houlsby 和 R. Soricut，“Pali-x：扩大多语言视觉和语言模型”，CoRR，卷。绝对 /2305.18565，2023。

- [366] M. Dehghani、J. Djolonga、B. Mustafa、P. Padlewski、J. Heek、J. Gilmer、A. Steiner、M. Caron、R. Geirhos、I. Alabdulmohsin、R. Jenatton、L. Beyer、M. Tschannen、A. Arnab、X. Wang、C. Riquelme、M. Minderer、J. Puigcerver、U. Evci、M. Kumar、S. van Steenkiste、G. F. Elsayed、A. Mahendran、F. Yu、A. Oliver、F. Huot、J. Bastings、M. P. Collier、A. A. Gritsenko、V. Birodkar、C. Vasconcelos、Y. Tay、T. Mensink、A. Kolesnikov、F. Pavetic、D. Tran、 T. Kipf、M. Lucic、X. Zhai、D. Keysers、J. Harmsen 和 N. Houlsby，“将视觉变换器扩展到 220 亿个参数”，CoRR，卷。绝对 /2302.05442，2023。

- [367] Y. Tay、M. Dehghani、V. Q. Tran、X. Garcia、J. Wei、X. Wang、H. W. Chung、D. Bahri、T. Schuster、H. S. Cheng、D. Zhou、N. Houlsby 和 D. Metzler，“UL2：统一语言学习范式”，载于 ICLR，2023 年。

- [368] R.Zhang、J.Han、A.Zhou、X.Hu、S.Yan、P.Lu、H.Li、P.Gao 和 Y.Qiao，“Llama-adapter：零初始化注意力的语言模型的高效微调”，CoRR，卷。 ABS/2303.16199，2023。

- [369] P. 高，J. 韩，R. 张，Z. 林，S. Geng，A. Zhou，W. 张，P. Lu，C. He，X. Yue，H. Li，and Y. Qiao，“Llama-adapter V2：参数高效视觉指令模型”，CoRR，卷。绝对/2304.15010，2023。

- [370] 小号

。 Huang、L. Dong、W. Wang、Y.hao、S. Singhal、S. Ma、T. Lv、L. Cui、O. K. Mohammed、B. Patra、Q. Liu、K. Aggarwal、Z. Chi、J. Bjorck、V. Chaudhary、S. Som、X. Song 和 F. Wei，“语言并不是您所需要的全部：使感知与语言模型保持一致”CoRR，卷。 ABS/2302.14045，2023。

- [371] Z. Peng、W. Wang、L. Dong、Y.hao、S. Huang、S. Ma 和 F. Wei，“Kosmos-2：面向世界的 落地关联（Grounding） 多模态大语言模型”，CoRR，卷。 ABS/2306.14824，2023。

- [372] H. Wang、S. Ma、S. Huang、L. Dong、W. Wang、Z. Peng、Y. Wu、P. Bajaj、S. Singhal、A. Benhaim、B. Patra、Z. Liu、V. Chaudhary、X. Song 和 F. Wei，《基础变形金刚》，CoRR，卷。绝对/2210.06423，2022。

- [373] W. Dai、J. Li、D. Li、A. M. H. Tiong、J. Zhu、W. Wang、B. Li、P. Fung 和 S. C. H. Hoi，“Instructblip：通过指令调整实现 通用（General-Purpose） 视觉语言模型”，CoRR，卷。绝对/2305.06500，2023。

- [374] D. Zhu、J. Chen、X. Shen、X. Li 和 M. Elhoseiny，“Minigpt-4：利用先进的大语言模型增强视觉语言理解”，CoRR，卷。 ABS/2304.10592，2023。

- [375] H.Zhang、X.Li 和 L.Bing，“Video-llama：用于视频理解的指令调整视听语言模型”，CoRR，卷。绝对/2306.02858，2023。

- [376] Y. Su、T. Lan、H. Li、J. Xu、Y. Wang 和 D. Cai，“Pandagpt：一种遵循所有指令的模型”，CoRR，卷。绝对/2305.16355，2023。

- [377] K. Li、Y. He、Y. Wang、Y. Li、W. Wang、P. Luo、Y. Wang、L. Wang 和 Y. Qiao，“视频聊天：以聊天为中心的视频理解”，CoRR，卷。绝对/2305.06355，2023。

- [378] S. 贡献者。 (2023) Stablelm：稳定性人工智能语言模型。 [在线的]。可用：https://github.com/stability-AI/stableLM

- [379] L.Zhao、E.Yu、Z.Ge、J.Yang、H.Wei、H.Zhou、J.Sun、Y.Peng、R.Dong、C.Han 和 X.Zhang，“Chatspot：通过精确参考指令调整引导多模态 LLMS”，CoRR，卷。绝对/2307.09474，2023。

- [380] Q. Ye、H. Xu、G. Xu、J. Ye、M. Yan、Y. Zhou、J. Wang、A. Hu、P. Shi、Y. Shi、C. Li、Y. Xu、H. Chen、J. Tian、Q. Qi、J. Zhang 和 F. Huang，“mplug-owl：模块化赋予大型语言模型多模态”，CoRR，卷。 ABS/2304.14178，2023。

- [381] Q. Ye、H. Xu、J. Ye、M. Yan、A. Hu、H. Liu、Q. Qiu、J. Zhang、F. Huang 和 J. Zhou，“mplug-owl2：通过模态协作彻底改变多模态大语言模型”，CoRR，卷。绝对/2311.04257，2023。

- [382] C. Xu、D.Guo、N. Duan 和 J. J. McAuley，“Baize：一种对自助聊天数据进行参数高效调整的开源聊天模型”，载于 EMNLP。计算语言学协会，2023 年，第 6268-6278 页。

- [383] C. Wu、S. Yin、W. Qi、X. Wang、Z. Tang 和 N. Duan，“视觉 chatgpt：用视觉 基础模型（Foundation Models） 进行说话、绘图和编辑”，CoRR，卷。绝对/2303.04671，2023。

- [384] F. Chen、M. Han、H. Zhu、Q. Zhu、J. Shi、S. Xu 和 B. Xu，“XLLM：通过将多模态视为外语来引导高级大语言模型”，CoRR，卷。绝对/2305.04160，2023。

- [385] X. Zhai、A. Kolesnikov、N. Houlsby 和 L. Beyer，“缩放视觉变换器”，载于 CVPR。 IEEE，2022 年，第 1204-1213 页。

- [386] 杜正、钱勇、刘旭、丁明、邱静、杨正、J

。 Tang，“GLM：使用自回归空白填充的通用语言模型预训练”，载于 ACL (1) 。计算语言学协会，2022 年，第 320-335 页。

- [387] F. Chen、D. 张、M. Han、X. Chen、J. Shi、S. Xu 和 B. Xu，“VLP：视觉语言预训练调查”，Int。 J.奥托姆。计算机.，卷。 20、没有。 1，第 38-56 页，2023 年。

- [388] P. Xu、X. Zhu 和 D. A. Clifton，“使用 Transformer 进行多模态学习：一项调查”，IEEE Trans。模式肛门。马赫。情报.，卷。 45，没有。 10，第 12 113-12 132 页，2023 年。

- [389] X. Wang、G. Chen、G. qian、P. Gau、X. Wei、Y. Wang、Y. Tian 和 W. Gau，“大规模多模态预训练模型：综合调查”，Mach。英特尔。 Res.，卷。 20、没有。 4，第 447-482 页，2023 年。

- [390] J. 张、J. 黄、S. Jin 和 S. Lu，“视觉任务的视觉语言模型：一项调查”，CoRR，卷。绝对 /2304.00685，2023。

- [391] C. Sun、A. Myers、C. Vondrick、K. Murphy 和 C. Schmid，“Videobert：视频和语言表示学习的联合模型”，载于_ICCV_。 IEEE，2019 年，第 7463-7472 页。

- [392] H. Bao、W. Wang、L. Dong、Q. Liu、O. K. Mohammed、K. Aggarwal、S. Som、S. Piao 和 F. Wei，“Vlmo：混合模态专家的统一视觉语言预训练”，《NeurIPS》，2022 年。

- [393] X. Zhai、X. Wang、B. Mustafa、A. Steiner、D. Keysers、A. Kolesnikov 和 L. Beyer，“Lit：带有锁定图像文本调整的零样本传输”，载于_CVPR_。 IEEE，2022 年，第 18 102-18 112 页。

- [394] M. Tsimpoukelli、J. Menick、S. Cabi、S. M. A. Eslami、O. Vinyals 和 F. Hill，“利用冻结语言模型进行多模态少样本学习”，载于 NeurIPS，2021 年，第 200-212 页。

- [395] J. Yu、Z. Wang、V. Vasudevan、L. Yeung、M. Seyedhosseini 和 Y. Wu，“Coca：对比字幕是图像文本 基础模型（Foundation Models）”，Trans。马赫。学习。 Res.，卷。 2022 年，2022 年。

- [396] P. Wang、A. Yang、R. Men、J. Lin、S. Bai、Z. Li、J. Ma、C. Zhou、J. Zhou 和 H. Yang，“OFA：通过简单的序列到序列学习框架统一架构、任务和模式”，《ICML》，第 1 卷。 162.PMLR，2022 年，第 23 318-23 340 页。

- [397] E. J. Hu、Y. Shen、P. Wallis、Z. Allen-Zhu、Y. Li、S. Wang、L. Wang 和 W. Chen，“Lora：大型语言模型的低阶适应”，载于 ICLR，2022 年。

- [398] P. Esser、R. Rombach 和 B. Ommer，“驯服用于高分辨率图像合成的变压器”，载于 CVPR。计算机视觉基金会/IEEE，2021 年，第 12 873-12 883 页。

- [399] C. Cheng、T. Vuong、J. Cai 和 D. Phung，“Movq：调制量化向量以生成高保真图像”，载于 NeurIPS，2022 年。

- [400] X. Zhai、B. Mustafa、A. Kolesnikov 和 L. Beyer，“语言图像预训练的 Sigmoid 损失”，载于 _ICCV_。 IEEE，2023 年，第 11 941-11 952 页。

- [401] X. Gu、T. Lin、W. Kuo 和 Y. Cui，“通过视觉和语言知识蒸馏进行开放词汇对象检测”，载于 ICLR，2022 年。

[402] A. Kamath, M. Singh, Y. LeCun, G. Synnaeve, I. Misra, and N. Carion, "MDETR - modulated detection for end-to-end multi-modal understanding," in *ICCV*. IEEE, 2021, pp. 1760–1770.

[403] V. Blukis, C. Paxton, D. Fox, A. Garg, and Y. Artzi, "A persistent spatial semantic representation for high-level natural language instruction execution," in *CoRL*, vol. 164. PMLR, 2021, pp. 706–717.

[404] K. He, X. Chen, S. Xie, Y. Li, P. Dollár, and R. B. Girshick, "Masked autoencoders are scalable vision learners," in *CVPR*. IEEE, 2022, pp. 15 979–15 988.

[405] M. Minderer, A. A. Gritsenko, A. Stone, M. Neumann, D. Weissenborn, A. Dosovitskiy, A. Mahendran, A. Arnab, M. Dehghani, Z. Shen, X. Wang, X. Zhai, T. Kipf, and N. Houlsby, "Simple open-vocabulary object detection with vision transformers," *CoRR*, vol. abs/2205.06230, 2022.

[406] Z. Liu, H. Mao, C. Wu, C. Feichtenhofer, T. Darrell, and S. Xie, "A convnet for the 2020s," in *CVPR*. IEEE, 2022, pp. 11 966–11 976.

[407] J. Ho, W. Chan, C. Saharia, J. Whang, R. Gao, A. A. Gritsenko, D. P. Kingma, B. Poole, M. Norouzi, D. J. Fleet, and T. Salimans, "Imagen video: High definition video generation with diffusion models," *CoRR*, vol. abs/2210.02303, 2022.

[408] R. Villegas, M. Babaeizadeh, P. Kindermans, H. Moraldo, H. Zhang, M. T. Saffar, S. Castro, J. Kunze, and D. Erhan, "Phenaki: Variable length video generation from open domain textual descriptions," in *ICLR*. OpenReview.net, 2023.

[409] M. S. M. Sajjadi, D. Duckworth, A. Mahendran, S. van Steenkiste, F. Pavetic, M. Lucic, L. J. Guibas, K. Greff, and T. Kipf, "Object scene representation transformer," in *NeurIPS*, 2022.

[410] C. R. Qi, H. Su, K. Mo, and L. J. Guibas, "Pointnet: Deep learning on point sets for 3d classification and segmentation," in *CVPR*. IEEE Computer Society, 2017, pp. 77–85.

[411] C. R. Qi, L. Yi, H. Su, and L. J. Guibas, "Pointnet++: Deep hierarchical feature learning on point sets in a metric space," in *NIPS*, 2017, pp. 5099–5108.

[412] Y. Hong, C. Lin, Y. Du, Z. Chen, J. B. Tenenbaum, and C. Gan, "3d concept learning and reasoning from multi-view images," in *CVPR*. IEEE, 2023, pp. 9202–9212.

[413] K. M. Jatavallabhula, A. Kuwajerwala, Q. Gu, M. Omama, G. Iyer, S. Saryazdi, T. Chen, A. Maalouf, S. Li, N. V. Keetha, A. Tewari, J. B. Tenenbaum, C. M. de Melo, K. M. Krishna, L. Paull, F. Shkurti, and A. Torralba, "Conceptfusion: Open-set multimodal 3d mapping," in *RSS*, 2023.

[414] N. Reimers and I. Gurevych, "Sentence-bert: Sentence embeddings using siamese bert-networks," in *EMNLP/IJCNLP (1)*. Association for Computational Linguistics, 2019, pp. 3980–3990.

[415] V. Sanh, L. Debut, J. Chaumond, and T. Wolf, "Distilbert, a distilled version of BERT: smaller, faster, cheaper and lighter," *CoRR*, vol. abs/1910.01108, 2019.

[416] M. N. Team. (2023) Introducing mpt-7b: A new standard for open-source, commercially usable llms. Accessed: 2023-05-05. [Online]. Available: www.mosaicml.com/blog/mpt-7b

[417] A. Awadalla, I. Gao, J. Gardner, J. Hessel, Y. Hanafy, W. Zhu, K. Marathe, Y. Bitton, S. Y. Gadre, S. Sagawa, J. Jitsev, S. Kornblith, P. W. Koh, G. Ilharco, M. Wortsman, and L. Schmidt, "Openflamingo: An open-source framework for training large autoregressive vision-language models," *CoRR*, vol. abs/2308.01390, 2023.

[418] S. Biderman, H. Schoelkopf, Q. G. Anthony, H. Bradley, K. O'Brien, E. Hallahan, M. A. Khan, S. Purohit, U. S. Prashanth, E. Raff, A. Skowron, L. Sutawika, and O. van der Wal, "Pythia: A suite for analyzing large language models across training and scaling," in *ICML*, vol. 202. PMLR, 2023, pp. 2397–2430.

[419] M. Javaheripi, S. Bubeck, M. Abdin, J. Aneja, S. Bubeck, C. C. T. Mendes, W. Chen, A. Del Giorno, R. Eldan, S. Gopi *et al.*, "Phi-2: The surprising power of small language models," *Microsoft Research Blog*, 2023.

[420] S. Black, S. Biderman, E. Hallahan, Q. Anthony, L. Gao, L. Golding, H. He, C. Leahy, K. McDonell, J. Phang, M. Pieler, U. S. Prashanth, S. Purohit, L. Reynolds, J. Tow, B. Wang, and S. Weinbach, "Gpt-neox-20b: An open-source autoregressive language model," *CoRR*, vol. abs/2204.06745, 2022.

[421] M. Chen, J. Tworek, H. Jun, Q. Yuan, H. P. de Oliveira Pinto, J. Kaplan, H. Edwards, Y. Burda, N. Joseph, G. Brockman, A. Ray, R. Puri, G. Krueger, M. Petrov, H. Khlaaf, G. Sastry, P. Mishkin, B. Chan, S. Gray, N. Ryder, M. Pavlov, A. Power, L. Kaiser, M. Bavarian, C. Winter, P. Tillet, F. P. Such, D. Cummings, M. Plappert, F. Chantzis, E. Barnes, A. Herbert-Voss, W. H. Guss, A. Nichol, A. Paino, N. Tezak, J. Tang, I. Babuschkin, S. Balaji, S. Jain, W. Saunders, C. Hesse, A. N. Carr, J. Leike, J. Achiam, V. Misra, E. Morikawa, A. Radford, M. Knight, M. Brundage, M. Murati, K. Mayer, P. Welinder, B. McGrew, D. Amodei, S. McCandlish, I. Sutskever, and W. Zaremba, "Evaluating large language models trained on code," *CoRR*, vol. abs/2107.03374, 2021.

[422] S. Karamcheti, S. Nair, A. Balakrishna, P. Liang, T. Kollar, and D. Sadigh, "Prismatic vlms: Investigating the design space of visually-conditioned language models," in *ICML*, 2024.

[423] L. Beyer, A. Steiner, A. S. Pinto, A. Kolesnikov, X. Wang, D. Salz, M. Neumann, I. Alabdulmohsin, M. Tschannen, E. Bugliarello, T. Unterthiner, D. Keysers, S. Koppula, F. Liu, A. Grycner, A. A. Gritsenko, N. Houlsby, M. Kumar, K. Rong, J. Eisenschlos, R. Kabra, M. Bauer, M. Bosnjak, X. Chen, M. Minderer, P. Voigtlaender, I. Bica, I. Balazevic, J. Puigcerver, P. Papalampidi, O. J. Hénaff, X. Xiong, R. Soricut, J. Harmsen, and X. Zhai, "Paligemma: A versatile 3b VLM for transfer," *CoRR*, vol. abs/2407.07726, 2024.

[424] A. Steiner, A. S. Pinto, M. Tschannen, D. Keysers, X. Wang, Y. Bitton, A. A. Gritsenko, M. Minderer, A. Sherbondy, S. Long, S. Qin, R. R. Ingle, E. Bugliarello, S. Kazemzadeh, T. Mesnard, I. Alabdulmohsin, L. Beyer, and X. Zhai, "Paligemma 2: A family of versatile vlms for transfer," *CoRR*, vol. abs/2412.03555, 2024.

[425] Chameleon-Team, "Chameleon: Mixed-modal early-fusion foundation models," *CoRR*, vol. abs/2405.09818, 2024.

[426] X. Wang, X. Zhang, Z. Luo, Q. Sun, Y. Cui, J. Wang, F. Zhang, Y. Wang, Z. Li, Q. Yu, Y. Zhao, Y. Ao, X. Min, T. Li, B. Wu, B. Zhao, B. Zhang, L. Wang, G. Liu, Z. He, X. Yang, J. Liu, Y. Lin, T. Huang, and Z. Wang, "Emu3: Next-token prediction is all you need," *CoRR*, vol. abs/2409.18869, 2024.

[427] H. Liu, W. Yan, M. Zaharia, and P. Abbeel, "World model on million-length video and language with blockwise ringattention," in *ICLR*. OpenReview.net, 2025.

[428] D. K. Misra, A. Bennett, V. Blukis, E. Niklasson, M. Shatkhin, and Y. Artzi, "Mapping instructions to actions in 3d environments with visual goal prediction," in *EMNLP*, 2018, pp. 2667–2678.

[429] C. Lynch, M. Khansari, T. Xiao, V. Kumar, J. Tompson, S. Levine, and P. Sermanet, "Learning latent plans from play," in *CoRL*, vol. 100. PMLR, 2019, pp. 1113–1132.

[430] M. Bhardwaj, B. Sundaralingam, A. Mousavian, N. D. Ratliff, D. Fox, F. Ramos, and B. Boots, "STORM: an integrated framework for fast joint-space model-predictive control for reactive manipulation," in *CoRL*, vol. 164. PMLR, 2021, pp. 750–759.

[431] A. Jaegle, S. Borgeaud, J. Alayrac, C. Doersch, C. Ionescu, D. Ding, S. Koppula, D. Zoran, A. Brock, E. Shelhamer, O. J. Hénaff, M. M. Botvinick, A. Zisserman, O. Vinyals, and J. Carreira, "Perceiver IO: A general architecture for structured inputs & outputs," in *ICLR*, 2022.

[432] E. Wijmans, A. Kadian, A. Morcos, S. Lee, I. Essa, D. Parikh, M. Savva, and D. Batra, "DD-PPO: learning near-perfect pointgoal navigators from 2.5 billion frames," in *ICLR*, 2020.

[433] M. Zhu, Y. Zhu, J. Li, J. Wen, Z. Xu, N. Liu, R. Cheng, C. Shen, Y. Peng, F. Feng, and J. Tang, "Scaling diffusion policy in transformer to 1 billion parameters for robotic manipulation," *CoRR*, vol. abs/2409.14411, 2024.

[434] J. W. Wei, L. Hou, A. K. Lampinen, X. Chen, D. Huang, Y. Tay, X. Chen, Y. Lu, D. Zhou, T. Ma, and Q. V. Le, "Symbol tuning improves in-context learning in language models," *CoRR*, vol. abs/2305.08298, 2023.

[435] W. Peebles and S. Xie, "Scalable diffusion models with transformers," in *ICCV*. IEEE, 2023, pp. 4172–4182.

[436] J. Ho, A. Jain, and P. Abbeel, "Denoising diffusion probabilistic models," in *NeurIPS*, 2020.

[437] Y. Song, J. Sohl-Dickstein, D. P. Kingma, A. Kumar, S. Ermon, and B. Poole, "Score-based generative modeling through stochastic differential equations," in *ICLR*, 2021.

[438] A. Zeng, P. Florence, J. Tompson, S. Welker, J. Chien, M. Attarian, T. Armstrong, I. Krasin, D. Duong, V. Sindhwani, and J. Lee, "Transporter networks: Rearranging the visual world for robotic manipulation," in *CoRL*, vol. 155. PMLR, 2020, pp. 726–747.

[439] R. Goyal, S. E. Kahou, V. Michalski, J. Materzynska, S. Westphal, H. Kim, V. Haenel, I. Fründ, P. Yianilos, M. Mueller-Freitag, F. Hoppe, C. Thurau, I. Bax, and R. Memisevic, "The "something something" video database for learning and evaluating visual common sense," in *ICCV*. IEEE Computer Society, 2017, pp. 5843–5851.

[440] X. Lin, J. So, S. Mahalingam, F. Liu, and P. Abbeel, "Spawnnet: Learning generalizable visuomotor skills from pre-trained networks," *CoRR*, vol. abs/2307.03567, 2023.

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- [402] A. Kamath、M. Singh、Y. LeCun、G. Synnaeve、I. Misra 和 N. Carion，“MDETR - 用于端到端多模态理解的调制检测”，载于_ICCV_。 IEEE，2021 年，第 1760-1770 页。

- [403] V. Blukis、C. Paxton、D. Fox、A. Garg 和 Y. Artzi，“高级自然语言指令执行的持久空间语义表示”，载于_CoRL_，卷。 164.PMLR，2021 年，第 706-717 页。

- [404] K. He、X. Chen、S. Xie、Y. Li、P. Doll´ar 和 R. B. Girshick，“蒙面自动编码器是可扩展的视觉学习器”，载于_CVPR_。 IEEE，2022 年，第 15 979-15 988 页。

- [405] M. Minderer、A. A. Gritsenko、A. Stone、M. Neumann、D. Weissenborn、A. Dosovitskiy、A. Mahendran、A. Arnab、M. Dehghani、Z. Shen、X. Wang、X. Zhai、T. Kipf 和 N. Houlsby，“使用视觉变换器进行简单的开放词汇对象检测”，CoRR，卷。绝对/2205.06230，2022。

- [406] Z. Liu、H. Mao、C. Wu、C. Feichtenhofer、T. Darrell 和 S. Xie，“2020 年代的卷积网络”，载于_CVPR_。 IEEE，2022 年，第 11 966-11 976 页。

- [407] J. Ho、W. Chan、C. Saharia、J. Whang、R. Gau、A. A. Gritsenko、D. P. Kingma、B. Poole、M. Norouzi、D. J. Fleet 和 T. Salimans，“Imagen 视频：利用扩散模型生成高清视频”，CoRR，卷。绝对/2210.02303，2022。

- [408] R. Villegas、M. Babaeizadeh、P. Kindermans、H. Moraldo、H. Zhang、M. T. Saffar、S. Castro、J. Kunze 和 D. Erhan，“Phenaki：根据开放域文本描述生成可变长度视频”，载于_ICLR_。 OpenReview.net，2023 年。

- [409] M. S. M. Sajjadi、D. Duckworth、A. Mahendran、S. van Steenkiste、F. Pavetic、M. Lucic、L. J. Guibas、K. Greff 和 T. Kipf，“对象场景表示变换器”，《NeurIPS》，2022 年。

- [410] C. R. Qi、H. Su、K. Mo 和 L. J. Guibas，“Pointnet：用于 3d 分类和分割的点集深度学习”，载于_CVPR_。 IEEE 计算机协会，2017 年，第 77-85 页。

- [411] C. R. Qi、L. Yi、H. Su 和 L. J. Guibas，“Pointnet++：度量空间中点集的深度分层特征学习”，载于 NIPS，2017 年，第 5099-5108 页。

- [412] Y. Hong、C. Lin、Y. Du、Z. Chen、J. B. Tenenbaum 和 C. Gan，“多视图图像的 3d 概念学习和推理”，载于_CVPR_。 IEEE，2023 年，第 9202-9212 页。

- [413] K. M. Jatavallabhula、A. Kuwajerwala、Q. Gu、M. Omama、G. Iyer、S. Saryazdi、T. Chen、A. Maalouf、S. Li、N. V. Keetha、A. Tewari、J. B. Tenenbaum、C. M. de Melo、K. M. Krishna、L. Paull、F. Shkurti 和A. Torralba，“概念融合：开放式多模态 3D 映射”，收录于 RSS，2023 年。

- [414] N. Reimers 和 I. Gurevych，“Sentence-bert：使用暹罗 bert 网络的句子嵌入”，见 EMNLP/IJCNLP (1)。计算语言学协会，2019 年，第 3980-3990 页。

- [415] V. Sanh、L. Debut、J. Chaumond 和 T. Wolf，“Distilbert，BERT 的精炼版本：更小、更快、更便宜、更轻”，CoRR，卷。绝对/1910.01108，2019。

- [416] M.N. 团队。 (2023) 推出 mpt-7b：开源、可商业使用的 llms 的新标准。访问时间：2023 年 5 月 5 日。 [在线的]。可用：www.mosaicml.com/blog/mpt-7b

- [417] A. Awadalla、I. Gau、J. Gardner、J. Hessel、Y. Hanafy、W. Zhu、K. Marathe、Y. Bitton、S. Y. Gadre、S. Sakawa、J. Jitsev、S. Kornblith、P. W. Koh、G. Ilharco、M. Wortsman 和 L

。 Schmidt，“Openflamingo：用于训练大型自回归视觉语言模型的开源框架”，CoRR，卷。绝对/2308.01390，2023。

- [418] S. Biderman、H. Schoelkopf、Q. G. Anthony、H. Bradley、K. O’Brien、E. Hallahan、M. A. Khan、S. Purohit、U. S. Prashanth、E. Raff、A. Skowron、L. Sutawika 和 O. van der Wal，“Pythia：用于分析跨训练和扩展的大型语言模型的套件”，载于 ICML ，卷。 202.PMLR，2023 年，第 2397-2430 页。

- [419] M. Javaheripi、S. Bubeck、M. Abdin、J. Aneja、S. Bubeck、C. C. T. Mendes、W. Chen、A. Del Giorno、R. Eldan、S. Gopi 等人，“Phi-2：小语言模型的惊人力量”，微软研究博客，2023 年。

- [420] S. Black、S. Biderman、E. Hallahan、Q. Anthony、L. Gau、L. Golding、H. He、C. Leahy、K. McDonell、J. Phang、M. Pieler、U. S. Prashanth、S. Purohit、L. Reynolds、J. Tow、B. Wang 和 S. Weinbach，“Gptneox-20b：开源自回归语言模型，”CoRR，卷。绝对/2204.06745，2022。

- [421] M. Chen、J. Tworek、H. Jun、Q. Yuan、H. P. de Oliveira Pinto、J. Kaplan、H. Edwards、Y. Burda、N. Joseph、G. Brockman、A. Ray、R. Puri、G. Krueger、M. Petrov、H. Khlaaf、G. Sastry、P. Mishkin、B. Chan、S. Gray、N. Ryder、M.巴甫洛夫、A. 鲍尔、L. 凯撒、M. 巴伐利亚、C. 温特、P. 蒂莱、F. P. 萨奇、D. 卡明斯、M. 普拉珀特、F. Chantzis、E. 巴恩斯、A. 赫伯特-沃斯、W. H. 格斯、A. 尼科尔、A. 佩诺、N. 特扎克、J. 唐、I. 巴布施金、S. 巴拉吉、S. Jain、W.桑德斯，C.黑塞，

- A. N. Carr、J. Leike、J. Achiam、V. Misra、E. Morikawa、A. Radford、M. Knight、M. Brundage、M. Murati、K. Mayer、P. Welinder、B. McGrew、D. Amodei、S. McCandlish、I. Sutskever 和 W. Zaremba，“评估在代码上训练的大型语言模型”，CoRR，卷。绝对/2107.03374，2021。

- [422] S. Karamcheti、S. Nair、A. Balakrishna、P. Liang、T. Kollar 和 D. Sadigh，“Prismatic vlms：调查视觉条件语言模型的设计空间”，载于 ICML，2024 年。

- [423] L. Beyer、A. Steiner、A. S. Pinto、A. Kolesnikov、X. Wang、D. Salz、M. Neumann、I. Alabdulmohsin、M. Tschannen、E. Bugliarello、T. Unterthiner、D. Keysers、S. Koppula、F. Liu、A. Grycner、A. A. Gritsenko、N. Houlsby， M. Kumar、K. Rong、J. Eisenschlos、R. Kabra、M. Bauer、M. Bosnjak、X. Chen、M. Minderer、P. Voigtlaender、I. Bica、I. Balazevic、J. Puigcerver、P. Papalampidi、O. J. H´enaff、X. Xiong、R. Soricut、J. Harmsen 和 X. Zhai，“Paligemma：用于传输的多功能 3b VLM”，CoRR，卷。绝对/2407.07726，2024。

- [424] A. Steiner、A. S. Pinto、M. Tschannen、D. Keysers、X. Wang、Y. Bitton、A. A. Gritsenko、M. Minderer、A. Sherbondy、S. Long、S. Qing、R. R. Ingle、E. Bugliarello、S. Kazemzadeh、T. Mesnard、I. Alabdulmohsin、L. Beyer、和 X. Zhai，“Paligemma 2：用于传输的多功能 vlm 系列”，CoRR，卷。绝对/2412.03555，2024。

- [425] Chameleon-Team，“Chameleon：混合模式早期融合 基础模型（Foundation Models）”，CoRR，卷。绝对/2405.09818，2024。

- [426] X. Wang、X. 张、Z. Luo、Q. Sun、Y. Cui、J. Wang、F. Zhu、Y. Wang、Z. Li、Q. Yu、Y. Zhu、Y. Ao、X. Min、T. Li、B. Wu、B. Zhu、B. 张、L. Wang、G. Liu、Z. He、X. Yang、J. Liu、Y. Lin、T. Huang 和 Z. Wang，“Emu3：下一个令牌预测就是您所需要的”，CoRR，卷。绝对/2409.18869，2024。

- [427] H. Liu、W. Yan、M. Zaharia 和 P

。 Abbeel，“世界模型（World Model）关于百万长度视频和语言的块环注意”，载于_ICLR_。 OpenReview.net，2025 年。

- [428] D. K. Misra、A. Bennett、V. Blukis、E. Niklasson、M. Shatkhin 和 Y. Artzi，“通过视觉目标预测将指令映射到 3d 环境中的动作”，载于 EMNLP，2018 年，第 2667-2678 页。

- [429] C. Lynch、M. Khansari、T. Xiao、V. Kumar、J. Tompson、S. Levine 和 P. Sermanet，“从游戏中学习潜在计划”，载于_CoRL_，卷。 100.PMLR，2019 年，第 1113-1132 页。

- [430] M. Bhardwaj、B. Sundaralingam、A. Mousavian、N. D. Ratliff、D. Fox、F. Ramos 和 B. Boots，“STORM：用于反应操作的快速联合空间模型预测控制的集成框架”，载于_CoRL_，卷。 164.PMLR，2021 年，第 750-759 页。

- [431] A. Jaegle、S. Borgeaud、J. Alayrac、C. Doersch、C. Ionescu、D. Ding、S. Koppula、D. Zoran、A. Brock、E. Shelhamer、O. J. H´enaff、M. M. Botvinick、A. Zisserman、O. Vinyals 和 J. Carreira，“Perceiver IO：结构化的通用架构”输入和输出”，载于 ICLR，2022 年。

- [432] E. Wijmans、A. Kadian、A. Morcos、S. Lee、I. Essa、D. Parikh、M. Savva 和 D. Batra，“DD-PPO：从 25 亿帧中学习近乎完美的点目标导航器”，载于 ICLR，2020 年。

- [433] M. Zhu、Y. Zhu、J. Li、J. Wen、Z. Xu、N. Liu、R. Cheng、C. Shen、Y. Peng、F. Feng 和 J. Tang，“将变压器中的 扩散策略（Diffusion Policy）扩展到 10 亿个参数以进行机器人操作”，CoRR，卷。绝对/2409.14411，2024。

- [434] J. W. Wei、L. Hou、A. K. Lampinen、X. Chen、D. Huang、Y. Tay、X. Chen、Y. Lu、D. Zhou、T. Ma 和 Q. V. Le，“符号调整改善了语言模型中的上下文学习”，CoRR，卷。绝对/2305.08298，2023。

- [435] W. Peebles 和 S. Xie，“带有变压器的可扩展扩散模型”，载于_ICCV_。 IEEE，2023 年，第 4172-4182 页。

- [436] J. Ho、A. Jain 和 P. Abbeel，“去噪扩散概率模型”，《NeurIPS》，2020 年。

- [437] Y. Song、J. Sohl-Dickstein、D. P. Kingma、A. Kumar、S. Ermon 和 B. Poole，“通过随机微分方程进行基于分数的生成建模”，载于 ICLR，2021 年。

- [438] A. Zeng、P. Florence、J. Tompson、S. Welker、J. Chien、M. Attarian、T. Armstrong、I. Krasin、D. Duong、V. Sindhwani 和 J. Lee，“传输网络：重新安排机器人操作的视觉世界”，载于_CoRL_，卷。 155.PMLR，2020 年，第 726-747 页。

- [439] R. Goyal、S. E. Kahou、V. Michalski、J. Materzynska、S. Westphal、H. Kim、V. Haenel、I. Fr´und、P. Yianilos、M. Mueller-Freitag、F. Hoppe、C. Thurau、I. Bax 和 R. Memisevic，“用于学习和评估视觉常识的“某事某事”视频数据库”，载于_ICCV_。 IEEE 计算机协会，2017 年，第 5843-5851 页。

- [440] X. Lin、J. So、S. Mahalingam、F. Liu 和 P. Abbeel，“Spawnnet：从预先训练的网络中学习通用视觉运动技能”，CoRR，卷。绝对/2307.03567，2023。

[441] S. P. Arunachalam, I. Güzey, S. Chintala, and L. Pinto, "Holo-dex: Teaching dexterity with immersive mixed reality," in *ICRA*. IEEE, 2023, pp. 5962–5969.

[442] N. M. M. Shafiullah, A. Rai, H. Etukuru, Y. Liu, I. Misra, S. Chintala, and L. Pinto, "On bringing robots home," *CoRR*, vol. abs/2311.16098, 2023.

[443] Y. Seo, D. Hafner, H. Liu, F. Liu, S. James, K. Lee, and P. Abbeel, "Masked world models for visual control," in *CoRL*, vol. 205. PMLR, 2022, pp. 1332–1344.

[444] R. Mendonca, S. Bahl, and D. Pathak, "Structured world models from human videos," in *RSS*, 2023.

[445] M. Pan, X. Zhu, Y. Wang, and X. Yang, "Iso-dream: Isolating and leveraging noncontrollable visual dynamics in world models," in *NeurIPS*, 2022.

[446] Z. Dai, Z. Yang, Y. Yang, J. G. Carbonell, Q. V. Le, and R. Salakhutdinov, "Transformer-xl: Attentive language models beyond a fixed-length context," in *ACL (1)*. Association for Computational Linguistics, 2019, pp. 2978–2988.

[447] A. Avetisyan, C. Xie, H. Howard-Jenkins, T. Yang, S. Aroudj, S. Patra, F. Zhang, D. P. Frost, L. Holland, C. Orme, J. Engel, E. Miller, R. A. Newcombe, and V. Balntas, "Scenescript: Reconstructing scenes with an autoregressive structured language model," *CoRR*, vol. abs/2403.13064, 2024.

[448] N. M. Shafiullah, Z. J. Cui, A. Altanzaya, and L. Pinto, "Behavior transformers: Cloning k modes with one stone," in *NeurIPS*, 2022.

[449] Z. J. Cui, Y. Wang, N. M. M. Shafiullah, and L. Pinto, "From play to policy: Conditional behavior generation from uncurated robot data," in *ICLR*, 2023.

[450] S. Lee, Y. Wang, H. Etukuru, H. J. Kim, N. M. M. Shafiullah, and L. Pinto, "Behavior generation with latent actions," in *ICML*, 2024.

[451] R. Sapkota, Y. Cao, K. I. Roumeliotis, and M. Karkee, "Vision-language-action models: Concepts, progress, applications and challenges," *CoRR*, vol. abs/2505.04769, 2025.

[452] Y. Zhong, F. Bai, S. Cai, X. Huang, Z. Chen, X. Zhang, Y. Wang, S. Guo, T. Guan, K. N. Lui, Z. Qi, Y. Liang, Y. Chen, and Y. Yang, "A survey on vision-language-action models: An action tokenization perspective," *CoRR*, vol. abs/2507.01925, 2025.

[453] M. U. Din, W. Akram, L. S. Saoud, J. Rosell, and I. Hussain, "Vision language action models in robotic manipulation: A systematic review," *CoRR*, vol. abs/2507.10672, 2025.

[454] D. Zhang, J. Sun, C. Hu, X. Wu, Z. Yuan, R. Zhou, F. Shen, and Q. Zhou, "Pure vision language action (VLA) models: A comprehensive survey," *CoRR*, vol. abs/2509.19012, 2025.

[455] S. Li, Y. Chen, L. Dong, S. Liu, D. Lan, L. Yu, and Z. Pang, "Transferring vision-language-action models to industry applications: Architectures, performance, and challenges," *CoRR*, vol. abs/2509.23121, 2025.

[456] K. Kawaharazuka, J. Oh, J. Yamada, I. Posner, and Y. Zhu, "Vision-language-action models for robotics: A review towards real-world applications," *IEEE Access*, vol. 13, pp. 162 467–162 504, 2025.

[457] Z. Yu, B. Wang, P. Zeng, H. Zhang, J. Zhang, L. Gao, J. Song, N. Sebe, and H. T. Shen, "A survey on efficient vision-language-action models," *CoRR*, vol. abs/2510.24795, 2025.

[458] S. Poria, N. Majumder, C. Hung, A. A. Bagherzadeh, C. Li, K. Kwok, Z. Wang, C. Tan, J. Wu, and D. Hsu, "10 open challenges steering the future of vision-language-action models," *CoRR*, vol. abs/2511.05936, 2025.

[459] C. Xu, S. Zhang, Y. Liu, B. Sun, W. Chen, B. Xu, Q. Liu, J. Wang, S. Wang, S. Luo, J. Peters, A. V. Vasilakos, S. Zafeiriou, and J. Deng, "An anatomy of vision-language-action models: From modules to milestones and challenges," *CoRR*, vol. abs/2512.11362, 2025.

[460] H. Deng, Z. Wu, H. Liu, W. Guo, Y. Xue, Z. Shan, C. Zhang, B. Jia, Y. Ling, G. Lu, and Z. Wang, "A survey on reinforcement learning of vision-language-action models for robotic manipulation," *TechRxiv*, 2025, preprint.

[461] T. Hu, X. Liu, S. Wang, Y. Zhu, A. Liang, L. Kong, G. Zhao, Z. Gong, J. Cen, Z. Huang, X. Hao, L. Li, H. Song, X. Li, J. Ma, S. Shen, J. Zhu, D. Tao, Z. Liu, and J. Liang, "Vision-language-action models for autonomous driving: Past, present, and future," *CoRR*, vol. abs/2512.16760, 2025.

[462] X. Li, C. Mata, J. Park, K. Kahatapitiya, Y. S. Jang, J. Shang, K. Ranasinghe, R. D. Burgert, M. Cai, Y. J. Lee, and M. S. Ryoo, "Llara: Supercharging robot learning data for vision-language policy," in *ICLR*. OpenReview.net, 2025.

[463] K. Pertsch, K. Stachowicz, B. Ichter, D. Driess, S. Nair, Q. Vuong, O. Mees, C. Finn, and S. Levine, "FAST: efficient action tokenization for vision-language-action models," *CoRR*, vol. abs/2501.09747, 2025.

[464] P. Intelligence, K. Black, N. Brown, J. Darpinian, K. Dhabalia, D. Driess, A. Esmail, M. Equi, C. Finn, N. Fusai, M. Y. Galliker, D. Ghosh, L. Groom, K. Hausman, B. Ichter, S. Jakubczak, T. Jones, L. Ke, D. LeBlanc, S. Levine, A. Li-Bell, M. Mothukuri, S. Nair, K. Pertsch, A. Z. Ren, L. X. Shi, L. M. Smith, J. T. Springenberg, K. Stachowicz, J. Tanner, Q. Vuong, H. Walke, A. Walling, H. Wang, L. Yu, and U. Zhilinsky, "$\pi_{0.5}$: a vision-language-action model with open-world generalization," *CoRR*, vol. abs/2504.16054, 2025.

[465] Z. Xu, H. L. Chiang, Z. Fu, M. G. Jacob, T. Zhang, T. E. Lee, W. Yu, C. Schenck, D. Rendleman, D. Shah, F. Xia, J. Hsu, J. Hoech, P. Florence, S. Kirmani, S. Singh, V. Sindhwani, C. Parada, C. Finn, P. Xu, S. Levine, and J. Tan, "Mobility VLA: multimodal instruction navigation with long-context vlms and topological graphs," in *CoRL*, ser. Proceedings of Machine Learning Research, vol. 270. PMLR, 2024, pp. 3866–3887.

[466] P. Ding, J. Ma, X. Tong, B. Zou, X. Luo, Y. Fan, T. Wang, H. Lu, P. Mo, J. Liu, Y. Wang, H. Zhou, W. Feng, J. Liu, S. Huang, and D. Wang, "Humanoid-vla: Towards universal humanoid control with visual integration," *CoRR*, vol. abs/2502.14795, 2025.

[467] P. Ding, H. Zhao, W. Zhang, W. Song, M. Zhang, S. Huang, N. Yang, and D. Wang, "QUAR-VLA: vision-language-action model for quadruped robots," in *ECCV (5)*, ser. Lecture Notes in Computer Science, vol. 15063. Springer, 2024, pp. 352–367.

[468] X. Tong, P. Ding, D. Wang, W. Zhang, C. Cui, M. Sun, Y. Fan, H. Zhao, H. Zhang, Y. Dang, S. Huang, and S. Lyu, "Quart-online: Latency-free large multimodal language model for quadruped robot learning," *CoRR*, vol. abs/2412.15576, 2024.

[469] H. Zhao, W. Song, D. Wang, X. Tong, P. Ding, X. Cheng, and Z. Ge, "More: Unlocking scalability in reinforcement learning for quadruped vision-language-action models," *CoRR*, vol. abs/2503.08007, 2025.

[470] Y. Zhong, X. Huang, R. Li, C. Zhang, Y. Liang, Y. Yang, and Y. Chen, "Dexgraspvla: A vision-language-action framework towards general dexterous grasping," *CoRR*, vol. abs/2502.20900, 2025.

[471] Y. Cui, Y. Zhang, L. Tao, Y. Li, X. Yi, and Z. Li, "End-to-end dexterous arm-hand VLA policies via shared autonomy: VR teleoperation augmented by autonomous hand VLA policy for efficient data collection," *CoRR*, vol. abs/2511.00139, 2025.

[472] NVIDIA, "Alpamayo-r1: Bridging reasoning and action prediction for generalizable autonomous driving in the long tail," *CoRR*, vol. abs/2511.00088, 2025.

[473] H. Arai, K. Miwa, K. Sasaki, K. Watanabe, Y. Yamaguchi, S. Aoki, and I. Yamamoto, "Covla: Comprehensive vision-language-action dataset for autonomous driving," in *WACV*. IEEE, 2025, pp. 1933–1943.

[474] Y. Li, S. Shang, W. Liu, B. Zhan, H. Wang, Y. Wang, Y. Chen, X. Wang, Y. An, C. Tang, L. Hou, L. Fan, and Z. Zhang, "Drivevla-w0: World models amplify data scaling law in autonomous driving," *CoRR*, vol. abs/2510.12796, 2025.

[475] J. Hwang, R. Xu, H. Lin, W. Hung, J. Ji, K. Choi, D. Huang, T. He, P. Covington, B. Sapp, Y. Zhou, J. Guo, D. Anguelov, and M. Tan, "EMMA: end-to-end multimodal model for autonomous driving," *Trans. Mach. Learn. Res.*, vol. 2025, 2025.

[476] P. Chen, P. Bu, Y. Wang, X. Wang, Z. Wang, J. Guo, Y. Zhao, Q. Zhu, J. Song, S. Yang, J. Wang, and B. Zheng, "Combatvla: An efficient vision-language-action model for combat tasks in 3d action role-playing games," *CoRR*, vol. abs/2503.09527, 2025.

[477] M. Li, Z. Wang, K. He, X. Ma, and Y. Liang, "JARVIS-VLA: post-training large-scale vision language models to play visual games with keyboards and mouse," in *ACL (Findings)*. Association for Computational Linguistics, 2025, pp. 17 878–17 899.

[478] N. G. Tsung-Wei Ke and and K. Fragkiadaki, "3d diffuser actor: Policy diffusion with 3d scene representations," in *CoRL*, ser. Proceedings of Machine Learning Research, vol. 270. PMLR, 2024, pp. 1949–1974.

[479] X. Li, L. Heng, J. Liu, Y. Shen, C. Gu, Z. Liu, H. Chen, N. Han, R. Zhang, H. Tang, S. Zhang, and H. Dong, "3ds-vla," *arXiv preprint*, 2025.

[480] Y. C. Jiahui Zhang and, Y. Xu, Z. Huang, Y. Zhou, Y. Yuan, X. Cai, G. Huang, X. Quan, H. Xu, and L. Zhang, "4d-vla: Spatiotemporal vision-language-action pretraining with cross-scene, calibration," *CoRR*, vol. abs/2506.22242, 2025.

[481] M. A. Kohei Sendai and, T. Matsushima, Y. Matsuo, and Y. Iwasawa, "Leave no observation behind: Real-time correction for VLA action, chunks," *CoRR*, vol. abs/2509.23224, 2025.

[482] H. C. Siyuan Huang and, Y. Liu, Y. Zhu, H. Dong, A. Boularias, P. Gao, and H. Li, "A3VLM: actionable articulation-aware vision language model," in *CoRL*, ser. Proceedings of Machine Learning Research, vol. 270. PMLR, 2024, pp. 1675–1690.

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- [441] S. P. Arunachalam、I. Gùuzey、S. Chintala 和 L. Pinto，“Holo-dex：通过沉浸式混合现实进行灵活性教学”，载于 ICRA。 IEEE，2023 年，第 5962-5969 页。

- [442] N.M.M. Shafiullah、A. Rai、H. Etukuru、Y. Liu、I. Misra、S. Chintala 和 L. Pinto，“关于将机器人带回家”，CoRR，卷。 ABS/2311.16098，2023。

- [443] Y. Seo、D. Hafner、H. Liu、F. Liu、S. James、K. Lee 和 P. Abbeel，“用于视觉控制的蒙版 世界模型（World Models）”，载于_CoRL_，卷。 205.PMLR，2022 年，第 1332-1344 页。

- [444] R. Mendonca、S. Bahl 和 D. Pathak，“根据人类视频构建 世界模型（World Models）”，载于 RSS，2023 年。

- [445] M. Pan、X. Zhu、Y. Wang 和 X. Yang，“Iso-dream：隔离和利用 世界模型（World Models） 中的不可控视觉动态”，《NeurIPS》，2022 年。

- [446] Z. Dai、Z. Yang、Y. Yang、J. G. Carbonell、Q. V. Le 和 R. Salakhutdinov，“Transformer-xl：超出固定长度上下文的注意力语言模型”，载于_ACL (1)_。计算语言学协会，2019 年，第 2978-2988 页。

- [447] A. Avetisyan、C. Xie、H. Howard-Jenkins、T. Yang、S. Aroudj、S. Patra、F. Zhu、D. P. Frost、L. Holland、C. Orme、J. Engel、E. Miller、R. A. Newcombe 和 V. Balntas，“Scenescript：使用自回归结构化语言模型重建场景”，CoRR，卷。绝对/2403.13064，2024。

- [448] N. M. Shafiullah、Z. J. Cui、A. Altanzaya 和 L. Pinto，“行为变压器：用一块石头克隆 k 模式”，载于 NeurIPS，2022 年。

- [449] Z. J. Cui、Y. Wang、N. M. M. Shafiullah 和 L. Pinto，“从游戏到策略：根据未经整理的机器人数据生成条件行为”，载于 ICLR，2023 年。

- [450] S. Lee、Y. Wang、H. Etukuru、H. J. Kim、N. M. M. Shafiullah 和 L. Pinto，“具有潜在动作的行为生成”，载于 ICML，2024 年。

- [451] R. Sapkota、Y. Cao、K. I. Roumeliotis 和 M. Karkee，“视觉语言-动作模型：概念、进展、应用和挑战”，CoRR，卷。绝对/2505.04769，2025。

- [452] Y.Zhong、F.Bai、S.Cai、X.Huang、Z.Chen、X.Zhang、Y.Wang、S.Guo、T.Guan、K.N.Lui、Z.Qi、Y.Liang、Y.Chen 和 Y.Yang，“视觉语言动作（Vision-Language-Action） 模型调查：动作标记化视角”，CoRR，卷。绝对/2507.01925，2025。

- [453] M. U. Din、W. Akram、L. S. Saoud、J. Rosell 和 I. Hussain，“机器人操作中的视觉语言动作模型：系统综述”，CoRR，卷。绝对/2507.10672，2025。

- [454] D. 张、J. Sun、C. Hu、X. Wu、Z. Yuan、R. Zhou、F. Shen 和 Q. Zhou，“纯视觉语言动作（VLA）模型：综合调查”，CoRR，卷。 ABS/2509.19012，2025。

- [455] S. Li、Y. Chen、L. Dong、S. Liu、D. Lan、L. Yu 和 Z. Pang，“将 视觉语言动作（Vision-Language-Action） 模型转移到行业应用：架构、性能和挑战”，CoRR，卷。绝对/2509.23121，

2025。

- [456] K. Kawaharazuka、J. Oh、J. Yamada、I. Posner 和 Y. Zhu，“机器人视觉语言动作模型：对现实世界应用的回顾”，IEEE Access，卷。 13，第 162 467-162 504 页，2025 年。

- [457] Z. Yu、B. Wang、P. Zeng、H. Zhang、J. Zhuang、L. Gau、J. Song、N. Sebe 和 H. T. Shen，“高效 视觉语言动作（Vision-Language-Action） 模型的调查”，CoRR，卷。绝对/2510.24795，2025。

- [458] S. Poria、N. Majumder、C. Hung、A. A

。 Bagherzadeh、C. Li、K. Kwok、Z. Wang、C. Tan、J. Wu 和 D. Hsu，“引导 视觉语言动作（Vision-Language-Action） 模型未来的 10 个开放挑战”，CoRR，卷。绝对/2511.05936，2025。

- [459] C. Xu、S. 张、Y. Liu、B. Sun、W. Chen、B. Xu、Q. Liu、J. Wang、S. Wang、S. Luo、J. Peters、A. V. Vasilakos、S. Zafeiriou 和 J

。 Deng，“视觉语言动作（Vision-Language-Action） 模型剖析：从模块到里程碑和挑战”，CoRR，卷。绝对/2512.11362，2025。

- [460] 邓浩、吴正浩、刘浩、郭伟、薛勇、单正正

。张，B. Jia，Y. Ling，G. Lu，和 Z. Wang，“对用于机器人操作的 视觉语言动作（Vision-Language-Action） 模型的 强化学习（Reinforcement Learning） 的调查”，TechRxiv，2025 年，预印本。

- [461] 胡涛，刘翔，S

。王玉

。朱阿良、孔丽、赵国、龚子、岑建、黄子、郝X、李丽、宋华、X

。 Li、J. Ma、S. Shen、J. Zhu、D.Tao、Z.Liu 和 J.Liang，“自动驾驶 视觉语言动作（Vision-Language-Action） 模型：过去、现在和未来”，CoRR，第 1 卷。绝对/2512.16760，2025。

- [462] X. Li、C. Mata、J. Park、K. Kahatapitiya、Y. S. Jang、J. Shang、K. Ranasinghe、R. D. Burgert、M. Cai、Y. J. Lee 和 M. S. Ryoo，“Llara：为视觉语言策略增压机器人学习数据”，载于_ICLR_。 OpenReview.net，2025 年。

- [463] K. Pertsch、K. Stachowicz、B. Ichter、D. Driess、S. Nair、Q. Vuong、O. Mees、C. Finn 和 S. Levine，“FAST：视觉语言动作（Vision-Language-Action） 模型的高效动作标记化”，CoRR，卷。绝对/2501.09747，2025。

- [464] P. Intelligence、K. Black、N. Brown、J. Darpinian、K. Dhabalia、D. Driess、A. Esmail、M. Equi、C. Finn、N. Fusai、M. Y. Galliker、D. Ghosh、L. Groom、K

。豪斯曼，B

。 Ichter、S. Jakubczak、T. Jones、L. Ke、D. LeBlanc、S. Levine、A. Li-Bell、M. Mothukuri、S. Nair、K. Pertsch、A. Z. Ren、L. X. Shi、L. M. Smith、J. T. Springenberg、K. Stachowicz、J. Tanner、Q. Vuong、H. Walke、A. Walling、H. Wang、L. Yu 和 U.zhilinsky，“π 0.5：具有开放世界泛化的 视觉语言动作（Vision-Language-Action） 模型”，CoRR，卷。绝对/2504.16054，2025。

- [465] Z. Xu、H. L. Jiang、Z. Fu、M. G. Jacob、T. E. Lee、W. Yu、C. Schenck、D. Rendleman、D. Shah、F. Xia、J. Hsu、J. Hoech、P. Florence、S. Kirmani、S. Singh、V. Sindhwani、C. Parada、C. Finn、P. Xu、S. Levine 和 J. Tan，“Mobility VLA：使用长上下文 vlm 和拓扑图的多模式指令导航”，载于_CoRL_，ser。机器学习研究论文集，卷。 270.PMLR，2024 年，第 3866-3887 页。

- [466] P. Ding，J. Ma，X. Tong，B. Zou，X. Luo，Y. Fan，T. Wang，H. Lu，P. Mo，J. Liu，Y. Wang，H. Zhou，W. Feng，J. Liu，S. Huang，and D. Wang，“Humanoid-vla：通过视觉集成实现通用人形控制”，CoRR，卷。绝对/2502.14795，2025。

- [467] P. Ding、H. Zhu、W. Zhang、W. Song、M. Zhu、S. Huang、N. Yang 和 D. Wang，“QUAR-VLA：四足机器人的 视觉语言动作（Vision-Language-Action） 模型”，载于 ECCV (5)，系列。计算机科学讲义，卷。 15063。施普林格，2024 年，第 352-367 页。

- [468] X. Tong、P. Ding、D. Wang、W. Zhu、C. Cui、M. Sun、Y. Fan、H. Zhao、H. Zhuang、Y. Dang、S. Huang 和 S. Lyu，“Quart-online：用于四足机器人学习的无延迟大型多模态语言模型”，CoRR，卷。绝对/2412.15576，2024。

- [469] 赵宏、宋文、王东、童晓、丁鹏、X

。程和Z

。 Ge，“更多：解锁四足动物 视觉语言动作（Vision-Language-Action） 模型的 强化学习（Reinforcement Learning） 的可扩展性”，CoRR，卷。绝对/2503.08007，2025。

- [470] Y.Zhong、X.Huang、R.Li、C.Zhang、Y.Liang、Y.Yang 和 Y.Chen，“Dexgraspvla：面向一般灵巧抓取的 视觉语言动作（Vision-Language-Action） 框架”，CoRR，卷。绝对/2502.20900，2025。

- [471] Y. Cui、Y. 张、L. Tao、Y. Li、X. Yi 和 Z. Li，“通过共享自主性实现端到端灵巧的手臂 VLA 策略：通过自主手 VLA 策略增强 VR 远程操作以实现高效数据收集”，CoRR，卷。绝对/2511.00139，2025。

- [472] NVIDIA，“Alpamayo-r1：长尾通用自动驾驶的桥接推理和动作预测”，CoRR，卷。绝对/2511.00088，2025。

- [473] H. Arai、K. Miwa、K. Sasaki、K. Watanabe、Y. Yamaguchi、S. Aoki 和 I. Yamamoto，“Covla：自动驾驶综合 视觉语言动作（Vision-Language-Action） 数据集”，载于_WACV_。 IEEE，2025 年，第 1933-1943 页。

- [474] Y. Li、S. Shang、W. Liu、B. Zhan、H. Wang、Y. Wang、Y. Chen、X. Wang、Y. An、C. Tang、L. Hou、L. Fan 和 Z. Zhang，“Drivevla-w0：世界模型（World Models） 放大自动驾驶中的数据缩放定律”，CoRR，卷。绝对/2510.12796，2025。

- [475] J. Hwang、R. Xu、H. Lin、W. Hung、J. Ji、K. Choi、D. Huang、T. He、P. Covington、B. Sapp、Y. Zhou、J.Guo、D. Anguelov 和 M. Tan，“EMMA：自动驾驶端到端多模式模型”，Trans。马赫。学习。 Res.，卷。 2025 年，2025 年。

- [476] P. Chen、P. Bu、Y. Wang、X. Wang、Z. Wang、J.Guo、Y. Zhao、Q. Zhu、J. Song、S. Yang、J. Wang 和 B. Cheng，“Combatvla：用于 3D 动作角色扮演游戏中战斗任务的高效 视觉语言动作（Vision-Language-Action） 模型”，CoRR，卷。绝对/2503.09527，2025。

- [477] M. Li、Z. Wang、K. He、X. Ma 和 Y. Liang，“JARVIS-VLA：后训练大型视觉语言模型以使用键盘和鼠标玩视觉游戏”，载于_ACL（调查结果）_。计算语言学协会，2025 年，第 17 878-17 899 页。

- [478] N. G. Tsung-Wei Ke 和 K. Fragkiadaki，“3d 扩散器演员：具有 3d 场景表示的策略扩散”，载于 _CoRL_，系列。机器学习研究论文集，卷。 270.PMLR，2024 年，第 1949-1974 页。

- [479] X. Li、L. Heng、J. Liu、Y. Shen、C. Gu、Z. Liu、H. Chen、N. Han、R. Zhang、H. Tang、S. Zhang 和 H. Dong，“3ds-vla”，arXiv 预印本，2025 年。

- [480] Y. C. Jiahui 张和 Y. Xu、Z. Huang、Y. Zhou、Y. Yuan、X. Cai、G. Huang、X. Quan、H. Xu 和 L. Zhang，“4d-vla：跨场景、校准的时空 视觉语言动作（Vision-Language-Action） 预训练”，CoRR，卷。绝对/2506.22242，2025。

- [481] M. A. Kohei Sendai 和 T. Matsushima、Y. Matsuo 和 Y. Iwasawa，“不留下任何观察：VLA 动作的实时校正，块”，CoRR，卷。绝对/2509.23224，2025。

- [482] H. C. Siyuan Huang 和 Y. Liu、Y. Zhu、H. Dong、A. Boularias、P. Gau 和 H. Li，“A3VLM：可操作的清晰度感知视觉语言模型”，载于_CoRL_，系列。机器学习研究论文集，卷。 270.PMLR，2024 年，第 1675-1690 页。

[483] K. K. Minho Park and, J. Hyung, H. Jang, H. Jin, J. Yun, H. Lee, and J. Choo, "ACG: action coherence guidance for flow-based VLA models," *CoRR*, vol. abs/2510.22201, 2025.

[484] T. W. Wencheng Ye and, L. Zhu, F. Li, and G. Yang, "Actdistill: General action-guided self-derived distillation for efficient, vision-language-action models," *CoRR*, vol. abs/2511.18082, 2025.

[485] H. G. Yuntao Dai and, T. Wang, Q. Cheng, Y. Zheng, Z. Qiu, L. Gong, W. Lou, and X. Zhou, "Actionflow: A pipelined action acceleration for vision language, models on edge," *CoRR*, vol. abs/2512.20276, 2025.

[486] Y. L. Weijie Shen and, Y. Wu, Z. Liang, S. Gu, D. Wang, T. Nian, L. Xu, Y. Qin, J. Pang, X. Guan, X. Yang, and Y. Mu, "Expertise need not monopolize: Action-specialized mixture of experts, for vision-language-action learning," *CoRR*, vol. abs/2510.14300, 2025.

[487] Y. C. Xiaohuan Pei and, S. Xu, Y. Wang, Y. Shi, and C. Xu, "Action-aware dynamic pruning for efficient vision-language-action, manipulation," *CoRR*, vol. abs/2509.22093, 2025.

[488] Z. W. Siyu Xu and, Y. Wang, C. Xia, T. Huang, and C. Xu, "Affordance field intervention: Enabling vlas to escape memory traps, in robotic manipulation," *CoRR*, vol. abs/2512.07472, 2025.

[489] Z. Y. Yizheng Zhang and, J. Lai, C. Lu, and L. Han, "Agentworld: An interactive simulation platform for scene construction, and mobile robotic manipulation," *CoRR*, vol. abs/2508.07770, 2025.

[490] H. Q. Irmak Guzey and, J. Urain, C. Wang, J. Yin, K. Bodduluri, M. Lambeta, L. Pinto, A. Rai, J. Malik, T. Wu, A. Sharma, and H. Bharadhwaj, "Dexterity from smart lenses: Multi-fingered robot manipulation with, in-the-wild human demonstrations," *CoRR*, vol. abs/2511.16661, 2025.

[491] B. L. Wenxin Zheng and, B. Xu, E. Feng, J. Gu, and H. Chen, "Leveraging os-level primitives for robotic action management," *CoRR*, vol. abs/2508.10259, 2025.

[492] Z. W. Yiyang Huang and, Z. Wan, Y. Tian, H. Xu, Y. Han, and Y. Gan, "ANNIE: be careful of your robots," *CoRR*, vol. abs/2509.03383, 2025.

[493] S. C. Yuhua Jiang and, Y. Ding, F. Gao, and B. Qi, "Asyncvla: Asynchronous flow matching for vision-language-action models," *CoRR*, vol. abs/2511.14148, 2025.

[494] C. W. Yang Zhang and, O. Lu, Y. Zhao, Y. Ge, Z. S. 0001, X. L. 0001, C. Z. 0012, C. Bai, and X. L. 0001, "Align-then-steer: Adapting the vision-language action models through unified latent guidance," 2025. [Online]. Available: https://dblp.org/rec/journals/corr/abs-2509-02055

[495] J. T. Zhenlong Yuan and, J. Luo, R. Chen, C. Qian, L. Sun, X. Chu, Y. Cai, D. Zhang, and S. Li, "Autodrive-r^2: Incentivizing reasoning and self-reflection, capacity for VLA model in autonomous driving," *CoRR*, vol. abs/2509.01944, 2025.

[496] Y. X. Hanshi Wang and, Z. Xu, J. Gao, Y. Liu, W. Hu, K. Wang, and Z. Zhang, "Autoprune: Each complexity deserves a pruning policy," *CoRR*, vol. abs/2509.23931, 2025.

[497] T. C. Zewei Zhou and, S. Z. Zhao, Y. Zhang, Z. Huang, B. Zhou, and J. Ma, "Autovla: A vision-language-action model for end-to-end autonomous, driving with adaptive reasoning and reinforcement fine-tuning," *CoRR*, vol. abs/2506.13757, 2025.

[498] J. L. Lei Xiao and, J. Gao, F. Ye, Y. Jin, J. Qian, J. Zhang, Y. Wu, and X. Yu, "Ava-vla: Improving vision-language-action models with active visual attention," 2025. [Online]. Available: https://dblp.org/rec/journals/corr/abs-2511-18960

[499] L. L. Harris Song and, "Avi: Action from volumetric inference," *CoRR*, vol. abs/2510.21746, 2025.

[500] G. T. Xueyang Zhou and, G. Zhang, H. Wang, P. Zhou, and L. Sun, "Badvla: Towards backdoor attacks on vision-language-action models, via objective-decoupled optimization," *CoRR*, vol. abs/2505.16640, 2025.

[501] Z. Z. Kechun Xu and, A. Chen, S. Zhao, Q. Huang, Y. Yang, H. Lu, R. Xiong, M. Tomizuka, and Y. Wang, "Seeing to act, prompting to specify: A bayesian factorization of, vision language action policy," *CoRR*, vol. abs/2512.11218, 2025.

[502] Y. F. Hao Luo and, W. Zhang, S. Zheng, Y. Wang, H. Yuan, J. Liu, C. Xu, Q. Jin, and Z. Lu, "Being-h0: Vision-language-action pretraining from large-scale human, videos," *CoRR*, vol. abs/2507.15597, 2025.

[503] J. F. Yanbo Mao and, R. Zhang, H. Xie, and M. Yao, "Beyond success: Refining elegant robot manipulation from mixed-quality, data via just-in-time intervention," *CoRR*, vol. abs/2511.22555, 2025.

[504] M. A. C. Koffivi Fidèle Gbagbe and, A. Alabbas, O. Alyunes, A. Lykov, and D. Tsetserukou, "Bi-vla: Vision-language-action model-based system for bimanual robotic, dexterous manipulations," in *SMC*. IEEE, 2024, pp. 2864–2869.

[505] T. B. Masato Kobayashi and, "Bi-vla: Bilateral control-based imitation learning via vision-language, fusion for action generation," *CoRR*, vol. abs/2509.18865, 2025.

[506] Y. C. Peiyan Li and, H. Wu, X. Ma, X. Wu, Y. Huang, L. Wang, T. Kong, and T. Tan, "Bridgevla: Input-output alignment for efficient 3d manipulation learning, with vision-language models," *CoRR*, vol. abs/2506.07961, 2025.

[507] A. Z. R. Asher J. Hancock and and A. Majumdar, "Run-time observation interventions make vision-language-action models, more visually robust," in *ICRA*. IEEE, 2025, pp. 9499–9506.

[508] Y. Y. Xiuxiu Qi and, J. Cao, L. Bai, C. Fan, C. Cao, and H. Wang, "Continuous vision-language-action co-learning with semantic-physical, alignment for behavioral cloning," *CoRR*, vol. abs/2511.14396, 2025.

[509] Y. Z. Zhongyi Zhou and, J. Wen, C. Shen, and Y. Xu, "Chatvla-2: Vision-language-action model with open-world embodied reasoning, from pretrained knowledge," *CoRR*, vol. abs/2505.21906, 2025.

[510] R. Y. Zijian An and, Y. Feng, and L. Zhou, "CLAW: A vision-language-action framework for weight-aware robotic, grasping," *CoRR*, vol. abs/2509.14143, 2025.

[511] J. K. Gi-Cheon Kang and, K. Shim, J. K. Lee, and B. Zhang, "CLIP-RT: learning language-conditioned robotic policies from natural, language supervision," *CoRR*, vol. abs/2411.00508, 2024.

[512] Q. Y. Zeyuan Chen and, Y. Chen, T. Wu, J. Zhang, Z. Ding, J. Li, Y. Yang, and H. Dong, "Clutterdexgrasp: A sim-to-real system for general dexterous grasping, in cluttered scenes," *CoRR*, vol. abs/2506.14317, 2025.

[513] F. S. Dapeng Zhang and, R. Zhao, Y. Chen, P. Zhi, C. Li, R. Zhou, and Q. Zhou, "Coc-vla: Delving into adversarial domain transfer for explainable, autonomous driving via chain-of-causality visual-language-action model," *CoRR*, vol. abs/2511.19914, 2025.

[514] R. Z. Wei Li and, R. Shao, J. He, and L. Nie, "Cogvla: Cognition-aligned vision-language-action model via instruction-driven, routing & sparsification," *CoRR*, vol. abs/2508.21046, 2025.

[515] X. C. Qihang Peng and, C. Yang, S. Shi, and H. Li, "Colavla: Leveraging cognitive latent reasoning for hierarchical parallel, trajectory planning in autonomous driving," *CoRR*, vol. abs/2512.22939, 2025.

[516] S. T. Yuhui Chen and, S. Liu, Y. Zhou, H. Li, and D. Zhao, "Conrft: A reinforced fine-tuning method for VLA models via consistency, policy," *CoRR*, vol. abs/2502.05450, 2025.

[517] Y. W. Puhao Li and, Z. Xi, W. Li, Y. Huang, Z. Zhang, Y. Chen, J. Wang, S. Zhu, T. Liu, and S. Huang, "Controlvla: Few-shot object-centric adaptation for pre-trained vision-language-action, models," *CoRR*, vol. abs/2506.16211, 2025.

[518] W. K. Tian Nian and, Y. Mu, T. Chen, S. Zhu, and B. Hu, "Control your robot: A unified system for robot control and policy, deployment," *CoRR*, vol. abs/2509.23823, 2025.

[519] Y. C. Shiyu Fang and, H. Liang, C. Lv, P. Hang, and J. Sun, "Corevla: A dual-stage end-to-end autonomous driving framework for, long-tail scenarios via collect-and-refine," *CoRR*, vol. abs/2509.15968, 2025.

[520] T. Y. Zhaohui Wang and and H. Tang, "Cot4ad: A vision-language-action model with explicit chain-of-thought, reasoning for autonomous driving," *CoRR*, vol. abs/2511.22532, 2025.

[521] W. D. Zhenghao "Mark" Peng and, Y. You, Y. Chen, W. Luo, T. Tian, Y. Cao, A. Sharma, D. Xu, B. Ivanovic, B. Li, B. Zhou, Y. Wang, and M. Pavone, "Counterfactual VLA: self-reflective vision-language-action model, with adaptive reasoning," *CoRR*, vol. abs/2512.24426, 2025.

[522] O. H. Daniel San José Pro and, R. Römer, M. Dösch, M. Schuck, and A. P. Schoellig, "CRISP - compliant ROS2 controllers for learning-based manipulation, policies and teleoperation," *CoRR*, vol. abs/2509.06819, 2025.

[523] H. L. Yi-Lin Wei and, Y. Lin, P. Wang, Z. Liang, G. Liu, and W. Zheng, "Cyclemanip: Enabling cyclic task manipulation via effective historical, perception and understanding," *CoRR*, vol. abs/2512.01022, 2025.

[524] H. L. Changyeon Kim and, Y. Seo, K. Lee, and Y. Zhu, "DEAS: detached value learning with action sequence for scalable, offline RL," *CoRR*, vol. abs/2510.07730, 2025.

[525] L. H. Haibo Hu and, N. Guan, and C. J. Xue, "Deead: Dynamic early exit of vision-language action for efficient, autonomous driving," *CoRR*, vol. abs/2511.20720, 2025.

[526] Y. L. Cheng Yin and, W. Xu, S. Tam, X. Zeng, Z. Liu, and Z. Yin, "Deepthinkvla: Enhancing reasoning capability of vision-language-action, models," *CoRR*, vol. abs/2511.15669, 2025.

[527] S. Wu, Y. Ji, Q. Li, Z. Zhang, Q. He, W. Xie, G. Zhang, B. Bayramli, Y. Ding, and H. Lu, "Dejavu: Towards experience feedback learning for embodied intelligence," *arXiv preprint*, 2025.

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- [483] K. K. Minho Park 和 J. Hyung、H. Jang、H. Jin、J. Yun、H. Lee 和 J. Choo，“ACG：基于流的 VLA 模型的动作一致性指导”，CoRR，卷。绝对/2510.22201，2025。

- [484] T. W. Wen Cheng Ye 和 L. Zhu、F. Li 和 G. Yang，“Actdistill：用于高效视觉语言动作模型的通用动作引导自导出蒸馏”，CoRR，卷。绝对/2511.18082，2025。

- [485] H. G. Yuntao Dai 和 T. Wang、Q. Cheng、Y. Cheng、Z. Qiu、L. Kong、W. Lou 和 X. Zhou，“Actionflow：视觉语言的流水线动作加速，边缘模型”，CoRR，卷。 ABS/2512.20276，2025。

- [486] Y. L. Weijie Shen and, Y. Wu, Z. Liang, S. Gu, D. Wang, T. Nian, L. Xu, Y.qin, J. Pang, X.guan, X. Yang, and Y. Mu,“专业知识不需要垄断：行动专业的专家混合体，用于视觉语言-行动学习，”CoRR，卷。绝对/2510.14300，2025。

- [487] Y.C.XiaohuanPei 和 S.Xu、Y.Wang、Y.Shi 和 C.Xu，“用于高效 视觉语言动作（Vision-Language-Action）操作的动作感知动态剪枝”，CoRR，卷。绝对/2509.22093，2025。

- [488] Z. W. Siyu Xu 和 Y. Wang、C. Xia、T. Huang 和 C. Xu，“可供性（Affordance）现场干预：在机器人操作中使 vlas 能够逃脱记忆陷阱”，CoRR，卷。绝对/2512.07472，2025。

- [489] Z. Y. Yizheng 张、J. Lai、C. Lu 和 L. Han，“Agentworld：用于场景构建和移动机器人操作的交互式仿真平台”，CoRR，卷。绝对/2508.07770，2025。

- [490] H. Q. Irmak Guzey 和 J. Urain、C. Wang、J. Yin、K. Bodduluri、M. Lambeta、L. Pinto、A. Rai、J. Malik、T. Wu、A. Sharma 和 H. Bharadhwaj，“智能镜头的灵活性：在野外人类演示中的多指机器人操作”，CoRR，卷。绝对/2511.16661，2025。

- [491] B. L. Wenxin Cheng 和 B. Xu、E. Feng、J. Gu 和 H. Chen，“利用操作系统级原语进行机器人动作管理”，CoRR，卷。绝对/2508.10259，2025。

- [492] Z. W. Yiyang Huang 和 Z. Wan、Y. Tian、H. Xu、Y. Han 和 Y. Gan，“ANNIE：小心你的机器人”，CoRR，卷。绝对/2509.03383，2025。

- [493] S. C. Yuhua Jiang 和 Y. Ding、F. Gau 和 B. Qi，“Asyncvla：用于 视觉语言动作（Vision-Language-Action）模型的异步 流匹配（Flow Matching）”，CoRR，卷。绝对/2511.14148，2025。

- [494] C. W. Yang Zhang 和 O. Lu、Y. Zhao、Y. Ge、Z. S. 0001、X. L. 0001、C. Z. 0012、C. Bai 和 X. L. 0001，“对齐然后转向：通过统一的潜在指导调整视觉语言动作模型”，2025 年。[在线]。可用：https://dblp.org/rec/journals/corr/abs-2509-02055

- [495] J.T.Zhenlong Yuan、J.Luo、R.Chen、C.Qian、L.Sun、X.Chu、Y.Cai、D.Zhang、S.Li，“Autodrive-r2：激励推理和自我反思，自动驾驶中 VLA 模型的能力”，CoRR，第 1 卷。绝对/2509.01944，2025。

- [496] Y. X. Hanshi Wang 和 Z. Xu、J. Gau、Y. Liu、W. Hu、K. Wang 和 Z. Zhang，“自动修剪：每种复杂性都需要修剪策略”，CoRR，卷。绝对/2509.23931，2025。

- [497] T. C. Zewei Zhou 和 S. Z. Zhu、Y. Zhang、Z. Huang、B. Zhou 和 J. Ma，“Autovla：用于端到端自主驾驶的 视觉语言动作（Vision-Language-Action） 模型，具有自适应推理和强化微调功能”，CoRR，第 1 卷。绝对/2506.13757，2025。

- [498] J.L.LeiXiao和J.Gao、F.Ye、Y.Jin、J.Qian、J.Zhang、Y.Wu和X.Yu，“Ava-vla：通过主动视觉注意改进视觉语言动作（Vision-Language-Action）模型”，2025年。[在线]。可用：https://dblp.org/rec/journals/corr/abs-2511-18960

- [499] L. L. Harris Song 和，“Avi：体积推理的动作”，CoRR，卷。绝对/2510.21746，2025。

- [500] G.T.Xueyang Zhou 和 G.Zhang、H.Wang、P.Zhou 和 L.Sun，“Badvla：通过目标解耦优化对 视觉语言动作（Vision-Language-Action） 模型进行后门攻击”，CoRR，第 1 卷。绝对/2505.16640，2025。

- [501] Z. Z. Kechun Xu 和 A. Chen、S. Zhao、Q. Huang、Y. Yang、H. Lu、R. Xiong、M. Tomizuka 和 Y. Wang，“看到行动，提示指定：视觉语言行动策略的贝叶斯分解”，CoRR，卷。绝对/2512.11218，2025。

- [502] Y.F.Hao Luo 和 W.Zhang、S.Zheng、Y.Wang、H.Yuan、J.Liu、C.Xu、Q.Jin 和 Z.Lu，“Being-h0：来自大规模人类视频的 视觉语言动作（Vision-Language-Action） 预训练”，CoRR，卷。绝对/2507.15597，2025。

- [503] J. F. Yanbo Mao 和 R. Zhang、H. Xie 和 M. Yao，“超越成功：通过即时干预从混合质量的数据中精炼优雅的机器人操作”，CoRR，卷。绝对/2511.22555，2025。

- [504] M. A. C. Koffivi Fidele Gbagbe 和 A. Alabbas、O. Alyunes、A. Lykov 和 D. Tsetserukou，“Bi-vla：视觉语言动作（Vision-Language-Action） 基于模型的双手机器人灵巧操作系统”，载于_SMC_。 IEEE，2024 年，第 2864-2869 页。

- [505] T. B. Masato Kobayashi 和，“Bi-vla：通过视觉语言基于双边控制的 模仿学习（Imitation Learning），动作生成融合”，CoRR，卷。绝对/2509.18865，2025。

- [506] Y. C. Peiyan Li 和 H. Wu、X. Ma、X. Wu、Y. Huang、L. Wang、T. Kong 和 T. Tan，“Bridgevla：使用视觉语言模型实现高效 3D 操作学习的输入输出对齐”，CoRR，卷。绝对/2506.07961，2025。

- [507] A. Z. R. Asher J. Hancock 和 A. Majumdar，“运行时观察干预使 视觉语言动作（Vision-Language-Action） 模型在视觉上更加稳健”，载于_ICRA_。 IEEE，2025 年，第 9499-9506 页。

- [508] Y. Y. Xiuxiu Qi 和 J. Cao、L. Bai、C. Fan、C. Cao 和 H. Wang，“持续 视觉语言动作（Vision-Language-Action） 协同学习与语义-物理、行为克隆对齐”，CoRR，卷。绝对/2511.14396，2025。

- [509] Y.Z.Zhongyi Zhou 和 J.Wen、C.Shen 和 Y.Xu，“Chatvla-2：来自预训练知识的具有开放世界具体推理的 视觉语言动作（Vision-Language-Action） 模型”，CoRR，卷。绝对/2505.21906，2025。

- [510] R. Y. Zijian An、Y. Feng 和 L. Zhou，“CLAW：用于重量感知机器人抓取的视觉语言动作框架”，CoRR，卷。绝对/2509.14143，2025。

- [511] J. K. Gi-Cheon Kang 和 K. Shim、J. K. Lee 和 B. Zhang，“CLIP-RT：从自然语言监督中学习语言条件机器人策略”，CoRR，卷。绝对/2411.00508，2024。

- [512] Q. Y. Zeyuan Chen 和 Y. Chen、T. Wu、J. Zhang、Z. Ding、J. Li、Y. Yang 和 H. Dong，“Clutterdexgrasp：用于在杂乱场景中进行一般灵巧抓取的 仿真到现实（Sim-to-Real） 系统”，CoRR，卷。绝对/2506.14317，2025。

- [513] F.S.张大鹏、R.Zhao、Y.Chen、P.Zhi、C.Li、R.Zhou 和 Q

。 Zhou，“Coc-vla：通过因果链视觉语言动作模型深入研究对抗性域转移以实现可解释的自动驾驶”，CoRR，卷。 ABS/2511.19914，2025。

- [514] R. Z. Wei Li 和 R. Shao、J. He 和 L. Nie，“Cogvla：通过指令驱动、路由和稀疏化实现认知对齐的 视觉语言动作（Vision-Language-Action） 模型”，CoRR，卷。绝对/2508.21046，2025。

- [515] X. C. Qihang Peng 和 C. Yang、S. Shi 和 H. Li，“Colavla：利用认知潜在推理实现自动驾驶中的分层并行、轨迹规划”，CoRR，卷。绝对/2512.22939，2025。

- [516] S. T. Yuhui Chen 和 S. Liu、Y. Zhou、H. Li 和 D. Zhu，“Conrft：通过一致性、策略对 VLA 模型进行强化微调方法”，CoRR，卷。绝对/2502.05450，2025。

- [517] Y. W. Puhao Li 和 Z. Xi、W. Li、Y. Huang、Z. 张、Y. Chen、J. Wang、S. Zhu、T. Liu 和 S. Huang，“Controlvla：针对预训练 视觉语言动作（Vision-Language-Action） 模型的少量对象中心自适应”，CoRR，卷。绝对/2506.16211，2025。

- [518] W. K. Tian Nian 和 Y. Mu、T. Chen、S. Zhu 和 B. Hu，“控制你的机器人：机器人控制和策略、部署的统一系统”，CoRR，卷。绝对/2509.23823，2025。

- [519] Y. C. Shiyu Fang 和 H. Liang、C. Lv、P. Hang 和 J. Sun，“Corevla：通过收集和细化实现长尾场景的双阶段端到端自动驾驶框架”，CoRR，第 1 卷。绝对/2509.15968，2025。

- [520] T. Y. Wang 和 H. Tang，“Cot4ad：具有明确思想链的视觉语言动作模型，自动驾驶推理”，CoRR，卷。绝对/2511.22532，2025。

- [521] W. D. Chenghao ”Mark” Peng 和 Y. You、Y. Chen、W. Luo、T. Tian、Y. Cao、A. Sharma、D. Xu、B. Ivanovic、B. Li、B. Zhou、Y. Wang 和 M. Pavone，“反事实（Counterfactual） VLA：具有自适应推理的自反射视觉语言动作模型”，CoRR，卷。绝对/2512.24426，2025。

- [522] O. H. Daniel San Jos´e Pro 和 R. R´omer、M. D´osch、M. Schuck 和 A. P. Schoellig，“CRISP - 兼容 ROS2 控制器，用于基于学习的操作、策略和远程操作”，CoRR，卷。绝对/2509.06819，2025。

- [523] H. L. Yi-Lin Wei 和 Y. Lin、P. Wang、Z. Liang、G. Liu 和 W. Cheng，“Cyclemanip：通过有效的历史、感知和理解实现循环任务操纵”，CoRR，卷。绝对/2512.01022，2025。

- [524] H. L. Changyeon Kim 和 Y. Seo、K. Lee 和 Y. Zhu，“DEAS：采用行动序列进行可扩展的离线强化学习的独立价值学习”，CoRR，卷。绝对/2510.07730，2025。

- [525] L. H. Haibo Hu、N.guan、C.J.Xue，“Deead：视觉语言动作的动态提前退出，实现高效自动驾驶”，CoRR，卷。绝对/2511.20720，2025。

- [526] Y. L. Cheng Yin 和 W. Xu、S. Tam、X. Zeng、Z. Liu 和 Z. Yin，“Deepthinkvla：增强视觉语言动作、模型的推理能力”，CoRR，卷。绝对/2511.15669，2025。

- [527] S. Wu、Y. Ji、Q. Li、Z.Zhang、Q.He、W.Xie、G.Zhang、B.Bayramli、Y.Ding 和 H.Lu，“Dejavu：面向 具身智能（Embodied Intelligence） 的体验反馈学习”，arXiv 预印本，2025 年。

[528] Y. L. Tianyuan Yuan and, C. Lu, Z. Chen, T. Jiang, and H. Zhao, "Depthvla: Enhancing vision-language-action models with depth-aware, spatial reasoning," *CoRR*, vol. abs/2510.13375, 2025.

[529] E. Z. Bin Xie and, F. Jia, H. Shi, H. Fan, H. Zhang, H. Li, J. Sun, J. Bin, J. Huang, K. Liu, K. Liu, K. Gu, L. Sun, M. Zhang, P. Han, R. Hao, R. Zhang, S. Huang, S. Xie, T. Wang, T. Liu, W. Tang, W. Zhu, Y. Chen, Y. Liu, Y. Zhou, Y. Liu, Y. Zhao, Y. Ma, Y. Wei, Y. Chen, Z. Chen, Z. Li, Z. Wu, Z. Zhang, Z. Liu, Z. Yan, and Z. Zhang, "Dexbotic: Open-source vision-language-action toolbox," *CoRR*, vol. abs/2510.23511, 2025.

[530] Z. X. Cheng Chi and, S. Feng, E. Cousineau, Y. Du, B. Burchfiel, R. Tedrake, and S. Song, "Diffusion policy: Visuomotor policy learning via action diffusion," *Int. J. Robotics Res.*, vol. 44, no. 10-11, pp. 1684–1704, 2025.

[531] M. Z. Junjie Wen and, Y. Zhu, Z. Tang, J. Li, Z. Zhou, C. Li, X. Liu, Y. Peng, C. Shen, and F. Feng, "Diffusion-vla: Scaling robot foundation models via unified diffusion, and autoregression," *CoRR*, vol. abs/2412.03293, 2024.

[532] A. J. Yu Gao and, Y. Wang, J. Wang, H. Jiang, Z. Sun, Y. Heng, W. Shuo, H. Zhao, and H. Sun, "Diffvla++: Bridging cognitive reasoning and end-to-end driving through, metric-guided alignment," *CoRR*, vol. abs/2510.17148, 2025.

[533] R. Liang, Y. Zheng, K. Zheng, T. Tan, J. Li, L. Mao, Z. Wang, G. Chen, H. Ye, J. Liu, J. Wang, and X. Zhan, "Dichotomous diffusion policy optimization," *arXiv preprint*, 2025.

[534] Z. F. Rushuai Yang and, T. Zhang, K. Wang, C. Zhang, L. Zhao, X. Su, Y. Chen, and J. Bian, "Discover, learn, and reinforce: Scaling vision-language-action pretraining, with diverse rl-generated trajectories," *CoRR*, vol. abs/2511.19528, 2025.

[535] M. K. Nikita Kachaev and, D. Zelezetsky, A. K. Kovalev, and A. I. Panov, "Don't blind your VLA: aligning visual representations for OOD, generalization," *CoRR*, vol. abs/2510.25616, 2025.

[536] J. K. ByungOk Han and and J. Jang, "A dual process VLA: efficient robotic manipulation leveraging VLM," *CoRR*, vol. abs/2410.15549, 2024.

[537] Z. Z. Guo Ye and, X. Zhao, S. Wu, H. Lu, S. Lu, and H. Liu, "Learning to feel the future: Dreamtacvla for contact-rich manipulation," *CoRR*, vol. abs/2512.23864, 2025.

[538] H. L. Wenyao Zhang and, Z. Qi, Y. Wang, X. Yu, J. Zhang, R. Dong, J. He, H. Wang, Z. Zhang, L. Yi, W. Zeng, and X. Jin, "Dreamvla: A vision-language-action model dreamed with comprehensive, world knowledge," *CoRR*, vol. abs/2507.04447, 2025.

[539] S. G. Jiacheng Ye and, J. Gao, J. Fan, S. Wu, W. Bi, H. Bai, L. Shang, and L. Kong, "Dream-vl & dream-vla: Open vision-language and vision-language-action, models with diffusion language model backbone," *CoRR*, vol. abs/2512.22615, 2025.

[540] Z. L. Zhen Fang and, J. Liu, H. Chen, Y. Zeng, S. Huang, Z. Chen, L. Chen, S. Zhang, and F. Zhao, "Dualvla: Building a generalizable embodied agent via partial decoupling, of reasoning and action," *CoRR*, vol. abs/2511.22134, 2025.

[541] H. Z. Teqiang Zou and, Y. Nong, Y. Li, K. Liu, H. Yang, X. Ling, X. Li, and L. Ma, "Asynchronous fast-slow vision-language-action policies for whole-body, robotic manipulation," *CoRR*, vol. abs/2512.20188, 2025.

[542] M. Z. Junjie Wen and, J. Liu, Z. Liu, Y. Yang, L. Zhang, S. Zhang, Y. Zhu, and Y. Xu, "dvla: Diffusion vision-language-action model with multimodal chain-of-thought," *CoRR*, vol. abs/2509.25681, 2025.

[543] Y. H. Travis Davies and, A. Gladstone, Y. Liu, X. Chen, H. Ji, H. Liu, and L. Hu, "Ebt-policy: Energy unlocks emergent physical reasoning capabilities," *CoRR*, vol. abs/2510.27545, 2025.

[544] W. C. Michal Zawalski and, K. Pertsch, O. Mees, C. Finn, and S. Levine, "Robotic control via embodied chain-of-thought reasoning," in *CoRL*, ser. Proceedings of Machine Learning Research, vol. 270. PMLR, 2024, pp. 3157–3181.

[545] S. B. William Chen and, S. Mirchandani, O. Mees, D. Driess, K. Pertsch, and S. Levine, "Training strategies for efficient embodied reasoning," *CoRR*, vol. abs/2505.08243, 2025.

[546] Y. W. Yantai Yang and, Z. Wen, L. Zhongwei, C. Zou, Z. Zhang, C. Wen, and L. Zhang, "Efficientvla: Training-free acceleration and compression for vision-language-action, models," *CoRR*, vol. abs/2506.10100, 2025.

[547] M. Z. S. Binjie Zhang and, "Ego-centric predictive model conditioned on hand trajectories," *CoRR*, vol. abs/2508.19852, 2025.

[548] R. C. Zefu Lin and, C. Hanning, X. Wang, J. Xu, X. Jin, C. Wenbo, H. Zhou, L. Fan, W. Li, and Z. Zhang, "Embodiedcoder: Parameterized embodied mobile manipulation via modern, coding model," *CoRR*, vol. abs/2510.06207, 2025.

[549] T. H. Nhat Chung and, T. Nguyen, H. Le, F. Bumgarner, D. M. H. Nguyen, K. Vo, K. Yamazaki, C. Rainwater, T. Kieu, A. Nguyen, and N. Le, "Rethinking progression of memory state in robotic manipulation: An, object-centric perspective," *CoRR*, vol. abs/2511.11478, 2025.

[550] X. W. Zhehao Dong and, Z. Zhu, Y. Wang, Y. Wang, Y. Zhou, B. Wang, C. Ni, R. Ouyang, W. Qin, X. Chen, Y. Ye, and G. Huang, "EMMA: generalizing real-world robot manipulation via generative, visual transfer," *CoRR*, vol. abs/2509.22407, 2025.

[551] P. H. Qi Sun and, P. T. Deep, V. Toh, U. Tan, D. Ghosal, and S. Poria, "Emma-x: An embodied multimodal action model with grounded chain of, thought and look-ahead spatial reasoning," in *ACL (1)*. Association for Computational Linguistics, 2025, pp. 14 199–14 214.

[552] L. B. Chikit Ng and, G. Wang, Y. Wang, H. Gao, K. Yuan, C. Jin, T. Zeng, and H. Ren, "Endovla: Dual-phase vision-language-action model for autonomous tracking, in endoscopy," *CoRR*, vol. abs/2505.15206, 2025.

[553] H. S. Delin Qu and, Q. Chen, Z. Chen, X. Gao, X. Ye, Q. Lv, M. Shi, G. Ren, C. Ruan, M. Yao, H. Yang, J. Bao, B. Zhao, and D. Wang, "Embodiedonevision: Interleaved vision-text-action pretraining for, general robot control," *CoRR*, vol. abs/2508.21112, 2025.

[554] S. W. Yi Liu and, D. Wei, X. Cai, L. Zhong, J. Yang, G. Ren, J. Zhang, M. Yao, C. Li, X. He, L. Chen, and J. Luo, "Unified embodied VLM reasoning with robotic action via autoregressive, discretized pre-training," *CoRR*, vol. abs/2512.24125, 2025.

[555] G. W. Chang Nie and, Z. Lie, and H. Wang, "ERMV: editing 4d robotic multi-view images to enhance embodied agents," *CoRR*, vol. abs/2507.17462, 2025.

[556] Y. P. Chengmeng Li and, "Embodiment transfer learning for vision-language-action models," *CoRR*, vol. abs/2511.01224, 2025.

[557] A. M. Samarth Chopra and, B. Carnovale, E. Sokolson, R. Kubendran, and S. Dickerson, "Everydayvla: A vision-language-action model for affordable robotic, manipulation," *CoRR*, vol. abs/2511.05397, 2025.

[558] Y. Z. Tao Lin and, Y. Du, J. Zhang, J. Liu, Y. Chen, E. Gu, Z. Liu, H. Cai, Y. Zou, L. Zou, Z. Zhou, G. Li, and B. Zhao, "Evo-1: Lightweight vision-language-action model with preserved semantic, alignment," *CoRR*, vol. abs/2511.04555, 2025.

[559] Y. A. Shahram Najam Syed and, A. Jakobsson, and J. Ichnowski, "Expres-vla: Specializing vision-language-action models through experience, replay and retrieval," *CoRR*, vol. abs/2511.06202, 2025.

[560] Y. H. Jiashu Yang and, Y. Xie, N. Guo, and W. Lian, "Look, zoom, understand: The robotic eyeball for embodied perception," *CoRR*, vol. abs/2511.15279, 2025.

[561] W. K. Qi Lv and, H. Li, J. Zeng, Z. Qiu, D. Qu, H. Song, Q. Chen, X. Deng, and J. Pang, "F1: A vision-language-action model bridging understanding and, generation to actions," *CoRR*, vol. abs/2509.06951, 2025.

[562] J. D. Zijun Lin and, H. Fang, D. Fox, R. Krishna, C. Tan, and B. Wen, "Failsafe: Reasoning and recovery from failures in vision-language-action, models," *CoRR*, vol. abs/2510.01642, 2025.

[563] Q. Z. Jiajun Cao and, P. Jia, X. Zhao, B. Lan, X. Zhang, X. Wei, S. Chen, Z. Li, Y. Wang, L. Li, X. Liu, M. Lu, and S. Zhang, "Fastdrivevla: Efficient end-to-end driving via plug-and-play reconstruction-based, token pruning," *CoRR*, vol. abs/2507.23318, 2025.

[564] H. Chen, J. Liu, C. Gu, Z. Liu, R. Zhang, X. Li, and X. He, "Fast-in-slow: A dual-system vla model unifying fast manipulation within slow reasoning," *arXiv preprint*, 2025.

[565] J. W. Ruijie Zheng and, S. Reed, J. Bjorck, Y. Fang, F. Hu, J. Jang, K. Kundalia, Z. Lin, L. Magne, A. Narayan, Y. L. Tan, G. Wang, Q. Wang, J. Xiang, Y. Xu, S. Ye, J. Kautz, F. Huang, Y. Zhu, and L. Fan, "FLARE: robot learning with implicit world modeling," *CoRR*, vol. abs/2505.15659, 2025.

[566] H. Z. Moritz Reuss and, M. Rühle, Ö. E. Yagmurlu, F. Otto, and R. Lioutikov, "FLOWER: democratizing generalist robot policies with efficient vision-language-action, flow policies," *CoRR*, vol. abs/2509.04996, 2025.

[567] Z. Zhong, H. Yan, J. Li, X. Liu, X. Gong, T. Zhang, W. Song, J. Chen, X. Zheng, H. Wang, and H. Li, "Flowvla: Visual chain of thought-based motion reasoning for vision-language-action models," *arXiv preprint*, 2025.

[568] H. L. Jiawen Yu and, Q. Yu, J. Ren, C. Hao, H. Ding, G. Huang, G. Huang, Y. Song, P. Cai, C. Lu, and W. Zhang, "Forcevla: Enhancing VLA models with a force-aware moe for contact-rich, manipulation," *CoRR*, vol. abs/2505.22159, 2025.

[569] S. L. Yanjia Huang and, S. Liu, Q. Xu, M. Wu, X. Gao, and Z. Tu, "Forge-tree: Diffusion-forcing tree search for long-horizon robot manipulation," *CoRR*, vol. abs/2510.21744, 2025.

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- [528] Y. L. Tianyuan Yuan 和 C. Lu、Z. Chen、T. Jiang 和 H. Zhao，“Depthvla：通过深度感知、空间推理增强 视觉语言动作（Vision-Language-Action） 模型”，CoRR，卷。绝对/2510.13375，2025。

- [529] E. Z. Bin Xie 和 F. Jia、H. Shi、H. Fan、H. 张、H. Li、J. Sun、J. Bin、J. Huang、K. Liu、K. Liu、K. Gu、L. Sun、M. 张、P. Han、R.Hao、R.Zhang、S. Huang、S. Xie、T. Wang、T. Liu、W. Tang、W. Zhu、Y. Chen、Y. Liu、Y. Zhou、 Y. Liu、Y. Zhu、Y. Ma、Y. Wei、Y. Chen、Z. Chen、Z. Li、Z. Wu、Z. Zhang、Z. Liu、Z. Yan 和 Z. Zhang，“Dexbotic：开源 视觉语言动作（Vision-Language-Action） 工具箱”，CoRR，卷。绝对/2510.23511，2025。

- [530] Z. X. Cheng Chi 和 S. Feng、E. Cousineau、Y. Du、B. Burchfiel、R. Tedrake 和 S. Song，“扩散策略（Diffusion Policy）：通过动作扩散进行视觉运动策略学习”，_Int。 J. 机器人研究，卷。 44，没有。 10-11，第 1684-1704 页，2025 年。

- [531] M. Z. Junjie Wen 和 Y. Zhu、Z. Tang、J. Li、Z. Zhou、C. Li、X. Liu、Y. Peng、C. Shen 和 F. Feng，“Diffusion-vla：通过统一扩散和自回归缩放机器人 基础模型（Foundation Models）”，CoRR，卷。绝对/2412.03293，2024。

- [532] A. J. Yu Gau 和 Y. Wang、J. Wang、H. Jiang、Z. Sun、Y. Heng、W. Shuo、H. Zhao 和 H. Sun，“Diffvla++：桥接认知推理和端到端驱动，度量引导对齐”，CoRR，卷。绝对/2510.17148，2025。

- [533] R.Liang、Y.Zheng、K.Zheng、T.Tan、J.Li、L.Mao、Z.Wang、G.Chen、H.Ye、J.Liu、J.Wang 和 X.Zhan，“二分法 扩散策略（Diffusion Policy） 优化”，arXiv 预印本，2025 年。

- [534] Z. F. Rushuai Yang 和 T. 张、K. Wang、C. 张、L. 赵、X. Su、Y. Chen 和 J. Bian，“发现、学习和强化：使用不同的 rl 生成轨迹扩展视觉语言动作预训练”，CoRR，第 1 卷。绝对/2511.19528，2025。

- [535] M. K. Nikita Kachaev 和 D. Zelezetsky、A. K. Kovalev 和 A. I. Panov，“不要蒙蔽您的 VLA：调整 OOD 的视觉表示、泛化”，CoRR，卷。绝对/2510.25616，2025。

- [536] J. K. ByungOk Han 和 J. Jang，“双进程 VLA：利用 VLM 的高效机器人操作”，CoRR，卷。绝对/2410.15549，2024。

- [537] Z.Z.Guo Ye 和 X.Zhao、S.Wu、H.Lu、S.Lu 和 H.Liu，“学习感受未来：用于接触丰富操作的 Dreamtacvla”，CoRR，卷。绝对/2512.23864，2025。

- [538] H. L. Wenyao Zhang 和 Z. Qi、Y. Wang、X. Yu、J. Zhu、R. Dong、J. He、H. Wang、Z. Zhang、L. Yi、W. Zeng 和 X. Jin，“Dreamvla：视觉语言动作（Vision-Language-Action） 模型梦想具有全面的世界知识”，CoRR，卷。绝对/2507.04447，2025。

- [539] S. G. Jia Cheng Ye 和 J. Gau、J. Fan、S. Wu、W. Bi、H. Bai、L. Shang 和 L. Kong，“Dream-vl 和 dream-vla：开放视觉语言和 视觉语言动作（Vision-Language-Action），具有扩散语言模型骨干的模型”，CoRR，卷。绝对/2512.22615，2025。

- [540]Z.L.Zhen Fang 和 J.Liu、H.Chen、Y.Zeng、S.Huang、Z.Chen、L.Chen、S.Zhang 和 F.Zhao，“Dualvla：通过推理和行动的部分解耦构建可泛化的体现智能体”，CoRR，卷。绝对/2511.22134，2025。

- [541] H. Z. 邹特强、Y. Nong、Y. Li、K. Liu、H. Yang、X. Ling、X. Li 和 L

。 Ma，“用于全身机器人操作的异步快慢 视觉语言动作（Vision-Language-Action） 策略”，CoRR，卷。 ABS/2512.20188，2025。

- [542] M. Z. Junjie Wen 和 J. Liu、Z. Liu、Y. Yang、L. Zhu、S. Zhu、Y. Zhu 和 Y. Xu，“dvla：具有多模式思想链的扩散 视觉语言动作（Vision-Language-Action） 模型”，CoRR，卷。绝对/2509.25681，2025。

- [543] Y. H. Travis Davies 和 A. Gladstone、Y. Liu、X. Chen、H. Ji、H. Liu 和 L. Hu，“Ebt 策略：能源解锁新兴物理推理能力”，CoRR，卷。绝对/2510.27545，2025。

- [544] W. C. Michal Zawalski 和 K. Pertsch、O. Mees、C. Finn 和 S. Levine，“通过体现的思想链推理进行机器人控制”，载于_CoRL_，系列。机器学习研究论文集，卷。 270.PMLR，2024 年，第 3157-3181 页。

- [545] S. B. William Chen 和 S. Mirchandani、O. Mees、D. Driess、K. Pertsch 和 S. Levine，“高效具身推理的训练策略”，CoRR，卷。绝对/2505.08243，2025。

- [546] Y. W. Yantai Yang 和 Z. Wen、L.Zhongwei、C. Zou、Z.Zhang、C.Wen 和 L.Zhang，“Efficientvla：视觉语言动作（Vision-Language-Action） 模型的免训练加速和压缩”，CoRR，卷。绝对/2506.10100，2025。

- [547] M. Z. S. Binjie Zhang 和，“以手部轨迹为条件的以自我为中心的预测模型”，CoRR，卷。绝对/2508.19852，2025。

- [548] R. C. Zefu Lin 和 C. Hanning、X. Wang、J. Xu、X. Jin、C. Wenbo、H. Zhou、L. Fan、W. Li 和 Z. Zhang，“Embodiedcoder：通过现代编码模型进行参数化体现移动操作”，CoRR，卷。绝对/2510.06207，2025。

- [549] T. H. Nhat Chung 和 T. Nguyen、H. Le、F. Bumgarner、D. M. H. Nguyen、K. Vo、K. Yamazaki、C. Rainwater、T. Kieu、A. Nguyen 和 N. Le，“重新思考机器人操作中记忆状态的进展：以对象为中心的视角”，CoRR，卷。绝对/2511.11478，2025。

- [550] X.W.Zhehao Dong 和 Z.Zhu、Y.Wang、Y.Wang、Y.Zhou、B.Wang、C.Ni、R.Ouyang、W.Qin、X.Chen、Y.Ye 和 G.Huang，“EMMA：通过生成、视觉迁移概括现实世界的机器人操作”，CoRR，卷。绝对/2509.22407，2025。

- [551] P. H. Qi Sun 和 P. T. Deep、V. Toh、U. Tan、D. Ghosal 和 S. Poria，“Emma-x：具有接地链、思想和前瞻空间推理的具体多模态行动模型”，载于_ACL (1)_。计算语言学协会，2025 年，第 14 199-14 214 页。

- [552] L. B. Chikit Ng 和 G. Wang、Y. Wang、H. Gau、K. Yuan、C. Jin、T. Zeng 和 H. Ren，“Endovla：内窥镜检查中用于自主跟踪的双阶段视觉语言动作模型”，CoRR，卷。绝对/2505.15206，2025。

- [553] H. S. Delin Qu 和 Q. Chen、Z. Chen、X. Gau、X. Ye、Q. Lv、M. Shi、G. Ren、C. Ruan、M. Yao、H. Yang、J. Bao、B. Zhu 和 D. Wang，“Embodiedonevision：通用机器人控制的交错视觉-文本-动作预训练”，CoRR，卷。绝对/2508.21112，2025。

- [554] S. W. Yi Liu 和 D. Wei、X. Cai、L.zhong、J. Yang、G. Ren、J.Zhang、M. Yao、C. Li、X. He、L. Chen 和 J. Luo，“通过自回归、离散预训练与机器人动作进行统一的体现 VLM 推理”，CoRR，卷。绝

对/2512.24125，2025。

- [555] G. W. Chang Nie、Z. Lie 和 H. Wang，“ERMV：编辑 4d 机器人多视图图像以增强具体智能体”，CoRR，卷。绝对/2507.17462，2025。

- [556] Y. P. Chengmeng Li 和，“视觉语言动作模型的具体迁移学习”，CoRR，卷。绝对/2511.01224，2025。

- [557] A. M. Samarth Chopra 和 B. Carnovale、E. Sokolson、R. Kubendran 和 S. Dickerson，“Everydayvla：用于经济实惠的机器人操作的 视觉语言动作（Vision-Language-Action） 模型”，CoRR，卷。绝对/2511.05397，2025。

- [558] Y.Z.Tao Lin 和 Y.Du、J.Zhang、J.Liu、Y.Chen、E.Gu、Z.Liu、H.Cai、Y.Zou、L.Zou、Z.Zhou、G.Li 和 B.Zhao，“Evo-1：保留语义、对齐的轻量级 视觉语言动作（Vision-Language-Action） 模型”，CoRR，卷。绝对/2511.04555，2025。

- [559] Y. A. Shahram Najam Syed、A. Jakobsson 和 J. Ichnowski，“Expres-vla：通过经验、重播和检索专业化 视觉语言动作（Vision-Language-Action） 模型”，CoRR，卷。绝对/2511.06202，2025。

- [560] Y.H.Jiashu Yang 和 Y.Xie、N.Guo 和 W.Lian，“观察、缩放、理解：用于具体感知的机器人眼球”，CoRR，卷。绝对/2511.15279，2025。

- [561] W. K. Qi Lv 和 H. Li、J. Zeng、Z. Qiu、D. Qu、H. Song、Q. Chen、X. Deng 和 J. Pang，“F1：连接理解和生成到行动的 视觉语言动作（Vision-Language-Action） 模型”，CoRR，卷。绝对/2509.06951，2025。

- [562] J. D. Zijun Lin 和 H. Fang、D. Fox、R. Krishna、C. Tan 和 B. Wen，“故障安全：视觉语言动作模型中的故障推理和恢复”，CoRR，卷。绝对/2510.01642，2025。

- [563]Q.Z.Jiajun Cao 和 P.Jia、X.Zhao、B.Lan、X.Zhang、X.Wei、S.Chen、Z.Li、Y.Wang、L.Li、X.Liu、M.Lu 和 S.Zhang，“Fastdrivevla：通过即插即用重建、令牌剪枝实现高效端到端驱动”，CoRR，卷。绝对/2507.23318，2025。

- [564] H. Chen、J. Liu、C. Gu、Z. Liu、R. Zhang、X. Li 和 X. He，“Fast-inslow：一种双系统 vla 模型，在慢速推理中统一快速操作”，arXiv 预印本，2025 年。

- [565] J. W. Ruijie Cheng 和，S. Reed，J. Bjorck，Y. Fang，F. Hu，J. Jang，K. Kundalia，Z. Lin，L. Magne，A. Narayan，Y. L. Tan，G. Wang，Q. Wang，J. Xiang，Y. Xu，S. Ye，J. Kautz，F. Huang，Y. Zhu 和 L. Fan，“FLARE：隐式世界的机器人学习建模，”CoRR，卷。绝对/2505.15659，2025。

- [566] H. Z. 莫里茨·罗伊斯 (H. Z. Moritz Reuss) 和 M. Rühle，“O. E. Yagmurlu、F. Otto 和 R. Lioutikov，“FLOWER：通过高效的 视觉语言动作（Vision-Language-Action）、流策略使通用机器人策略民主化”，CoRR，卷。绝对/2509.04996，2025。

- [567] Z.zhong、H.Yan、J.Li、X.Liu、X.Gong、T.Zhang、W.Song、J.Chen、X.Zheng、H.Wang 和 H.Li，“Flowvla：视觉语言动作（Vision-Language-Action） 模型基于思想的运动推理的视觉链”，arXiv 预印本，2025 年。

- [568] H. L. Jiawen Yu 和 Q. Yu、J. Ren、C.Hao、H.Ding、G.Huang、G.Huang、Y.Song、P.Cai、C.Lu 和 W.Zhang，“Forcevla：通过力感知萌部增强 VLA 模型以实现丰富的接触操作”，CoRR，卷。绝对/2505.22159，2025。

- [569] S. L. Yanjia Huang 和 S. Liu、Q. Xu、M. Wu、X. Gau 和 Z. Tu，“Forge-tree：用于 长时程（Long-Horizon） 机器人操作的扩散强迫树搜索”，CoRR，卷。绝对/2510.21744，2025。

[570] Z. D. Yifan Yang and, T. Xie, F. Cao, P. Shen, P. Song, P. Jin, G. Sun, S. Xu, Y. You, and J. Liu, "FPC-VLA: A vision-language-action framework with a supervisor, for failure prediction and correction," *CoRR*, vol. abs/2509.04018, 2025.

[571] J. L. Xin Wang and, Z. Weng, Y. Wang, Y. Gao, T. Pang, C. Du, Y. Teng, Y. Wang, Z. Wu, X. Ma, and Y. Jiang, "Freezevla: Action-freezing attacks against vision-language-action, models," *CoRR*, vol. abs/2509.19870, 2025.

[572] Q. Z. Weiqi Li and, R. Zhai, L. Lin, and G. Wang, "VLA models are more generalizable than you think: Revisiting physical, and spatial modeling," *CoRR*, vol. abs/2512.02902, 2025.

[573] T. Y. Tao Jiang and, Y. Liu, C. Lu, J. Cui, X. Liu, S. Cheng, J. Gao, H. Xu, and H. Zhao, "Galaxea open-world dataset and G0 dual-system VLA model," *CoRR*, vol. abs/2509.00576, 2025.

[574] H. N. Arjun Vaithilingam Sudhakar and, M. Reymond, M. Liu, J. Rajendran, and S. Chandar, "A generalist hanabi agent," in *ICLR*. OpenReview.net, 2025.

[575] G. A. Team, "Gen-0: Embodied foundation models that scale with physical interaction," *arXiv preprint*, 2025.

[576] M. M. Ali Abouzeid and, Z. Sun, and D. Song, "Geoaware-vla: Implicit geometry aware vision-language-action model," *CoRR*, vol. abs/2509.14117, 2025.

[577] H. Z. Wenxuan Song and, P. Ding, C. Cui, S. Lyu, Y. Fan, and D. Wang, "Germ: A generalist robotic model with mixture-of-experts for quadruped, robot," in *IROS*. IEEE, 2024, pp. 11 879–11 886.

[578] P. D. Hongyin Zhang and, S. Lyu, Y. Peng, and D. Wang, "GEVRM: goal-expressive video generation model for robust visual, manipulation," in *ICLR*. OpenReview.net, 2025.

[579] L. G. Shunlei Li and, J. Wang, C. Che, X. Xiao, J. Cao, Y. Hu, and H. R. Karimi, "Information-theoretic graph fusion with vision-language-action model, for policy reasoning and dual robotic control," *CoRR*, vol. abs/2508.05342, 2025.

[580] M. C. Minghao Guo and, J. Tao, R. Xu, Y. Yan, X. Liang, I. Laptev, and X. Chang, "Glad: Geometric latent distillation for vision-language-action models," *CoRR*, vol. abs/2512.09619, 2025.

[581] J. Jabbour, D.-K. Kim, M. Smith, J. Patrikar, R. Ghosal, Y. Wang, A. Agha, V. J. Reddi, and S. Omidshafiei, "Dont run with scissors," *arXiv preprint*, 2025.

[582] A. A. Gemini Robotics Team and, S. Abeyruwan, J. Ainslie, J. Alayrac, M. G. Arenas, A. Balakrishna, N. Batchelor, A. Bewley, J. T. Bingham, M. Bloesch, K. Bousmalis, P. Brakel, A. Brohan, T. Buschmann, A. Byravan, S. Cabi, K. Caluwaerts, F. Casarini, C. Chan, O. Chang, L. Chappellet-Volpini, J. E. Chen, X. Chen, H. L. Chiang, K. Choromanski, A. Collister, D. B. D'Ambrosio, S. Dasari, T. Davchev, M. K. Dave, C. Devin, N. D. Palo, T. Ding, C. Doersch, A. Dostmohamed, Y. Du, D. Dwibedi, S. T. Egambaram, M. Elabd, T. Erez, X. Fang, C. Fantacci, C. Fong, E. Frey, C. Fu, R. Gao, M. Giustina, K. Gopalakrishnan, L. Graesser, O. Groth, A. Gupta, R. Hafner, S. Hansen, L. Hasenclever, S. Haves, N. Heess, B. Hernaez, A. Hofer, J. Hsu, L. Huang, S. H. Huang, A. Iscen, M. G. Jacob, D. Jain, S. Jesmonth, A. Jindal, R. Julian, D. Kalashnikov, M. E. Karagozler, S. Karp, M. Kecman, J. C. Kew, D. Kim, F. Kim, J. Kim, T. Kipf, S. Kirmani, K. Konyushkova, L. Y. Ku, Y. Kuang, T. Lampe, A. Laurens, T. A. Le, I. Leal, A. X. Lee, T. E. Lee, G. Lever, J. Liang, L. Lin, F. Liu, S. Long, C. Lu, S. Maddineni, A. Majumdar, K. Maninis, A. Marmon, S. Martinez, A. H. Michaely, and N. Milonopoulos, "Gemini robotics 1.5: Pushing the frontier of generalist robots with, advanced embodied reasoning, thinking, and motion transfer," *CoRR*, vol. abs/2510.03342, 2025.

[583] K. Z. Zijian Zhang and, Z. Chen, J. Jang, Y. Li, C. Wang, M. Ding, D. Fox, and H. Yao, "GRAPE: generalizing robot policy via preference alignment," *CoRR*, vol. abs/2411.19309, 2024.

[584] M. C. Helong Huang and, K. Tan, X. Quan, G. Huang, and H. Zhang, "Graphcot-vla: A 3d spatial-aware reasoning vision-language-action, model for robotic manipulation with ambiguous instructions," *CoRR*, vol. abs/2508.07650, 2025.

[585] M. Y. Shengliang Deng and, S. Wei, H. Ma, Y. Yang, J. Chen, Z. Zhang, T. Yang, X. Zhang, H. Cui, Z. Zhang, and H. Wang, "Graspvla: a grasping foundation model pre-trained on billion-scale, synthetic action data," *CoRR*, vol. abs/2505.03233, 2025.

[586] Z. F. Maëlic Neau and, P. E. Santos, A. Bosser, and C. Buche, "Graspvla: Graph-based symbolic action representation for long-horizon, planning with VLA policies," *CoRR*, vol. abs/2511.04357, 2025.

[587] G. C. Ruoshi Wen and, Z. Cui, M. Du, Y. Gou, Z. Han, L. Huang, M. Lei, Y. Li, Z. Li, W. Liu, Y. Liu, X. Ma, H. Niu, Y. Ouyang, Z. Ren, H. Shi, W. Xu, H. Zhang, J. Zhang, X. Zhang, L. Zheng, W. Zhong, Y. Zhou, Z. Zhu, and H. Li, "Gr-dexter technical report," *CoRR*, vol. abs/2512.24210, 2025.

[588] X. M. Yunfei Li and, J. Xu, Y. Cui, Z. Cui, Z. Han, L. Huang, T. Kong, Y. Liu, H. Niu, W. Peng, J. Qiao, Z. Ren, H. Shi, Z. Su, J. Tian, Y. Xiao, S. Zhang, L. Zheng, H. Li, and Y. Wu, "GR-RL: going dexterous and precise for long-horizon robotic manipulation," *CoRR*, vol. abs/2512.01801, 2025.

[589] Y. D. Yi Li and, J. Zhang, J. Jang, M. Memmel, C. R. Garrett, F. Ramos, D. Fox, A. Li, A. Gupta, and A. Goyal, "HAMSTER: hierarchical action models for open-world robot manipulation," in *ICLR*. OpenReview.net, 2025.

[590] FIGURE, "Helix: A vision-language-action model for generalist humanoid control," *arXiv preprint*, 2025.

[591] P. D. Minghui Lin and, S. Wang, Z. Zhuang, Y. Liu, X. Tong, W. Song, S. Lyu, S. Huang, and D. Wang, "Hif-vla: Hindsight, insight and foresight through motion representation, for vision-language-action models," *CoRR*, vol. abs/2512.09928, 2025.

[592] B. L. Zhiying Du and, Y. Liang, Y. Shen, H. Cao, X. Zheng, Z. Feng, Z. Wu, J. Yang, and Y. Jiang, "Himoe-vla: Hierarchical mixture-of-experts for generalist vision-language-action, policies," *CoRR*, vol. abs/2512.05693, 2025.

[593] R. Z. Guanxing Lu and, H. Lin, H. Zhang, and Y. Tang, "Human-in-the-loop online rejection sampling for robotic manipulation," *CoRR*, vol. abs/2510.26406, 2025.

[594] B. I. Lucy Xiaoyang Shi and, M. R. Equi, L. Ke, K. Pertsch, Q. Vuong, J. Tanner, A. Walling, H. Wang, N. Fusai, A. Li-Bell, D. Driess, L. Groom, S. Levine, and C. Finn, "Hi robot: Open-ended instruction following with hierarchical vision-language-action, models," in *ICML*. OpenReview.net, 2025.

[595] Y. G. Jianke Zhang and, X. Chen, Y. Wang, Y. Hu, C. Shi, and J. Chen, "Hirt: Enhancing robotic control with hierarchical robot transformers," in *CoRL*, ser. Proceedings of Machine Learning Research, vol. 270. PMLR, 2024, pp. 933–946.

[596] Y. Z. Xinyu Xu and, Y. Li, L. Han, and C. Lu, "Humanvla: Towards vision-language directed object rearrangement by, physical humanoid," in *NeurIPS*, 2024.

[597] C. X. Yuan Zhang and, W. Xu, C. Ji, J. Wu, and J. Pan, "iflybot-vla technical report," *CoRR*, vol. abs/2511.01914, 2025.

[598] W. G. Dekun Lu and and K. Jia, "Imaginationpolicy: Towards generalizable, precise and reliable end-to-end, policy for robotic manipulation," *CoRR*, vol. abs/2509.20841, 2025.

[599] H. G. Haohan Chi and, Z. Liu, J. Liu, C. Liu, J. Li, K. Yang, Y. Yu, Z. Wang, W. Li, L. Wang, X. Hu, H. Sun, H. Zhao, and H. Zhao, "Impromptu VLA: open weights and open data for driving vision-language-action, models," *CoRR*, vol. abs/2505.23757, 2025.

[600] Z. S. Ulas Berk Karli and and T. Fitzgerald, "INSIGHT: inference-time sequence introspection for generating help, triggers in vision-language-action models," *CoRR*, vol. abs/2510.01389, 2025.

[601] H. L. Shuai Yang and, Y. Chen, B. Wang, Y. Tian, T. Wang, H. Wang, F. Zhao, Y. Liao, and J. Pang, "Instructvla: Vision-language-action instruction tuning from understanding, to manipulation," *CoRR*, vol. abs/2507.17520, 2025.

[602] K. G. Yandu Chen and, Y. Wen, Y. Zhao, T. Wang, and L. Nie, "Intentionvla: Generalizable and efficient embodied intention reasoning, for human-robot interaction," *CoRR*, vol. abs/2510.07778, 2025.

[603] B. M. Nan Sun and, Y. Li, C. Wang, D. Guo, and H. Liu, "Transforming monolithic foundation models into embodied multi-agent, architectures for human-robot collaboration," *CoRR*, vol. abs/2512.00797, 2025.

[604] Y. C. Xinyi Chen and, Y. Fu, N. Gao, J. Jia, W. Jin, H. Li, Y. Mu, J. Pang, Y. Qiao, Y. Tian, B. Wang, B. Wang, F. Wang, H. Wang, T. Wang, Z. Wang, X. Wei, C. Wu, S. Yang, J. Ye, J. Yu, J. Zeng, J. Zhang, J. Zhang, S. Zhang, F. Zheng, B. Zhou, and Y. Zhu, "Internvla-m1: A spatially guided vision-language-action framework, for generalist robot policy," *CoRR*, vol. abs/2510.13778, 2025.

[605] J. Z. Yanjiang Guo and, X. Chen, X. Ji, Y. Wang, Y. Hu, and J. Chen, "Improving vision-language-action model with online reinforcement learning," in *ICRA*. IEEE, 2025, pp. 15 665–15 672.

[606] Y. G. Anqing Jiang and, Y. Wang, Z. Sun, S. Wang, Y. Heng, H. Sun, S. Tang, L. Zhu, J. Chai, J. Wang, Z. Gu, H. Jiang, and L. Sun, "IRL-VLA: training an vision-language-action policy via reward world, model," *CoRR*, vol. abs/2508.06571, 2025.

[607] W. Xu, L. Zhuang, and L. Shan, "Kv-efficient vla: A method to speed up vision language models with rnn-gated chunked kv cache," *arXiv preprint*, 2025.

[608] X. G. Zuolei Li and, X. Wang, and J. Fu, "Latbot: Distilling universal latent actions for vision-language-action, models," *CoRR*, vol. abs/2511.23034, 2025.

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- [570] Z. D. Yifan Yang 和 T. Xie、F. Cao、P. Shen、P. Song、P. Jin、G. Sun、S. Xu、Y. You 和 J. Liu，“FPC-VLA：带有监督者的 视觉语言动作（Vision-Language-Action） 框架，用于故障预测和纠正”，CoRR，卷。绝对/2509.04018，2025。

- [571] J. L. Xin Wang 和 Z. Weng、Y. Wang、Y. Gau、T. Pang、C. Du、Y. Teng、Y. Wang、Z. Wu、X. Ma 和 Y. Jiang，“Freezevla：针对 视觉语言动作（Vision-Language-Action） 模型的 Actionfreezing 攻击”，CoRR，卷。绝对/2509.19870，2025。

- [572] Q. Z. Weiqi Li 和 R. Zhai、L. Lin 和 G. Wang，“VLA 模型比您想象的更通用：重新审视物理和空间建模”，CoRR，卷。绝对/2512.02902，2025。

- [573] T.Y.Tao Jiang 和 Y.Liu、C.Lu、J.Cui、X.Liu、S.Cheng、J.Gao、H.Xu 和 H.Zhao，“Galaxea 开放世界数据集和 G0 双系统 VLA 模型”，CoRR，卷。绝对/2509.00576，2025。

- [574] H. N. Arjun Vaithilingam Sudhakar 和 M. Reymond、M. Liu、J. Rajendran 和 S. Chandar，“多面手 hanabi 特工”，载于 ICLR。 OpenReview.net，2025 年。

- [575] G. A. 团队，“Gen-0：体现 基础模型（Foundation Models），可通过物理交互进行扩展”，arXiv 预印本，2025 年。

- [576] M. M. Ali Abouzeid、Z. Sun 和 D. Song，“Geoaware-vla：隐式几何感知 视觉语言动作（Vision-Language-Action） 模型”，CoRR，卷。绝对/2509.14117，2025。

- [577] H. Z. Wenxuan Song 和 P. Ding、C. Cui、S. Lyu、Y. Fan 和 D. Wang，“Germ：四足机器人专家混合的通才机器人模型”，载于_IROS_。 IEEE，2024 年，第 11 879-11 886 页。

- [578] P. D. Hongyin Zhu 和 S. Lyu、Y. Peng 和 D. Wang，“GEVRM：用于鲁棒视觉和操作的目标表达视频生成模型”，载于 ICLR。 OpenReview.net，2025 年。

- [579] L. G. Shunlei Li 和 J. Wang、C. Che、X. Xiao、J. Cao、Y. Hu 和 H. R. Karimi，“信息理论图融合与视觉语言动作模型，用于策略推理和双机器人控制”，CoRR，卷。绝对/2508.05342，2025。

- [580] M. C. Minghao Mao 和 J. Tao、R. Xu、Y. Yan、X. Liang、I. Laptev 和 X. Chang，“Glad：视觉语言动作模型的几何潜在蒸馏”，CoRR，卷。绝对/2512.09619，2025。

- [581] J.贾布尔，D.-K。 Kim、M. Smith、J. Patrikar、R. Ghosal、Y. Wang、A. Agha、V. J. Reddi 和 S. Omidshafiei，“不要拿着剪刀奔跑”，arXiv 预印本，2025 年。

- [582] A. A. Gemini 机器人团队和 S. Abeyruwan、J. Ainslie、J. Alayrac、M. G. Arenas、A. Balakrishna、N. Batchelor、A. Bewley、J. T. Bingham、M. Bloesch、K. Bousmalis、P. Brakel、A. Brohan、T. Buschmann、A. Byravan、S. Cabi、K. Caluwaerts、F. Casarini、C. Chan、O. Chang、L. Chappellet-Volpini、J. E. Chen、X. Chen、H. L. Jiang、K. Choromanski、A. Collister、D. B. D’Ambrosio、S. Dasari、T. Davchev、M. K. Dave、C. Devin、N. D. Palo、T. Ding、C. Doersch、A. Dostmohamed、Y. Du、D. Dwibedi、S. T. Egambaram、M. Elabd、T. Erez、X. Fang、C. Fantacci、C. Fong、E. Frey、C. Fu、R. 高、M. Giustina、K. Gopalakrishnan、L. Graesser、O. Groth、A. Gupta、

R. Hafner、S. Hansen、L. Hasenclever、 S. Haves、N. Heess、B. Hernaez、A. Hofer、J. Hsu、L.Huang、S. H. Huang、A. Iscen、M. G. Jacob、D. Jain、S. Jesmonth、A. Jindal、R. Julian、D.Kalashnikov、M. E. Karagozler、S. Karp、M. Kecman、J. C. Kew、D. Kim、F. Kim、J. Kim、T

。 Kipf、S. Kirmani、K. Konyushkova、L. Y. Ku、Y. Kuang、T. Lampe、A. Laurens、T. A. Le、I. Leal、A. X. Lee、T. E. Lee、G. Lever、J. Liang、L. Lin、F. Liu、S. Long、C. Lu、S. Maddineni、A. Majumdar、K. Maninis、A. Marmon、S. Martinez、A. H. Michaely 和 N. Milonopoulos，“Gemini 机器人 1.5：通过先进的具身推理、思维和动作转移推动通用机器人的前沿”，CoRR，卷。绝对/2510.03342，2025。

- [583] K. Z. Zijian Zhang 和 Z. Chen、J. Jang、Y. Li、C. Wang、M. Ding、D. Fox 和 H. Yao，“GRAPE：通过偏好对齐概括机器人策略”，CoRR，卷。 ABS/2411.19309，2024。

- [584] M. C. Helong Huang 和 K. Tan、X. Quan、G. Huang 和 H. Zhang，“Graphcot-vla：3d 空间感知推理 视觉语言动作（Vision-Language-Action），具有模糊指令的机器人操作模型”，CoRR，卷。绝对/2508.07650，2025。

- [585] M. Y. Shenliang Deng 和 S. Wei、H. Ma、Y. Yang、J. Chen、Z. Zhang、T. Yang、X. Zhang、H. Cui、Z. Zhang 和 H. Wang，“Graspvla：在十亿级合成动作数据上预训练的抓取 基础模型（Foundation Model）”，CoRR，卷。绝对/2505.03233，2025。

- [586] Z. F. Maélic Neau 和 P. E. Santos、A. Bosser 和 C. Buche，“Graspvla：长时程（Long-Horizon）的基于图的符号动作表示，使用 VLA 策略进行规划”，CoRR，卷。绝对/2511.04357，2025。

- [587] G. C. Ruoshi Wen、Z. Cui、M. Du、Y. Gou、Z. Han、L. Huang、M. Lei、Y. Li、Z. Li、W. Liu、Y. Liu、X. Ma、H. Niu、Y. Ouyang、Z. Ren、H. Shi、W. Xu、H. Zhang、J. Zhang、X. Zhang、L. Cheng、W.zhong、

- Y. Zhou、Z. Zhu 和 H. Li，“Gr-dexter 技术报告”，CoRR，卷。绝对/2512.24210，2025。

- [588] X. M. Yunfei Li and, J. Xu, Y. Cui, Z. Cui, Z. Han, L. Huang, T. Kong, Y. Liu, H. Niu, W. Peng, J. Qiao, Z. Ren, H. Shi, Z. Su, J. Tian, Y. Shaw, S.Zhang, L. Cheng, H. Li, and Y. Wu,“GR-RL：灵巧而精确长时程（Long-Horizon） 机器人操作，”CoRR，卷。绝对/2512.01801，2025。

- [589] Y. D. Yi Li 和 J. 张、J. Jang、M. Memmel、C. R. Garrett、F. Ramos、D. Fox、A. Li、A. Gupta 和 A. Goyal，“HAMSTER：开放世界机器人操作的分层动作模型”，载于_ICLR_。 OpenReview.net，2025 年。

- [590] 图，“Helix：用于通用人形控制的 视觉语言动作（Vision-Language-Action） 模型”，arXiv 预印本，2025 年。

- [591] P. D. Minghui Lin 和 S. Wang、Z. Zhuang、Y. Liu、X. Tong、W. Song、S. Lyu、S. Huang 和 D. Wang，“Hif-vla：针对 视觉语言动作（Vision-Language-Action） 模型的后见之明、洞察力和远见，通过运动表示”，CoRR，卷。绝对/2512.09928，2025。

- [592] B.L.Zhiying Du 和 Y.Liang、Y.Shen、H.Cao、X.Zheng、Z.Feng、Z.Wu、J.Yang 和 Y.Jiang，“Himoe-vla：通才 视觉语言动作（Vision-Language-Action） 的专家分层混合，策略”，CoRR，卷。绝对/2512.05693，2025。

- [593] R.Z.GuanxingLu 和 H.Lin、H.Zhang 和 Y.Tang，“用于机器人操作的人在环在线拒绝采样”，CoRR，卷。绝对/2510.26406，2025。

- [594] B. I. Lucy Xiaoyang Shi 和 M. R. Equi、L. Ke、K. Pertsch、Q. Vuong、J. Tanner、A. Walling、H. Wang、N. Fusai、A. Li-Bell、D. Driess、L. Groom、S. Levine 和 C. Finn，“嗨机器人：采用分层 视觉语言动作（Vision-Language-Action） 模型的开放式指令”，载于_ICML_。 OpenReview.net，2025 年。

- [595] Y.G

。张建科、X. Chen、Y. Wang、Y. Hu、C. Shi 和 J. Chen，“Hirt：利用分层机器人变压器增强机器人控制”，载于_CoRL_，系列。机器学习研究论文集，卷。 270.PMLR，2024 年，第 933-946 页。

- [596] Y. Z. Xinyu Xu 和 Y. Li、L. Han 和 C. Lu，“Humanvla：通过物理人形物体实现视觉语言引导的对象重排”，《NeurIPS》，2024 年。

- [597] C. X. Yuan 张、W. Xu、C. Ji、J. Wu 和 J. Pan，“iflybot-vla 技术报告”，CoRR，卷。绝对/2511.01914，2025。

- [598] W. G. Dekun Lu 和 K. Jia，“Imaginationpolicy：走向可推广、精确和可靠的端到端机器人操作策略”，CoRR，卷。绝对/2509.20841，2025。

- [599] H.G.Haohan Chi 和 Z.Liu、J.Liu、C.Liu、J.Li、K.Yang、Y.Yu、Z.Wang、W.Li、L.Wang、X.Hu、H.Sun、H.Zhao 和 H.Zhao，“即兴 VLA：用于驱动视觉语言动作、模型的开放权重和开放数据”，CoRR，卷。绝对/2505.23757，2025。

- [600] Z. S. Ulas Berk Karli 和 T. Fitzgerald，“洞察力：用于生成帮助的推理时间序列内省，视觉语言动作模型中的触发器”，CoRR，卷。绝对/2510.01389，2025。

- [601] H. L. Shuai Yang 和 Y. Chen、B. Wang、Y. Tian、T. Wang、H. Wang、F. Zhu、Y. Liao 和 J. Pang，“Instructvla：视觉语言动作（Vision-Language-Action） 从理解到操作的指令调优”，CoRR，卷。绝对/2507.17520，2025。

- [602] K. G. Yandu Chen 和 Y. Wen、Y. Zhao、T. Wang 和 L. Nie，“Intentionvla：用于人机交互的可概括且高效的体现意图推理”，CoRR，卷。绝对/2510.07778，2025。

- [603] B. M. Nan Sun 和 Y. Li、C. Wang、D.Guo 和 H.Liu，“将整体 基础模型（Foundation Models） 转变为具体的多智能体、人机协作架构”，CoRR，卷。绝对/2512.00797，2025。

- [604] Y.C.Xinyi Chen 和 Y.C.Xinyi Chen 和 Y.Fu、N.Gao、J.Jia、W.Jin、H.Li、Y.Mu、J.Pang、Y.Qiao、Y.Tian、B.Wang、B.Wang、F.Wang、H.Wang、T.Wang、Z.Wang、X.Wei、C.Wu、S.Yang、J.Ye、J.Yu、J.Zeng、J.Zhang、J.Zhang、S.Zhang、F.郑 B. Zhou 和 Y. Zhu，“Internvla-m1：用于通用机器人策略的空间引导 视觉语言动作（Vision-Language-Action） 框架”，CoRR，卷。绝对/2510.13778，2025。

- [605] J. Z. Yanjia Gui 和 X. Chen、X. Ji、Y. Wang、Y. Hu 和 J. Chen，“利用在线 强化学习（Reinforcement Learning） 改进 视觉语言动作（Vision-Language-Action） 模型”，载于_ICRA_。 IEEE，2025 年，第 15 665-15 672 页。

- [606] Y. G. Anqing Jiang 和 Y. Wang、Z. Sun、S. Wang、Y. Heng、H. Sun、S. Tang、L. Zhu、J. Chai、J. Wang、Z. Gu、H. Jiang 和 L. Sun，“IRL-VLA：通过奖励世界模型训练 视觉语言动作（Vision-Language-Action） 策略”，CoRR，卷。绝对/2508.06571，2025。

- [607] W. Xu、L. Zhuang 和 L. Shan，“Kv 高效 vla：一种利用 rnn 门控分块 kv 缓存加速视觉语言模型的方法”，arXiv preprint，2025 年。

- [608] X. G. Zuolei Li 和 X. Wang 以及 J. Fu，“Latbot：提取 视觉语言动作（Vision-Language-Action）模型的通用潜在动作”，CoRR，卷。绝对/2511.23034，2025。

[609] Y. N. Bahey Tharwat and, A. Abouzeid, and I. Reid, "Latent action pretraining through world modeling," *CoRR*, vol. abs/2509.18428, 2025.

[610] K. C. Shuhan Tan and, Y. Chen, R. Tian, Y. You, Y. Wang, W. Luo, Y. Cao, P. Krähenbühl, M. Pavone, and B. Ivanovic, "Latent chain-of-thought world modeling for end-to-end driving," *CoRR*, vol. abs/2512.10226, 2025.

[611] W. C. Ameesh Shah and, A. Godbole, F. Mora, S. A. Seshia, and S. Levine, "Learning affordances at inference-time for vision-language-action, models," *CoRR*, vol. abs/2510.19752, 2025.

[612] K. D. G. Justin Williams and, R. George, and M. Sarkar, "Lite VLA: efficient vision-language-action control on cpu-bound, edge robots," *CoRR*, vol. abs/2511.05642, 2025.

[613] J. S. Yi Yang and, S. Kou, Y. Wang, and Z. Deng, "Lohovla: A unified vision-language-action model for long-horizon, embodied tasks," *CoRR*, vol. abs/2506.00411, 2025.

[614] X. G. Xiaofan Wang and, J. Fu, Z. Li, D. Fortier, G. Mullins, A. Kolobov, and B. Guo, "Lola: Long horizon latent action learning for general robot manipulation," *CoRR*, vol. abs/2512.20166, 2025.

[615] P. D. Yiguo Fan and, S. Bai, X. Tong, Y. Zhu, H. Lu, F. Dai, W. Zhao, Y. Liu, S. Huang, Z. Fan, B. Chen, and D. Wang, "Long-vla: Unleashing long-horizon capability of vision language action, model for robot manipulation," *CoRR*, vol. abs/2508.19958, 2025.

[616] T. Z. Boyuan Chen and, H. Geng, K. Song, C. Zhang, P. Li, W. T. Freeman, J. Malik, P. Abbeel, R. Tedrake, V. Sitzmann, and Y. Du, "Large video planner enables generalizable robot control," *CoRR*, vol. abs/2512.15840, 2025.

[617] K. G. Yi Yang and, Y. Wen, H. Li, Y. Zhao, T. Wang, and X. Liu, "Maniagent: An agentic framework for general robotic manipulation," *CoRR*, vol. abs/2510.11660, 2025.

[618] X. L. Yi Yang and, Y. Chen, J. Song, Y. Wang, Z. Xiao, J. Su, Y. Qiaoben, P. Liu, and Z. Deng, "Mantis: A versatile vision-language-action model with disentangled, visual foresight," *CoRR*, vol. abs/2511.16175, 2025.

[619] J. L. Chenyang Gu and, H. Chen, R. Huang, Q. Wuwu, Z. Liu, X. Li, Y. Li, R. Zhang, P. Jia, P. Heng, and S. Zhang, "Manualvla: A unified VLA model for chain-of-thought manual generation, and robotic manipulation," *CoRR*, vol. abs/2512.02013, 2025.

[620] W. G. Runhao Li and, Z. Wu, C. Wang, H. Deng, Z. Weng, Y. Tan, and Z. Wang, "MAP-VLA: memory-augmented prompting for vision-language-action model, in robotic manipulation," *CoRR*, vol. abs/2511.09516, 2025.

[621] J. Z. Zhihao Zhan and, L. Zhang, Q. Lv, H. Liu, J. Zhang, W. Li, Z. Chen, T. Chen, K. Wang, L. Lin, and G. Wang, "$\mathscr{E}_0$: Enhancing generalization and fine-grained, control in VLA models via continuized discrete diffusion," *CoRR*, vol. abs/2511.21542, 2025.

[622] Ö. E. Y. Moritz Reuss and, F. Wenzel, and R. Lioutikov, "Multimodal diffusion transformer: Learning versatile behavior from, multimodal goals," in *Robotics: Science and Systems*, 2024.

[623] J. P. Ajay Sridhar and, S. Sharma, and C. Finn, "Memer: Scaling up memory for robot control via experience retrieval," *CoRR*, vol. abs/2510.20328, 2025.

[624] B. X. Hao Shi and, Y. Liu, L. Sun, F. Liu, T. Wang, E. Zhou, H. Fan, X. Zhang, and G. Huang, "Memoryvla: Perceptual-cognitive memory in vision-language-action models, for robotic manipulation," *CoRR*, vol. abs/2508.19236, 2025.

[625] Z. Y. Chen Li and, H. Zhang, F. Chen, C. Zhu, A. Bolimera, and M. Savvides, "Metavla: Unified meta co-training for efficient embodied adaption," *CoRR*, vol. abs/2510.05580, 2025.

[626] D. K. Suhyeok Jang and, C. Kim, Y. Kim, and J. Shin, "Verifier-free test-time sampling for vision language action models," *CoRR*, vol. abs/2510.05681, 2025.

[627] I. Z. Haoyun Li and, R. Ouyang, X. Wang, Z. Zhu, Z. Yang, Z. Zhang, B. Wang, C. Ni, W. Qin, X. Chen, Y. Ye, G. Huang, Z. Song, and X. Wang, "Mimicdreamer: Aligning human and robot demonstrations for scalable, VLA training," *CoRR*, vol. abs/2509.22199, 2025.

[628] D. Z. Haoyu Fu and, Z. Zhao, J. Cui, H. Xie, B. Wang, G. Chen, D. Liang, and X. Bai, "Minddrive: A vision-language-action model for autonomous driving, via online reinforcement learning," *CoRR*, vol. abs/2512.13636, 2025.

[629] S. X. Peijun Tang and, B. Sun, B. Huang, K. Luo, H. Yang, W. Jin, and J. Wang, "Mind to hand: Purposeful robotic control via embodied reasoning," *CoRR*, vol. abs/2512.08580, 2025.

[630] X. W. Zhenhan Yin and, J. Jiang, K. Deng, P. Chen, S. Li, C. Liu, X. Xu, J. Song, L. Gao, and H. T. Shen, "Mivla: Towards generalizable vision-language-action model with human-robot, mutual imitation pre-training," *CoRR*, vol. abs/2512.15411, 2025.

[631] J. L. Zhuoyang Liu and, J. Xu, N. Han, C. Gu, H. Chen, K. Zhou, R. Zhang, K. Hsieh, K. Wu, Z. Che, J. Tang, and S. Zhang, "MLA: A multisensory language-action model for multimodal understanding, and forecasting in robotic manipulation," *CoRR*, vol. abs/2509.26642, 2025.

[632] X. C. Haotian Liang and, B. Wang, M. Chen, Y. Liu, Y. Zhang, Z. Chen, T. Yang, Y. Chen, J. Pang, D. Liu, X. Yang, Y. Mu, W. Shao, and P. Luo, "MM-ACT: learn from multimodal parallel generation to act," *CoRR*, vol. abs/2512.00975, 2025.

[633] G. W. Dong Jing and, J. Liu, W. Tang, Z. Sun, Y. Yao, Z. Wei, Y. Liu, Z. Lu, and M. Ding, "Mixture of horizons in action chunking," *CoRR*, vol. abs/2511.19433, 2025.

[634] N. S. Dmytro Kuzmenko and, "Moira: Modular instruction routing architecture for multi-task robotics," *CoRR*, vol. abs/2507.01843, 2025.

[635] M. X. Chengshu Li and, A. Bahety, H. Yin, Y. Jiang, H. Huang, J. Wong, S. Garlanka, C. Gokmen, R. Zhang, W. Liu, J. Wu, R. Martín-Martín, and L. Fei-Fei, "Momagen: Generating demonstrations under soft and hard constraints, for multi-step bimanual mobile manipulation," *CoRR*, vol. abs/2510.18316, 2025.

[636] Y. Z. Zhenyu Wu and, X. Xu, Z. Wang, and H. Yan, "Momanipvla: Transferring vision-language-action models for general, mobile manipulation," in *CVPR*. Computer Vision Foundation / IEEE, 2025, pp. 1714–1723.

[637] Y. W. Shuo Wang and, W. Li, Y. Wang, M. Chen, K. Wang, Z. Su, X. Cai, Y. Jin, D. Li, and Z. Fan, "Monodream: Monocular vision-language navigation with panoramic dreaming," *CoRR*, vol. abs/2508.02549, 2025.

[638] T. X. Austin Stone and, Y. Lu, K. Gopalakrishnan, K. Lee, Q. Vuong, P. Wohlhart, S. Kirmani, B. Zitkovich, F. Xia, C. Finn, and K. Hausman, "Open-world object manipulation using pre-trained vision-language models," in *CoRL*, ser. Proceedings of Machine Learning Research, vol. 229. PMLR, 2023, pp. 3397–3417.

[639] W. S. Han Zhao and, D. Wang, X. Tong, P. Ding, X. Cheng, and Z. Ge, "More: Unlocking scalability in reinforcement learning for quadruped, vision-language-action models," in *ICRA*. IEEE, 2025, pp. 11 212–11 218.

[640] R. Z. Chengbo Yuan and, M. Liu, Y. Hu, S. Wang, L. Yi, C. Wen, S. Zhang, and Y. Gao, "Motiontrans: Human VR data enable motion-level learning for robotic, manipulation policies," *CoRR*, vol. abs/2509.17759, 2025.

[641] A. M. Zhenyu Wu and, X. Xu, H. Yin, Y. Liang, Z. Wang, J. Lu, and H. Yan, "Moto: A zero-shot plug-in interaction-aware navigation for general, mobile manipulation," *CoRR*, vol. abs/2509.01658, 2025.

[642] J. Z. Alexander Spiridonov and, N. Nikolov, L. V. Gool, and D. P. Paudel, "Generalist robot manipulation beyond action labeled data," *CoRR*, vol. abs/2509.19958, 2025.

[643] H. T. Hongzhe Bi and, S. Xie, Z. Wang, S. Huang, H. Liu, R. Zhao, Y. Feng, C. Xiang, Y. Rong, H. Zhao, H. Liu, Z. Su, L. Ma, H. Su, and J. Zhu, "Motus: A unified latent action world model," *CoRR*, vol. abs/2512.13030, 2025.

[644] X. S. Pu Zhao and, Z. Kong, Y. Shen, S. Chang, A. Akbari, T. Rupprecht, L. Lu, E. Nan, C. Yang, Y. He, W. Shi, X. Xu, Y. Huang, W. Jiang, W. Wang, Y. Chen, Y. He, and Y. Wang, "Open-source multimodal moxin models with moxin-vlm and moxin-vla," *CoRR*, vol. abs/2512.22208, 2025.

[645] F. J. Peilong Han and, M. Zhang, Y. Qiu, H. Tang, Y. Zheng, T. Wang, and J. Hao, "MUVLA: learning to explore object navigation via map understanding," *CoRR*, vol. abs/2509.25966, 2025.

[646] Y. J. An-Chieh Cheng and, Z. Yang, X. Zou, J. Kautz, E. Biyik, H. Yin, S. Liu, and X. Wang, "Navila: Legged robot vision-language-action model for navigation," *CoRR*, vol. abs/2412.04453, 2024.

[647] M. P. Sajjad Pakdamansavoji and, A. Sigal, Z. Li, R. H. Yang, and A. Rasouli, "Improving robotic manipulation robustness via NICE scene surgery," *CoRR*, vol. abs/2511.22777, 2025.

[648] L. Magne, A. Awadalla, G. Wang, Y. Xu, J. Belofsky, F. Hu, J. Kim, L. Schmidt, G. Gkioxari, J. Kautz, Y. Yue, Y. Choi, Y. Zhu, and L. J. Fan, "Nitrogen: An open foundation model for generalist gaming agents," *arXiv preprint*, 2026.

[649] D. D. Rokas Bendikas and, M. Peschl, S. Haresh, and P. Mazzaglia, "Focusing on what matters: Object-agent-centric tokenization for vision, language action models," *CoRR*, vol. abs/2509.23655, 2025.

[650] T. H. Khoa Vo and, Y. Ikebe, T. Pham, N. Chung, M. N. Vu, D. N. H. Minh, A. Nguyen, A. Gunderman, C. Rainwater, and N. Le, "Clutter-resistant vision-language-action models through object-centric, and geometry grounding," *CoRR*, vol. abs/2512.22519, 2025.

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- [609] Y. N. Bahey Tharwat 和 A. Abouzeid 以及 I. Reid，“通过世界建模进行潜在动作预训练”，CoRR，卷。绝对/2509.18428，2025。

- [610] K. C. Shuhan Tan 和 Y. Chen、R. Tian、Y. You、Y. Wang、W. Luo、Y. Cao、P. Krěnbùhl、M. Pavone 和 B. Ivanovic，“端到端驾驶的潜在思想链世界建模”，CoRR，卷。 ABS/2512.10226，2025。

- [611] W. C. Ameesh Shah 和 A. Godbole、F. Mora、S. A. Seshia 和 S. Levine，“在视觉语言动作、模型的推理时间学习 可供性（Affordances）”，CoRR，卷。绝对/2510.19752，2025。

- [612] K. D. G. Justin Williams、R. George 和 M. Sarkar，“Lite VLA：对 CPU 限制的边缘机器人进行高效 视觉语言动作（Vision-Language-Action） 控制”，CoRR，卷。绝对/2511.05642，2025。

- [613] J. S. Yi Yang 和 S. Kou、Y. Wang 和 Z. Deng，“Lohovla：长时程（Long-Horizon） 的统一 视觉语言动作（Vision-Language-Action） 模型，具体任务”，CoRR，卷。绝对/2506.00411，2025。

- [614] X.G.Xiaofan Wang 和 J.Fu、Z.Li、D.Fortier、G.Mullins、A.Kolobov 和 B.Guo，“Lola：通用机器人操作的长视野潜在动作学习”，CoRR，卷。 ABS/2512.20166，2025。

- [615] P. D. Yiguo Fan、S. Bai、X. Tong、Y. Zhu、H. Lu、F. Dai、W. Zhao、Y. Liu、S. Huang、Z. Fan、B. Chen 和 D. Wang，“Long-vla：释放 长时程（Long-Horizon） 视觉语言动作能力，机器人操作模型”，CoRR，卷。绝对/2508.19958，2025。

- [616] T. Z. Boyuan Chen 和 H. Geng、K. Song、C. Zhu、P. Li、W. T. Freeman、J. Malik、P. Abbeel、R. Tedrake、V. Sitzmann 和 Y. Du，“大型视频规划器可实现通用机器人控制”，CoRR，卷。绝对/2512.15840，2025。

- [617] K. G. Yi Yang 和 Y. Wen、H. Li、Y. Zhao、T. Wang 和 X. Liu，“Maniagent：通用机器人操作的智能体框架”，CoRR，卷。绝对/2510.11660，2025。

- [618] X. L. Yi Yang 和 Y. Chen、J. Song、Y. Wang、Z. Xiao、J. Su、Y. Qiaoben、P. Liu 和 Z. Deng，“Mantis：一种具有解开、视觉远见的多功能视觉语言-动作模型”，CoRR，卷。绝对/2511.16175，2025。

- [619] J. L. Chenyang Gu 和 H. Chen、R. Huang、Q. Wuwu、Z. Liu、X. Li、Y. Li、R. Zhu、P. Jia、P. Heng 和 S. Zhang，“Manualvla：用于思想链手动生成和机器人操作的统一 VLA 模型”，CoRR，卷。绝对/2512.02013，2025。

- [620] W. G. Runhao Li 和 Z. Wu、C. Wang、H. Deng、Z. Weng、Y. Tan 和 Z. Wang，“MAP-VLA：机器人操作中 视觉语言动作（Vision-Language-Action） 模型的记忆增强提示”，CoRR，卷。绝对/2511.09516，2025。

- [621] J.Z.Zhihao Zhan 和 L.Zhang、Q.Lv、H.Liu、J.Zhang、W.Li、Z.Chen、T.Chen、K.Wang、L.Lin 和 G.Wang，“_E_0：通过连续离散扩散增强 VLA 模型的泛化和细粒度控制”，CoRR，卷。绝对/2511.21542，2025。

- [622] ¡O。 E. Y. Moritz Reuss、F. Wenzel 和 R. Lioutikov，“多模态扩散变压器：从多模态目标中学习多种行为”，《机器人学：科学与系统》，2024 年。

- [623] J. P. Ajay Sridhar、S. Sharma 和 C. Finn，“Memer：通过经验检索扩展机器人控制的内存”，CoRR，卷。绝对/2510.20328，2025。

- [624] B.X.郝石和，Y.Liu，L.Sun，F.Liu，T.Wang，E.Zhou，H

。 Fan、X.Zhang 和 G.Huang，“Memoryvla：视觉语言动作（Vision-Language-Action） 模型中的感知认知记忆，用于机器人操作”，CoRR，卷。绝对/2508.19236，2025。

- [625] Z. Y. Chen Li 和 H. Zhang、F. Chen、C. Zhu、A. Bolimera 和 M. Savvides，“Metavla：用于高效体现适应的统一元协同训练”，CoRR，卷。绝对/2510.05580，2025。

- [626] D. K. Suhyeok Jang 和 C. Kim、Y. Kim 和 J. Shin，“视觉语言动作模型的无验证测试时间采样”，CoRR，卷。绝对/2510.05681，2025。

- [627] I.Z.Haoyun Li 和 R.Ouyang、X.Wang、Z.Zhu、Z.Yang、Z.Zhang、B.Wang、C.Ni、W.Qin、X.Chen、Y.Ye、G.Huang、Z.Song 和 X.Wang，“Mimicdreamer：调整人类和机器人演示以进行可扩展的 VLA 训练”，CoRR，卷。绝对/2509.22199，2025。

- [628] D.Z.Haoyu Fu 和 Z.Zhao、J.Cui、H.Xie、B.Wang、G.Chen、D.Liang 和 X.Bai，“Minddrive：用于自动驾驶的 视觉语言动作（Vision-Language-Action） 模型，通过在线 强化学习（Reinforcement Learning）”，CoRR，卷。绝对/2512.13636，2025。

- [629] S. X. Peijun Tang 和 B. Sun、B. Huang、K. Luo、H. Yang、W. Jin 和 J. Wang，“心手：通过具身推理进行有目的的机器人控制”，CoRR，卷。绝对/2512.08580，2025。

- [630] X.W.Zhenhan Yin 和 J.Jiang、K.Deng、P.Chen、S.Li、C.Liu、X.Xu、J.Song、L.Gao 和 H.T.Shen，“Mivla：利用人机、相互模仿预训练实现可推广的 视觉语言动作（Vision-Language-Action） 模型”，CoRR，卷。绝对/2512.15411，2025。

- [631] J. L. Zhuoyang Liu 和 J. Xu、N. Han、C. Gu、H. Chen、K. Zhou、R. Zhang、K. Hsieh、K. Wu、Z. Che、J. Tang 和 S. Zhang，“MLA：用于多模态理解和机器人操作预测的多感官语言动作模型”，CoRR，卷。绝对/2509.26642，2025。

- [632] X.C.Haotian Liang 和 B.Wang、M.Chen、Y.Liu、Y.Zhang、Z.Chen、T.Yang、Y.Chen、J.Pang、D.Liu、X.Yang、Y.Mu、W.Shao 和 P.Luo，“MM-ACT：从多模态并行生成中学习到行动”，CoRR，卷。绝对/2512.00975，2025。

- [633] G. W. Dong Jing、J. Liu、W. Tang、Z. Sun、Y. Yao、Z. Wei、Y. Liu、Z. Lu 和 M. Ding，“动作分块（Action Chunking） 中的地平线混合”，CoRR，卷。绝对/2511.19433，2025。

- [634] N. S. Dmytro Kuzmenko 和，“Moira：多任务机器人的模块化指令路由架构”，CoRR，卷。绝对/2507.01843，2025。

- [635] M. X. Chengshu Li 和 A. Bahety、H. Yin、Y. Jiang、H. Huang、J. Wong、S. Garlanka、C. Gokmen、R. Zhu、W. Liu、J. Wu、R. Mart´ınMart´ın 和 L. Fei-Fei，“Momagen：在软约束和硬约束下生成演示，用于多步双手移动操作”，CoRR，卷。绝对/2510.18316，2025。

- [636] Y.Z.Zhenyu Wu 和 X.Xu、Z.Wang 和 H.Yan，“Momanipvla：传输 视觉语言动作（Vision-Language-Action） 模型进行一般移动操作”，载于 CVPR。计算机视觉基金会/IEEE，2025 年，第 1714-1723 页。

- [637] Y. W. Shuo Wang 和 W. Li、Y. Wang、M. Chen、K. Wang、Z. Su、X. Cai、Y. Jin、D. Li 和 Z. Fan，“Monodream：单眼视觉语言导航与全景梦”，CoRR，卷。绝对/2508.02549，2025。

- [638] T. X. Austin Stone 和 Y. Lu、K. Gopalakrishnan、K. Lee、Q. Vuong、P. Wohlhart、S. Kirmani、B. Zitkovich、F. Xia、C. Finn 和 K

。 Hausman，“使用预先训练的视觉语言模型进行开放世界对象操作”，载于_CoRL_，ser。机器学习研究论文集，卷。 229.PMLR，2023 年，第 3397-3417 页。

- [639] W. S. Han Zhu 和 D. Wang、X. Tong、P. Ding、X. Cheng 和 Z. Ge，“更多：为四足动物、视觉语言动作（Vision-Language-Action） 模型解锁 强化学习（Reinforcement Learning） 的可扩展性”，载于_ICRA_。 IEEE，2025 年，第 11 212-11 218 页。

- [640] R. Z. Chengbo Yuan 和 M. Liu、Y. Hu、S. Wang、L. Yi、C. Wen、S. Zhang 和 Y. Gau，“Motiontrans：人类 VR 数据支持机器人操作策略的运动级学习”，CoRR，卷。绝对/2509.17759，2025。

- [641] A.M.Zhenyu Wu 和 X.Xu、H.Yin、Y.Liang、Z.Wang、J.Lu 和 H.Yan，“Moto：用于一般移动操作的零样本插件交互感知导航”，CoRR，卷。绝对/2509.01658，2025。

- [642] J. Z. Alexander Spiridonov 和 N. Nikolov、L. V. Gool 和 D. P. Paudel，“超越动作标记数据的通才机器人操纵”，CoRR，卷。绝对/2509.19958，2025。

- [643] H. T. Hongzhe Bi and, S. Xie, Z. Wang, S. Huang, H. Liu, R. Zhu, Y. Feng, C. Xiang, Y. Rong, H. Zhao, H. Liu, Z. Su, L. Ma, H. Su, and J. Zhu, “Motus: 统一的潜在动作 世界模型（World Model）,” CoRR , vol.绝对/2512.13030，2025。

- [644] X. S. Pu Zhao 和 Z. Kong、Y. Shen、S. Chang、A. Akbari、T. Rupprecht、L. Lu、E. Nan、C. Yang、Y. He、W. Shi、X. Xu、Y. Huang、W. Jiang、W. Wang、Y. Chen、Y. He 和 Y. Wang，“使用 moxin-vlm 和 moxin-vla 的开源多模态 Moxin 模型”，CoRR ，卷。绝对/2512.22208，2025。

- [645] F. J. Peilong Han 和 M.Zhang、Y.Qiu、H.Tang、Y.Zheng、T.Wang 和 J.Hao，“MUVLA：通过地图理解学习探索对象导航”，CoRR，卷。绝对/2509.25966，2025。

- [646] Y. J. An-Chieh Cheng 和 Z. Yang、X. Zou、J. Kautz、E. Biyik、H. Yin、S. Liu 和 X. Wang，“Navila：用于导航的腿式机器人 视觉语言动作（Vision-Language-Action） 模型”，CoRR，卷。绝对/2412.04453，2024。

- [647] M. P. Sajjad Pakdamansavoji 和 A. Sigal、Z. Li、R. H. Yang 和 A. Rasouli，“通过 NICE 场景手术提高机器人操作稳健性”，CoRR，卷。绝对/2511.22777，2025。

- [648] L. Magne、A. Awadalla、G. Wang、Y. Xu、J. Belofsky、F. Hu、J. Kim、L. Schmidt、G. Gkioxari、J. Kautz、Y. Yue、Y. Choi、Y. Zhu 和 L. J. Fan，“Nitrogen：面向通才游戏智能体的开放 基础模型（Foundation Model）”，arXiv 预印本， 2026 年。

- [649] D. D. Rokas Bendikas 和 M. Peschl、S. Haresh 和 P. Mazzaglia，“关注重要的事情：以对象智能体为中心的视觉标记化、语言动作模型”，CoRR，卷。绝对/2509.23655，2025。

- [650] T. H. Khoa Vo 和 Y. Ikebe、T. Pham、N. Chung、M. N. Vu、D. N. H. Minh、A. Nguyen、A. Gunderman、C. Rainwater 和 N. Le，“通过以对象为中心的抗干扰 视觉语言动作（Vision-Language-Action） 模型和几何 落地关联（Grounding）”，CoRR，卷。绝对/2512.22519，2025。

[651] S. Y. Julong Wei and, P. Li, Q. Hu, Z. Gan, and W. Ding, "Occllama: An occupancy-language-action generative world model for, autonomous driving," *CoRR*, vol. abs/2409.03272, 2024.

[652] L. K. Ruixun Liu and, D. Li, and H. Zhao, "Occvla: Vision-language-action model with implicit 3d occupancy supervision," *CoRR*, vol. abs/2509.05578, 2025.

[653] H. D. Tianyi Zhang and, H. Hao, Y. Qiao, J. Dai, and Z. Hou, "Grounding actions in camera space: Observation-centric vision-language-action, policy," *CoRR*, vol. abs/2508.13103, 2025.

[654] S. C. Zihao Wang and, Z. Mu, H. Lin, C. Zhang, X. Liu, Q. Li, A. Liu, X. S. Ma, and Y. Liang, "Omnijarvis: Unified vision-language-action tokenization enables open-world, instruction following agents," in *NeurIPS*, 2024.

[655] Q. N. Pei Liu and, X. Lu, H. Liu, W. Ma, D. She, P. Jia, X. Lang, and J. Ma, "Omnireason: A temporal-guided vision-language-action framework for, autonomous driving," *CoRR*, vol. abs/2509.00789, 2025.

[656] C. C. Huaihai Lyu and, S. Xie, P. Wang, X. Chen, S. Zhang, and C. Xu, "Omnisat: Compact action token, faster auto regression," *CoRR*, vol. abs/2510.09667, 2025.

[657] C. G. Noriaki Hirose and, D. Shah, and S. Levine, "Omnivla: An omni-modal vision-language-action model for robot navigation," *CoRR*, vol. abs/2509.19480, 2025.

[658] R. N. Fanqi Lin and, Y. Hu, J. You, J. Zhao, and Y. Gao, "Onetwovla: A unified vision-language-action model with adaptive, reasoning," *CoRR*, vol. abs/2505.11917, 2025.

[659] M. L. Zihao Wang and, K. He, X. Wang, Z. Mu, A. Liu, and Y. Liang, "Openha: A series of open-source hierarchical agentic models in minecraft," *CoRR*, vol. abs/2509.13347, 2025.

[660] K. P. Moo Jin Kim and, S. Karamcheti, T. Xiao, A. Balakrishna, S. Nair, R. Rafailov, E. P. Foster, P. R. Sanketi, Q. Vuong, T. Kollar, B. Burchfiel, R. Tedrake, D. Sadigh, S. Levine, P. Liang, and C. Finn, "Openvla: An open-source vision-language-action model," in *CoRL*, ser. Proceedings of Machine Learning Research, vol. 270. PMLR, 2024, pp. 2679–2713.

[661] F. L. Huang Huang and, L. Fu, T. Wu, M. Mukadam, J. Malik, K. Goldberg, and P. Abbeel, "OTTER: A vision-language-action model with text-aware visual feature, extraction," in *ICML*. OpenReview.net, 2025.

[662] S. L. Xiaopeng Lin and, B. Yu, R. Yang, C. Wu, Y. Miao, Y. Jin, Y. Shi, C. Huang, B. Cheng, and K. Chen, "Physbrain: Human egocentric data as a bridge from vision language, models to physical intelligence," *CoRR*, vol. abs/2512.16793, 2025.

[663] J. L. Zhihao Wang and, J. Zheng, W. Zhang, D. Liu, Y. Zheng, H. Niu, J. Yu, and X. Zhan, "Physiagent: An embodied agent framework in physical world," *CoRR*, vol. abs/2509.24524, 2025.

[664] K. Chen, Z. Liu, T. Zhang, Z. Guo, S. Xu, H. Lin, H. Zang, Q. Zhang, Z. Yu, G. Fan, T. Huang, Y. Wang, and C. Yu, "π_{rl}: Online RL fine-tuning for flow-based vision-language-action models," *CoRR*, vol. abs/2510.25889, 2025.

[665] G. S. Wenqi Liang and, Y. He, J. Dong, S. Dai, I. Laptev, S. H. Khan, and Y. Cong, "Pixelvla: Advancing pixel-level understanding in vision-language-action, model," *CoRR*, vol. abs/2511.01571, 2025.

[666] E. L. H. Yi Zhang and, K. Chao, N. Petrovic, Y. Song, C. Wu, and A. Knoll, "A unified perception-language-action framework for adaptive autonomous, driving," *CoRR*, vol. abs/2507.23540, 2025.

[667] J. Z. Hang Yu and, Y. Liu, K. Li, C. Ma, D. Zhang, Y. Hu, G. Chen, J. Xie, J. Guo, J. Zhao, and Y. Gao, "Point what you mean: Visually grounded instruction policy," *CoRR*, vol. abs/2512.18933, 2025.

[668] X. W. Ziwen Li and, H. Zhang, R. Chen, R. Lin, X. He, H. Huang, Y. Guo, F. Karray, T. Liu, and M. Gong, "Posa-vla: Enhancing action generation via pose-conditioned anchor, attention," *CoRR*, vol. abs/2512.03724, 2025.

[669] Z. H. Jiahui Zhang and, C. Gu, Z. Ma, and L. Zhang, "Reinforcing action policies by prophesying," *CoRR*, vol. abs/2511.20633, 2025.

[670] H. K. Seongmin Park and, W. Jeon, J. Yang, B. Jeon, Y. Oh, and J. Choi, "Quantization-aware imitation-learning for resource-efficient robotic, control," *CoRR*, vol. abs/2412.01034, 2024.

[671] Y. C. Yixuan Li and, M. Zhou, and H. Li, "Qdepth-vla: Quantized depth prediction as auxiliary supervision for, vision-language-action models," *CoRR*, vol. abs/2510.14836, 2025.

[672] Q. V. Yevgen Chebotar and, K. Hausman, F. Xia, Y. Lu, A. Irpan, A. Kumar, T. Yu, A. Herzog, K. Pertsch, K. Gopalakrishnan, J. Ibarz, O. Nachum, S. A. Sontakke, G. Salazar, H. T. Tran, J. Peralta, C. Tan, D. Manjunath, J. Singh, B. Zitkovich, T. Jackson, K. Rao, C. Finn, and S. Levine, "Q-transformer: Scalable offline reinforcement learning via autoregressive, q-functions," in *CoRL*, ser. Proceedings of Machine Learning Research, vol. 229. PMLR, 2023, pp. 3909–3928.

[673] P. D. Xinyang Tong and, Y. Fan, D. Wang, W. Zhang, C. Cui, M. Sun, H. Zhao, H. Zhang, Y. Dang, S. Huang, and S. Lyu, "Quart-online: Latency-free multimodal large language model for quadruped, robot learning," in *ICRA*. IEEE, 2025, pp. 9533–9539.

[674] L. F. Justin Yu and, H. Huang, K. El-Refai, R. A. Ambrus, R. Cheng, M. Z. Irshad, and K. Goldberg, "Real2render2real: Scaling robot data without dynamics simulation or, robot hardware," *CoRR*, vol. abs/2505.09601, 2025.

[675] L. W. Songming Liu and, B. Li, H. Tan, H. Chen, Z. Wang, K. Xu, H. Su, and J. Zhu, "RDT-1B: a diffusion foundation model for bimanual manipulation," in *ICLR*. OpenReview.net, 2025.

[676] Z. Y. Dapeng Zhang and, Z. Chen, C. Liao, Y. Chen, F. Shen, Q. Zhou, and T. Chua, "Reasoning-vla: A fast and general vision-language-action reasoning, model for autonomous driving," *CoRR*, vol. abs/2511.19912, 2025.

[677] Y. Z. Pengxiang Li and, Y. Wang, H. Wang, H. Zhao, J. Liu, X. Zhan, K. Zhan, and X. Lang, "Discrete diffusion for reflective vision-language-action models in, autonomous driving," *CoRR*, vol. abs/2509.20109, 2025.

[678] S. D. Kaustubh Sridhar and, D. Jayaraman, and I. Lee, "REGENT: A retrieval-augmented generalist agent that can act in-context, in new environments," in *ICLR*. OpenReview.net, 2025.

[679] C. W. Wenlong Huang and, Y. Li, R. Zhang, and L. Fei-Fei, "Rekep: Spatio-temporal reasoning of relational keypoint constraints, for robotic manipulation," in *CoRL*, ser. Proceedings of Machine Learning Research, vol. 270. PMLR, 2024, pp. 4573–4602.

[680] Z. Z. Yajat Yadav and, A. Wagenmaker, K. Pertsch, and S. Levine, "Robust finetuning of vision-language-action robot policies via parameter, merging," *CoRR*, vol. abs/2512.08333, 2025.

[681] T. C. Jiyeon Koo and, H. Kang, E. Pyo, T. G. Oh, T. Kim, and A. J. Choi, "Retovla: Reusing register tokens for spatial reasoning in vision-language-action, models," *CoRR*, vol. abs/2509.21243, 2025.

[682] J. Z. Sombit Dey and, N. Nikolov, L. V. Gool, and D. P. Paudel, "Revla: Reverting visual domain limitation of robotic foundation models," in *ICRA*. IEEE, 2025, pp. 8679–8686.

[683] S. D. Kaustubh Sridhar and, D. Jayaraman, and I. Lee, "RICL: adding in-context adaptability to pre-trained vision-language-action, models," *CoRR*, vol. abs/2508.02062, 2025.

[684] G. V. Konstantinos Bousmalis and, D. Rao, C. M. Devin, A. X. Lee, M. B. Villalonga, T. Davchev, Y. Zhou, A. Gupta, A. Raju, A. Laurens, C. Fantacci, V. Dalibard, M. Zambelli, M. F. Martins, R. Pevceviciute, M. Blokzijl, M. Denil, N. Batchelor, T. Lampe, E. Parisotto, K. Zolna, S. E. Reed, S. G. Colmenarejo, J. Scholz, A. Abdolmaleki, O. Groth, J. Regli, O. Sushkov, T. Rothörl, J. E. Chen, Y. Aytar, D. Barker, J. Ortiz, M. A. Riedmiller, J. T. Springenberg, R. Hadsell, F. Nori, and N. Heess, "Robocat: A self-improving generalist agent for robotic manipulation," *Trans. Mach. Learn. Res.*, vol. 2024, 2024.

[685] C. Y. Zongzheng Zhang and, H. Xu, M. Liao, X. Qi, H. Gao, Z. Wang, and H. Zhao, "Robochemist: Long-horizon and safety-compliant robotic chemical experimentation," *CoRR*, vol. abs/2509.08820, 2025.

[686] H. L. Qingwen Bu and, L. Chen, J. Cai, J. Zeng, H. Cui, M. Yao, and Y. Qiao, "Towards synergistic, generalized, and efficient dual-system for robotic, manipulation," *CoRR*, vol. abs/2410.08001, 2024.

[687] M. L. Xinghang Li and, H. Zhang, C. Yu, J. Xu, H. Wu, C. Cheang, Y. Jing, W. Zhang, H. Liu, H. Li, and T. Kong, "Vision-language foundation models as effective robot imitators," in *ICLR*. OpenReview.net, 2024.

[688] W. Z. Weixin Mao and, Z. Jiang, D. Fang, Z. Zhang, Z. Lan, F. Jia, T. Wang, H. Fan, and O. Yoshie, "Robomatrix: A skill-centric hierarchical framework for scalable, robot task planning and execution in open-world," *CoRR*, vol. abs/2412.00171, 2024.

[689] H. X. Weifan Guan and, C. Zhang, A. Li, Q. Hu, and J. Cheng, "Roboneuron: A modular framework linking foundation models and ROS, for embodied AI," *CoRR*, vol. abs/2512.10394, 2025.

[690] J. W. Shunlei Li and, R. Dai, W. Ma, W. Y. Ng, Y. Hu, and Z. Li, "Robonurse-vla: Robotic scrub nurse system based on vision-language-action, model," in *IROS*. IEEE, 2025, pp. 3986–3993.

[691] J. F. Siyin Wang and, F. Liu, X. He, H. Wu, J. Shi, K. Huang, Z. Fei, J. Gong, Z. Wu, Y. Jiang, S. Ng, T. Chua, and X. Qiu, "Roboomni: Proactive robot manipulation in omni-modal context," *CoRR*, vol. abs/2510.23763, 2025.

[692] C. C. Huajie Tan and, X. Chen, Y. Ji, Z. Zhao, X. Hao, Y. Lyu, M. Cao, J. Zhao, H. Lyu, E. Zhou, N. Chen, Y. Fu, C. Peng, W. Guo, D. Liang, Z. Chen, M. Lyu, C. He, Y. Ao, Y. Lin, P. Wang, Z. Wang, and S. Zhang, "Roboos-next: A unified memory-based framework for

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- [651] S. Y. Julong Wei 和 P. Li、Q. Hu、Z. Gan 和 W. Ding，“Occllama：自动驾驶的占用语言动作生成 世界模型（World Model）”，CoRR，卷。绝对/2409.03272，2024。

- [652] L.K. Ruixun Liu 和 D. Li 和 H. Zhu，“Occvla：具有隐式 3D 占用监督的视觉语言动作模型”，CoRR，卷。绝对/2509.05578，2025。

- [653] H.D. Tianyi 张和 H.Hao、Y.Qiao、J.Dai 和 Z.Hou，“相机空间中的 落地关联（Grounding） 动作：以观察为中心的视觉语言动作、策略”，CoRR，卷。绝对/2508.13103，2025。

- [654] S.C. Zihao Wang 和 Z. Mu、H. Lin、C. Zhu、X. Liu、Q. Li、A. Liu、X. S. Ma 和 Y. Liang，“Omnijarvis：统一视觉-语言动作标记化实现开放世界、指令跟踪智能体”，载于 NeurIPS，2024 年。

- [655] Q. N. Pei Liu 和 X. Lu、H. Liu、W. Ma、D. She、P. Jia、X. Lang 和 J. Ma，“Omnireason：自动驾驶的时间引导 视觉语言动作（Vision-Language-Action） 框架”，CoRR，卷。绝对/2509.00789，2025。

- [656] C. C. Huaihai Lyu 和 S. Xie、P. Wang、X. Chen、S. Zhang 和 C. Xu，“Omnisat：紧凑动作令牌，更快的自动回归”，CoRR，卷。绝对/2510.09667，2025。

- [657] C. G. Noriaki Hirose 和 D. Shah 以及 S. Levine，“Omnivla：用于机器人导航的全模态 视觉语言动作（Vision-Language-Action） 模型”，CoRR，卷。绝对/2509.19480，2025。

- [658] R. N. Fanqi Lin 和 Y. Hu、J. You、J. Zhao 和 Y. Gau，“Onetwovla：具有自适应推理功能的统一视觉语言动作（Vision-Language-Action） 模型”，CoRR，卷。绝对/2505.11917，2025。

- [659] M. L. Zihao Wang 和 K. He、X. Wang、Z. Mu、A. Liu 和 Y. Liang，“Openha：minecraft 中的一系列开源分层智能体模型”，CoRR，卷。绝对/2509.13347，2025。

- [660] K. P. Moo Jin Kim 和 S. Karamcheti、T. Xiao、A. Balakrishna、S. Nair、R. Rafailov、E. P. Foster、P. R. Sanketi、Q. Vuong、T. Kollar、B. Burchfiel、R. Tedrake、D. Sadigh、S. Levine、P. Liang 和 C. Finn，“Openvla：一个开源软件视觉语言动作（Vision-Language-Action） 模型，”in CoRL，系列。机器学习研究论文集，卷。 270.PMLR，2024 年，第 2679-2713 页。

- [661] F. L. Huang Huang 和 L. Fu、T. Wu、M. Mukadam、J. Malik、K. Goldberg 和 P. Abbeel，“OTTER：具有文本感知视觉特征和提取功能的 视觉语言动作（Vision-Language-Action） 模型”，载于 _ICML_。 OpenReview.net，2025 年。

- [662] S.L.Xiaopeng Lin 和 B.Yu、R.Yang、C.Wu、Y.Miao、Y.Jin、Y.Shi、C.Huang、B.Cheng 和 K.Chen，“Physbrain：人类以自我为中心的数据作为从视觉语言、模型到物理智能的桥梁”，CoRR，卷。绝对/2512.16793，2025。

- [663] J. L. 王志豪、J. 郑、W. 张、D. 刘、Y. 郑、H. Niu、J. Yu 和 X. Zhan，“Physiagent：物理世界中的体现智能体框架”，CoRR，卷。绝对/2509.24524，2025。

- [664] K. Chen, Z. Liu, T.Zhang, Z.Guo, S.Xu, H.Lin, H.Zang, Q.Zhang, Z.Yu, G.Fan, T.Huang, Y.Wang, and C.Yu, “ π rl: 基于流的 视觉语言动作（Vision-Language-Action） 模型的在线 RL 微调”，CoRR，卷。

绝对/2510.25889，2025。

- [665] G. S. Wenqi Liang 和 Y. He、J. Dong、S. Dai、I. Laptev、S. H. Khan 和 Y. Cong，“Pixelvla：推进视觉语言动作、模型中的像素级理解”，CoRR，卷。绝对/2511.01571，2025。

- [666] E. L. H. 张毅和 K

。 Chao、N. Petrovic、Y. Song、C. Wu 和 A. Knoll，“自适应自主驾驶的统一感知-语言-动作框架”，CoRR，卷。绝对/2507.23540，2025。

- [667] J. Z. Hang Yu 和 Y. Liu、K. Li、C. Ma、D. Zhang、Y. Hu、G. Chen、J. Xie、J.Guo、J. Zhao 和 Y. Gau，“指出你的意思：基于视觉的教学策略”，CoRR，卷。绝对/2512.18933，2025。

- [668] X.W. Ziwen Li 和 H.Zhang、R.Chen、R.Lin、X.He、H.Huang、Y.Guo、F.Karray、T.Liu 和 M.Gong，“Posa-vla：通过姿势条件锚点、注意力增强动作生成”，CoRR，卷。绝对/2512.03724，2025。

- [669] Z. H. Jiahui 张和 C. Gu、Z. Ma 和 L. 张，“通过预言强化行动策略”，CoRR，卷。绝对/2511.20633，2025。

- [670] H.K. Seongmin Park 和 W. Jeon、J. Yang、B. Jeon、Y. Oh 和 J. Choi，“资源高效型机器人控制的量化感知模仿学习”，CoRR，卷。绝对/2412.01034，2024。

- [671] Y. C. Yixuan Li、M. Zhou 和 H. Li，“Q深度-vla：量化深度预测作为 视觉语言动作（Vision-Language-Action）模型的辅助监督”，CoRR，卷。绝对/2510.14836，2025。

- [672] Q. V. Yevgen Chebotar 和 K. Hausman、F. Xia、Y. Lu、A. Irpan、A. Kumar、T. Yu、A. Herzog、K. Pertsch、K. Gopalakrishnan、J. Ibarz、O. Nachum、S. A. Sontakke、G. Salazar、H. T. Tran、J. Peralta、C. Tan、D. Manjunath、J. Singh、B. Zitkovich、T. Jackson、K. Rao、C. Finn 和 S. Levine，“Q-transformer：可扩展离线 强化学习（Reinforcement Learning）

- 通过自回归，q 函数，”in CoRL ，ser。机器学习研究论文集，卷。 229.PMLR，2023 年，第 3909-3928 页。

- [673] P. D. Xinyang Tong 和 Y. Fan、D. Wang、W. Zhuang、C. Cui、M. Sun、H. Zhu、H. Zhuang、Y. Dang、S. Huang 和 S. Lyu，“Quart-online：用于四足机器人学习的无延迟多模态大语言模型”，载于_ICRA_。 IEEE，2025 年，第 9533-9539 页。

- [674] L. F. Justin Yu 和 H. Huang、K. El-Refai、R. A. Ambrus、R. Cheng、M. Z. Irshad 和 K. Goldberg，“Real2render2real：无需动态模拟或机器人硬件即可缩放机器人数据”，CoRR，卷。绝对/2505.09601，2025。

- [675] L. W. Songming Liu 和 B. Li、H. Tan、H. Chen、Z. Wang、K. Xu、H. Su 和 J. Zhu，“RDT-1B：用于双手操作的扩散 基础模型（Foundation Model）”，载于_ICLR_。 OpenReview.net，2025 年。

- [676] Z. Y. Dapeng Zhang 和 Z. Chen、C. Liao、Y. Chen、F. Shen、Q. Zhou 和 T. Chua，“Reasoning-vla：快速通用的 视觉语言动作（Vision-Language-Action） 推理，自动驾驶模型”，CoRR，卷。 ABS/2511.19912，2025。

- [677] Y. Z. Pengxiang Li 和 Y. Wang、H. Wang、H. Zhu、J. Liu、X. Zhan、K. Zhan 和 X. Lang，“自动驾驶中反射 视觉语言动作（Vision-Language-Action） 模型的离散扩散”，CoRR，卷。 ABS/2509.20109，2025。

- [678] S. D. Kaustubh Sridhar 和 D. Jayaraman 以及 I. Lee，“REGENT：一种检索增强型多面手智能体，可以在新环境中根据上下文进行操作”，载于_ICLR_。 OpenReview.net，2025 年。

- [679] C. W. Wenlong Huang 和 Y. Li、R.Zhang 和 L. Fei-Fei，“Rekep：用于机器人操作的关系关键点约束的时空推理”，载于 CoRL，ser。机器学习研究论文集，卷。 270.PMLR，2024 年，第 4573-4602 页。

- [680] Z. Z. Yajat Yadav 和 A. Wagenmaker、K. Pertsch 和 S

。 Levine，“通过参数、合并对 视觉语言动作（Vision-Language-Action） 机器人策略进行鲁棒微调”，CoRR，卷。绝对/2512.08333，2025。

- [681] T. C. Jiyeon Koo 和 H. Kang、E. Pyo、T. G. Oh、T. Kim 和 A. J. Choi，“Retovla：在视觉语言动作、模型中重用寄存器标记进行空间推理”，CoRR，卷。绝对/2509.21243，2025。

- [682] J. Z. Sombit Dey 和 N. Nikolov、L. V. Gool 和 D. P. Paudel，“Revla：恢复机器人 基础模型（Foundation Models） 的视觉域限制”，载于_ICRA_。 IEEE，2025 年，第 8679-8686 页。

- [683] S. D. Kaustubh Sridhar 和 D. Jayaraman 以及 I. Lee，“RICL：为预先训练的 视觉语言动作（Vision-Language-Action） 模型添加上下文适应性”，CoRR，卷。绝对/2508.02062，2025。

- [684] G. V. Konstantinos Bousmalis 和 D. Rao、C. M. Devin、A. X. Lee、M. B. Villalonga、T. Davchev、Y. Zhou、A. Gupta、A. Raju、A. Laurens、C. Fantacci、V. Dalibard、M. Zambelli、M. F. Martins、R. Pevceviciute、M. Blokzijl、 M. Denil、N. Batchelor、T. Lampe、E. Parisotto、K. Zolna、S. E. Reed、S. G. Colmenarejo、J. Scholz、A. Abdolmaleki、O. Groth、J. Regli、O. Sushkov、T. Rothorl、J. E. Chen、Y. Aytar、D. Barker、J. Ortiz、M. A. Riedmiller、J. T. Springenberg、R. Hadsell、F. Nori 和 N. Heess，“Robocat：用于机器人操作的 自我改进（Self-Improving） 多面手智能体”，Trans。马赫。学习。 Res.，卷。 2024 年，2024 年。

- [685] C. Y. Zongzheng 张和 H. Xu、M. Liao、X. Qi、H. Gau、Z. Wang 和 H. Zhu，“机器人化学家：长时程（Long-Horizon） 和符合安全的机器人化学实验”，CoRR，卷。绝对/2509.08820，2025。

- [686] H. L. Qingwen Bu 和 L. Chen、J. Cai、J. Zeng、H. Cui、M. Yao 和 Y. Qiao，“走向协同、广义和高效的机器人操纵双系统”，CoRR，卷。绝对/2410.08001，2024。

- [687] M. L. Xinghang Li 和 H.Zhang、C.Yu、J.Xu、H.Wu、C.Cheang、Y.Jing、W.Zhang、H.Liu、H.Li 和 T.Kong，“视觉语言 基础模型（Foundation Models） 作为有效的机器人模仿者”，载于_ICLR_。 OpenReview.net，2024 年。

- [688] W. Z. Weixin Mao 和 Z. Jiang、D. Fang、Z. Zhang、Z. Lan、F. Jia、T. Wang、H. Fan 和 O. Yoshie，“Robomatrix：开放世界中可扩展的机器人任务规划和执行的以技能为中心的分层框架”，CoRR，卷。绝对/2412.00171，2024。

- [689] H.X.WeifanGuan 和 C.Zhang、A.Li、Q.Hu 和 J.Cheng，“Roboneuron：连接 基础模型（Foundation Models） 和 ROS 的模块化框架，适用于 具身智能（Embodied AI）”，CoRR，卷。绝对/2512.10394，2025。

- [690] J. W. Shunlei Li 和 R. Dai、W. Ma、W. Y. Ng、Y. Hu 和 Z. Li，“Robonurse-vla：基于视觉语言动作模型的机器人擦洗护士系统”，载于_IROS_。 IEEE，2025 年，第 3986-3993 页。

- [691] J. F. Siyin Wang 和 F. Liu、X. He、H. Wu、J. Shi、K. Huang、Z. Fei、J. Kong、Z. Wu、Y. Jiang、S. Ng、T. Chua 和 X. Qiu，“Roboomni：全模态环境中的主动机器人操作”，CoRR，卷。绝对

/2510.23763，2025。

- [692] C. C. Huajie Tan 和 X. Chen、Y. Ji、Z. Zhu、X.hao、Y. Lyu、M. Cao、J. Zhao、H. Lyu、E. Zhou、N. Chen、Y. Fu、C. Peng、W.Guo、D. Liang、Z. Chen、M. Lyu、C. He、Y. Ao、Y. Lin、P. Wang、Z. Wang 和 S. Zhang，“Roboos-next：一个基于内存的统一框架

lifelong, scalable, and robust multi-robot collaboration," *CoRR*, vol. abs/2510.26536, 2025.

[693] J. D. Wentao Yuan and, V. Blukis, W. Pumacay, R. Krishna, A. Murali, A. Mousavian, and D. Fox, "Robopoint: A vision-language model for spatial affordance prediction, in robotics," in *CoRL*, ser. Proceedings of Machine Learning Research, vol. 270. PMLR, 2024, pp. 4005–4020.

[694] C. Y. Boshi An and and R. K. Katzschmann, "Robotic assistant: Completing collaborative tasks with dexterous vision-language-action, models," *CoRR*, vol. abs/2510.25713, 2025.

[695] P. L. Xinghang Li and, M. Liu, D. Wang, J. Liu, B. Kang, X. Ma, T. Kong, H. Zhang, and H. Liu, "Towards generalist robot policies: What matters in building vision-language-action, models," *CoRR*, vol. abs/2412.14058, 2024.

[696] Z. W. Jianing Guo and, C. Tu, Y. Ma, X. Kong, Z. Liu, J. Ji, S. Zhang, Y. Chen, K. Chen, Q. Dou, Y. Yang, X. Liu, H. Zhao, W. Lv, and S. Li, "On robustness of vision-language-action model against multi-modal, perturbations," *CoRR*, vol. abs/2510.00037, 2025.

[697] L. L. Mingtong Dai and, Y. Bai, Y. Liu, Z. Wang, R. SU, C. Chen, L. Lin, and X. Wu, "Rover: Robot reward model as test-time verifier for vision-language-action, model," *CoRR*, vol. abs/2510.10975, 2025.

[698] W. B. Tobias Jülg and and F. Walter, "Refined policy distillation: From VLA generalists to RL experts," in *IROS*. IEEE, 2025, pp. 11 677–11 684.

[699] J. L. Taeyoung Kim and, M. Koo, D. Kim, K. Lee, C. Kim, Y. Seo, and J. Shin, "Contrastive representation regularization for vision-language-action, models," *CoRR*, vol. abs/2510.01711, 2025.

[700] T. Y. Brianna Zitkovich and, S. Xu, P. Xu, T. Xiao, F. Xia, J. Wu, P. Wohlhart, S. Welker, A. Wahid, Q. Vuong, V. Vanhoucke, H. T. Tran, R. Soricut, A. Singh, J. Singh, P. Sermanet, P. R. Sanketi, G. Salazar, M. S. Ryoo, K. Reymann, K. Rao, K. Pertsch, I. Mordatch, H. Michalewski, Y. Lu, S. Levine, L. Lee, T. E. Lee, I. Leal, Y. Kuang, D. Kalashnikov, R. Julian, N. J. Joshi, A. Irpan, B. Ichter, J. Hsu, A. Herzog, K. Hausman, K. Gopalakrishnan, C. Fu, P. Florence, C. Finn, K. A. Dubey, D. Driess, T. Ding, K. M. Choromanski, X. Chen, Y. Chebotar, J. Carbajal, N. Brown, A. Brohan, M. G. Arenas, and K. Han, "RT-2: vision-language-action models transfer web knowledge to robotic, control," in *CoRL*, ser. Proceedings of Machine Learning Research, vol. 229. PMLR, 2023, pp. 2165–2183.

[701] S. K. Soroush Nasiriany and, T. Ding, L. Smith, Y. Zhu, D. Driess, D. Sadigh, and T. Xiao, "Rt-affordance: Affordances are versatile intermediate representations, for robot manipulation," in *ICRA*. IEEE, 2025, pp. 8249–8257.

[702] T. D. Suneel Belkhale and, T. Xiao, P. Sermanet, Q. Vuong, J. Tompson, Y. Chebotar, D. Dwibedi, and D. Sadigh, "RT-H: action hierarchies using language," in *Robotics: Science and Systems*, 2024.

[703] J. W. K. Samuel Schmidgall and, A. Kuntz, A. E. Ghazi, and A. Krieger, "General-purpose foundation models for increased autonomy in robot-assisted, surgery," *Nat. Mac. Intell.*, vol. 6, no. 11, pp. 1275–1283, 2024.

[704] S. K. Jiayuan Gu and, P. Wohlhart, Y. Lu, M. G. Arenas, K. Rao, W. Yu, C. Fu, K. Gopalakrishnan, Z. Xu, P. Sundaresan, P. Xu, H. Su, K. Hausman, C. Finn, Q. Vuong, and T. Xiao, "Rt-trajectory: Robotic task generalization via hindsight trajectory, sketches," in *ICLR*. OpenReview.net, 2024.

[705] A. R. Abby O'Neill and, A. Maddukuri, A. Gupta, A. Padalkar, A. Lee, A. Pooley, A. Gupta, A. Mandlekar, A. Jain, A. Tung, A. Bewley, A. Herzog, A. Irpan, A. Khazatsky, A. Rai, A. Gupta, A. E. Wang, A. Singh, A. Garg, A. Kembhavi, A. Xie, A. Brohan, A. Raffin, A. Sharma, A. Yavary, A. Jain, A. Balakrishna, A. Wahid, B. Burgess-Limerick, B. Kim, B. Schölkopf, B. Wulfe, B. Ichter, C. Lu, C. Xu, C. Le, C. Finn, C. Wang, C. Xu, C. Chi, C. Huang, C. Chan, C. Agia, C. Pan, C. Fu, C. Devin, D. Xu, D. Morton, D. Driess, D. Chen, D. Pathak, D. Shah, D. Büchler, D. Jayaraman, D. Kalashnikov, D. Sadigh, E. Johns, E. P. Foster, F. Liu, F. Ceola, F. Xia, F. Zhao, F. Stulp, G. Zhou, G. S. Sukhatme, G. Salhotra, G. Yan, G. Feng, G. Schiavi, G. Berseth, G. Kahn, G. Wang, H. Su, H. Fang, H. Shi, H. Bao, H. B. Amor, H. I. Christensen, H. Furuta, H. Walke, H. Fang, H. Ha, I. Mordatch, I. Radosavovic, I. Leal, J. Liang, J. Abou-Chakra, J. Kim, J. Drake, J. Peters, J. Schneider, J. Hsu, J. Bohg, J. T. Bingham, J. Wu, J. Gao, J. Hu, J. Wu, J. Wu, J. Sun, J. Luo, J. Gu, J. Tan, J. Oh, J. Wu, J. Lu, J. Yang, J. Malik, J. Silvério, J. Hejna, J. Booher, J. Tompson, J. Yang, J. Salvador, J. J. Lim, J. Han, K. Wang, K. Rao, K. Pertsch, K. Hausman, K. Go, K. Gopalakrishnan, K. Goldberg, K. Byrne, K. Oslund, K. Kawaharazuka, K. Black, K. Lin, K. Zhang, K. Ehsani, K. Lekkala, K. Ellis, K. Rana, K. Srinivasan, K. Fang, K. P. Singh, K. Zeng, K. Hatch, K. Hsu, L. Itti, L. Y. Chen, L. Pinto, L. Fei-Fei, L. Tan, L. J. Fan, L. Ott, L. Lee, L. Weihs, M. Chen, M. Lepert, M. Memmel, M. Tomizuka, M. Itkina, M. G. Castro, M. Spero, M. Du, M. Ahn, M. C. Yip, M. Zhang, M. Ding, M. Heo, M. K. Srirama, M. Sharma, M. J. Kim, N. Kanazawa, N. Hansen, N. Heess, N. J. Joshi, N. Sünderhauf, N. Liu, N. D. Palo, N. M. M. Shafiullah, O. Mees, O. Kroemer, O. Bastani, P. R. Sanketi, P. T. Miller, P. Yin, P. Wohlhart, P. Xu, P. D. Fagan, P. Mitrano, P. Sermanet, P. Abbeel, P. Sundaresan, Q. Chen, Q. Vuong, R. Rafailov, R. Tian, R. Doshi, R. Martín-Martín, R. Baijal, R. Scalise, R. Hendrix, R. Lin, R. Qian, R. Zhang, R. Mendonca, R. Shah, R. Hoque, R. Julian, S. Bustamante-Gomez, S. Kirmani, S. Levine, S. Lin, S. Moore, S. Bahl, S. Dass, S. D. Sonawani, S. Song, S. Xu, S. Haldar, S. Karamcheti, S. Adebola, S. Guist, S. Nasiriany, S. Schaal, S. Welker, S. Tian, S. Ramamoorthy, S. Dasari, S. Belkhale, S. Park, S. Nair, S. Mirchandani, T. Osa, T. Gupta, T. Harada, T. Matsushima, T. Xiao, T. Kollar, T. Yu, T. Ding, T. Davchev, T. Z. Zhao, T. Armstrong, T. Darrell, T. Chung, V. Jain, V. Vanhoucke, W. Zhan, W. Zhou, W. Burgard, X. Chen, X. Wang, X. Zhu, X. Geng, X. Liu, L. Xu, X. Li, Y. Lu, Y. J. Ma, Y. Kim, Y. Chebotar, Y. Zhou, Y. Zhu, Y. Wu, Y. Xu, Y. Wang, Y. Bisk, Y. Cho, Y. Lee, Y. Cui, Y. Cao, Y. Wu, Y. Tang, Y. Zhu, Y. Zhang, Y. Jiang, Y. Li, Y. Li, Y. Iwasawa, Y. Matsuo, Z. Ma, Z. Xu, Z. J. Cui, Z. Zhang, and Z. Lin, "Open x-embodiment: Robotic learning datasets and RT-X models : Open, x-embodiment collaboration," in *ICRA*. IEEE, 2024, pp. 6892–6903.

[706] V. B. Ankit Goyal and, J. Xu, Y. Guo, Y. Chao, and D. Fox, "RVT-2: learning precise manipulation from few demonstrations," in *Robotics: Science and Systems*, 2024.

[707] S. H. Yuming Jiang and, S. Xue, Y. Zhao, J. Cen, S. Leng, K. Li, J. Guo, K. Wang, M. Chen, F. Wang, D. Zhao, and X. Li, "Rynnvla-001: Using human demonstrations to improve robot manipulation," *CoRR*, vol. abs/2509.15212, 2025.

[708] S. H. Jun Cen and, Y. Yuan, K. Li, H. Yuan, C. Yu, Y. Jiang, J. Guo, X. Li, H. Luo, F. Wang, D. Zhao, and H. Chen, "Rynnvla-002: A unified vision-language-action and world model," *CoRR*, vol. abs/2511.17502, 2025.

[709] B. Zhang, Y. Zhang, J. Ji, Y. Lei, J. Dai, Y. Chen, and Y. Yang, "Safevla: Towards safety alignment of vision-language-action model via constrained learning," *arXiv preprint*, 2025.

[710] K. C. Isabel Leal and, D. Jain, A. Dubey, J. Varley, M. S. Ryoo, Y. Lu, F. Liu, V. Sindhwani, Q. Vuong, T. Sarlós, K. Oslund, K. Hausman, and K. Rao, "SARA-RT: scaling up robotics transformers with self-adaptive robust, attention," in *ICRA*. IEEE, 2024, pp. 6920–6927.

[711] Y. Z. Minjie Zhu and, J. Li, J. Wen, Z. Xu, N. Liu, R. Cheng, C. Shen, Y. Peng, F. Feng, and J. Tang, "Scaling diffusion policy in transformer to 1 billion parameters for, robotic manipulation," in *ICRA*. IEEE, 2025, pp. 10 838–10 845.

[712] C. Li, J. Liu, G. Wang, X. Li, S. Chen, L. Heng, C. Xiong, J. Ge, R. Zhang, K. Zhou, and S. Zhang, "A self-correcting vision-language-action model for fast and slow system manipulation," *arXiv preprint*, 2024.

[713] A. L. Yilin Wu and, T. Hermans, F. Ramos, A. Bajcsy, and C. Pérez-D'Arpino, "Do what you say: Steering vision-language-action models via runtime, reasoning-action alignment verification," *CoRR*, vol. abs/2510.16281, 2025.

[714] J. Z. Beichen Wang and, S. Dong, I. Fang, and C. Feng, "VLM see, robot do: Human demo video to robot action plan via vision, language model," in *IROS*. IEEE, 2025, pp. 17 215–17 222.

[715] Z. A. Ran Yang and, L. Zhou, and Y. Feng, "Seqvla: Sequential task execution for long-horizon manipulation with, completion-aware vision-language-action model," *CoRR*, vol. abs/2509.14138, 2025.

[716] L. L. Kevin Qinghong Lin and, D. Gao, Z. Yang, S. Wu, Z. Bai, S. W. Lei, L. Wang, and M. Z. Shou, "Showui: One vision-language-action model for GUI visual agent," in *CVPR*. Computer Vision Foundation / IEEE, 2025, pp. 19 498–19 508.

[717] L. Wang, "Sigma: The key for vision-language-action models toward telepathic, alignment," *CoRR*, vol. abs/2512.00783, 2025.

[718] Y. Z. Haozhan Li and, J. Yu, Y. Zhang, Z. Yang, K. Zhang, X. Zhu, Y. Zhang, T. Chen, G. Cui, D. Wang, D. Luo, Y. Fan, Y. Sun, J. Zeng, J. Pang, S. Zhang, Y. Wang, Y. Mu, B. Zhou, and N. Ding, "Simplevla-rl: Scaling VLA training via reinforcement learning," *CoRR*, vol. abs/2509.09674, 2025.

[719] H. S. Ayudh Saxena and, S. Routray, R. R. Shah, and E. Pahwa, "SITCOM: scaling inference-time compute for vlas," *CoRR*, vol. abs/2510.04041, 2025.

[720] D. A. Mustafa Shukor and, F. Capuano, P. Kooijmans, S. Palma, A. Zouitine, M. Aractingi, C. Pascal, M. Russi, A. Marafioti, S. Alibert, M. Cord, T. Wolf, and R. Cadène, "Smolvla: A vision-language-

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终身、可扩展且强大的多机器人协作”，CoRR，第 1 卷，abs/2510.26536，2025 年。

- [693] J. D. Wentao Yuan 和 V. Blukis、W. Pumacay、R. Krishna、A. Murali、A. Mousavian 和 D. Fox ，“Robopoint：机器人技术中用于空间 可供性（Affordance） 预测的视觉语言模型”，载于_CoRL_，系列。机器学习研究论文集，卷。 270.PMLR，2024 年，第 4005-4020 页。

- [694] C. Y. Boshi An 和 R. K. Katzschmann，“机器人助手：使用灵巧的 视觉语言动作（Vision-Language-Action） 模型完成协作任务”，CoRR，卷。绝对/2510.25713，2025。

- [695] P. L. Xinghang Li 和 M. Liu、D. Wang、J. Liu、B. Kang、X. Ma、T. Kong、H. Zhang 和 H. Liu，“迈向通才机器人策略：构建 视觉语言动作（Vision-Language-Action） 模型的重要因素”，CoRR，卷。绝对/2412.14058，2024。

- [696] Z.W.JianingGuo 和 C.Tu、Y.Ma、X.Kong、Z.Liu、J.Ji、S.Zhang、Y.Chen、K.Chen、Q.Dou、Y.Yang、X.Liu、H.Zhao、W.Lv 和 S.Li，“论 视觉语言动作（Vision-Language-Action） 模型对多模态扰动的鲁棒性”，CoRR，卷。绝对/2510.00037，2025。

- [697] L. L. Mingtong Dai 和 Y. Bai、Y. Liu、Z. Wang、R. SU、C. Chen、L. Lin 和 X. Wu，“Rover：机器人奖励模型作为 视觉语言动作（Vision-Language-Action） 模型的测试时间验证器”，CoRR，卷。绝对/2510.10975，2025。

- [698] W. B. Tobias Jìulg 和 F. Walter，“精致的策略蒸馏：从 VLA 通才到 RL 专家”，载于 IROS。 IEEE ，2025 年，第 11 677-11 684 页。

- [699] J. L. Taeyoung Kim 和 M. Koo、D. Kim、K. Lee、C. Kim、Y. Seo 和 J. Shin，“视觉语言动作的对比表示正则化，模型”，CoRR，卷。绝对/2510.01711，2025。

- [700] T. Y. Brianna Zitkovich 和 S. Xu、P. Xu、T. Xiao、F. Xia、J. Wu、P. Wohlhart、S. Welker、A. Wahid、Q. Vuong、V. Vanhoucke、H. T. Tran、R. Soricut、A. Singh、J. Singh、P. Sermanet、P. R. Sanketi、G. Salazar、M. S. Ryoo、K. Reymann、K. Rao、K. Pertsch、I. Mordatch、H. Michalewski、Y. Lu、S. Levine、L. Lee、T. E. Lee、I. Leal、Y. Kuang、D. Kalashnikov、R. Julian、N. J. Joshi、A. Irpan、B. Ichter、J. Hsu、A. Herzog、K. Hausman、K. Gopalakrishnan、C. Fu、P. Florence、C. Finn、K. A. Dubey、D. Driess、T. Ding、K. M. Choromanski、X. Chen、Y. Chebotar、J. Carbajal、N. Brown、A. Brohan、M. G. Arenas 和 K. Han，“RT-2：视觉语言动作（Vision-Language-Action） 模型将网络知识传输到机器人控制”，载于_CoRL_，爵士。机器学习研究论文集，卷。 229.PMLR，2023 年，第 2165-2183 页。

- [701] S. K. Soroush Nasiriany 和 T. Ding、L. Smith、Y. Zhu、D. Driess、D. Sadigh 和 T. Xiao，“Rt-可供性（Affordance）：可供性（Affordances） 是用于机器人操作的多功能中间表示”，载于_ICRA_。IEEE，2025 年，第 8249-8257 页。

- [702] T.D.Suneel Belkhale 和 T.Xiao、P.Sermanet、Q.Vuong、J.Tompson、Y.Chebotar、D.Dwibedi 和 D.Sadigh，“RT-H：使用语言的动作层次结构”，《机器人：科学与系统》，2024 年。

- [703] J. W. K. Samuel Schmidgall 和 A. Kuntz、A. E. Ghazi 和 A. Krieger，“通用（General-Purpose）基础模型（Foundation Models）提高机器人辅助手术的自主性”，Nat。麦克.情报.，卷。 6、没有。 11，第 1275-1283 页，2024 年。

- [704] S.K. Jiayuan Gu 和 P. Wohlhart、Y. Lu、M. G. Arenas、K. Rao、W. Yu、C. Fu、K. Gopalakrishnan、Z

。 Xu、P. Sundaresan、P. Xu、H. Su、K. Hausman、C. Finn、Q. Vuong 和 T. Xiao，“Rt 轨迹：通过事后轨迹、草图进行机器人任务泛化”，载于 ICLR。 OpenReview.net，2024 年。

- [705] A. R. Abby O’Neill 和 A. Maddukuri、A. Gupta、A. Padalkar、A. Lee、A. Pooley、A. Gupta、A. Mandlekar、A. Jain、A. Tung、A. Bewley、A. Herzog、A. Irpan、A. Khazatsky、A. Rai、A. Gupta、A. E. Wang、A. Singh、 A. Garg、A. Kembhavi、A. Xie、A. Brohan、A. Raffin、A. Sharma、A. Yavary、A. Jain、A. Balakrishna、A. Wahid、B. BurgessLimerick、B. Kim、B. Scholkopf、B. Wulfe、B. Ichter、C. Lu、C. Xu、C. Le、C. Finn、C. Wang、C. Xu、 C. Chi、C. Huang、C. Chan、C. Agia、C. Pan、C. Fu、C. Devin、D. Xu、D. Morton、D. Driess、D. Chen、D. Pathak、D. Shah、D. Buchler、D. Jayaraman、D. Kalashnikov、D. Sadigh、E. Johns、E. P. Foster、F. Liu、F. Ceola、F. Xia、F. Zhao、F. Stulp、G. Zhou、G. S. Sukhatme、G. Salhotra、G. Yan、G. Feng、G. Schiavi、G. Berseth、G. Kahn、G. Wang、H. Su、H. Fang、H. Shi、H. Bao、H. B. Amor、H. I. Christensen、H. Furuta、H. Walke、H. Fang、H. Ha、I. Mordatch、I. Radosavovic、I. Leal、J. Liang、J. Abou-Chakra、J. Kim、J. Drake、J. Peters、J. Schneider、J. Hsu、J. Bohg、J. T. Bingham、J. Wu、J. Gau、J. Hu、J. Wu、J. Wu、J. Sun、J. Luo、J. Gu、J. Tan、J. Oh、J. Wu、J. Lu、J. Yang、J. Malik、J. Silv´erio、J. Hejna、J. Booher、J. Tompson、J. Yang、J. Salvador、J. J. Lim、J. Han、K. Wang、K. Rao、K. Pertsch、K. Hausman、K. Go、K. Gopalakrishnan、K. Goldberg、K. Byrne、K. Oslund、K. Kawaharazuka、K. Black、K. Lin、K. 张、K. Ehsani、K. Lekkala、K. Ellis、K. Rana、K. Srinivasan、K. Fang、K. P. Singh、K. Zeng、K. Hatch、K. Hsu、L. Itti、L. Y. Chen、L. Pinto、L. FeiFei、L. Tan、L. J. Fan、L. Ott、L. Lee、L. Weihs、M. Chen、M. Lepert、

- M. Memmel、M. Tomizuka、M. Itkina、M. G. Castro、M. Spero、M. Du、M. Ahn、M. C. Yip、M. Zhu、M. Ding、M. Heo、M. K. Srirama、M. Sharma、M. J. Kim、N. Kanazawa、N. Hansen、N. Heess、N. J. Joshi、N. Süunderhauf、N. Liu、 N. D. Palo、N. M. M. Shafiullah、O. Mees、O. Kroemer、O. Bastani、P. R. Sanketi、P. T. Miller、P. Yin、P. Wohlhart、P. Xu、P. D. Fagan、P. Mitrano、P. Sermanet、P. Abbeel、P. Sundaresan、Q. Chen、Q. Vuong、R. Rafailov、R. Tian， R. Doshi、R. Mart´ın-Mart´ın、R. Baijal、R. Scalise、R. Hendrix、R. Lin、R. 钱、R. 张、R. Mendonca、R. Shah、R. Hoque、R. Julian、S. BustamanteGomez、S. Kirmani、S. Levine、S. Lin、S. Moore、S. Bahl、S. Dass、S. D. Sonawani、S.宋、S. Xu、S. Haldar、S. Karamcheti、S. Adebola、S. Guist、S. Nasiriany、S. Schaal、S. Welker、S. Tian、S. Ramamoorthy、S. Dasari、S. Belkhale、S. Park、S. Nair、S. Mirchandani、T. Osa、T. Gupta、T. Harada、T. Matsushima、T. Xiao、T. Kollar、T. Yu、T. Ding、T. Davchev、T. Z. 赵、T. Armstrong、T. Darrell、T. Chung、V. Jain、V. Vanhoucke、W. Zhan、W. Zhou、W. Burgard、X. Chen、X. Wang、X. Zhu、X. Geng、X. Liu、L. Xu、X. Li、Y. Lu、Y. J. Ma、Y. Kim、Y. Chebotar、 Y. Zhou、Y. Zhu、Y. Wu、Y. Xu、Y. Wang、Y. Bisk、Y. Cho、Y. Lee、Y. Cui、Y. Cao、Y. Wu、Y. Tang、Y. Zhu、Y. Zhang、Y. Jiang、Y. Li、Y. Li、Y. Iwasawa、Y. Matsuo、Z. Ma、Z. Xu、Z. J. Cui、Z. Zhang 和 Z. Lin，“开放 x 实施例：机器人学习数据集和 RT-X 模型：开放、x 实施例协作”，参见 ICRA。 IEEE，2024 年，第 6892-6903 页。

- [706] V.B

。 Ankit Goyal 和 J. Xu、Y.Guo、Y.Chao 和 D.Fox，“RVT-2：从少量演示中学习精确操作”，《机器人：科学与系统》，2024 年。

- [707] S. H. Yuming Jiang 和 S.薛、Y.Zhao、J.Cen、S.Leng、K.Li、J.Guo、K.Wang、M.Chen、F.Wang、D.Zhao 和 X.Li，“Rynnvla-001：利用人类演示来改进机器人操作”，CoRR，卷。ABS/2509.15212，2025。

- [708] S. H. Jun Cen 和 Y. Yuan、K. Li、H. Yuan、C. Yu、Y. Jiang、J.Guo、X. Li、H. Luo、F. Wang、D. Zhao 和 H. Chen，“Rynnvla-002：统一的 视觉语言动作（Vision-Language-Action） 和 世界模型（World Model）”，CoRR，卷。绝对/2511.17502，2025。

- [709] B. 张、Y. 张、J. Ji、Y. Lei、J. Dai、Y. Chen 和 Y. Yang，“Safevla：通过约束学习实现 视觉语言动作（Vision-Language-Action） 模型的安全对齐”，arXiv 预印本，2025 年。

- [710] K. C. Isabel Leal 和 D. Jain、A. Dubey、J. Varley、M. S. Ryoo、Y. Lu、F. Liu、V. Sindhwani、Q. Vuong、T. Sarl´os、K. Oslund、K. Hausman 和 K. Rao，“SARA-RT：通过自适应稳健、注意力来扩展机器人变压器”，载于_ICRA_。 IEEE，2024 年，第 6920-6927 页。

- [711] Y. Z. Minjie Zhu 和 J. Li、J. Wen、Z. Xu、N. Liu、R. Cheng、C. Shen、Y. Peng、F. Feng 和 J. Tang，“将变压器中的 扩散策略（Diffusion Policy） 扩展到 10 亿个参数以进行机器人操作”，载于 ICRA。 IEEE，2025 年，第 10 838-10 845 页。

- [712] C. Li、J. Liu、G. Wang、X. Li、S. Chen、L. Heng、C. Xiong、J. Ge、R. Zhang、K. Zhou 和 S. Zhang，“用于快速和慢速系统操作的自校正视觉语言动作模型”，arXiv 预印本，2024 年。

- [713] A. L. Yilin Wu 和 T. Hermans、F. Ramos、A. Bajcsy 和 C. P´erezD’Arpino，“照你说的做：通过运行时引导 视觉语言动作（Vision-Language-Action） 模型、推理-动作对齐验证”，CoRR，卷。绝对/2510.16281，2025。

- [714] J. Z. Beichen Wang 和 S. Dong、I. Fang 和 C. Feng，“VLM 看，机器人做：通过视觉、语言模型将人类演示视频转为机器人行动计划”，载于 IROS。 IEEE，2025 年，第 17 215-17 222 页。

- [715] Z. A. Ran Yang、L. Zhou 和 Y. Feng，“Seqvla：使用完成感知 视觉语言动作（Vision-Language-Action） 模型进行 长时程（Long-Horizon） 操作的顺序任务执行”，CoRR，卷。绝对/2509.14138，2025。

- [716] L.L.Kevin Qinghong Lin 和 D.Gao、Z.Yang、S.Wu、Z.Bai、S.W.Lei、L.Wang 和 M.Z.Shou，“Showui：一个用于 GUI 视觉智能体的 视觉语言动作（Vision-Language-Action） 模型”，载于_CVPR_。计算机视觉基金会/IEEE，2025 年，第 19 498-19 508 页。

- [717] L. Wang，“Sigma：视觉语言动作（Vision-Language-Action） 模型实现心灵感应、对齐的关键”，CoRR，卷。绝对/2512.00783，2025。

- [718] Y.Z.Haozhan Li 和 J.Yu、Y.Zhang、Z.Yang、K.Zhang、X.Zhu、Y.Zhang、T.Chen、G.Cui、D.Wang、D.Luo、Y.Fan、Y.Sun、J.Zeng、J.Pang、S.Zhang、Y.Wang、Y.Mu、B.Zhou 和 N.Ding，《Simplevlarl：通过扩展 VLA 训练》 强化学习（Reinforcement Learning），”CoRR，卷。绝对/2509.09674，2025。

- [719] H. S. Ayudh Saxena 和 S. Routray、R. R. Shah 和 E. Pahwa，“SITCOM：vlas 的缩放推理时间计算”，CoRR，卷。绝对/2510.04041，2025。

- [720] D. A. Mustafa Shukor 和 F. Capuano、P. Kooijmans、S. Palma、A. Zouitine、M. Aractingi、C. Pascal、M. Russi、A. Marafioti、S. Alibert、M. Cord、T. Wolf 和 R. Cadene，“Smolvla：一种视觉语言-

action model for affordable and efficient, robotics," *CoRR*, vol. abs/2506.01844, 2025.

[721] W. X. Jianping Jiang and, Z. Lin, H. Zhang, T. Ren, Y. Gao, Z. Lin, Z. Cai, L. Yang, and Z. Liu, "SOLAMI: social vision-language-action modeling for immersive interaction, with 3d autonomous characters," in *CVPR*. Computer Vision Foundation / IEEE, 2025, pp. 26 887–26 898.

[722] W. S. Fuhao Li and, H. Zhao, J. Wang, P. Ding, D. Wang, L. Zeng, and H. Li, "Spatial forcing: Implicit spatial representation alignment for vision-language-action, model," *CoRR*, vol. abs/2510.12276, 2025.

[723] J. X. Hanzhen Wang and, J. Pan, Y. Zhou, and G. Dai, "Specprune-vla: Accelerating vision-language-action models via action-aware, self-speculative pruning," *CoRR*, vol. abs/2509.05614, 2025.

[724] R. Y. Songsheng Wang and, Z. Yuan, C. Yu, F. Gao, Y. Wang, and D. F. Wong, "Spec-vla: Speculative decoding for vision-language-action models with, relaxed acceptance," *CoRR*, vol. abs/2507.22424, 2025.

[725] Y. L. Hengyu Fang and, Y. Du, L. Du, and H. Yang, "SQAP-VLA: A synergistic quantization-aware pruning framework for, high-performance vision-language-action models," *CoRR*, vol. abs/2509.09090, 2025.

[726] Y. Y. Zhejia Cai and, X. Chang, S. Liang, R. Chen, F. Xiong, M. Xu, and R. Huang, "Seeing space and motion: Enhancing latent actions with spatial and, dynamic awareness for VLA," *CoRR*, vol. abs/2509.26251, 2025.

[727] G. Z. Feng Xu and, X. Kong, T. Fu, D. F. N. Gordon, X. An, and B. Busam, "STARE-VLA: progressive stage-aware reinforcement for fine-tuning, vision-language-action models," *CoRR*, vol. abs/2512.05107, 2025.

[728] M. Y. Shengliang Deng and, Y. Zheng, J. Su, W. Zhang, X. Zhao, H. Cui, Z. Zhang, and H. Wang, "Stereovla: Enhancing vision-language-action models with stereo vision," *CoRR*, vol. abs/2512.21970, 2025.

[729] J. Z. Wenjun Lin and, K. Cai, and K. Wang, "STORM: search-guided generative world models for robotic manipulation," *CoRR*, vol. abs/2512.18477, 2025.

[730] P. G. Yufan He and, M. Xu, Z. Li, A. Myronenko, D. Imans, B. Liu, D. Yang, M. Gu, Y. Ji, Y. Jin, R. Zhao, B. Shen, and D. Xu, "Surgworld: Learning surgical robot policies from videos via world, modeling," *CoRR*, vol. abs/2512.23162, 2025.

[731] C. C. Chaojun Ni and, X. Wang, Z. Zhu, W. Zheng, B. Wang, T. Chen, G. Zhao, H. Li, Z. Dong, Q. Zhang, Y. Ye, Y. Wang, G. Huang, and W. Mei, "Swiftvla: Unlocking spatiotemporal dynamics for lightweight VLA, models at minimal overhead," *CoRR*, vol. abs/2512.00903, 2025.

[732] X. Z. Zonghuan Xu and, X. Ma, and Y. Jiang, "Tabvla: Targeted backdoor attacks on vision-language-action models," *CoRR*, vol. abs/2510.10932, 2025.

[733] S. M. Jonas Geiping and, N. Jain, J. Kirchenbauer, S. Singh, B. R. Bartoldson, B. Kailkhura, A. Bhatele, and T. Goldstein, "Scaling up test-time compute with latent reasoning: A recurrent, depth approach," *CoRR*, vol. abs/2502.05171, 2025.

[734] H. Z. Shuaijun Wang and, D. Xiang, and Y. You, "Tacrefinenet: Tactile-only grasp refinement between arbitrary in-hand, object poses," *CoRR*, vol. abs/2509.25746, 2025.

[735] S. W. Jialei Huang and, F. Lin, Y. Hu, C. Wen, and Y. Gao, "Tactile-vla: Unlocking vision-language-action model's physical knowledge, for tactile generalization," *CoRR*, vol. abs/2507.09160, 2025.

[736] H. X. Zongzheng Zhang and, Z. Yang, C. Yue, Z. Lin, H. Gao, Z. Wang, and H. Zhao, "TA-VLA: elucidating the design space of torque-aware vision-language-action, models," *CoRR*, vol. abs/2509.07962, 2025.

[737] C. Huang, Y. Wu, M. Chen, Y. F. Wang, and F. Yang, "Thinkact: Vision-language-action reasoning via reinforced visual latent planning," *CoRR*, vol. abs/2507.16815, 2025.

[738] Z. W. Guanxing Lu and, C. Liu, J. Lu, and Y. Tang, "Thinkbot: Embodied instruction following with thought chain reasoning," in *ICLR*. OpenReview.net, 2025.

[739] J. Z. Shaoan Wang and, M. Li, J. Liu, A. Li, K. Wu, F. Zhong, J. Yu, Z. Zhang, and H. Wang, "Trackvla: Embodied visual tracking in the wild," *CoRR*, vol. abs/2505.23189, 2025.

[740] P. D. Jiacheng Liu and, Q. Zhou, Y. Wu, D. Huang, Z. Peng, W. Xiao, W. Zhang, L. Yang, C. Lu, and D. Wang, "Trajbooster: Boosting humanoid whole-body manipulation via trajectory-centric, learning," *CoRR*, vol. abs/2509.11839, 2025.

[741] Y. G. Zhenyang Liu and, S. Zheng, X. Xue, and Y. Fu, "Trivla: A triple-system-based unified vision-language-action model, for general robot control," *CoRR*, vol. abs/2507.01424, 2025.

[742] Y. Bai, Z. Wang, Y. Liu, K. Luo, Y. Wen, M. Dai, W. Chen, Z. Chen, L. Liu, G. Li, and L. Lin, "Learning to see and act: Task-aware virtual view exploration for robotic manipulation," *arXiv preprint*, 2025.

[743] W. S. Jiayi Chen and, P. Ding, Z. Zhou, H. Zhao, F. Tang, D. Wang, and H. Li, "Unified diffusion VLA: vision-language-action model via joint discrete, denoising diffusion process," *CoRR*, vol. abs/2511.01718, 2025.

[744] Y. Z. Zhangyuan Wang and, Y. Yan, X. Tian, X. Shao, M. Li, W. Li, G. Su, W. Cui, and D. Fan, "Underwatervla: Dual-brain vision-language-action architecture for, autonomous underwater navigation," *CoRR*, vol. abs/2509.22441, 2025.

[745] Y. H. Jianke Zhang and, Y. Guo, X. Chen, Y. Liu, W. Chen, C. Lu, and J. Chen, "Unicod: Enhancing robot policy via unified continuous and discrete, representation learning," *CoRR*, vol. abs/2510.10642, 2025.

[746] C. C. Jiasen Lu and, S. Lee, Z. Zhang, S. Khosla, R. Marten, D. Hoiem, and A. Kembhavi, "Unified-io 2: Scaling autoregressive multimodal models with vision, language, audio, and action," in *CVPR*. IEEE, 2024, pp. 26 429–26 445.

[747] K. W. Jiazhao Zhang and, S. Wang, M. Li, H. Liu, S. Wei, Z. Wang, Z. Zhang, and H. Wang, "Uni-navid: A video-based vision-language-action model for unifying, embodied navigation tasks," *CoRR*, vol. abs/2412.06224, 2024.

[748] Z. L. Hao Lu and, G. Jiang, Y. Luo, S. Chen, Y. Zhang, and Y. Chen, "Uniugp: Unifying understanding, generation, and planing for end-to-end, autonomous driving," *CoRR*, vol. abs/2512.09864, 2025.

[749] Y. Y. Hui Lu and, Y. Yang, C. Yi, Q. Zhang, B. Shen, A. C. Kot, and X. Jiang, "When robots obey the patch: Universal transferable patch attacks on, vision-language-action models," *CoRR*, vol. abs/2511.21192, 2025.

[750] Y. G. Jianke Zhang and, Y. Hu, X. Chen, X. Zhu, and J. Chen, "UP-VLA: A unified understanding and prediction model for embodied, agent," in *ICML*. OpenReview.net, 2025.

[751] Z. W. Anqi Li and, J. Zhang, M. Li, Y. Qi, Z. Chen, Z. Zhang, and H. Wang, "Urbanvla: A vision-language-action model for urban micromobility," *CoRR*, vol. abs/2510.23576, 2025.

[752] Z. W. Junwen Gu and, P. Si, S. Qiu, Y. Feng, L. Sun, L. Luo, L. Yu, J. Wang, and Z. Wu, "USIM and U0: A vision-language-action dataset and model for, general underwater robots," *CoRR*, vol. abs/2510.07869, 2025.

[753] R. Y. Chuning Zhu and, S. Feng, B. Burchfiel, P. Shah, and A. Gupta, "Unified world models: Coupling video and action diffusion for pre-training, on large robotic datasets," *CoRR*, vol. abs/2504.02792, 2025.

[754] L. A. Jonas Pai and, V. Montesinos, B. Forrai, O. Mees, and E. Nava, "mimic-video: Video-action models for generalizable robot control beyond, vlas," *CoRR*, vol. abs/2512.15692, 2025.

[755] J. L. Yueru Jia and, S. Liu, R. Zhou, W. Yu, Y. Yan, X. Chi, Y. Guo, B. Shi, and S. Zhang, "Video2act: A dual-system video diffusion policy with robotic spatio-motional, modeling," *CoRR*, vol. abs/2512.03044, 2025.

[756] F. W. Yichao Shen and, Z. Du, Y. Liang, Y. Lu, J. Yang, N. Zheng, and B. Guo, "Videovla: Video generators can be generalizable robot manipulators," *CoRR*, vol. abs/2512.06963, 2025.

[757] J. Y. Jeongeun Park and, B. Jeon, J. Park, J. Shin, N. Cho, K. Lee, S. Yun, and S. Choi, "Hierarchical vision language action model using success and failure, demonstrations," *CoRR*, vol. abs/2512.03913, 2025.

[758] M. W. Guangyan Chen and, Q. Shao, Z. Zhou, W. Mao, T. Cui, M. Zhu, Y. Deng, L. Yang, Z. Zhang, Y. Yang, H. Chen, and Y. Yue, "See once, then act: Vision-language-action model with task learning, from one-shot video demonstrations," *CoRR*, vol. abs/2512.07582, 2025.

[759] H. H. Ankit Goyal and, X. Yang, V. Blukis, and F. Ramos, "VLA-0: building state-of-the-art vlas with zero modification," *CoRR*, vol. abs/2510.13054, 2025.

[760] C. M. Hanyu Zhou and and G. H. Lee, "VLA-4D: embedding 4d awareness into vision-language-action models, for spatiotemporally coherent robotic manipulation," *CoRR*, vol. abs/2511.17199, 2025.

[761] P. D. Yihao Wang and, L. Li, C. Cui, Z. Ge, X. Tong, W. Song, H. Zhao, W. Zhao, P. Hou, S. Huang, Y. Tang, W. Wang, R. Zhang, J. Liu, and D. Wang, "Vla-adapter: An effective paradigm for tiny-scale vision-language-action, model," *CoRR*, vol. abs/2509.09372, 2025.

[762] M. Z. Yuze Wu and, X. Li, Y. Du, Y. Fan, W. Li, Z. Han, X. Zhou, and F. Gao, "VLA-AN: an efficient and onboard vision-language-action framework, for aerial navigation in complex environments," *CoRR*, vol. abs/2512.15258, 2025.

[763] Y. W. Siyu Xu and, C. Xia, D. Zhu, T. Huang, and C. Xu, "Vla-cache: Towards efficient vision-language-action model via adaptive, token caching in robotic manipulation," *CoRR*, vol. abs/2502.02175, 2025.

原文第 35 页 | 中文翻译初稿

经济实惠且高效的机器人行动模型，”CoRR，第 1 卷，abs/2506.01844，2025 年。

- [721] W.X.Jianping Jiang 和 Z.Lin、H.Zhang、T.Ren、Y.Gao、Z.Lin、Z.Cai、L.Yang 和 Z.Liu，“SOLAMI：具有 3D 自主角色的沉浸式交互的社交 视觉语言动作（Vision-Language-Action） 建模”，载于_CVPR_。计算机视觉基金会/IEEE，2025 年，第 26 887-26 898 页。

- [722] W. S. Fuhao Li 和 H. Zhao、J. Wang、P. Ding、D. Wang、L. Zeng 和 H. Li，“空间强迫：视觉语言动作（Vision-Language-Action） 模型的隐式空间表示对齐”，CoRR，卷。绝对/2510.12276，2025。

- [723] J. X. Hanzhen Wang 和 J. Pan、Y. Zhou 和 G. Dai，“Specprunevla：通过动作感知、自我推测修剪加速 视觉语言动作（Vision-Language-Action） 模型”，CoRR，卷。绝对/2509.05614，2025。

- [724] R. Y. Songsheng Wang 和 Z. Yuan、C. Yu、F. Gau、Y. Wang 和 D. F. Wong，“Spec-vla：轻松接受的视觉语言动作模型的推测解码”，CoRR，卷。绝对/2507.22424，2025。

- [725] Y. L. Hengyu Fang 和 Y. Du、L. Du 和 H. Yang，“SQAP-VLA：高性能 视觉语言动作（Vision-Language-Action） 模型的协同量化感知剪枝框架”，CoRR，卷。绝对/2509.09090，2025。

- [726] Y. Y. Zhejia Cai 和 X. Chang、S. Liang、R. Chen、F. Xiong、M. Xu 和 R. Huang，“观察空间和运动：通过空间和动态意识增强 VLA 的潜在动作”，CoRR，卷。绝对/2509.26251，2025。

- [727] G. Z. Feng Xu 和 X. Kong、T. Fu、D. F. N. Gordon、X. An 和 B. Busam，“STARE-VLA：用于微调的渐进阶段感知强化，视觉语言动作（Vision-Language-Action） 模型”，CoRR，卷。绝对/2512.05107，2025。

- [728] M. Y. Shenliang Deng 和 Y. Cheng、J. Su、W.Zhang、X.Zhao、H.Cui、Z.Zhang 和 H.Wang，“Stereovla：增强具有立体视觉的 视觉语言动作（Vision-Language-Action） 模型”，CoRR，卷。绝对/2512.21970，2025。

- [729] J. Z. Wenjun Lin、K. Cai 和 K. Wang，“STORM：用于机器人操作的搜索引导生成 世界模型（World Models）”，CoRR，卷。绝对/2512.18477，2025。

- [730] P. G. Yufan He 和 M. Xu、Z. Li、A. Myronenko、D. Imans、B. Liu、D. Yang、M. Gu、Y. Ji、Y. Jin、R. Zhu、B. Shen 和 D. Xu，“Surgworld：通过世界、建模从视频中学习手术机器人策略”，CoRR，卷。绝对/2512.23162，2025。

- [731] C. C. Chaojun Ni 和 X. Wang、Z. Zhu、W. Cheng、B. Wang、T. Chen、G. Zhu、H. Li、Z. Dong、Q. Zhang、Y. Ye、Y. Wang、G. Huang 和 W. Mei，“Swiftvla：解锁轻量级 VLA 的时空动力学，以最小开销建立模型”，CoRR，卷。绝对/2512.00903，2025。

- [732] X. Z. Zonghuan Xu 和 X. Ma 和 Y. Jiang，“Tabvla：针对 视觉语言动作（Vision-Language-Action） 模型的针对性后门攻击”，CoRR，卷。绝对/2510.10932，2025。

- [733] S. M. Jonas Geiping 和 N. Jain、J. Kirchenbauer、S. Singh、B. R. Bartoldson、B. Kailkhura、A. Bhatele 和 T. Goldstein，“利用潜在推理扩展测试时计算：一种循环的深度方法”，CoRR，卷。绝对

/2502.05171，2025。

- [734] H. Z. Shuaijun Wang 和 D. Xiang 和 Y. You，“Tacrefinenet：仅触觉掌握任意手持物体姿势之间的细化”，CoRR，卷。绝对/2509.25746，2025。

- [735] S.W. Jialei Huang 和 F. Lin、Y. Hu、C. Wen 和 Y

。高，“Tactilevla：解锁 视觉语言动作（Vision-Language-Action） 模型的物理知识，用于触觉泛化”，CoRR，卷。绝对/2507.09160，2025。

- [736] H.X.ZongzhengZhang 和 Z.Yang、C.Yue、Z.Lin、H.Gao、Z.Wang 和 H.Zhao，“TA-VLA：阐明扭矩感知 视觉语言动作（Vision-Language-Action） 模型的设计空间”，CoRR，卷。绝对/2509.07962，2025。

- [737] C. Huang、Y. Wu、M. Chen、Y. F. Wang 和 F. Yang，“Thinkact：通过强化视觉潜在规划进行视觉语言-动作推理”，CoRR，卷。绝对/2507.16815，2025。

- [738] Z.W.Guanxing Lu 和 C.Liu、J.Lu 和 Y.Tang，“Thinkbot：遵循思想链推理的具体指令”，载于 ICLR。 OpenReview.net，2025 年。

- [739] J. Z. Shaoan Wang 和 M. Li、J. Liu、A. Li、K. Wu、F.zhong、J. Yu、Z.Zhang 和 H.Wang，“Trackvla：野外体现的视觉跟踪”，CoRR，卷。绝对/2505.23189，2025。

- [740] P. D. Jia Cheng Liu 和 Q. Zhou、Y. Wu、D. Huang、Z. Peng、W.iao、W.Zhang、L. Yang、C. Lu 和 D. Wang，“Trajbooster：通过以轨迹为中心的学习促进人形全身操纵”，CoRR，卷。绝对/2509.11839，2025。

- [741] Y.G.Zhenyang Liu 和 S.Zheng、X.Xue 和 Y.Fu，“Trivla：基于三系统的统一 视觉语言动作（Vision-Language-Action） 模型，用于通用机器人控制”，CoRR，卷。绝对/2507.01424，2025。

- [742] Y. Bai、Z. Wang、Y. Liu、K. Luo、Y. Wen、M. Dai、W. Chen、Z. Chen、L. Liu、G. Li 和 L. Lin，“学习观察和行动：机器人操作的任务感知虚拟视图探索”，arXiv 预印本，2025 年。

- [743] W. S. Jiayi Chen 和 P. Ding、Z. Zhou、H. Zhu、F. Tang、D. Wang 和 H. Li，“统一扩散 VLA：通过联合离散去噪扩散过程的 视觉语言动作（Vision-Language-Action） 模型”，CoRR，卷。绝对/2511.01718，2025。

- [744] Y.Z.Zhangyuan Wang 和 Y.Yan、X.Tian、X.Shao、M.Li、W.Li、G.Su、W.Cui 和 D.Fan，“Underwatervla：自主水下导航的双脑视觉语言动作架构”，CoRR，卷。绝对/2509.22441，2025。

- [745] Y.H.JiankeZhang 和 Y.Guo、X.Chen、Y.Liu、W.Chen、C.Lu 和 J.Chen，“Unicod：通过统一的连续和离散表示学习增强机器人策略”，CoRR，卷。绝对/2510.10642，2025。

- [746] C. C. Jiasen Lu 和 S. Lee、Z. Zhang、S. Khosla、R. Marten、D. Hoiem 和 A. Kembhavi，“Unified-io 2：利用视觉、语言、音频和动作扩展自回归多模态模型”，载于 CVPR。 IEEE，2024 年，第 26 429-26 445 页。

- [747] K. W. Jiazhao 张和 S. Wang、M. Li、H. Liu、S. Wei、Z. Wang、Z. 张和 H. Wang，“Uni-navid：用于统一具体导航任务的基于视频的视觉语言动作模型”，CoRR，卷。绝对/2412.06224，2024。

- [748] Z.L.Hao Lu 和 G.Jiang、Y.Luo、S.Chen、Y.Zhang 和 Y.Chen，“Uniugp：统一端到端自动驾驶的理解、生成和规划”，CoRR，卷。绝对/2512.09864，2025。

- [749] Y. Y. Hui Lu 和 Y. Yang、C. Yi、Q. Zhang、B. Shen、A. C. Kot 和 X. Jiang，“当机器人遵守补丁时：针对 视觉语言动作（Vision-Language-Action） 模型的通用可转移补丁攻击”，CoRR，卷。绝对/2511.21192，2025。

- [750] Y. G. 张建科、Y. Hu、X. Chen、X. Zhu 和 J

。 Chen，“UPVLA：体现主体的统一理解和预测模型”，载于_ICML_。 OpenReview.net，2025 年。

- [751] Z. W. Anqi Li 和 J.Zhang、M.Li、Y.Qi、Z.Chen、Z.Zhang 和 H.Wang，“Urbanvla：城市微交通的 视觉语言动作（Vision-Language-Action） 模型”，CoRR，卷。绝对/2510.23576，2025。

- [752] Z. W. Junwen Gu 和 P. Si、S. Qiu、Y. Feng、L. Sun、L. Luo、L. Yu、J. Wang 和 Z. Wu，“USIM 和 U0：通用水下机器人的 视觉语言动作（Vision-Language-Action） 数据集和模型”，CoRR，卷。绝对/2510.07869，2025。

- [753] R. Y. Chuning Zhu 和 S. Feng、B. Burchfiel、P. Shah 和 A. Gupta，“统一 世界模型（World Models）：在大型机器人数据集上耦合视频和动作扩散进行预训练”，CoRR，卷。绝对/2504.02792，2025。

- [754] L. A. Jonas Pai 和 V. Montesinos、B. Forrai、O. Mees 和 E. Nava，“模仿视频：超越 vlas 的通用机器人控制视频动作模型”，CoRR，卷。绝对/2512.15692，2025。

- [755] J. L. Yueru Jia 和 S. Liu、R. Zhou、W. Yu、Y. Yan、X. Chi、Y.Guo、B. Shi 和 S.Zhang，“Video2act：具有机器人空间运动、建模的双系统视频 扩散策略（Diffusion Policy）”，CoRR，卷。绝对/2512.03044，2025。

- [756] F. W. Yichao Shen 和 Z. Du、Y. Liang、Y. Lu、J. Yang、N. Cheng 和 B.Guo，“Videovla：视频生成器可以是通用的机器人操纵器”，CoRR，卷。绝对/2512.06963，2025。

- [757] J. Y. Jeongeun Park 和 B. Jeon、J. Park、J. Shin、N. Cho、K. Lee、S. Yun 和 S. Choi，“使用成功和失败的分层视觉语言动作模型，演示”，CoRR，卷。绝对/2512.03913，2025。

- [758] M.W.Guangyan Chen 和 Q.Shao、Z.Zhou、W.Mao、T.Cui、M.Zhu、Y.Deng、L.Yang、Z.Zhang、Y.Yang、H.Chen 和 Y.Yue，“看一次，然后行动：带有任务学习的 视觉语言动作（Vision-Language-Action） 模型，来自一次性视频演示”，CoRR，卷。绝对/2512.07582，2025。

- [759] H. H. Ankit Goyal 和 X. Yang、V. Blukis 和 F. Ramos，“VLA0：以零修改构建最先进的 vlas”，CoRR，卷。绝对/2510.13054，2025。

- [760] C. M. Hanyu Zhou 和 G. H. Lee，“VLA-4D：将 4d 意识嵌入到 视觉语言动作（Vision-Language-Action） 模型中，用于时空相干的机器人操作”，CoRR，卷。 ABS/2511.17199，2025。

- [761] P. D. Yihao Wang 和 L. Li、C. Cui、Z. Ge、X. Tong、W. Song、H. Zhu、W. Zhua、P. Hou、S. Huang、Y. Tang、W. Wang、R. Zhang、J. Liu 和 D. Wang，“Vla-adapter：微小尺度视觉语言动作的有效范式，模型”，CoRR，卷。绝对/2509.09372，2025。

- [762] M. Z. Yuze Wu 和 X. Li、Y. Du、Y. Fan、W. Li、Z. Han、X. Zhou 和 F. Gau，“VLA-AN：一种高效的机载 视觉语言动作（Vision-Language-Action） 框架，用于复杂环境中的空中导航”，CoRR，第 1 卷。绝对/2512.15258，2025。

- [763] Y. W. Siyu Xu 和 C. Xia、D. Zhu、T. Huang 和 C. Xu，“Vlacache：通过机器人操作中的自适应令牌缓存实现高效的 视觉语言动作（Vision-Language-Action） 模型”，CoRR，卷。绝对/2502.02175，2025。

[764] Y. X. Yuping Yan and, Y. Zhang, L. Lyu, H. Wang, and Y. Jin, "When alignment fails: Multimodal adversarial attacks on vision-language-action, models," *CoRR*, vol. abs/2511.16203, 2025.

[765] Z. L. Chongkai Gao and, Z. Chi, J. Huang, X. Fei, Y. Hou, Y. Zhang, Y. Lin, Z. Fang, Z. Jiang, and L. Shao, "VLA-OS: structuring and dissecting planning representations and, paradigms in vision-language-action models," *CoRR*, vol. abs/2506.17561, 2025.

[766] J. L. Zhuo Li and, Z. Dong, T. Teng, Q. Rouxel, D. G. Caldwell, and F. Chen, "Towards deploying VLA without fine-tuning: Plug-and-play inference-time, VLA policy steering via embodied evolutionary diffusion," *CoRR*, vol. abs/2511.14178, 2025.

[767] Y. C. Ziyan Liu and, H. Cai, T. Lin, S. Yang, Z. Liu, and B. Zhao, "Vla-pruner: Temporal-aware dual-level visual token pruning for efficient, vision-language-action inference," *CoRR*, vol. abs/2511.16449, 2025.

[768] O. G. Y. Cyrus Neary and, A. Kuramshin, O. Aslan, and G. Berseth, "Improving pre-trained vision-language-action policies with model-based, search," *CoRR*, vol. abs/2508.12211, 2025.

[769] S. M. Hyunki Seong and, H. Ahn, J. Kang, and D. H. Shim, "Vla-r: Vision-language action retrieval toward open-world end-to-end autonomous driving," 2025. [Online]. Available: https://dblp.org/rec/journals/corr/abs-2511-12405

[770] Z. Z. Angen Ye and, B. Wang, X. Wang, D. Zhang, and Z. Zhu, "Vla-r1: Enhancing reasoning in vision-language-action models," 2025. [Online]. Available: https://dblp.org/rec/journals/corr/abs-2510-01623

[771] L. Z. Yongsheng Zhao and, B. Cheng, G. Yao, X. Wen, and H. Gao, "VLA-RAIL: A real-time asynchronous inference linker for VLA, models and robots," *CoRR*, vol. abs/2512.24673, 2025.

[772] P. D. Hengtao Li and, R. Suo, Y. Wang, Z. Ge, D. Zang, K. Yu, M. Sun, H. Zhang, D. Wang, and W. Su, "VLA-RFT: vision-language-action reinforcement fine-tuning with verified, rewards in world simulators," *CoRR*, vol. abs/2510.00406, 2025.

[773] P. D. Wei Zhao and, M. Zhang, Z. Gong, S. Bai, H. Zhao, and D. Wang, "VLAS: vision-language-action model with speech instructions for, customized robot manipulation," in *ICLR*. OpenReview.net, 2025.

[774] T. Z. Ganlin Yang and, H. Hao, W. Wang, Y. Liu, D. Wang, G. Chen, Z. Cai, J. Chen, W. Su, W. Zhou, Y. Qiao, J. Dai, J. Pang, G. Luo, W. Wang, Y. Mu, and Z. Hou, "Vlaser: Vision-language-action model with synergistic embodied reasoning," *CoRR*, vol. abs/2510.11027, 2025.

[775] Y. S. Jiaming Tang and, Y. Zhao, S. Yang, Y. Lin, Z. Zhang, J. Hou, Y. Lu, Z. Liu, and S. Han, "VLASH: real-time vlas via future-state-aware asynchronous inference," *CoRR*, vol. abs/2512.01031, 2025.

[776] X. W. Asher J. Hancock and, L. Zha, O. Russakovsky, and A. Majumdar, "Actions as language: Fine-tuning vlms into vlas without catastrophic, forgetting," *CoRR*, vol. abs/2509.22195, 2025.

[777] M. W. Guangyan Chen and, T. Cui, Y. Mu, H. Lu, T. Zhou, Z. Peng, M. Hu, H. Li, L. Yuan, Y. Yang, and Y. Yue, "Vlmimic: Vision language models are visual imitation learner for fine-grained, actions," in *NeurIPS*, 2024.

[778] C. W. Wenlong Huang and, R. Zhang, Y. Li, J. Wu, and L. Fei-Fei, "Voxposer: Composable 3d value maps for robotic manipulation with language, models," in *CoRL*, ser. Proceedings of Machine Learning Research, vol. 229. PMLR, 2023, pp. 540–562.

[779] M. W. Mingjie Xu and, Y. Zhao, J. C. L. Li, and W. Ou, "Llava-spacesgg: Visual instruct tuning for open-vocabulary scene graph, generation with enhanced spatial relations," in *WACV*. IEEE, 2025, pp. 6362–6372.

[780] J. C. Yifang Xu and, F. Cai, Z. Zhu, H. Shang, S. Luan, M. Xu, N. Zhang, Y. Li, J. Cai, and S. Zhu, "Wam-flow: Parallel coarse-to-fine motion planning via discrete flow, matching for autonomous driving," *CoRR*, vol. abs/2512.06112, 2025.

[781] J. C. Haoran Jiang and, Q. Bu, L. Chen, M. Shi, Y. Zhang, D. Li, C. Suo, C. Wang, Z. Peng, and H. Li, "Wholebodyvla: Towards unified latent VLA for whole-body loco-manipulation, control," *CoRR*, vol. abs/2512.11047, 2025.

[782] Z. Y. Fangqi Zhu and, Z. Hong, Q. Shou, X. Ma, and S. Guo, "WMPO: world model-based policy optimization for vision-language-action, models," *CoRR*, vol. abs/2511.09515, 2025.

[783] X. C. Zezhong Qian and, Y. Li, S. Wang, Z. Qin, X. Ju, S. Han, and S. Zhang, "Wristworld: Generating wrist-views via 4d world models for robotic, manipulation," *CoRR*, vol. abs/2510.07313, 2025.

[784] H. C. Pei Yang and, Y. Song, and M. Z. Shou, "X-humanoid: Robotize human videos to generate humanoid videos at scale," *CoRR*, vol. abs/2512.04537, 2025.

[785] K. W. Shichao Fan and, Z. Che, X. Wang, D. Wu, F. Liao, N. Liu, Y. Zhang, Z. Zhao, Z. Xu, M. Li, Q. Liu, S. Zhang, M. Wan, and J. Tang, "XR-1: towards versatile vision-language-action models via learning, unified vision-motion representations," *CoRR*, vol. abs/2511.02776, 2025.

[786] J. L. Jinliang Zheng and, Z. Wang, D. Liu, X. Kang, Y. Feng, Y. Zheng, J. Zou, Y. Chen, J. Zeng, Y. Zhang, J. Pang, J. Liu, T. Wang, and X. Zhan, "X-VLA: soft-prompted transformer as scalable cross-embodiment vision-language-action, model," *CoRR*, vol. abs/2510.10274, 2025.

原文第 36 页 | 中文翻译初稿

- [764] Y. X. Yuping Yan 和 Y.Zhang、L.Lyu、H.Wang 和 Y.Jin，“当对齐失败时：对视觉语言动作、模型的多模态对抗攻击”，CoRR，卷。绝对/2511.16203，2025。

- [765] Z. L. 高崇凯、Z. Chi、J. Huang、X. Fei、Y. Hou、Y. Zhuang、Y. Lin、Z. Fang、Z. Jiang 和 L. Shao，“VLA-OS：构建和剖析视觉语言动作模型中的规划表示和范式”，CoRR，卷。绝对/2506.17561，2025。

- [766] J. L. Zhuo Li 和 Z. Dong、T. Teng、Q. Rouxel、D. G. Caldwell 和 F. Chen，“在不进行微调的情况下部署 VLA：即插即用推理时间，通过体现进化扩散进行 VLA 策略控制”，CoRR，卷。绝对/2511.14178，2025。

和 J. Tang，“XR-1：通过学习、统一视觉运动表示实现多功能 视觉语言动作（Vision-Language-Action）模型”，CoRR，卷。绝对/2511.02776，2025。

- [786] J. L. Jinliang Cheng 和 Z. Wang、D. Liu、X. Kang、Y. Feng、Y. Cheng、J. Zou、Y. Chen、J. Zeng、Y. Zhang、J. Pang、J. Liu、T. Wang 和 X. Zhan，“X-VLA：作为可扩展跨实施例 视觉语言动作（Vision-Language-Action）模型的软提示变压器”，CoRR，卷。绝对/2510.10274，2025。

- [767] Y. C. Ziyan Liu 和 H. Cai、T. Lin、S. Yang、Z. Liu 和 B. Zhu，“Vlapruner：用于高效 视觉语言动作（Vision-Language-Action）推理的时间感知双层视觉标记修剪”，CoRR，卷。绝对/2511.16449，2025。

- [768] O.G.Y. Cyrus Neary 和 A. Kuramshin、O. Aslan 和 G. Berseth，“通过基于模型的搜索改进预训练的 视觉语言动作（Vision-Language-Action）策略”，CoRR，卷。绝对/2508.12211，2025。

- [769] S. M. Hyunki Seong 和 H. Ahn、J. Kang 和 D. H. Shim，“Vla-r：面向开放世界端到端自动驾驶的视觉语言动作检索”，2025 年。[在线]。可用：https://dblp.org/rec/journals/corr/abs-2511-12405

- [770] Z. Z. Angen Ye 和 B. Wang、X. Wang、D. Zhang 和 Z. Zhu，“Vlar1：增强 视觉语言动作（Vision-Language-Action）模型中的推理”，2025 年。[在线]。可用：https://dblp.org/rec/journals/corr/abs-2510-01623

- [771] L.Z.YongshengZhao和B.Cheng、G.Yao、X.Wen和H.Gao，“VLA-RAIL：用于VLA、模型和机器人的实时异步推理链接器”，CoRR，卷。绝对/2512.24673，2025。

- [772] P. D. Hengtao Li 和 R. Suo、Y. Wang、Z. Ge、D. Zang、K. Yu、M. Sun、H. Zhang、D. Wang 和 W. Su，“VLA-RFT：视觉语言动作（Vision-Language-Action）强化微调，并在世界模拟器中进行验证和奖励”，CoRR，卷。绝对/2510.00406，2025。

- [773] P.D.WeiZhao 和 M.Zhang、Z.Gong、S.Bai、H.Zhao 和 D.Wang，“VLAS：带有语音指令的 视觉语言动作（Vision-Language-Action）模型，用于定制机器人操作”，载于 ICLR。 OpenReview.net，2025 年。

- [774] T. Z. Ganlin Yang 和 H. Hai、W. Wang、Y. Liu、D. Wang、G. Chen、Z. Cai、J. Chen、W. Su、W. Zhou、Y. Qiao、J. Dai、J. Pang、G. Luo、W. Wang、Y. Mu 和 Z. Hou，“Vlaser：具有协同体现推

理的 视觉语言动作（Vision-Language-Action） 模型”， CoRR，卷。绝对/2510.11027，2025。

- [775] Y. S. Jiaming Tang 和 Y. Zhu、S. Yang、Y. Lin、Z. Zhang、J. Hou、Y. Lu、Z. Liu 和 S. Han，“VLASH：通过未来状态感知异步推理的实时 vlas”，CoRR，卷。绝对/2512.01031，2025。

- [776] X.W. Asher J. Hancock 和 L. Zha, O

。 Russakovsky 和 A. Majumdar，“作为语言的动作：将 vlm 微调为 vlas，而不会发生灾难性的遗忘”，CoRR，卷。绝对/2509.22195，2025。

- [777] M.W.Guangyan Chen 和 T.Cui、Y.Mu、H.Lu、T.Zhou、Z.Peng、M.Hu、H.Li、L.Yuan、Y.Yang 和 Y.Yue，“Vlmimic：视觉语言模型是细粒度动作的视觉模仿学习器”，《NeurIPS》，2024 年。

- [778] C. W. Wenlong Huang 和 R. Zhang、Y. Li、J. Wu 和 L. Fei-Fei，“Voxposer：用于使用语言、模型进行机器人操作的可组合 3D 值图”，载于_CoRL_，系列。机器学习研究论文集，卷。 229.PMLR，2023 年，第 540-562 页。

- [779] M. W. Mingjie Xu 和 Y. Zhao、J. C. L. Li 和 W. Ou，“Llavaspacesgg：开放词汇场景图的视觉指令调整，增强空间关系的生成”，载于_WACV_。 IEEE，2025 年，第 6362-6372 页。

- [780] J. C. Yifang Xu 和 F. Cai、Z. Zhu、H. Shang、S. Luan、M. Xu、N. Zhang、Y. Li、J. Cai 和 S. Zhu，“Wam-flow：通过离散流的并行粗到细运动规划，自动驾驶匹配”，CoRR，卷。绝对/2512.06112，2025。

- [781] J.C.Haoran Jiang 和 Q.Bu、L.Chen、M.Shi、Y.Zhang、D.Li、C.Suo、C.Wang、Z.Peng 和 H.Li，“Wholebodyvla：迈向全身局部操作、控制的统一潜在 VLA”，CoRR，卷。绝对/2512.11047，2025。

- [782] Z. Y. Fangqi Zhu 和 Z. Hong、Q. Shou、X. Ma 和 S.Guo，“WMPO：基于 世界模型（World Model） 的视觉语言动作策略优化，模型”，CoRR，卷。绝对/2511.09515，2025。

- [783] X. C. Zezhong Qi 和 Y. Li、S. Wang、Z.qin、X. Ju、S. Han 和 S.Zhang，“手腕世界：通过 4d 世界模型（World Models） 为机器人操作生成手腕视图”，CoRR，卷。绝对/2510.07313，2025。

- [784] H. C. Pei Yang 和 Y. Song 以及 M. Z. Shou，“X-人形机器人：机器人化人类视频以大规模生成人形视频”，CoRR，卷。绝对/2512.04537，2025。

- [785] K. W. Shichao Fan 和 Z. Che、X. Wang、D. Wu、F. Liao、N. Liu、Y. 张、Z. 赵、Z. Xu、M. Li、Q. Liu、S. Zhang、M. Wan、

APPENDIX

A. Background

1) Unimodal Models: Vision-language-action models integrate three modalities, often relying on existing unimodal models. The transition from convolutional neural networks (e.g., ResNet [220]) to visual Transformers (e.g., ViT [221], SAM [222]) in computer vision has facilitated the development of vision foundation models (VFMs). In natural language processing, the evolution from recurrent neural networks (e.g., LSTM [223], GRU [224]) to Transformers [3] initially led to the pretrain-finetune paradigm (e.g., BERT [27], ChatGPT [225]), followed by the recent success of prompt tuning driven by large language models. Reinforcement learning (e.g., DQN [4], AlphaGo [226], PPO [227], Dactyl [228]) has also witnessed a shift towards employing Transformers to model the Markov Decision Process as autoregressive sequential data. DPO [229] directly trains LLMs on human preferences, simplifying RLHF.

2) Vision-Language Models: Vision-language tasks, encompassing image captioning [230], visual question answering [231], visual grounding [232], require the fusion of computer vision and natural language processing models. Early efforts, such as Show and Tell [233], leveraged the success of early CNNs and RNNs. The advent of high-capacity language models, including BERT [27] and GPT [234], ushered in a new era of Transformer-based VLMs. One of the pioneering self-supervised pretraining methods is ViLBERT [235], while CLIP [23] popularized contrastive pretraining approaches for multimodal alignment. The recent emergence of large language models has driven the development of multimodal LLMs (MLLMs) or large multimodal models (LMMs), which achieve state-of-the-art performance on multimodal instruction-following tasks. Representative MLLMs include Flamingo [6], BLIP-2 [7], and LLaVA [8]. VLMs share a close relationship with VLAs, as the multimodal architectures of VLMs can be readily adopted for VLAs. Additionally, VLMs can function as high-level task planners if they possess sufficient reasoning capabilities.

B. Background (Extended Version): Unimodal Models

Vision-language-action models involve three modalities, and consequently, many VLAs depend on existing unimodal models for processing inputs from different modalities. Therefore, it is crucial to summarize representative developments in unimodal models, as they often serve as integral components in VLAs. Specifically, for the vision modality, we collect models designed for image classification, object detection, and image segmentation, as these tasks are particularly relevant for robotic learning. Natural language processing models play a crucial role in enabling VLAs to understand language instructions or generate language responses. Reinforcement learning is a foundational component for obtaining optimal policies, facilitating the generation of appropriate actions in a given environment and condition. A brief timeline of the development of unimodal models is depicted in Figure 7. Additionally, Figure 8 highlights the progressive increase in model size within these fields.

1) Computer Vision: Computer vision witnessed the inception of modern neural networks. In robotics, object classification models can be used to inform a policy about which objects are of interest, and models for object detection or image segmentation can help precisely locate objects. Therefore, we mainly summarize approaches for these tasks, but numerous excellent surveys on visual models, ranging from convolutional neural networks (CNNs) [236] to Transformers [237], offer more detailed insights. Interested readers are directed to these surveys for a more comprehensive introduction. Here, we will briefly touch upon some of the key developments in the field of computer vision.

a) Convolutional Neural Network: Early developments in computer vision (CV) were primarily focused on the image classification task. LeNet [238] was among the first convolutional neural networks, designed for identifying handwritten digits in zip codes. In 2012, AlexNet [2] emerged as a breakthrough by winning the ImageNet challenge, showcasing the potential of neural networks. VGG [239] demonstrated the benefits of increasing the depth of CNNs. GoogLeNet [240], also known as Inception-V1, introduced the concept of blocks. ResNet [220] introduced skip connections or residual connections. Inception-ResNet [241], as the name suggests, combines residual connects and inception blocks. ResNeXt [242] explored the concept of split, transform, and merge. SENet [243] introduces the squeeze-and-excitation blocks, utilizing a type of attention mechanism. EfficientNet [244] studied the width, depth, and resolution of CNN models with "compound scaling," highlighting the trade-off between efficiency and performance.

Alongside image classification, object detection became an integral component in many applications. Building upon the success of image classification backbone networks, a series of works optimized region-based methods: R-CNN [245], Fast R-CNN [246], Faster R-CNN [247], and Mask R-CNN [248]. Grid-based methods like YOLO [249] are also widely adopted. Bottom-up, top-down is also a popular strategy, employed by FPN [250], RetinaNet [251], BUTD [252], etc. In scenarios requiring more detailed and precise object detection, image segmentation aims to determine the exact outline of objects. Many popular models adopt an "encoder-decoder" architecture, where the encoder understands both the global and local context of the image, and the decoder produces a segmentation map based on this context information. Representative works following this idea include FCN [253], SegNet [254], Mask R-CNN [248], and U-Net [255].

b) Vision Transformer: Convolutional neural networks (CNNs) have historically been the foundation of computer vision models. However, the landscape shifted with the introduction of the Transformer architecture in the seminal work by [3]. This paradigm shift was initiated by ViT [221]. It revolutionizes image processing by breaking down images into 16-by-16 pixel patches, treating each as a token akin to those in NLP; leveraging a BERT-like model, ViT encodes these patches and has exhibited superior performance over many traditional CNN models in image classification tasks.

The transformative power of the Transformer extends beyond classification. DETR [256] employs an encoder-decoder

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附录
A。背景

1) 单峰模型： 视觉语言动作（Vision-Language-Action） 模型集成了三种模态，通常依赖于现有的单峰模型。计算机视觉中从卷积神经网络（例如，ResNet [220]）到视觉 Transformers（例如，ViT [221]，SAM [222]）的转变促进了视觉 基础模型（Foundation Models）（VFM）的发展。在自然语言处理中，从循环神经网络（例如 LSTM [223]、GRU [224]）到 Transformers [3] 的演变最初导致了预训练微调范式（例如 BERT [27]、ChatGPT [225]），随后由大型语言模型驱动的即时调优取得了最近的成功。 强化学习（Reinforcement Learning）（例如，DQN [4]、AlphaGo [226]、PPO [227]、Dactyl [228]）也见证了使用 Transformer 将马尔可夫决策过程建模为自回归序列数据的转变。 DPO [229] 直接根据人类偏好训练法学硕士，简化了 RLHF。

_2）视觉语言模型：_视觉语言任务，包括图像字幕[230]、视觉问答[231]、视觉落地关联（Grounding）[232]，需要计算机视觉和自然语言处理模型的融合。早期的努力，例如 Show and Tell [233]，利用了早期 CNN 和 RNN 的成功。高容量语言模型（包括 BERT [27] 和 GPT [234]）的出现，开创了基于 Transformer 的 VLM 的新时代。 ViLBERT [235] 是开创性的自监督预训练方法之一，而 CLIP [23] 则推广了用于多模态对齐的对比预训练方法。最近出现的大型语言模型推动了多模态 LLM (MLLM) 或大型多模态模型 (LMM) 的发展，它们在多模态指令跟踪任务上实现了最先进的性能。代表性的 MLLM 包括 Flamingo [6]、BLIP-2 [7] 和 LLaVA [8]。 VLM 与 VLA 有着密切的关系，因为 VLM 的多模式架构可以很容易地用于 VLA。此外，如果 VLM 具有足够的推理能力，则可以充当高级任务规划器。
B。背景（扩展版）：单峰模型

视觉语言动作（Vision-Language-Action） 模型涉及三种模态，因此，许多 VLA 依赖于现有的单模态模型来处理来自不同模态的输入。因此，总结单峰模型的代表性发展至关重要，因为它们通常作为 VLA 的组成部分。具体来说，对于视觉模态，我们收集专为图像分类、对象检测和图像分割而设计的模型，因为这些任务与机器人学习特别相关。自然语言处理模型在使 VLA 理解语言指令或生成语言响应方面发挥着至关重要的作用。 强化学习（Reinforcement Learning） 是获得最佳策略的基础组件，有助于在给定环境和条件下生成适当的操作。图 7 描绘了单峰模型发展的简要时间表。此外，图 8 突出显示了这些领域内模型大小的逐步增加。

_1）计算机视觉：_计算机视觉见证了现代神经网络的诞生

。在机器人技术中，对象分类模型可用于告知有关哪些对象感兴趣的策略，而对象检测或图像分割模型可以帮助精确定位对象。因此，我们主要总结这些任务的方法，但从卷积神经网络（CNN）[236]到 Transformers [237]，许多关于视觉模型的优秀调查提供了更详细的见解。有兴趣的读者可以阅读这些调查以获得更全面的介绍。在这里，我们将简要介绍计算机视觉领域的一些关键发展。

_a）卷积神经网络：_计算机视觉（CV）的早期发展主要集中在图像分类任务上。 LeNet [238] 是最早的卷积神经网络之一，旨在识别邮政编码中的手写数字。 2012 年，AlexNet [2] 取得了突破，赢得了 ImageNet 挑战，展示了神经网络的潜力。 VGG [239] 证明了增加 CNN 深度的好处。 GoogLeNet [240]，也称为 Inception-V1，引入了块的概念。 ResNet [220]引入了跳跃连接或剩余连接。 Inception-ResNet [241]，顾名思义，结合了残差连接和初始块。 ResNeXt [242]探索了分割、变换和合并的概念。 SENet

[243] 利用一种注意力机制引入了挤压和激励块。 EfficientNet [244] 通过“复合缩放”研究了 CNN 模型的宽度、深度和分辨率，强调了效率和性能之间的权衡。

除了图像分类之外，目标检测也成为许多应用中不可或缺的组成部分。基于图像分类主干网络的成功，一系列基于区域的优化方法的工作：R-CNN [245]、Fast R-CNN [246]、Faster R-CNN [247] 和 Mask R-CNN [248]。 YOLO [249] 等基于网格的方法也被广泛采用。自下而上、自上而下也是一种流行的策略，FPN[250]、RetinaNet[251]、BUTD[252]等都采用这种策略。在需要更详细和精确的对象检测的场景中，图像分割的目的是确定对象的精确轮廓。许多流行的模型采用“编码器-解码器”架构，其中编码器了解图像的全局和局部上下文，解码器根据该上下文信息生成分割图。遵循这一思想的代表作品包括FCN[253]、SegNet[254]、Mask R-CNN[248]和U-Net[255]。

b) Vision Transformer： 卷积神经网络 (CNN) 历来是计算机视觉模型的基础。然而，随着 [3] 的开创性工作中 Transformer 架构的引入，情况发生了变化。这种范式转变是由 ViT [221] 发起的。它通过将图像分解为 16×16 像素块，将每个块视为类似于 NLP 中的标记，彻底改变了图像处理； ViT 利用类似 BERT 的模型对这些补丁进行编码，并在图像分类任务中表现出优于许多传统 CNN 模型的性能。

Transformer 的变革力量超越了分类。 DETR [256]采用编码器-解码器

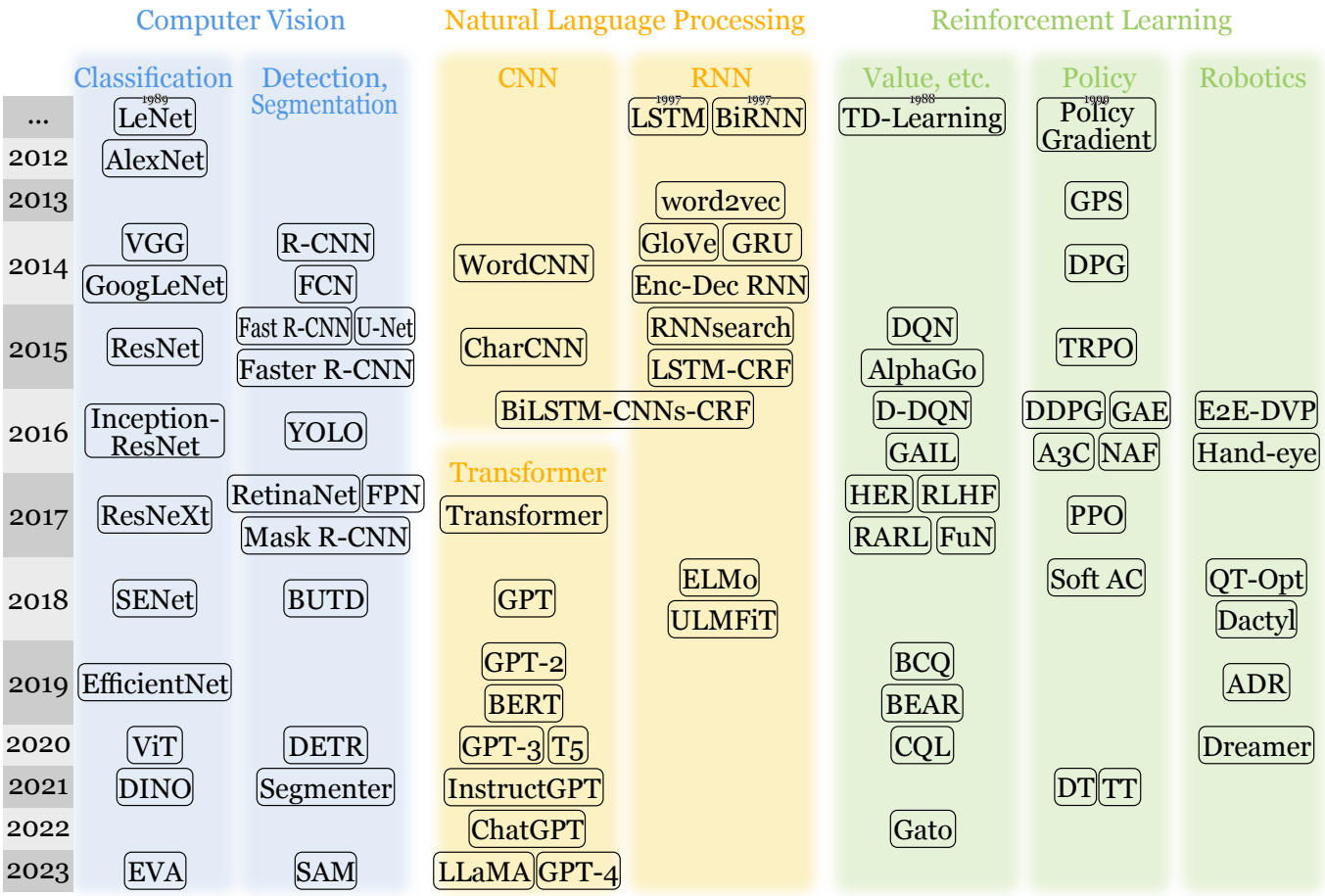


Fig. 7. A brief timeline of pivotal unimodal models leading to the development of vision-language-action models, organized by their publication years. Details can be found in Appendix:B.

Transformer architecture to tackle object detection. The encoder processes the input image, and its output embeddings are fed into the decoder through cross-attention. Notably, DETR introduces learnable object queries to the decoder, facilitating the extraction of crucial object-wise information from the encoder's output. Venturing into image segmentation, Segmenter [257] was the first to utilize Transformer on this task. The Segment Anything model (SAM) [222] achieves remarkable milestones in promptable segmentation, zero-shot performance, and versatile architecture, further underlining the transformative impact of Vision Transformer models in various computer vision domains.

c) Vision in 3D: Aside from the most common RGB data, other types of visual inputs are widely used [258], [259]. In robotics, depth maps are useful since they provide essential 3D information that is not explicitly stored in RGB images. Depth maps can be captured with Microsoft Kinect [1] or Intel RealSense [2] or recovered from pure RGB images. Point clouds [260] are also popular visual input types due to the widespread adoption of LiDARs and 3D scanners; depth maps can be easily converted to point clouds. Volumetric data [261], such as voxels or octrees, is usually more information-rich than depth maps and is suitable for representing rigid objects. Despite the widespread use of 3D meshes as the default data format in computer graphics, their irregular nature poses challenges for neural networks [262].

2) Natural Language Processing: Natural Language Processing (NLP) plays a pivotal role in VLA, serving as a vital component for understanding user instructions or even generating appropriate textual responses. The recent surge in NLP owes much to the success of Transformer models [3]. In the landscape of contemporary NLP, there is a noticeable shift towards implicit learning of language syntax and semantics, a departure from the previous paradigms. To provide context, this subsection will commence with a concise overview of fundamental yet enduring concepts before delving into the noteworthy advancements in contemporary NLP. For an in-depth exploration of the progress in the NLP domain, readers are directed to comprehensive surveys by [263] and [264], which meticulously review the trajectory of advancements in NLP.

a) Early Developments: The field of NLP, which was more frequently referred to as Computational Linguistics (CL) in the early days, tries to solve various tasks regarding to natural language. In CL, natural languages used to be processed in a hierarchical way: word, syntax, and semantics. Firstly, on the

[1]https://azure.microsoft.com/en-us/products/kinect-dk/
[2]https://www.intelrealsense.com

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计算机视觉 自然语言处理 强化学习（Reinforcement Learning）
分类检测、CNN RNN 值等 策略机器人
... LeNet 分割 LSTM BiRNN TD-学习策略
梯度
2012 AlexNet
2013 word2vec GPS
VGG R-CNN GloVe GRU
2014 WordCNN DPG
GoogLeNet FCN Enc-Dec RNN
2013 Fast R-CNN U-Net RNNsearch DQN
2015 ResNet CharCNN TRPO
更快 R-CNN LSTM-CRF AlphaGo
Inception- BiLSTM-CNNs-CRF D-DQN DDPG GAE E2E-DVP
2016 YOLO
ResNet GAIL A3C NAF手眼变压器
2013 RetinaNet FPN HER RLHF
2017 ResNeXt Transformer PPO
Mask R-CNN RARL FuN
ELMo Soft AC QT-Opt
2018 SENet BUTD GPT
ULMFiT Dactyl
2013 GPT-2 BCQ
2019 EfficientNet ADR
BERT BEAR
2020 ViT DETR GPT-3 T5 CQL Dreamer
20 2113 DINO Segmenter InstructGPT DT TT
2022 ChatGPT Gato
2023 EVA SAM LLaMA GPT-4

图 7. 导致 视觉语言动作（Vision-Language-Action）模型开发的关键单峰模型的简短时间表，按其发布年份组织。详细信息请参见附录：B。

用于解决对象检测问题的 Transformer 架构。编码器处理输入图像，其输出嵌入通过交叉注意力输入解码器。值得注意的是，DETR 向解码器引入了可学习的对象查询，有助于从编码器的输出中提取关键的对象信息。 Segmenter [257] 涉足图像分割，是第一个在这项任务中使用 Transformer 的人。 Segment Anything 模型（SAM）[222] 在快速分割、零样本性能和多功能架构方面实现了显着的里程碑，进一步强调了 Vision Transformer 模型在各种计算机视觉领域的变革性影响。

c) 3D 视觉： 除了最常见的 RGB 数据之外，其他类型的视觉输入也被广泛使用 [258]、[259]。在机器人技术中，深度图非常有用，因为它们提供了未明确存储在 RGB 图像中的基本 3D 信息。深度图可以使用 Microsoft Kinect1 或 Intel RealSense2 捕获，也可以从纯 RGB 图像中恢复。由于 LiDAR 和 3D 扫描仪的广泛采用，点云 [260] 也是流行的视觉输入类型；深度图可以轻松转换为点云。体积数据 [261]，例如体素或八叉树，通常比深度信息更丰富

> 1https://azure.microsoft.com/en-us/products/kinect-dk/

> 2https://www.intelrealsense.com

映射并适合表示刚性物体。尽管 3D 网格广泛用作计算机图形学中的默认数据格式，但它们的不规则性质给神经网络带来了挑战[262]。

2) 自然语言处理： 自然语言处理 (NLP) 在 VLA 中发挥着关键作用，是理解用户指令甚至生成适当文本响应的重要组成部分。最近 NLP 的激增很大程度上归功于 Transformer 模型的成功 [3]。在当代 NLP 领域，语言语法和语义的内隐学习发生了明显的转变，这与之前的范式背道而驰

。为了提供背景信息，本小节将首先对基本但持久的概念进行简要概述，然后再深入研究当代 NLP 中值得注意的进展。为了深入探索 NLP 领域的进展，读者可以阅读 [263] 和 [264] 的综合调查，这些调查仔细回顾了 NLP 的进展轨迹。

a) 早期发展： NLP 领域早期更常被称为计算语言学 (CL)，试图解决与自然语言相关的各种任务。在 CL 中，自然语言过去以分层方式进行处理：单词、语法和语义。首先，关于

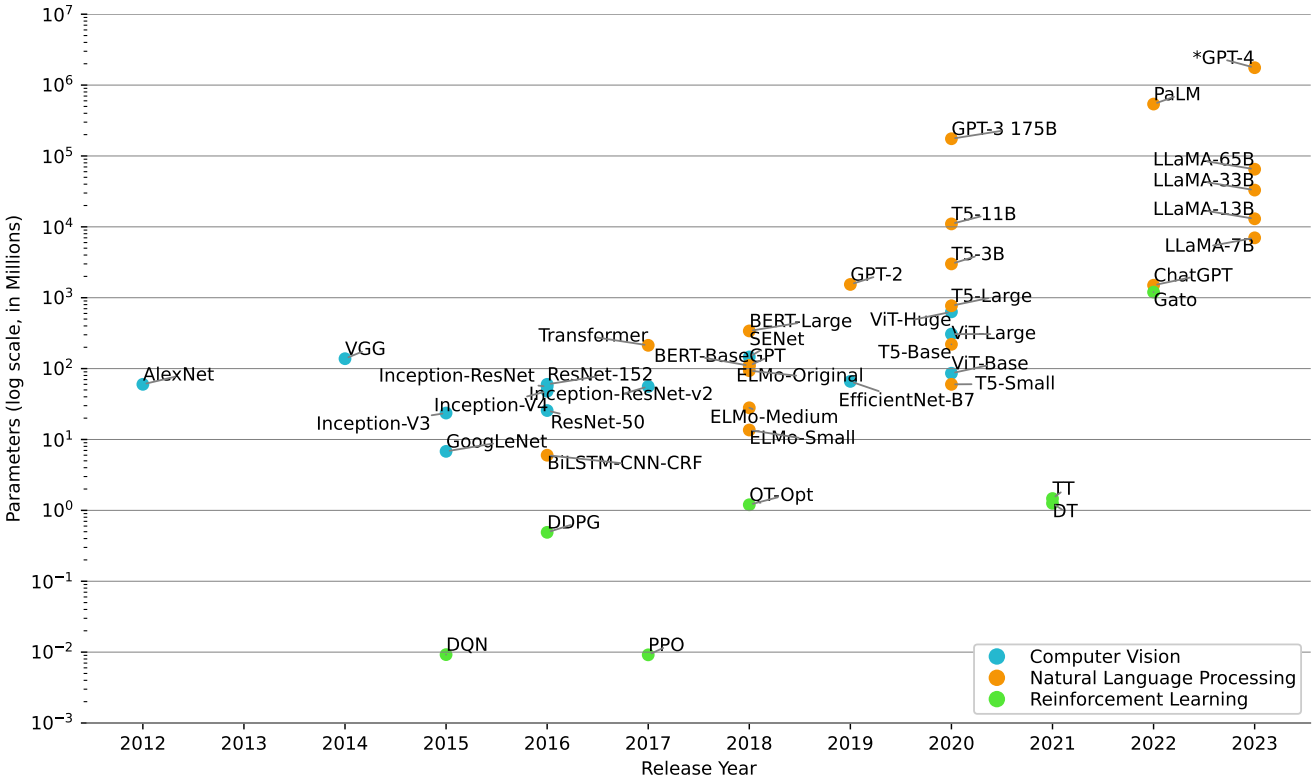


Fig. 8. The growing scale of unimodal models over the years. ∗GPT-4's model size is estimated since it has not been officially disclosed.

word level, many aspects need to be accounted for, including morphology, lexicology, phonology, etc. This leads to problems like tokenization, lemmatization, stemming, semantic relations, word sense disambiguation, and Zipf's Law problem. Then, in terms of syntax, natural language, in contrast to formal language, has much less restricted grammar and thus is more challenging to parse: in the Chomsky hierarchy, natural language is generally considered to follow context-sensitive grammar, while programming languages are covered under context-free grammar. Syntactic parsing includes tasks such as part-of-speech tagging, constituency parsing, dependency parsing, and named entity recognition. Finally, to understand the semantics of a sentence in written language or an utterance in spoken language, the following tasks were studied: semantic role labeling, frame semantic parsing, abstract meaning representation, logical form parsing, etc.

b) Recurrent Neural Network & Convolutional Neural Network: In the initial stages of NLP, rudimentary models relied on simple feed-forward neural networks to tackle various tasks [265]. After the introduction of word embeddings like word2vec [266], [267] and GloVe [268], NLP techniques embraced recurrent neural networks (RNNs) [269], such as LSTM [223], GRU [224], RNNsearch [270], LSTM-CRF [271], etc. Examples of representative RNNs in NLP include ELMo [272] and ULMFiT [273]. While RNNs played a significant role, alternative models utilizing convolutional neural networks also emerged. WordCNN [274] employed CNNs at the word level, with word2vec features serving as input to the CNNs. Another approach, CharCNN [275], focused on modeling language at the character level with CNN. Subsequent research [276] highlighted that character-level CNNs excel at capturing word morphology, and their combination with a word-level LSTM backbone can significantly enhance performance.

c) Transformer & Large Language Model: The groundbreaking Transformer model, introduced by [3], revolutionized natural language processing through the introduction of the self-attention mechanism, inspiring a cascade of subsequent works. BERT [27] leverages the Transformer encoder stack, excelling in natural language understanding. On the other hand, the GPT family [234], [277], [278], [279] is built upon Transformer decoder blocks, showcasing prowess in natural language generation tasks. A line of work strives to refine the original BERT, including RoBERTa [280], ALBERT [281], ELECTRA [282]. Simultaneously, a parallel line of research following the GPT paradigm has given rise to models like XLNet [283], OPT [284]. BART [285], an encoder-decoder Transformer, distinguishes itself through pretraining using the denoising sequence-to-sequence task. Meanwhile, T5 [286] introduces modifications to the original Transformer, maintaining the encoder-decoder architecture. T5 unifies various NLP tasks through a shared text-to-text format, exhibiting enhanced performance in transfer learning. These diverse models collectively showcase the versatility and ongoing evolution within the NLP landscape.

Over the past few years, there has been a remarkable expansion in the size of language models, driven by the

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10 7
*GPT-4
10 6
PaLM
GPT-3 175B
10 5 LLaMA-65B
LLaMA-33B
T5-11B LLaMA-13B
10 4
LLaMA-7B
T5-3B
GPT-2 ChatGPT
10 3 T5-Large Gato
BERT-Large ViT-Huge
变压器 SENet ViT-Large
10 2 AlexNet VGG Inception-ResNet ResNet-152 BERT-Base ELMo-Original GPT T5-Base ViT-Base
T5-Small
Inception-V4Inception-ResNet-v2 EfficientNet-B7
Inception-V3 ResNet-50 ELMo-Medium
10 1 GoogLeNet ELMo-Small
BiLSTM-CNN-CRF
QT-Opt TT
10 0 DT
DDPG
10 1
10 2 DQN PPO 计算机视觉
自然语言处理
强化学习（Reinforcement Learning）
10 3
2012 2013 2014 2015 2016 2017 2018 2019 2020 2021 2022 2023
发布年份
参数（对数刻度，单位：百万）

图 8. 多年来单峰模型的规模不断扩大。 __ GPT-4 的模型大小是估计值，因为尚未正式披露。

在单词层面，需要考虑很多方面，包括形态学、词汇学、音韵学等。这会导致诸如标记化、词形还原、词干提取、语义关系、词义消歧和齐夫定律问题等问题。然后，在语法方面，与形式语言相比，自然语言的

语法限制要少得多，因此解析起来更具挑战性：在乔姆斯基层次结构中，自然语言通常被认为遵循上下文相关语法，而编程语言则涵盖上下文无关语法。句法解析包括词性标记、选区解析、依存解析和命名实体识别等任务。最后，为了理解书面语言句子或口语话语的语义，研究了以下任务：语义角色标记、框架语义解析、抽象意义表示、逻辑形式解析等。

b) 递归神经网络和卷积神经网络： 在 NLP 的初始阶段，基本模型依靠简单的前馈神经网络来处理各种任务 [265]。在引入 word2vec [266]、[267] 和 GloVe [268] 等词嵌入之后，NLP 技术开始拥抱循环神经网络 (RNN) [269]，例如 LSTM [223]、GRU [224]、RNNsearch [270]、LSTM-CRF [271] 等。 NLP 中代表性 RNN 的例子包括 ELMo [272] 和ULMFiT [273]。虽然 RNN 发挥了重要作用，但利用卷积神经网络的替代模型也出现了。 WordCNN [274] 在单词级别使用 CNN，并将 word2vec 特征作为 CNN 的输入。另一个

CharCNN [275] 方法专注于使用 CNN 在字符级别对语言进行建模。随后的研究[276]强调字符级 CNN 擅长捕获单词形态，并且它们与单词级 LSTM 主干的结合可以显着提高性能。

c) Transformer 和大型语言模型： [3] 引入的开创性 Transformer 模型通过引入自注意力机制彻底改变了自然语言处理，激发了一系列后续工作。 BERT [27] 利用 Transformer 编码器堆栈，在自然语言理解方面表现出色

。另一方面，GPT 系列 [234]、[277]、[278]、[279] 是基于 Transformer 解码器块构建的，展示了在自然语言生成任务中的实力。一系列工作致力于完善原始的 BERT，包括 RoBERTa [280]、ALBERT [281]、ELECTRA [282]。同时，遵循 GPT 范式的并行研究产生了 XLNet [283]、OPT [284] 等模型。 BART [285] 是一种编码器-解码器 Transformer，它通过使用去噪序列到序列任务进行预训练而脱颖而出。同时，T5[286]引入了对原始Transformer的修改，保持了编码器-解码器架构。 T5 通过共享的文本到文本格式统一了各种 NLP 任务，展示了迁移学习方面的增强性能。这些不同的模型共同展示了 NLP 领域的多功能性和持续发展。

在过去的几年里，在

scalability of the Transformer architecture. This trend has given rise to a series of large language models (LLMs) that have demonstrated breakthrough performance and capabilities not achievable with smaller models. A landmark model in this evolution is ChatGPT [225], which has sparked considerable interest and inspired a series of works in this domain, such as GPT-4 [279], PaLM [287], PaLM-2 [288], LLaMA [289], LLaMA 2 [290], ERNIE 3.5 [291]. Notably, LLaMA stands out as one of the few open-source LLMs, fostering interesting developments. The introduction of "instruction-tuning" has allowed efficient finetuning of a pretrained LLM to become an instruction-following model. This technique is popularized by InstructGPT [292] and FLAN [293], [294]. Recent advancements in instruction-following models include Alpaca [295], employing self-instruction, and Vicuna [296], leveraging conversations from ShareGPT. As LLMs grow in scale and power, there is a shift away from the need for finetuning on downstream tasks. With appropriate prompts, LLMs can produce accurate outputs without task-specific training, a paradigm known as *prompt engineering*. This approach differs from the traditional *pretrain-finetune* paradigm. DPO [229] introduces a method to train LLMs directly on human preferences without requiring explicit reward modeling or reinforcement learning, simplifying upon previous RLHF methods.

3) Reinforcement Learning: Reinforcement learning (RL) seeks to acquire a policy capable of taking optimal actions based on observations of the environment. Numerous vision-language-action models are constructed based on paradigms such as imitation learning or Temporal Difference (TD) learning within RL. Many challenges faced in the development of robotic policies can be effectively addressed through insights gained from the field of RL. Consequently, delving into RL methods presents a valuable avenue for enhancing robotic learning. For a deeper exploration of RL methods, readers can refer to more comprehensive reviews provided by [297], [298], and [299].

a) Deep Reinforcement Learning: The advent of deep reinforcement learning can be attributed to the success of pioneering models, Deep Q-Network [4] and AlphaGo [226]. Deep learning, with its ability to provide low-dimensional representations, proved instrumental in overcoming traditional computational and memory complexity challenges in reinforcement learning. In recent years, a multitude of value-function-based approaches have surfaced. Double DQN (D-DQN) [300] addresses the action value overestimation issue of DQN. Hindsight experience replay (HER) [301] focuses on the sparse reward issue. Batch-Constrained deep Q-learning (BCQ) [302] presents an approach aimed at enhancing off-policy learning by constraining the action space. BEAR [303] endeavors to alleviate instability arising from bootstrapping errors in off-policy RL. [304] introduces conservative Q-learning (CQL) to address the overestimation of values by standard off-policy RL methods.

Another paradigm within reinforcement learning is policy search, encompassing methods such as policy gradient and actor-critic techniques. These approaches aim to combat persistent challenges, including instability, slow convergence, and data inefficiency. Guided policy search (GPS) [305] learns a policy with importance sampling guided by another controller toward a local optimum. Deterministic policy gradient (DPG) [306], deep deterministic policy gradient (DDPG) [307], and asynchronous advantage actor-critic (A3C) [308] improve efficiency without compromising stability. Normalized advantage functions (NAF) [309] are a continuous variant of Q-learning, allowing for Q-learning with experience replay. Soft actor-critic (Soft AC) [310] takes advantage of the maximum entropy RL framework to lower sample complexity and improve convergence properties. Trust region policy optimization (TRPO) [311] and proximal policy optimization (PPO) [227] utilize trust region methods to stabilize policy gradients, with PPO additionally incorporating a truncated generalized advantage estimation (GAE) [312].

Beyond these, various other RL methodologies exist, such as imitation learning and hierarchical reinforcement learning. Generative adversarial imitation learning (GAIL) [313] is an imitation learning method that uses a generative adversarial framework to discriminate expert trajectories against generated trajectories. Robust adversarial reinforcement learning (RARL) [314] incorporates adversarial agents to enhance generalization. RLHF [315] utilizes human preferences without access to the reward function. FeUdal Networks (FuN) [316] introduce a hierarchical reinforcement learning architecture featuring a Manager module and a Worker module.

b) Robotics: In the field of RL, robotics stands out as one of the most prevalent and impactful applications. A noteworthy contribution to this field is E2E-DVP [317]. This pioneering model represents one of the first end-to-end solutions for robotic control. Its neural network is designed to take raw images as input and generate motor torques as output. [5] curated a substantial real-world dataset and developed a CNN that predicts grasps based on monocular input images. Building upon these foundations, QT-Opt [318] further expands the dataset and model scale, introducing closed-loop control capabilities to enhance robotic control systems. Then Dreamer [53] addresses long-horizon tasks. OpenAI also developed a dexterous robot hand that can solve the Rubik's cube [228], [319].

4) Graph: Graph is ubiquitous in many scenarios, such as social networks, molecule structures, 3D object meshes, etc. Even images and text can be modeled as a 2D grid and a linear graph (path graph), respectively. To process graph-structured data, recurrent graph neural networks [320] were first introduced, which were later optimized by convolutional graph neural networks. The review of graph neural networks (GNN) [321] can be referred to for more in-depth details.

Convolutional graph neural networks can be generally divided into two categories: spectral-based and spatial-based. Spectral-based convolutional GNNs draw inspiration from graph signal processing, which provides theoretical support for the design of the networks. However, spatial-based convolutional GNNs have an advantage in terms of their efficiency and flexibility. Spectral CNN [322] is one of the first convolutional GNNs, but it is not robust to changes in graph structure and has a high computational cost. ChebNet [323] and GCN [324] significantly reduced the cost: ChebNet used an approximation

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Transformer 架构的可扩展性。这种趋势催生了一系列大型语言模型（LLM），这些模型展示了较小模型无法实现的突破性性能和功能。这一演变过程中的一个里程碑式的模型是ChatGPT [225]，它引起了人们极大的兴趣，并启发了该领域的一系列工作，例如GPT-4 [279]、PaLM [287]、PaLM-2 [288]、LLaMA [289]、LLaMA 2 [290]、ERNIE 3.5 [291]。值得注意的是，LLaMA 是为数不多的开源法学硕士之一，促进了有趣的发展。 “指令调整”的引入允许对预训练的 LLM 进行有效的微调，使其成为指令跟踪模型。该技术由 InstructGPT [292] 和 FLAN [293]、[294] 推广。指令遵循模型的最新进展包括采用自我指令的 Alpaca [295] 和利用 ShareGPT 对话的 Vicuna [296]。随着法学硕士规模和实力的增长，对下游任务进行微调的需求发生了转变。通过适当的提示，法学硕士无需特定任务的训练即可产生准确的输出，这种范例称为“提示工程”。这种方法不同于传统的_pretrain-finetune_范例。 DPO [229] 引入了一种直接根据人类偏好训练 LLM 的方法，无需显式奖励模型或 强化学习（Reinforcement Learning），简化了之前的 RLHF 方法。

3) 强化学习（Reinforcement Learning）: 强化学习（Reinforcement Learning） (RL) 寻求获得一种能够根据对环境的观察采取最佳行动的策略。许多视觉语言动作模型都是基于 模仿学习（Imitation Learning）或 RL 中的时间差异 (TD) 学习等范式构建的。机器人策略制定中面临的许多挑战可以通过从强化学习领域获得的见解来有效解决。因此，深入研究强化学习方法为增强机器人学习提供了一条宝贵的途径。为了更深入地探索 RL 方法，读者可以参考[297]、[298]和[299]提供的更全面的评论。

a) 深度 强化学习（Reinforcement Learning）： 深度 强化学习（Reinforcement Learning） 的出现可以归因于开创性模型、深度 Q-Network [4] 和 AlphaGo [226] 的成功。深度学习具有提供低维表示的能力，在 强化学习（Reinforcement Learning） 中被证明有助于克服传统的计算和内存复杂性挑战。近年来，出现了大量基于价值函数的方法。双DQN（DDQN）[300]解决了DQN的动作值高估问题。事后经验重播（HER）[301]关注稀疏奖励问题。批量约束深度 Q 学习（BCQ）[302]提出了一种旨在通过约束动作空间来增强离策略学习的方法。 BEAR [303] 致力于减轻离策略强化学习中自举错误引起的不稳定性。 [304]引入保守 Q 学习（CQL）来解决标准离策略 RL 方法对值的高估问题。

强化学习（Reinforcement Learning） 中的另一个范例是策略搜索，包括策略梯度和行动者批评技术等方法

。这些方法旨在应对持续存在的挑战，包括不稳定、收敛缓慢和

数据效率低下。引导策略搜索（GPS）[305]学习由另一个控制器引导的重要采样策略以实现局部最优。确定性策略梯度（DPG）[306]、深度确定性策略梯度（DDPG）[307]和异步优势参与者-批评者（A3C）[308]在不影响稳定性的情况下提高了效率。归一化优势函数（NAF）[309]是 Q 学习的连续变体，允许进行经验回放的 Q 学习。 Soft actorcritic（Soft AC）[310]利用最大熵强化学习框架来降低样本复杂性并提高收敛性能。信任域策略优化（TRPO）[311]和近端策略优化（PPO）[227]利用信任域方法来稳定策略梯度，PPO还结合了截断的广义优势估计（GAE）[312]。

除此之外，还存在各种其他 RL 方法，例如 模仿学习（Imitation Learning） 和分层 强化学习（Reinforcement Learning）。生成对抗 模仿学习（Imitation Learning） (GAIL) [313] 是一种 模仿学习（Imitation Learning） 方法，它使用生成对抗框架来区分专家轨迹和生成轨迹。强大的对抗性 强化学习（

Reinforcement Learning ）(RARL) [314] 结合了对抗性智能体来增强泛化。 RLHF [315]利用人类偏好而无需访问奖励函数。 FeUdal Networks (FuN) [316] 引入了一种分层 强化学习（Reinforcement Learning） 架构，具有管理器模块和工作器模块。

b) 机器人技术： 在强化学习领域，机器人技术是最流行和最有影响力的应用之一。 E2E-DVP [317]对该领域的一个值得注意的贡献。这一开创性模型代表了最早的机器人控制端到端解决方案之一。其神经网络旨在将原始图像作为输入并生成电机扭矩作为输出。 [5] 策划了一个大量的现实世界数据集，并开发了一个 CNN，可以根据单眼输入图像预测抓取情况。在此基础上，QT-Opt [318]进一步扩展了数据集和模型规模，引入闭环控制功能来增强机器人控制系统。然后 Dreamer [53] 解决了 长时程（Long-Horizon） 任务。 OpenAI还开发了一种灵巧的机器人手，可以解魔方[228]、[319]。

_4）图：_图在许多场景中无处不在，例如社交网络、分子结构、3D 对象网格等。甚至图像和文本也可以分别建模为 2D 网格和线性图（路径图）。为了处理图结构数据，首先引入了循环图神经网络[320]，随后通过卷积图神经网络对其进行了优化。更深入的细节可以参考图神经网络（GNN）[321]的综述。

卷积图神经网络一般可以分为两类：基于谱的和基于空间的。基于谱的卷积 GNN 从图信号处理中汲取灵感，为网络的设计提供了理论支持。然而，基于空间的卷积 GNN 在效率和灵活性方面具有优势

。 Spectral CNN [322] 是最早的卷积 GNN 之一，但它对图结构的变化不具有鲁棒性，并且计算成本较高。 ChebNet [323] 和 GCN [324] 显着降低了成本：ChebNet 使用了近似值

based on Chebyshev polynomials, and GCN is its first-order approximation. Neural Network for Graphs (NN4G) [325] is the first spatial network. MPNN [326] introduced a general framework of spatial-based networks under which most existing GNNs can be covered. But the drawback of MPNN is that it does not embed graph structure information, which is later solved by GIN [327]. GraphSage [328] improves efficiency by sampling a fixed number of neighbors. Graph Attention Network (GAT) [329] incorporates the attention mechanism.

Besides recurrent and convolutional graph neural networks, there are graph autoencoders [330], [331] and spatial-temporal graph neural networks [332]. Equivariant message passing networks are recently introduced to handle 3D graphs, including E(n)- and SE(3)- Equivariant GNN [333], [334]. Graph Transformer models make use of the power of Transformers to process graph data. There are already over 20 such graph Transformer models, such as GROVER [335] and SE(3)-Transformers [336].

a) Graph and Vision: Graph structures also exist in some computer vision tasks. Scene graph [337] can be used to express object relationships in most visual inputs. In addition to detecting objects in an image, scene graph generation necessitates the understanding of the relationships between detected objects. For example, a model needs to detect a person and a cup, and then understand that the person is holding the cup, which is the relationship between the two objects. Knowledge graphs often contain visual illustrations, such as WikiData. Those graphs can be helpful in downstream computer vision tasks.

b) Graph and Language: Graph structures are ubiquitous in language data [338]. Word-level graphs include dependency graphs, constituency graphs, AMR graphs, etc. Word-level means each node of those graphs corresponds to a word in the original text. These graphs can be used to explicitly represent the syntax or semantics of the raw sentence. Sentence-level graphs can be useful in dialog tracking [339], fact checking [340], etc. Document-level graphs include knowledge graphs [341], citation graphs [342], etc. They can be used in document-level tasks, such as document retrieval, document clustering, etc. Different types of language graphs are often processed using the aforementioned GNNs to facilitate downstream tasks.

C. Background (Extended Version): Vision-Language Models

Comprehensive surveys on VLMs exist, covering early BERT-based VLMs [387], [388] (Section C1), as well as more recent VLMs with contrastive pretraining [389], [390] (Section C2). Given the rapid evolution of this field and the emergence of new VLMs based on large language models, commonly known as large multimodal models (LMMs), we also compile the latest developments of LMMs (Section C3). To compare the most representative VLMs, we include their specifications in Table VIII.

1) Self-Supervised Pretraining: The evolution of the Transformer architecture to accommodate various modalities has given rise to robust multimodal Transformer models. Initial VLMs based on BERT can be broadly categorized into two types: single-stream and multi-stream [387]. Single-stream models employ a single stack of Transformer blocks to process both visual and linguistic inputs, whereas multi-stream models utilize a separate Transformer stack for each modality, with Transformer cross-attention layers exchanging multimodal information. To enhance alignment among modalities, these models incorporate various pretraining tasks aimed at absorbing knowledge from out-of-domain data. ViLBERT [235] stands as the pioneer in this line of work, featuring a multi-stream Transformer architecture. Text input undergoes standard processing in the language Transformer; image input is first processed using Faster R-CNN, and the output embeddings of all objects are then passed into the vision stream. The two Transformer outputs—language embeddings and vision embeddings—are combined using a novel co-attention Transformer layer. VL-BERT [345] adopts a single-stream multimodal Transformer, simply concatenating vision and language tokens into a single input sequence. VideoBERT [391] adapts mulitmodal Transformer models to video inputs. UNITER [346] proposes the word-region alignment loss to explicitly align word with image regions. ViLT [347] uses ViT-style [221] image patch projection to embed images, deviating from previous region or grid features.

SimVLM [348] opts for a streamlined approach, relying solely on a single prefix language modeling objective to reduce training costs. VLMo [392] and BEiT-3 [352] both introduce mixture-of-modality-experts Transformers to effectively handle multimodal inputs.

2) Contrastive Pretraining: Vision-language pretraining in the initial series of BERT-based VLMs has evolved, with refinements such as curating larger-scale pretraining datasets, leveraging multimodal contrastive learning, and exploring specialized multimodal architectures. CLIP [23] is one of the earliest attempts in vision-language contrastive pretraining. By contrastive pretraining on a large-scale image-text pair dataset, CLIP exhibits the capability to be transferred to downstream tasks in a zero-shot fashion. In the same line of work, other few-/zero-shot learners have emerged. FILIP [353] concentrates on finer-grained multimodal interactions with a token-wise contrastive objective. ALIGN [354] focuses on learning from noisy datasets collected without filtering and post-processing. "Locked-image Tuning" (LiT) [393] posits that only training the text model while freezing the image model yields the best results on new tasks. In Frozen [394], the pretrained language model is frozen and a vision encoder is trained to produce image embeddings as a part of language model prompts, exemplifying an instance of prompt tuning.

Unlike the two-tower frameworks of CLIP, FILIP, and ALIGN, which solely train unimodal encoders (image encoder and text encoder), ALBEF [356] additionally incorporates training a multimodal encoder on top of the unimodal encoders, with FLAVA [357] sharing a similar idea. In contrast to contrastive pretraining methods, CoCa [395] seeks to amalgamate the strengths of CLIP's contrastive learning and SimVLM's generative objective. Florence [351] generalizes representations from coarse, scene-level to fine, object-level, expands from images to videos, and encompasses modalities beyond RGB channels. OFA [396] draws inspiration from T5

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基于切比雪夫多项式，GCN 是其一阶近似。图神经网络（NN4G）[325]是第一个空间网络。 MPNN [326]引入了基于空间的网络的通用框架，在该框架下可以覆盖大多数现有的 GNN。但MPNN的缺点是它没有嵌入图结构信息，后来GIN[327]解决了这个问题。 GraphSage [328]通过对固定数量的邻居进行采样来提高效率。图注意力网络（GAT）[329]结合了注意力机制。

除了循环和卷积图神经网络之外，还有图自动编码器[330]、[331]和时空图神经网络[332]。最近引入了等变消息传递网络来处理 3D 图，包括 E(n)- 和 SE(3)- 等变 GNN [333]、[334]。 Graph Transformer 模型利用 Transformer 的强大功能来处理图形数据。已经有超过 20 个这样的图 Transformer 模型，例如 GROVER [335] 和 SE(3)Transformers [336]。

_a) 图和视觉：_图结构也存在于一些计算机视觉任务中。场景图[337]可用于表达大多数视觉输入中的对象关系。除了检测图像中的对象之外，场景图生成还需要了解检测到的对象之间的关系。例如，模型需要检测一个人和一个杯子，然后了解这个人拿着杯子，这就是两个物体之间的关系。知识图谱通常包含视觉插图，例如 WikiData。这些图表对于下游计算机视觉任务很有帮助。

_b）图和语言：_图结构在语言数据中无处不在[338]。词级图包括依赖图、选区图、AMR图等。词级是指这些图的每个节点对应于原文中的一个词。这些图可用于明确表示原始句子的语法或语义。句子级图可用于对话跟踪[339]、事实检查[340]等。文档级图包括知识图[341]、引文图[342]等。它们可用于文档级任务，例如文档检索、文档聚类等。不同类型的语言图通常使用上述GNN进行处理，以促进下游任务。

C。背景（扩展版）：视觉语言模型

对 VLM 进行了全面的调查，涵盖早期基于 BERT 的 VLM [387]、[388]（C1 节）以及较新的具有对比预训练的 VLM [389]、[390]（C2 节）。鉴于该领域的快速发展以及基于大型语言模型（通常称为大型多模态模型（LMM））的新 VLM 的出现，我们还汇编了 LMM 的最新进展（C3 节）。为了比较最具代表性的 VLM，我们在表 VIII 中列出了它们的规格。

1) 自监督预训练： Transformer 架构的演变以适应各种模式，从而产生了强大的多模式 Transformer 模型。基于 BERT 的初始 VLM 可大致分为两类

类型：单流和多流[387]

。单流模型采用单个 Transformer 块堆栈来处理视觉和语言输入，而多流模型对每种模态使用单独的 Transformer 堆栈，并通过 Transformer 交叉注意层交换多模态信息。为了增强模式之间的一致性，这些模型结合了各种预训练任务，旨在从域外数据中吸收知识。 ViLBERT [235] 是这一领域的先驱，具有多流 Transformer 架构。文本输入在语言 Transformer 中经过标准处理；首先使用 Faster R-CNN 处理图像输入，然后将所有对象的输出嵌入传递到视觉流中。两个 Transformer 输出（语言嵌入和视觉嵌入）使用新颖的共同关注 Transformer 层进行组合。 VL-BERT [345]采用单流多模态 Transformer，简单地将视觉和语言标记连接成单个输入序列。 VideoBERT [391]使多模态 Transformer 模型适应视频输入。 UNITER [346]提出了单词区域对齐损失来明确地将单词与图像区域对齐。 ViLT [347] 使用 ViTstyle [221] 图像块投影来嵌入图像，偏离先前的区域或网格特征。

SimVLM [348] 选择一种简化的方法，仅依靠单一前缀语言建模目标来降低训练成本。 VLMo [392] 和 BEiT-3 [352] 都引入了混合模态专家 Transformer 来有效处理多模态输入。

2) 对比预训练： 基于 BERT 的 VLM 初始系列中的视觉语言预训练已经发展，并进行了改进，例如整理更大规模的预训练数据集、利用多模态对比学习以及探索专门的多模态架构。 CLIP [23] 是视觉语言对比预训练的最早尝试之一。通过对大规模图像文本对数据集进行对比预训练，CLIP 展现了以零样本方式转移到下游任务的能力。在同一行工作中，还出现了其他少样本/零样本学习器。 FILIP [353] 专注于具有标记对比目标的更细粒度的多模式交互。 ALIGN [354]专注于从未经过滤和后处理收集的噪声数据集中学习。 “锁定图像调整”（LiT）[393] 认为，仅在冻结图像模型的同时训练文本模型才能在新任务中产生最佳结果。在 Frozen [394] 中，预训练的语言模型被冻结，视觉编码器被训练以生成图像嵌入作为语言模型提示的一部分，举例说明了提示调整的实例。

与仅训练单模态编码器（图像编码器和文本编码器）的 CLIP、FILIP 和 ALIGN 两塔框架不同，ALBEF [356] 另外在单模态编码器之上训练多模态编码器，FLAVA [357] 也有类似的想法。与对比预训练方法相比，CoCa [395] 寻求融合 CLIP 对比学习和 SimVLM 生成目标的优势。 Florence [351]概括了从粗略、场景级到精细、对象级的表示，从图像扩展到视频，并包含 RGB 通道之外的模式

。 OFA [396]从T5汲取灵感

Table VIII

VISION-LANGUAGE MODELS. IN THE "OBJECTIVE" COLUMN: "MLM": MASKED LANGUAGE MODELING. "MVM": MASKED VISION MODELING, RECONSTRUCTING MASKED IMAGE REGIONS. "VLM": BINARY CLASSIFICATION OF WHETHER VISION AND LANGUAGE INPUTS MATCH. "LM": AUTOREGRESSIVE LANGUAGE MODELING. "VLCL": VISION-LANGUAGE CONTRASTIVE LEARNING. WE ONLY INCLUDE REPRESENTATIVE MULTIMODAL DATASETS DUE TO LIMITED SPACE. "MM" INCLUDES MULTIMODAL TASKS SUCH AS VISUAL QUESTION ANSWERING, IMAGE CAPTIONING, AND VISION-LANGUAGE RETRIEVAL. "VISION" REPRESENTS COMPUTER VISION TASKS, SUCH AS IMAGE CLASSIFICATION. "LANGUAGE" REPRESENTS NATURAL LANGUAGE PROCESSING TASKS

		Vision Encoder		Language Encoder					
	Model	**Name**	**Params**	**Name**	**Params**	**VL-Fusion**	**Objectives**	**Datasets**	**Tasks**
Self-supervised	ViLBERT [235]	Faster R-CNN [247], [252]	44M	Dual-stream BERT-base [27]	221M	Dual-stream	MLM, MVM, VLM	COCO, VG	MM
	LXMERT [343]	Faster R-CNN [247], [252]	44M	Dual-stream BERT-base [27]	183M	Dual-stream	MLM, MVM, VLM, VQA	COCO, VG, VQA, GQA, VGQA	MM
	VisualBERT [344]	Faster R-CNN [247], [252]	60M	BERT-base [27]	110M	Single-stream	MLM, VLM	COCO	MM
	VL-BERT [345]	Faster R-CNN [247], [252]	44M	BERT-base [27]	110M	Single-stream	MLM, MVM	CC	MM
	UNITER [346]	Faster R-CNN [247], [252]	44M	BERT-base/ BERT-large [27]	86M/ 303M	Single-stream	MLM, VLM, MVM, WRA	COCO, VG, SBU, CC	MM
	ViLT [347]	Linear projection [221]	2.4M	BERT-base [27]	85M	Single-stream	MLM, VLM	COCO, VG, SBU, CC	MM
	SimVLM [348]	ViT/CoAtNet-huge (shared encoder) [349]	632M	Shared encoder	632M	Single-stream	PrefixLM	ALIGN dataset	MM
	GIT [350]	Florence [351]	637M	Transformer [3]	60M	Single-stream	LM	COCO, VG, SBU, CC, etc.	MM
	BEiT-3 [352]	V-FFN (+Shared Attn)	692M (+317M)	L-FFN (+Shared Attn)	692M (+317M)	Modality experts	MLM, VLM	COCO, VG, SBU, CC	MM, Vision
Contrastive	CLIP [23]	ViT [221]	428M	GPT-2 [277]	63M	Two-tower	VLCL	WTI	Vision
	FILIP [353]	ViT-L/14 [221]	428M	GPT [277]	117M	Two-tower	VLCL	FILIP300M (Self-collect)	MM, Vision
	ALIGN [354]	EfficientNet-L2 [355]	480M	BERT-large [27]	336M	Two-tower	VLCL	ALIGN dataset (Self-collect)	MM, Vision
	ALBEF [356]	ViT-B/16 [221]	87M	BERT-base [27]	85M	Dual-stream	MLM, VLM, VLCL	COCO, VG, CC, SBU	MM
	FLAVA [357]	ViT-B/16 [221]	87M	RoBERTa-base [280]	125M	Dual-stream	MLM, MVM, VLM, VLCL	COCO, VG, CC, SBU, etc. (PMD)	MM, Vision, Language
	Florence [351]	Hierarchical Vision Transformers [358], [359]	637M	RoBERTa [280]	125M	Two-tower	VLCL	FLD-900M (Self-collect)	Vision
Large Multimodal Model	Flamingo [6]	NFNet-F6 [360]	438M	Chinchilla [361]	70B	Dual-stream	LM	M3W, ALIGN dataset, LTIP, VTP	MM
	BLIP-2 [7]	CLIP ViT-L/14 [23], EVA ViT-G/14 [362] + Q-Former	428M, 1B	OPT [284] Flan-T5 [294]	6.7B 3B/11B	Single-stream	BLIP, LM	COCO, VG, CC, SBU, LAION	MM
	PaLI [363]	ViT-e [363]	4B	mT5-XXL [364]	13B	Single-stream	Mixed	WebLI, etc	MM
	PaLI-X [365]	ViT-22B [366]	22B	UL2 [367]	32B	Single-stream	Mixed	WebLI, etc	MM
	LLaMA-Adapter [368]	CLIP ViT-B/16 [23]	87M	LLaMA [289]	7B	Single-stream	LM	Self-instruct	Instruction-following
	LLaMA-Adapter-V2 [369]	CLIP ViT-L/14 [23]	428M	LLaMA [289]	7B	Single-stream	LM	GPT-4-LLM, COCO, ShareGPT	Instruction-following
	Kosmos-1 [370], Kosmos-2 [371]	CLIP ViT-L/14 [23]	428M	Magneto [372]	1.3B	Single-stream	LM	LAION, COYO, CC; Unnatural Instructions, FLANv2	Instruction-following (Kosmos-2 w/ grounding, referring)
	InstructBLIP [373]	EVA ViT-G/14 [362]	1B	Flan-T5 [294] Vicuna [296]	3B/11B 7B/13B	Single-stream	BLIP, LM	COCO, VQA, LLaVA-Instruct-150K, etc. (26 datasets)	Instruction-following
	LLaVA [8]	CLIP ViT-L/14 [23]	428M	LLaMA [289]	13B	Single-stream	LM	CC, (FT: GPT-assisted Visual Instruction Data Generation)	Instruction-following
	MiniGPT-4 [374]	EVA ViT-G/14 [362] + Q-Former [7]	1B	Vicuna [296] LLaMA2 [290]	7B/13B 7B	Single-stream	LM	CC, SBU, LAION (FT: SC)	Instruction-following
	Video-LLaMA [375]	EVA ViT-G/14 [362] + Q-Former [7]	1B	LLaMA [289]	7B/13B	Single-stream	BLIP, LM	CC595k	Instruction-following
	PandaGPT [376]	ImageBind ViT-H [211]	632M	Vicuna [296]	13B	Single-stream	LM	(FT: LLaVA data, MiniGPT-4 data)	Instruction-following
	VideoChat [377]	EVA ViT-G/14 [362] + Q-Former [7]	1B	StableVicuna [378]	13B	Single-stream	LM	COCO, VG, CC, SBU (FT: SC, MiniGPT-4, LLaVA data)	Instruction-following
	ChatSpot [379]	CLIP ViT-L/14 [23]	428M	Vicuna [296]	7B	Single-stream	LM	MGVLID, RegionChat	Instruction-following, Vision
	mPLUG-Owl [380], mPLUG-Owl2 [381]	CLIP ViT-L/14 [23] + Visual Abstractor	428M	LLaMA [289]	7B	Single-stream	LM	LAION, COYO, CC, COCO (FT: Alpaca, Vicuna, Baize [382] data)	Instruction-following
	Visual ChatGPT [383]	(22 different models)	-	ChatGPT [225]	-	Prompt Manager	-	Add image understanding and generation to ChatGPT	
	X-LLM [384]	ViT-G [385] + Q-Former [7] + Adapter	1.8B	ChatGLM [386]	6B	Single-stream	Three-stage training	MiniGPT-4 data, AISHELL-2, ActivityNet, VSDial-CN (SC)	Instruction-following

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表八

视觉语言模型。在“目标”栏中：“MLM”：掩码语言建模。 “MVM”：蒙蔽视觉建模，重建蒙蔽图像区域。 “VLM”：视觉和语言输入是否匹配的二元分类。 “LM”：自回归语言建模。 “VLCL”：视觉语言对比学习。由于空间有限，我们仅包含代表性的多模态数据集。 “MM”包括多模式任务，例如视觉问答、图像字幕和视觉语言检索。 “VISION”代表计算机视觉任务，例如图像分类。 “语言”代表自然语言处理任务

|模型|视觉编码器
名称|参数|语言 En
名称|编码器
参数|VL-Fusion|目标|数据集|任务|
|---|---|---|---|---|---|---|---|---|---|
|ViLBERT [235]|更快的 R-CNN [247]，
[252]|44M|双流
BERT-base [27]|221M|双流|MLM、MVM、
VLM|COCO、VG|MM|
|LXMERT [343]|更快的 R-CNN [247]，
[252]|44M|双流
BERT-base [27]|183M|双流|MLM、MVM、
VLM、VQA|COCO、VG、VQA、GQA、VGQA|MM|
|VisualBERT [344]|更快的 R-CNN [247]，
[252]|60M|BERT-base [27]|110M|单流|MLM、VLM|COCO|MM|
|监督
VL-BERT [345]|更快的 R-CNN [247]，
[252]|44M|BERT-base [27]|110M|单流|MLM、MVM|CC|MM|
|自支持
UNITER [346]|更快的R-CNN [247]，
[252]|44M|BERT-base/
BERT-large [27]|86M/
303M|单流|MLM、VLM、
MVM、WRA|COCO、VG、SBU、CC|MM|
|ViLT [347]|线性投影 [221]|2.4M|BERT-base [27]|85M|单流|MLM、VLM|COCO、VG、SBU、CC|MM|
|SimVLM [348]|ViT/CoAtNet-huge
（共享编码器）[349]|632M|共享编码器|632M|单流|PrefixLM|ALIGN 数据集|MM|
|GIT[350]|佛罗伦萨[351]|637M|变压器[3]|60M|单码流|LM|COCO、VG、SBU、CC等|MM|
|BEiT-3 [352]|V-FFN
(+共享收件人)|692M
(+317M)|L-FFN
(+共享收件人)|692M

(+317M)|模态专家|MLM、VLM|COCO、VG、SBU、CC|MM、Vision|
|CLIP [23]|ViT [221]|428M|GPT-2 [277]|63M|双塔|VLCL|WTI|Vision|
|FILIP [353]|ViT-L/14 [221]|428M|GPT [277]|117M|双塔|VLCL|FILIP300M（自取）|MM、远景|
|tive
ALIGN [354]|EfficientNet-L2 [355]|480M|BERT-large [27]|336M|双塔|VLCL|ALIGN 数据集（自采集）
|MM ，视觉|
|ontras
ALBEF [356]|ViT-B/16 [221]|87M|BERT-base [27]|85M|双流|MLM、VLM、
VLCL|COCO、VG、CC、SBU|MM|
|C
FLAVA [357]|ViT-B/16 [221]|87M|RoBERTa-base
[280]|125M|双码流|MLM、MVM、
VLM、VLCL|COCO、VG、CC、SBU等
(PMD)|MM、视觉、
语言|
|佛罗伦萨[351]|分层视觉
变形金刚[358]、[359]|637M|RoBERTa [280]|125M|双塔|VLCL|FLD-900M(自取)|视觉|
|Flamingo [6]|NFNet-F6 [360]|438M|Chinchilla [361]|70B|双流|LM|M3W、ALIGN 数据集、LTIP、
VTP|MM|
|BLIP-2 [7]|CLIP ViT-L/14 [23]，
EVA ViT-G/14 [362]
+ Q-Former|428M，
1B|OPT [284]
Flan-T5 [294]|6.7B
3B/11B|单流|BLIP、LM|COCO、VG、CC、SBU、莱昂|MM|
|PaLI [363]|ViT-e [363]|4B|mT5-XXL [364]|13B|单流|混合|WebLI等|MM|
|PaLI-X [365]|ViT-22B [366]|22B|UL2 [367]|32B|单流|混合|WebLI 等|MM|
|LLaMA 适配器
[368]|CLIP ViT-B/16 [23]|87M|LLaMA [289]|7B|单流|LM|自我指示|指示-
以下|
|LLaMA-适配器-V2
[369]|CLIP ViT-L/14 [23]|428M|LLaMA

[289]|7B|单流|LM|GPT-4-LLM、COCO、ShareGPT|说明-
以下|
|el
Kosmos-1 [370]、
Kosmos-2 [371]|CLIP ViT-L/14 [23]|428M|Magneto [372]|1.3B|单流|LM|LAION、COYO、CC；不自然
指令，FLANv2|指令-
以下
(Kosmos-2|
|I Mod|||||||w/ 落地关联（Grounding），
参考）|

|ltimoda
InstructBLIP [373]|EVA ViT-G/14 [362]|1B|Flan-T5 [294]
Vicuna [296]|3B/11B
7B/13B|单流|BLIP、LM|COCO、VQA、LLaVA-Instruct-
150K 等 (26数据集）|说明-
以下|
|rge Mu
LLaVA [8]|CLIP ViT-L/14 [23]|428M|LLaMA [289]|13B|单流|LM|CC，（FT：GPT 辅助视觉
指令数据生成）|指令-
以下|
|La
MiniGPT-4 [374]|EVA ViT-G/14 [362]
+ Q-Former [7]|1B|Vicuna [296]
LLaMA2 [290]|7B/13B
7B|单流|LM|CC、SBU、LAION (FT: SC)|说明-
以下|
|视频-LLaMA [375]|EVA ViT-G/14 [362]
+ Q-Former [7]|1B|LLaMA [289]|7B/13B|单流|BLIP、LM|CC595k|说明-
以下|
|PandaGPT [376]|ImageBind ViT-H [211]|632M|Vicuna [296]|13B|单流|LM|（FT：LLaVA 数据，
MiniGPT-4
数据）|说明-
以下|
|VideoChat [377]|EVA ViT-G/14 [362]
+ Q-Former [7]|1B|StableVicuna
[378]|13B|单流|LM|COCO、VG、CC、SBU（FT：SC、
MiniGPT-4、LLaVA 数据）|说明-
以下|
|ChatSpot [379]|CLIP ViT-L/14 [23]|428M|Vicuna [296]|7B|单流|LM|MGVLID、RegionChat|说明-
以下、视觉|
|mPLUG-Owl [380]、
mPLUG-Owl2 [381]|CLIP ViT-L/14 [23]
+ Visual Abstractor|428M|LLaMA [289]|7B|单流|LM|LAION、COYO、CC、COCO（FT：
羊驼毛、骆马毛、白泽 [382]数据）|说明-
以下|
|Visual ChatGPT
[383]|(22 种不同模型)|-|ChatGPT [225]|-|提示管理器|-|为 ChatGPT 添加图像理解和生成|
|X-LLM [384]|ViT-G [385] + Q-Former
[7] + 适配器|1.8B|ChatGLM [386]|6B|单流|三阶段
训练|MiniGPT-4 数据、AISHELL-2、
ActivityNet、VSDial-CN (SC)|说明-
以下|

[286] and proposes unifying diverse unimodal and multimodal tasks under a sequence-to-sequence learning framework.

3) Large Multimodal Model: Large language models (LLMs) encapsulate extensive knowledge, and efforts have been made to transfer this knowledge to multimodal tasks. Given the resource-intensive nature of finetuning entire LLMs on multimodal tasks due to their large size, various techniques have been explored to effectively connect frozen LLMs with vision encoders, enabling the combined model to acquire multimodal capabilities. Flamingo [6] connects a pretrained vision encoder, NFNet [360], and a large language model, Chinchilla [361], by inserting trainable gated cross-attention layers while keeping the rest of the model frozen. BLIP-2 [7] introduces Q-Former, bootstrapping vision-language representation learning first from a frozen CLIP ViT [23] and then from a frozen LLM, OPT [284] or Flan-T5 [294]. PaLI [363] and PaLI-X [365] investigate the advantages of jointly scaling up the vision and language components using large-scale multilingual image-text data.

Similar to developments in NLP, instruction-following has become a crucial aspect of VLMs, prompting the exploration of various multimodal instruction-tuning methods. LLaMA-Adapter [368], [369] employs a parameter-efficient finetuning (PEFT) technique, enabling LLaMA [289] to process visual inputs. Kosmos-1 [370] introduces a less restrictive input format that accommodates interleaved image and text. Its Magneto LLM [372] serves as a "general-purpose interface" for docking with perception modules [23]. Kosmos-2 [371] adds additional grounding and referring capabilities. InstructBLIP [373] achieves instruction-following using an instruction-aware Q-Former based on BLIP-2's Q-Former [7]. Comparable to Kosmos-2, ChatSpot [379] excels at following precise referring instructions, utilizing CLIP ViT [23] and Vicuna [296]. X-LLM [384] converts multimodality data into LLM inputs using X2L interfaces and treats them as foreign languages, where the X2L interface is inspired by the Q-Former from BLIP-2 [7]. mPLUG-Owl [380], [381] introduces a two-stage training paradigm that establishes a connection between a pretrained LLM with a visual encoder and a visual abstractor, thereby endowing LLMs with multimodality abilities. Visual ChatGPT [383] proposes a prompt manager that manages the interaction between ChatGPT and 22 visual foundation models, with the goal of equipping ChatGPT with the capability to understand and generate images.

Rather than employing intricate mechanisms to connect components for different modalities, both LLaVA [8] and MiniGPT-4 [374] propose connecting vision encoders with LLMs through a single linear layer. LLaVA adopts a two-stage instruction-tuning approach, pretraining the CLIP ViT vision encoder [23] in the first stage and finetuning the linear layer and the LLaMA LLM [289] in the second stage. In contrast, MiniGPT-4 freezes both the vision encoder (BLIP-2's ViT + Q-Former [7]) and the Vicuna LLM [296], only training the linear layer. Following MiniGPT-4, Video-LLaMA [375] handles videos by incorporating two branches for video and audio, each comprising a video/audio encoder and a BLIP-2-style Q-Former [7]. PandaGPT [376] leverages ImageBind [211] to encode vision/text/audio/depth/thermal/IMU data, feeding them to the Vicuna model [296] also through a linear layer. PandaGPT diverges from MiniGPT-4 by using LoRA [397] to train Vicuna alongside the linear layer.

D. Supplementary Related Work

Due to page limitations, we were unable to provide citations for all the mentioned models in the main text. Here is a list of supplementary related works:

Computer Vision:

- Mask R-CNN [248]
- YOLOv8 [249]
- EfficientNet [244]
- UNet [255]
- VQ-GAN [398]
- MoVQ [399]
- SigLIP [400]
- ViLD [401]
- MDETR [402]
- HLSM object detector [403]
- ResNet [220]
- ViT, ViT-B, ViT-L [26], [221]
- MAE [404]
- ViT-22B [366]
- EVA ViT-G/14 [362]
- OWL-ViT [405]
- ConvNeXt [406]
- SAM [222]
- Imagen video [407]
- C-ViViT [408]
- OSRT [409]
- PointNet [410]
- PointNet++ [411]
- 3D-CLR [412]
- ConceptFusion [413]

Natural Language Processing, LLM, VLM:

- Long Short-Term Memory (LSTM) [223]
- Transformer [3]
- Sentence-BERT / Sent.-BERT [414]
- DistilBERT [415]
- T5-small, T5-base, T5-XXL [286]
- LLaMA [289]
- LLaMA 2 [290]
- Vicuna-V1.5 13B [296]
- MPT [416]
- FLAN [293]
- Flamingo [6], [417]
- Pythia [418]
- Phi-2 [419]
- GPT-NeoX [420]
- Codex [421]
- InstructGPT [376]
- ChatGPT [225]
- GPT-2 [277]
- GPT-3 [278]
- GPT-4 [279]

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[286]并提出在序列到序列学习框架下统一不同的单模态和多模态任务。 _3）大型多模态模型：_大型语言模型（LLM）封装了广泛的知识，并且已经努力将这些知识转移到多模态任务中。考虑到在多模态任务上微调整个 LLM 的资源密集型性质，由于其规模较大，人们已经探索了各种技术来有效地将冻结的 LLM 与视觉编码器连接起来，使组合模型能够获得多模态能力。 Flamingo [6] 通过插入可训练的门控交叉注意力层，同时保持模型的其余部分冻结，将预训练的视觉编码器 NFNet [360] 和大型语言模型 Chinchilla [361] 连接起来。 BLIP-2 [7] 引入了 Q-Former，首先从冻结的 CLIP ViT [23] 中引导视觉语言表示学习，然后从冻结的 LLM、OPT [284] 或 Flan-T5 [294] 中学习。 PaLI [363] 和 PaLIX [365] 研究了使用大规模多语言图像文本数据联合扩展视觉和语言组件的优势。

与 NLP 的发展类似，指令跟踪已成为 VLM 的一个重要方面，促进了对各种多模式指令调整方法的探索。 LLaMAAdapter [368]、[369] 采用参数高效微调（PEFT）技术，使 LLaMA [289] 能够处理视觉输入。 Kosmos-1 [370] 引入了一种限制较少的输入格式，可以容纳交错的图像和文本。其Magneto LLM [372]充当“通用（General-Purpose）接口”，用于与感知模块对接[23]。 Kosmos-2 [371] 添加了额外的 落地关联（Grounding） 和引用功能。 InstructBLIP [373] 使用基于 BLIP-2 的 Q-Former [7] 的指令感知 Q-Former 来实现指令跟踪。与 Kosmos-2 相比，ChatSpot [379] 擅长利用 CLIP ViT [23] 和 Vicuna [296] 遵循精确的参考指令。 X-LLM [384] 使用 X2L 接口将多模态数据转换为 LLM 输入，并将它们视为外语，其中 X2L 接口的灵感来自于 BLIP-2 [7] 的 Q-Former。 mPLUG-Owl [380]、[381]引入了一种两阶段训练范式，该范式在带有视觉编码器的预训练法学硕士和视觉抽象器之间建立了联系，从而赋予法学硕士多模态能力。 Visual ChatGPT [383]提出了一个提示管理器，用于管理 ChatGPT 和 22 个视觉 基础模型（Foundation Models） 之间的交互，目的是使 ChatGPT 具有理解和生成图像的能力。

LLaVA [8] 和 MiniGPT-4 [374] 都建议通过单个线性层将视觉编码器与 LLM 连接起来，而不是采用复杂的机制来连接不同模态的组件。 LLaVA 采用两阶段指令调整方法，在第一阶段预训练 CLIP ViT 视觉编码器 [23]，在第二阶段微调线性层和 LLaMA LLM [289]。相比之下，MiniGPT-4 冻结了视觉编码器（BLIP-2 的 ViT + Q-Former [7]）和 Vicuna LLM [296]，仅训练线性层

。继 MiniGPT-4 之后，Video-LLaMA [375] 通过合并视频和音频的两个分支来处理视频，每个分支都包含一个视频/音频编码器和一个 BLIP-2 型 Q-Former [7]。 PandaGPT [376] 利用 ImageBind [211] 编码视觉/文本/音频/深度/热/IMU 数据，馈送

它们也通过线性层转换为 Vicuna 模型 [296]。 PandaGPT 与 MiniGPT-4 不同，它使用 LoRA [397] 沿着线性层训练 Vicuna。
D。补充相关工作

由于页面限制，我们无法为正文中提到的所有模型提供引用。以下是补充相关作品列表：
计算机视觉：

- 掩模 R-CNN [248]

- YOLOv8 [249]

- 高效网络 [244]

- UNet [255]

- VQ-GAN [398]

- 莫维Q [399]

- 西格利普 [400]

- 维德 [401]

- MDETR [402]

- HLSM 物体检测器 [403]

- 残差网络 [220]

- ViT、ViT-B、ViT-L [26]、[221]

- 梅[404]

- ViT-22B [366]

- EVA ViT-G/14 [362]

- 猫头鹰-ViT [405]

-ConvNeXt [406]

- 萨姆 [222]

- 图像视频 [407]

- C-ViViT [408]

- OSRT [409]

- 点网 [410]

- PointNet++ [411]

- 3D-CLR [412]

- 概念融合 [413]

自然语言处理，法学硕士，VLM：

- 长短期记忆（LSTM）[223]

- 变压器 [3]

- 句子-BERT / Sent.-BERT [414]

- DistillBERT [415]

- T5-小号、T5-底座、T5-XXL [286]

- 骆驼 [289]

- 美洲驼 2 [290]

- 骆驼毛-V1.5 13B [296]

- MPT [416]
- 馅饼 [293]
- 火烈鸟 [6], [417]
- 皮提亚 [418]
- Phi-2 [419]
- GPT-NeoX [420]
- 法典 [421]
- 指导GPT [376]
- 聊天GPT [225]
- GPT-2 [277]
- GPT-3 [278]
- GPT-4 [279]

- PaLI-X [365]
- Prismatic-7B VLM [422]
- PaliGemma [423]
- PaliGemma 2 [424]
- Chameleon [425]
- Emu3 [426]
- LWM-Chat-1M [427]
- LLaMA-Adapter [368]

Action Modules:

- LingUNet [428]
- Latent Motor Plans (LMP) [429]
- STORM [430]
- PerceiverIO [431]
- DD-PPO [432]
- ScaleDP [433]
- Symbol-tuning [434]
- DiT [435]

Others:

- DDPM [436]
- DDIM [437]
- Ravens [438]
- RLBench [180]
- Something V2: Something-something V2 [439]

Abbreviations:

- TFM: Transformer
- MCIL: multicontext imitation learning
- MLE: maximum likelihood estimation
- MPC: model predictive control
- CVAE: conditional variational autoencoder
- EDR: Everyday Robots

E. Additional VLA-Related Work

We include additional VLA-related work that could not be included in the main text due to the page limit.

1) Components of VLA: Pretraining: Value-Implicit Pretraining (VIP) [25] capitalizes on the temporal sequences present in videos, distinguishing itself from R3M [24] by attracting both the initial and target frames while simultaneously repelling successive frames. This objective aims to capture long-range temporal relationships and uphold local smoothness. While their own experiments demonstrate superior performance compared to R3M in specific tasks, subsequent research presents conflicting findings through more comprehensive evaluations [32].

SpawnNet [440] utilizes a two-stream architecture incorporating adapter layers to fuse features from both a pretrained vision encoder and features learned from scratch. This innovative approach eliminates the necessity of training the pretrained vision encoders while surpassing the performance of parameter-efficient finetuning (PEFT) methods, as evidenced by experimental results in robot manipulation tasks.

Holo-Dex [441] proposes learning low-dimensional representations for visual policies using self-supervised learning (SSL). In Dobb·E [442], SSL is also explored for 3D visual inputs, through a method termed "Home Pretrained Representations," using data collected from 22 homes in NYC. The authors extend SSL-pretrained representations beyond visual inputs. AuRL [45] introduces learning dynamic behaviors from audio, an often-overlooked yet important information source. Additionally, visual inputs were found to be insufficient for multi-fingered robotic hands, leading to T-Dex [213], a tactile-based dexterous policy.

2) Components of VLA: World Models: Masked World Model (MWM) [443] innovates by modifying the vision encoder of DreamerV2 to a hybrid composition of convolutional neural network and vision Transformer. In training this novel Convolutional-ViT vision encoder, MWM draws inspiration from the approach proposed in MAE [404]. Through the incorporation of an auxiliary reward prediction loss, the resulting learned latent dynamics model exhibits a notable performance improvement across various visual robotic tasks.

SWIM [444] advocates for the use of human videos in training a world model due to the availability of large-scale human-centric data. However, acknowledging the substantial gap between human and robot data, SWIM addresses this disparity by grounding actions in visual affordance maps. This approach involves inferring target poses based on the affordance maps, facilitating an effective transfer of knowledge from human data to enhance robot control.

Iso-Dream [445] introduces two key enhancements to the Dreamer framework. The first enhancement focuses on optimizing inverse dynamics by decoupling controllable and noncontrollable dynamics. The second enhancement involves optimizing agent behavior based on the decoupled latent imaginations. This approach is particularly advantageous, as the noncontrollable state transition branch can be rolled out independently of actions, yielding benefits for long-horizon decision-making processes.

Transformer-based World Model (TWM) [58] shares the same motivation as Dreamer but adopts a different approach by constructing a world model based on the Transformer-XL architecture [446]. This Transformer-based world model trains a model-free agent in latent imagination by generating new trajectories. Additionally, TWM suggests a modification to the balanced KL divergence loss from DreamerV2 and introduces a novel thresholded entropy loss tailored for the advantage actor-critic framework.

SceneScript [447] introduces a set of structured language commands that can define an entire scene, specifying the layout and objects. This language-based scene representation distinguishes itself from previous methods—such as meshes, voxel grids, point clouds, or radiance fields—by generating scenes in an autoregressive, token-based fashion. It achieves competitive results against state-of-the-art methods in layout estimation and object detection.

3) Components of VLA: Imitation Learning: BeT [448], an imitation learning variant of the Trajectory Transformer, addresses the noisy and multimodal nature of non-expert human demonstration data by using k-means-based action discretization coupled with continuous action correction. C-BeT [449] extends BeT by adding a target frame or demonstration as the goal specification. VQ-BeT [450] further improves

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- 巴利-X [365]

- Prismatic-7B VLM [422]

- 巴利杰玛 [423]

- 巴利语杰玛 2 [424]

- 变色龙 [425]

- 鹌鹑3 [426]

- LWM-聊天-1M [427]

- LLaMA 适配器 [368]

动作模块：

- LingUNet [428]

- 潜在运动计划 (LMP) [429]

- 风暴 [430]

- 感知器IO [431]

- DD-PPO [432]

- 规模DP [433]

- 符号调整 [434]

- 迪特 [435]

其他：

- DDPM [436]

- DDIM [437]

- 乌鸦 [438]

- RLBench [180]

- 某事V2：某事-某事V2 [439]

缩写：

- TFM：变压器

- MCIL：多上下文 模仿学习（Imitation Learning）

- MLE：最大似然估计

- MPC：模型预测控制

- CVAE：条件变分自动编码器

- EDR：日常机器人
E。其他 VLA 相关工作

我们包含了由于页数限制而无法包含在正文中的其他 VLA 相关工作。

1) VLA 的组成部分：预训练： 值隐式预训练 (VIP) [25] 利用视频中存在的时间序列，通过吸引初始帧和目标帧同时排斥连续帧，与 R3M [24] 区分开来。该目标旨在捕获长期时间关系并保持局部平滑性。虽然他们自己的实验证明在特定任务中与 R3M 相比具有优越的性能，但随后的研究通过更全面的评估提出了相互矛盾的发现 [32]。

SpawnNet [440]利用包含适配器层的双流架构来融合来自预训练视觉编码器的特征和从头开始学习的特征。这种创新方法消除了训练预训练视觉编码器的必要性，同时超越了参数高效微调（PEFT）方法的性能，机器人操作任务的实验结果证明了这一点。

Holo-Dex [441]提出使用自我监督学习（SSL）来学习视觉策略的低维表示。在 Dobb · E [442] 中，SSL 也被探索用于 3D 视觉

通过一种名为“家庭预训练表示”的方法，使用从纽约市 22 个家庭收集的数据进行输入。作者将 SSL 预训练表示扩展到视觉输入之外。 AuRL [45] 引入了从音频中学习动态行为，音频是一种经常被忽视但重要的信息源。此外，人们发现视觉输入对于多指机械手来说是不够的，因此出现了 T-Dex [213]，一种基于触觉的灵巧策略。

2）VLA的组成部分：世界模型（World Models）： Masked 世界模型（World Model）（MWM）[443]通过将DreamerV2的视觉编码器修改为卷积神经网络和视觉Transformer的混合组合进行创新。在训练这种新颖的卷积-ViT 视觉编码器时，MWM 从 MAE [404] 中提出的方法中汲取了灵感。通过结合辅助奖励预测损失，所得的学习潜在动力学模型在各种视觉机器人任务中表现出显着的性能改进。

由于大规模以人为中心的数据的可用性，SWIM [444] 提倡使用人类视频来训练 世界模型（World Model）。然而，认识到人类和机器人数据之间存在巨大差距，SWIM 通过视觉 可供性（Affordance） 地图中的 落地关联（Grounding） 操作来解决这一差距

。该方法涉及基于 可供性（Affordance） 地图推断目标位姿，有助于有效地传输人类数据中的知识以增强机器人控制。

Iso-Dream [445] 为 Dreamer 框架引入了两个关键增强功能。第一个增强功能侧重于通过解耦可控和不可控动力学来优化逆动力学。第二个增强涉及基于解耦的潜在想象力来优化智能体行为。这种方法特别有利，因为不可控状态转换分支可以独立于操作而推出，从而为 长时程（Long-Horizon） 决策过程带来好处。

基于 Transformer 的 世界模型（World Model）(TWM) [58] 与 Dreamer 具有相同的动机，但采用不同的方法，基于 Transformer-XL 架构构建 世界模型（World Model） [446]。这种基于 Transformer 的 世界模型（World Model） 通过生成新的轨迹来训练潜在想象力中的无模型智能体。此外，TWM 建议对 DreamerV2 的平衡 KL 散度损失进行修改，并引入专为优势演员-评论家框架量身定制的新型阈值熵损失。

SceneScript [447]引入了一组结构化语言命令，可以定义整个场景，指定布局和对象。这种基于语言的场景表示通过以自回归、基于标记的方式生成场景，与以前的方法（例如网格、体素网格、点云或辐射场）区分开来。它在布局估计和对象检测方面取得了与最先进的方法相媲美的结果。

3) VLA 的组件：模仿学习（Imitation Learning）: BeT [448]，轨迹变换器的 模仿学习（Imitation Learning）变体，通过使用基于 k 均值的动作离散化和连续动作校正来解决非专家人类演示数据的噪声和多模态性质。 C-BeT [449] 通过添加目标框架或演示作为目标规范来扩展 BeT。 VQ-BeT[450]进一步改进

BeT by incorporating vector quantization instead of k-means, enabling better modeling of long-range actions.

4) Subsequent Surveys on VLA: Since the initial release of this survey was made public, the field of embodied AI has experienced a proliferation of VLA models, leading to the emergence of several related surveys [451], [452], [453], [454], [455], [456], [457], [458], [459], [460], [461].

5) Latest Developments of VLA: LLaRA [462] introduces a data-centric method that uses a Vision-Language Model to automatically relabel existing robot demonstration videos with rich, descriptive language, significantly boosting the performance of downstream vision-language policies.

π_0 [118] undergoes further enhancements, including FAST [463] and $\pi_{0.5}$ [464]. FAST is a novel robot action tokenization method designed to prioritize dexterous skills. $\pi_{0.5}$ pushes the boundaries of in-the-wild generalization by incorporating co-training with diverse data sources.

Mobility VLA [465] proposes a hierarchical policy that integrates the commonsense reasoning capabilities of VLMs with a topological-graph-based low-level navigation policy.

While many VLAs focus primarily on robotic arms, recent advancements have extended their application to more complex embodiments, including humanoid and quadruped robots.

Humanoid Robot:

- GR00T N1 [135]
- Humanoid-VLA [466]

Quadruped Robot:

- QUAR-VLA [467]
- QUART-Online [468]
- MoRE [469]

Dexterous Hand:

- DexGraspVLA [470]
- DexGrasp-VLA [471]

Autonomous Vehicle:

- Alpamayo-R1 [472]
- CoVLA [473]
- DriveVLA-W0 [474]
- EMMA [475]

Additionally, some VLAs are designed to function as virtual assistants capable of taking actions within operating systems, video games, and other environments. Notable examples include:

- CombatVLA [476]
- JARVIS-VLA [477]

We utilize various charts to visualize key aspects of VLA developments from 2020 to 2025, including Figure 9, Figure 10, Figure 11, and Figure 12. To supplement the VLAs discussed in the main text, we employed a hybrid approach combining automated scripting and manual searching to retrieve VLA-related papers published between January 2020 and December 2025. We queried the keywords "VLA", "Vision-language-action", and "Vision language action", filtering false positives based on their relevance to "embodied AI" and "robotics". This pipeline yielded approximately 400 VLA-related papers. Acknowledging the potential for automated errors, we welcome feedback and requests for corrections regarding the included data.

Abbreviations of institutes appearing in Figure 9:

- **AI2**: Allen Institute for AI
- **AIBD**: Advanced Institute of Big Data
- **ANU**: Australian National University
- **AgiBot**: AgiBot Research
- **Alexandria Univ.**: Alexandria University
- **Alibaba**: Alibaba Group, DAMO Academy
- **Anyverse**: Anyverse Intelligence
- **BAAI**: Beijing Academy of Artificial Intelligence
- **BACFBC**: Innovation Center for Future Blockchain and Privacy Computing, Beijing
- **BIGAI**: State Key Lab of General Artificial Intelligence
- **BIT**: Beijing Institute of Technology
- **BJUT**: Beijing University of Technology
- **BKLSISMI**: Beijing Key Laboratory of Super Intelligent Security of Multi-Modal Information
- **BNU**: Beijing Normal University
- **BUAA**: Beihang University
- **BUPT**: Beijing University of Posts and Telecommunications
- **Bosch**: Bosch Center for Artificial Intelligence, Bosch Corporate Research, Bosch Mobility Solutions, Bosch Research
- **Bretagne**: Bretagne INP - ENIB
- **ByteDance**: ByteDance Research, ByteDance Seed
- **CAS**: Chinese Academy of Sciences
- **CAU**: Clark Atlanta University
- **CMU**: Carnegie Mellon University
- **CNRS**: CNRS IRL 2010 CROSSING
- **CSU**: Central South University
- **CUHK**: The Chinese University of Hong Kong
- **CUHK(SZ)**: The Chinese University of Hong Kong (Shenzhen)
- **CWRU**: Case Western Reserve University
- **Caltech**: California Institute of Technology
- **Cambridge**: Cambridge University
- **China Mobile**: China Mobile Information Technology Co., Ltd.,
- **China Telecom**: China Telecom
- **CityU**: City University of Hong Kong
- **Cognitive AI**: Cognitive AI Lab
- **Columbia**: Columbia University
- **DSO**: DSO National Laboratories
- **DUT**: Dalian University of Technology
- **Didi Chuxing**: Didi Chuxing
- **Drexel**: Drexel University
- **Duke**: Duke University
- **ECNU**: East China Normal University
- **EIT**: Eastern Institute of Technology
- **EKUT**: University of Tübingen
- **ENS**: École Normale Supérieure
- **ER**: Everyday Robots
- **ETH**: ETH Zurich

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BeT 通过结合矢量量化而不是 k 均值，可以更好地对远程动作进行建模。

4) VLA 的后续调查： 自该调查首次公开发布以来，具身智能（Embodied AI） 领域经历了 VLA 模型的激增，导致出现了多项相关调查 [451]、[452]、[453]、[454]、[455]、[456]、[457]、[458]、[459]、 [460]，[461]。

5) VLA 的最新发展： LLaRA [462] 引入了一种以数据为中心的方法，该方法使用视觉语言模型以丰富的描述性语言自动重新标记现有的机器人演示视频，从而显着提高下游视觉语言策略的性能。

π 0 [118] 经历了进一步的增强，包括 FAST [463] 和 π 0 . 5 [464]。 FAST 是一种新颖的机器人动作标记化方法，旨在优先考虑灵巧技能。 π 0 . 5 通过将协同训练与不同的数据源相结合，突破了野外泛化的界限。

Mobility VLA [465]提出了一种分层策略，它将 VLM 的常识推理能力与基于拓扑图的低级导航策略相结合。

虽然许多 VLA 主要关注机器人手臂，但最近的进展已将其应用扩展到更复杂的实施例，包括人形机器人和四足机器人。
人形机器人：

- GR00T N1 [135]

- 人形-VLA [466]

- 四足机器人：

- 夸尔-VLA [467]

- QUART-在线 [468]

- 更多 [469]
灵巧的手：

- DexGraspVLA [470]

- DexGrasp-VLA [471]
自动驾驶汽车：

- 阿尔帕玛约-R1 [472]

- 冠状病毒 [473]

- 驱动器VLA-W0 [474]

- 艾玛 [475]

此外，一些 VLA 被设计为虚拟助手，能够在操作系统、视频游戏和其他环境中执行操作。值得注意的例子包括：

- 战斗VLA [476]

- 贾维斯-VLA [477]

“机器人”。该流程产生了大约 400 篇 VLA 相关论文。认识到存在自动错误的可能性，我们欢迎针对所包含的数据提供反馈和更正请求。

图 9 中出现的机构缩写：

- AI2：艾伦人工智能研究所

- AIBD：大数据高级研究所

- ANU：澳大利亚国立大学

- AgiBot：AgiBot 研究

- 亚历山大大学。：亚历山大大学

- 阿里巴巴：阿里巴巴集团、达摩院

- 任何宇宙：任何宇宙情报

- BAAI：北京人工智能研究院

- BACFBC：未来区块链和隐私计算创新中心，北京

- BIGAI：通用人工智能国家重点实验室

- BIT：北京理工大学

- BJUT：北京工业大学

- BKLSISMI：多模态信息超智能安全北京市重点实验室

- BNU：北京师范大学

- BUAA：北京航空航天大学

- BUPT：北京邮电大学

- 博世：博世人工智能中心、博世企业研究、博世移动解决方案、博世研究

- 布列塔尼：布列塔尼 INP - ENIB

- 字节跳动：字节跳动研究院、字节跳动种子

- CAS：中国科学院

- CAU：克拉克亚特兰大大学

- CMU：卡内基梅隆大学

- CNRS：CNRS IRL 2010 穿越

- CSU：中南大学

- CUHK：香港中文大学

- 香港中文大学（深圳）：香港中文大学（深圳）

- CWRU：凯斯西储大学

- Caltech：加州理工学院

- 剑桥：剑桥大学

- 中国移动：中移信息技术有限公司

- 中国电信：中国电信

- CityU：香港城市大学

- 认知人工智能：认知人工智能实验室

- 哥伦比亚：哥伦比亚大学

- DSO：DSO国家实验室

- DUT：大连理工大学

我们利用各种图表来可视化 2020 年至 2025 年 VLA 发展的关键方面，包括图 9、图 10、图 11 和图 12。为了补充正文中讨论的 VLA，我们采用了自动脚本和手动搜索相结合的混合方法来检索 2020 年 1 月至 2025 年 12 月之间发表的 VLA 相关论文。我们查询了关键词“VLA”、“Visionlanguage-action”和“Vision language”动作”，根据与“具身智能（Embodied AI）”的相关性过滤误报，以及

- 滴滴出行：滴滴出行

- 德雷塞尔：德雷塞尔大学

- 杜克大学：杜克大学

- ECNU：华东师范大学

- EIT：东部理工学院

- EKUT：图宾根大学

- ENS：高等师范学院´

- ER：日常机器人

- ETH：苏黎世联邦理工学院

- **EdUHK**: The Education University of Hong Kong
- **Fudan**: Fudan University
- **GDUT**: Guangdong University of Technology
- **Gachon Univ.**: Gachon University
- **Galaxea**: Galaxea AI
- **Georgia Tech**: Georgia Institute of Technology
- **Google**: Google, Google DeepMind, Robotics at Google, Google Research, Brain Team, Gemini Robotics Team
- **HBUT**: Hebei University of Technology
- **HIT**: Harbin Institute of Technology
- **HIT(SZ)**: Harbin Institute of Technology (Shenzhen)
- **HKU**: The University of Hong Kong
- **HKUST**: Hong Kong University of Science and Technology
- **HKUST(GZ)**: Hong Kong University of Science and Technology (Guangzhou)
- **HUST**: Huazhong University of Science and Technology
- **Hanyang Univ.**: Hanyang University
- **Harvard**: Harvard University
- **Horizon**: Horizon Robotics
- **Huawei**: Huawei Technologies, Huawei Noah's Ark Lab
- **HuggingFace**: Hugging Face
- **Hyundai**: Hyundai Motor Company
- **IC**: Imperial College London
- **IEIT**: IEIT SYSTEMS Co., Ltd.
- **IIIT-H**: IIIT Hyderabad
- **IIT**: Istituto Italiano di Tecnologia
- **IMT**: IMT Atlantique
- **Infinigence**: Infinigence AI
- **JHU**: Johns Hopkins University
- **JLU**: Jilin University
- **KAIST**: KAIST AI
- **KIT**: Karlsruhe Institute of Technology
- **Kobe Univ.**: Kobe University
- **Korea Univ.**: Korea University
- **LZU**: Lanzhou University
- **Lehigh Univ.**: Lehigh University
- **LiAuto**: LiAuto Inc.
- **LimX**: LimX Dynamics
- **Logos**: Logos Robotics
- **Lumina**: Lumina Group
- **Lumos**: Lumos Robotics
- **MBZUAI**: Mohamed bin Zayed University of Artificial Intelligence
- **MEGVII**: MEGVII Technology
- **MIPT**: IAI MIPT
- **MIT**: Massachusetts Institute of Technology
- **MMLab**: MMLab @ CUHK, MMLab @ HKU
- **Macalester**: Macalester College
- **Meta**: Meta AI, FAIR-MetaAI, Meta Reality Labs
- **Microsoft**: Microsoft Research, Microsoft Research Asia, Microsoft Zurich
- **Midea**: Midea Group
- **Mila**: Mila — Quebec AI Institute
- **Monash Univ.**: Monash University
- **Moxin**: Moxin (Huzhou) Technology Co., Ltd.
- **NAVER**: NAVER AI Lab
- **NEU**: Northeastern University
- **NJU**: Nanjing University
- **NJUPT**: Nanjing University of Posts and Telecommunications
- **NKU**: Nankai University
- **NPU**: Northwestern Polytechnical University
- **NTU**: Nanyang Technological University
- **NUS**: National University of Singapore
- **NVIDIA**: NVIDIA Research
- **NWU**: Northwestern University
- **NYU**: New York University
- **NaUKMA**: National University of Kyiv-Mohyla Academy
- **Noematrix**: Noematrix Intelligence
- **OXE Team**: Open X-Embodiment Collaboration
- **OpenHelix**: OpenHelix Team
- **PCL**: Peng Cheng Laboratory
- **PI**: Physical Intelligence
- **PKU**: Peking University
- **PSL**: PSL Research University
- **Penn State**: Pennsylvania State University
- **Pitt**: University of Pittsburgh
- **PolyU**: The Hong Kong Polytechnic University
- **Princeton**: Princeton University
- **QMUL**: Queen Mary University of London
- **Qi Zhi**: Shanghai Qi Zhi Institute, Shanghai Qizhi Institute
- **Qualcomm**: Qualcomm AI Research
- **RU**: Rutgers University
- **RUC**: Renmin University of China
- **SBU**: Stony Brook University
- **SCUT**: South China University of Technology
- **SH AI Lab**: Shanghai AI Laboratory
- **SHU**: Shanghai University
- **SII**: Shanghai Innovation Institute
- **SJTU**: Shanghai Jiao Tong University
- **SKKU**: Sungkyunkwan University
- **SNU**: Seoul National University
- **STU**: ShanghaiTech University
- **SUST**: Southern University of Science and Technology
- **SUTD**: Singapore University of Technology and Design
- **SYSU**: Sun Yat-sen University
- **SZU**: Shenzhen University
- **Sea AI**: Sea AI Lab
- **SenseTime**: SenseTime Research
- **Siemens**: Siemens Corporation
- **Simplexity**: Simplexity Robotics
- **Sofia Univ.**: Sofia University "St. Kliment Ohridski"
- **Sony**: Sony Interactive Entertainment, Sony Research
- **Sorbonne Univ.**: Sorbonne University
- **Spirit**: Spirit AI
- **Stanford**: Stanford University
- **Syracuse**: Syracuse University
- **T-Stone**: T-Stone Robotics Institute
- **TJU**: Tianjin University
- **TUM**: Technical University of Munich
- **Taiwan Univ.**: National Taiwan University
- **Tencent**: Tencent Inc., Tencent Robotics X, Wechat AI
- **Tongji**: Tongji University

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- EdUHK：香港教育大学

- 复旦大学：复旦大学

- GDUT：广东工业大学

- 嘉泉大学: 嘉泉大学

- Galaxea：Galaxea AI

- 佐治亚理工学院：佐治亚理工学院

- 谷歌：谷歌、谷歌 DeepMind、谷歌机器人、谷歌研究、Brain 团队、Gemini 机器人团队

- HBUT：河北工业大学

- HIT：哈尔滨工业大学

- 哈尔滨工业大学（深圳）：哈尔滨工业大学（深圳）

- HKU：香港大学

- HKUST：香港科技大学

- HKUST(GZ)：香港科技大学（广州）

- HUST：华中科技大学

- 汉阳大学: 汉阳大学

- 哈佛大学：哈佛大学

- 地平线：地平线机器人

- 华为：华为技术有限公司、华为诺亚方舟实验室

- HuggingFace ：拥抱脸

- 现代：现代汽车公司

- IC：伦敦帝国理工学院

- IEIT：IEIT SYSTEMS 有限公司

- IIIT-H ：IIIT 海得拉巴

- IIT：意大利理工学院

- IMT ：IMT 大西洋

- Infinigence：Infinigence AI

- JHU：约翰·霍普金斯大学

- JLU : 吉林大学

- KAIST : KAIST AI

- KIT：卡尔斯鲁厄理工学院

- 神户大学: 神户大学

- 韩国大学: 高丽大学

- LZU : 兰州大学

- 里哈伊大学: 理海大学

- 力车 : 力车公司

- LimX：LimX 动力学

- 标志：标志机器人

- Lumina：Lumina 集团

- Lumos：Lumos 机器人

- MBZUAI：穆罕默德·本·扎耶德人工智能大学

- MEGVII：旷视科技

- MIPT：IAI MIPT

- MIT：麻省理工学院

- MMLab : MMLab@CUHK, MMLab@HKU

- 麦卡莱斯特：麦卡莱斯特学院

- Meta：Meta AI、FAIR-MetaAI、Meta Reality Labs

- 微软：微软研究院、微软亚洲研究院、微软苏黎世

- 美的：美的集团

- Mila : Mila - 魁北克人工智能研究所

- 莫纳什大学：莫纳什大学

- 魔心：魔心（湖州）科技有限公司

- NAVER：NAVER 人工智能实验室

- NEU：东北大学

- NJU : 南京大学

- NJUPT : 南京邮电大学

- NKU : 南开大学

- NPU：西北工业大学

- NTU：南洋理工大学
- NUS：新加坡国立大学
- NVIDIA：NVIDIA 研究部
- NWU：西北大学
- NYU：纽约大学
- NaUKMA ：基辅国立大学莫希拉学院
- Noematrix：Noematrix 智能
- OXE 团队：开放 X-Embodiment 协作
- OpenHelix ：OpenHelix 团队
- PCL：鹏程实验室
- PI：身体智力
- PKU：北京大学
- PSL：PSL 研究大学
- 宾夕法尼亚州立大学：宾夕法尼亚州立大学
- 皮特：匹兹堡大学
- PolyU：香港理工大学
- 普林斯顿：普林斯顿大学
- QMUL：伦敦玛丽女王大学
- 启智：上海启智研究院、上海启智研究院
- 高通：高通人工智能研究中心
- RU：罗格斯大学
- RUC：中国人民大学
- SBU：石溪大学
- SCUT：华南理工大学
- SH AI Lab：上海人工智能实验室
- SHU：上海大学
- SII：上海创新研究院
- SJTU：上海交通大学
- SKKU：成均馆大学
- SNU：首尔国立大学

- STU：上海科技大学
- SUST：南方科技大学
- SUTD：新加坡科技设计大学
- 中山大学：中山大学
- SZU：深圳大学
- 海洋人工智能：海洋人工智能实验室
- 商汤科技：商汤科技研究院
- 西门子：西门子公司
- Simplexity：Simplexity 机器人
- 索非亚大学：索非亚大学“圣克莱门特奥赫里德斯基”
- 索尼：索尼互动娱乐、索尼研究院
- 索邦大学：索邦大学
- 精神：精神AI
- 斯坦福大学：斯坦福大学
- 锡拉丘兹：锡拉丘兹大学
- T-Stone：T-Stone 机器人研究所
- TJU：天津大学
- TUM：慕尼黑工业大学
- 台湾大学: 国立台湾大学
- 腾讯：腾讯公司、腾讯Robotics X、微信AI
- 同济大学：同济大学

- **Toyota**: Toyota Research Institute
- **Tsinghua**: Tsinghua University
- **Turing**: Turing Inc.
- **U of U**: University of Utah
- **UATX**: The University of Austin at Texas
- **UAlberta**: University of Alberta
- **UArizona**: University of Arizona
- **UArkansas**: University of Arkansas
- **UBC**: The University of British Columbia
- **UCAS**: University of Chinese Academy of Sciences
- **UCB**: University of California, Berkeley
- **UCL**: University College London
- **UCLA**: University of California, Los Angeles
- **UCSD**: University of California San Diego
- **UCSI**: UCSI University
- **UCSP**: Universidad Católica San Pablo
- **UChicago**: University of Chicago
- **UESTC**: University of Electronic Science and Technology of China
- **UIC**: University of Illinois Chicago
- **UIUC**: University of Illinois Urbana-Champaign
- **UM**: University of Macau
- **UMD**: University of Maryland, College Park
- **UMU**: Umeå University
- **UNSW**: University of New South Wales
- **UOsaka**: The University of Osaka
- **UPenn**: University of Pennsylvania
- **UQ**: University of Queensland
- **USC**: University of Southern California
- **USST**: University of Shanghai for Science and Technology
- **USTC**: University of Science and Technology of China
- **USYD**: The University of Sydney
- **UT Austin**: The University of Texas at Austin
- **UT Dallas**: University of Texas at Dallas
- **UTN**: University of Technology Nuremberg
- **UTS**: University of Technology Sydney
- **UTokyo**: The University of Tokyo
- **UW**: University of Washington
- **UWarsaw**: University of Warsaw
- **UW–Madison**: University of Wisconsin-Madison
- **UdeM**: Université de Montréal
- **Uni Freiburg**: University of Freiburg
- **UofT**: University of Toronto
- **VT**: Virginia Tech
- **VinUni**: VinUniversity
- **WHU**: Wuhan University
- **WMU**: Wenzhou Medical University
- **Waseda**: Waseda University
- **Westlake Univ.**: Westlake University
- **X-Era**: X-Era AI Lab
- **X-Humanoid**: Beijing Innovation Center of Humanoid Robotics
- **XDU**: Xidian University
- **XJTU**: Xi'an Jiaotong University
- **XMU**: Xiamen University Malaysia
- **XPeng**: XPeng Motors
- **Xiaomi**: Beijing Xiaomi Robot Technology Co., Ltd., Xiaomi EV, Xiaomi Robotics
- **Yale**: Yale University
- **Yinwang**: Yinwang Intelligent Technology Co., Ltd.
- **Yuandao**: Yuandao AI
- **ZGCA**: Zhongguancun Academy
- **ZGCI**: Zhongguancun Institute of Artificial Intelligence
- **ZJU**: Zhejiang University
- **ZZU**: Zhengzhou University
- **ZhiCheng**: ZhiCheng AI
- **Zhongke Huiling**: Beijing Zhongke Huiling Robot Technology Co
- **iFlyTek**: iFlyTek Research and Development Group
- **mimic**: mimic robotics
- **valeo**: valeo.ai

List of VLAs appearing in Figure 9:

- 3D Diffuser Actor [478]
- 3DS-VLA [479]
- 4D-VLA [480]
- A2C2 [481]
- A3VLM [482]
- ACG [483]
- ACT [100]
- Act3D [91]
- ActDistill [484]
- ActionFlow [485]
- AdaMoE [486]
- ADP [487]
- AFI [488]
- AgentWorld [489]
- AINA [490]
- Alpamayo-R1 [472]
- AMS [491]
- ANNIE [492]
- AsyncVLA [493]
- ATE [494]
- AutoDrive-R2 [495]
- AutoPrune [496]
- AutoVLA [497]
- AVA-VLA [498]
- Avi [499]
- BadVLA [500]
- BayesVLA [501]
- BC-Z [82]
- Being-H0 [502]
- Beyond Success [503]
- Bi-VLA [504]
- Bi-VLA 2025 [505]
- BridgeVLA [506]
- BYOVLA [507]
- CCoL [508]
- ChatVLA-2 [509]
- CLAW [510]
- CLIPort [29]
- CLIP-RT [511]
- ClutterDexGrasp [512]
- CoC-VLA [513]
- CogACT [122]

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- 丰田：丰田研究所

- 清华大学：清华大学

- 图灵：图灵公司

- • U of U：犹他大学

- UATX：德克萨斯州奥斯汀大学

- UAlberta：阿尔伯塔大学

- UArizona：亚利桑那大学

- UArkansas ：阿肯色大学

- UBC：不列颠哥伦比亚大学

- UCAS：中国科学院大学

- UCB：加州大学伯克利分校

- UCL：伦敦大学学院

- UCLA：加州大学洛杉矶分校

- UCSD：加州大学圣地亚哥分校

小米电动车、小米机器人

- 耶鲁大学：耶鲁大学

- 银王：银王智能科技有限公司

- 元道：元道AI

- ZGCA：中关村学院

- ZGCI：中关村人工智能研究院

- ZJU：浙江大学

- ZZU：郑州大学

- 志成：志成AI

- 中科慧灵：北京中科慧灵机器人技术有限公司

- 科大讯飞：科大讯飞研发集团

- 模仿：模仿机器人

- 法雷奥：valeo.ai

- UCSI : UCSI大学
- UCSP：圣巴勃罗天主教大学
- 芝加哥大学：芝加哥大学
- UESTC : 中国电子科技大学
- UIC：伊利诺伊大学芝加哥分校
- UIUC：伊利诺伊大学厄巴纳-香槟分校
- UM : 澳门大学
- UMD：马里兰大学学院公园分校
- UMU : 于默埃大学
- UNSW : 新南威尔士大学
- UOsaka : 大阪大学
- UPenn : 宾夕法尼亚大学
- UQ：昆士兰大学
- USC：南加州大学
- USST : 上海理工大学
- USTC：中国科学技术大学
- USYD：悉尼大学
- UT Austin：德克萨斯大学奥斯汀分校
- UT Dallas：德克萨斯大学达拉斯分校
- UTN：纽伦堡科技大学
- UTS：悉尼科技大学
- UTokyo : 东京大学
- UW：华盛顿大学
- 华沙大学：华沙大学
- 威斯康星大学麦迪逊分校：威斯康星大学麦迪逊分校
- UdeM : 蒙特利尔大学
- Uni Freiburg : 弗莱堡大学
- UofT：多伦多大学
- VT：弗吉尼亚理工大学
- VinUni : Vin大学

- WHU : 武汉大学

- WMU : 温州医科大学

- 早稻田：早稻田大学

- 西湖大学: 西湖大学

- X-Era：X-Era人工智能实验室

- X-Humanoid : 北京仿人机器人创新中心

- XDU : 西安电子科技大学

- XJTU : 西安交通大学

- XMU : 厦门大学马来西亚分校

- 小鹏汽车：小鹏汽车

- 小米：北京小米机器人技术有限公司

图 9 中显示的 VLA 列表：

- 3D 扩散器演员 [478]

- 3DS-VLA [479]

- 4D-VLA [480]

- A2C2 [481]

- A3VLM [482]

- 二次元 [483]

- 行动 [100]

- Act3D [91]

- ActDistill [484]

- 动作流 [485]

- 阿达莫伊 [486]

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6) Beyond VLA: Recent research has expanded beyond standard VLA architectures in several key directions:

- **VLA + World Model:** Architectures that unify VLAs with world models.
  - WorldVLA [126]
  - UniVLA [127]
  - NORA-1.5 [136]
  - RynnVLA-002 [708]
  - Motus [643]
- **PLA, MLA:** Models incorporating perception modalities beyond vision.
  - Perception-Language-Action framework (PLA) [666]
  - Multisensory Language–Action model (MLA) [631]
- **VAM:** Video-Action Models that jointly capture semantics and dynamics during video pretraining, enabling more efficient post-training for low-level robot control.
  - mimic-video (VAM) [754]

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- • TriVLA [741] - UniVLA [127]

- • TVVE [742] - NORA-1.5 [136]

- • UD-VLA [743] - RynnVLA-002

- • 水下VLA [744] - Motus [643]

- • UniCoD [745] • PLA、MLA：

- • Unified-IO 2 [746] 超出视野。

- • Uni-NaVid [747] - 感知-语言-行动

- • UniPi [86] [666]

- • UniUGP [748] - 多感官

- • UniVLA [127] • UPA-RFAS [749] • VAM：视频动作 • UP-VLA [750] 抽动和动态 • UrbanVLA [751] 更高效 • USIM 和 U0 [752] - 模拟视频 • UWM [753] • VAM [754]

- VST [779]

- • WAM-Flow [780] • WholeBodyVLA [781] • WMPO [782]

- 6) 超越 VLA： 最近的研究已经扩展到超越

- 几个关键方向的标准VLA架构：

- VLA + 世界模型（World Model）：将 VLA 与 世界模型（World Models）统一的架构。

- WorldVLA [126]

- 诺拉-1.5 [136]

- 瑞恩VLA-002 [708]

- - 莫图斯 [643]

- PLA、MLA：包含视觉之外的感知模式的模型。

- 感知-语言-行动框架（PLA）[666]

- 多感官语言-动作模型（MLA）[631]

- VAM：视频动作模型，在视频预训练期间联合捕获语义和动态，从而为低级机器人控制提供更高效的后期训练。

- 模拟视频（VAM）[754]

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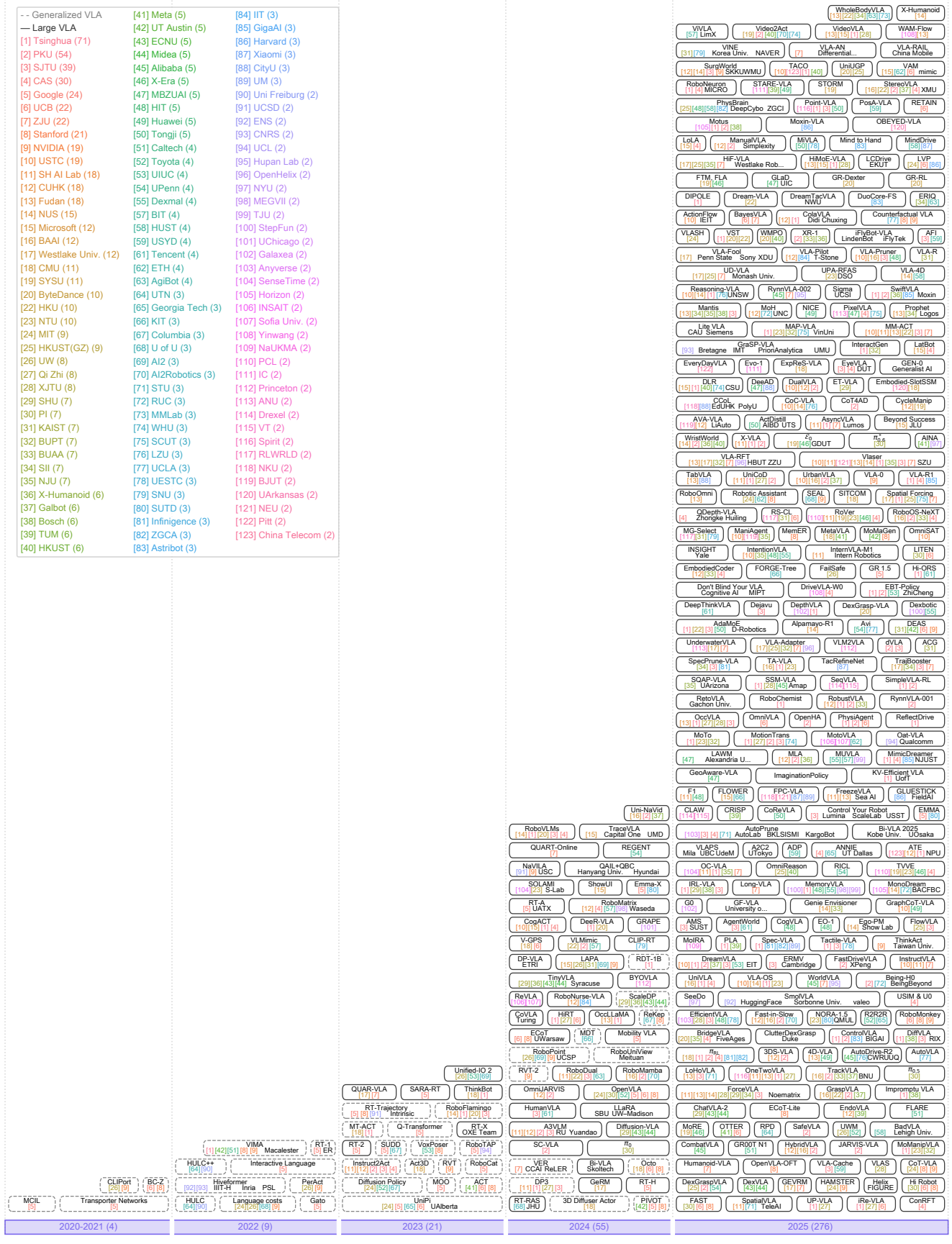


Fig. 9. Timeline of VLA models from 2020 to 2025. Bracketed numbers indicate the publication count for the corresponding year or institute.

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[47]GLaDUICDreamTacVLANWU GR-Dexter[20] DuoCore-FS[83] GR-RL[20][34]ERIQ[63]
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Chushing 反事实（Counterfactual）VLA[77] [8] [9]
[15] Microsoft (12)[16] BAAI (12) [58] HUST (4)[59] USYD (4) [100] StepFun (2)[101]芝加哥大学 (2)
VLASH[24] [1] [20]VST[22] [20]WMPO[40] [2] [33]XR-1[36] LindenBotiFlyBot-VLAiFlyTek [3]AFI [59]
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[12][84]VLA-PilotT-Stone [10]VLA-Pruner[16] [3] [48] VLA-R[31]
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(2)[104] SenseTime (2)[105] Horizon (2) [10]推理-VLA[17][14][25] [1] [76] [7]UD-VLAUNSWMonash
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NICE[49][113]PixelVLA[47][4][75][13]Prophet[34]徽标
[23] NTU (10) [66] KIT (3) [107] 索非亚大学

。 (2) Lite VLA MAP-VLA MM-ACT
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UMU InteractGen[1] [32] [15]LatBot [4]
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EveryDayVLA[122]DLR Evo-1[111]DeeAD ExpReS-VLADualVLA[18] ET-VLA[3]EyeVLA[4] DUT
Embodied-SlotSSMGeneralist AIGEN-0
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CycleManip[12][19]
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LiAuto [50]ActDistillAIBD UTS [11] [1]AsyncVLA[7] Lumos Beyond成功[15]吉林大学
[32]北京邮电大学(7)[75]华南理工大学(3)[116]精神(2)[14]手腕世界[2][36][40][11]X-VLA[1][2][19][46]□
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[9] SITCOM[18] [17]空间强迫 [1] [25][75] [7]
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MemER[8] MetaVLA[18][41] MoMaGen[42] [8] OmniSAT[10]
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Robotics [30]LITEN [6]
EmbodiedCoder FORGE-Tree FailSafe GR 1.5 Hi-ORS
[12][33] [4] [66] [26] [5] [1] [61]
不要蒙蔽您的 VLACognitive AI MIPT DriveVLA-W0[108][4] [1] [2]EBT 策略 [53] 志成
DeepThinkVLA Dejavu DepthVLA DexGrasp-VLA Dexbotic
[61] [3] [102][1] [20] [100][55]
[1] [22] [3] [50]AdaMoED-Robotics Alpamayo-R1[14] [54]Avi[77] [31][42]DEAS [6] [9]
水下VLA VLA-适配器 VLM2VLA dVLA ACG
[113][17] [7] [17][25][32] [7] [96] [112] [2] [3] [31]
SpecPrune-VLA TA-VLA TacRefineNet TrajBooster
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SQAP-VLA SSM-VLA SeqVLA SimpleVLA-RL
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Gachon Univ.RetoVLA RoboChemist[1] [12]RobustVLA [1] [2] [33] RynnVLA-001[2]
OccVLA OmniVLA OpenHA PhysiAgent ReflectDrive
[13] [1] [27][28] [3] [6] [2] [1] [2] [6] [1]
MoTo MotionTrans MotoVLA Oat-VLA
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LAWM MLA MUVLA MimicDreamer
[47] 亚历山大大学..

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GeoAware-VLA[47] ImaginationPolicy KV-Efficient VLA[1] UofT
F1 FLOWER FPC-VLA FreezeVLA GLUESTICK
[11][48] [15][66] [118][121][87][89] [11][13] Sea AI [86] FieldAI
Uni-NaVid CLAW CRISP CoReVLA 控制您的机器人 EMMA
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RoboVLMs TraceVLA AutoPrune Bi-VLA 2025
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QUART-Online[7] REGENT[54] MilaVLAPSUBC UdeM UTokyoA2C2 ADP[59] [4] [65]ANNIEUT Dallas
[123][12]ATE [1] NPU
NaVILA QAIL+QBC OC-VLA OmniReason RICL TVVE
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[5]RT-A UATX [12] [4]RoboMatrix [57][98] 早稻田 [102]G0 大学...GF-VLA Genie Envisioner[14][33]
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CambridgeERMV FastDriveVLA[2] XPeng InstructVLA[10][11] [7]
[29][36][43]TinyVLA[44] 雪城大学 BYOVLA[112] [16]UniVLA [1] [4] [10]VLA-OS[14] [1] [23]
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ReVLA RoboNurse-VLA ScaleDP SeeDo SmolVLA USIM & U0
[106][107] [12][84] [29][36][43][44] [97] [92] HuggingFace 索邦大学

。 valeo [4]
CoVLA HiRT OccLLaMA ReKep EfficientVLA 快中慢 NORA-1.5 R2R2R RoboMonkey
图灵 [1] [27] [6] [13] [1] [67] [8] [103][28] [3] [48][78] [12][16] [2] [70] [23][80]QMUL [52][65] [6] [8] [9]
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[1] ControlVLA[2] [83] BIGAI [1] [38]DiffVLA [3] RIX
[26]RoboPoint[69][9] UCSP RoboUniViewMeituan [18][1][2]π
RL[4][81][82]3DS-VLA[12][2]4D-VLA[13][49][45]AutoDrive-R2[76]CWRUUQ AutoVLA[77]
Unified-IO 2 RVT-2 RoboDual RoboMamba LoHoVLA OneTwoVLA TrackVLA π 0.5
[26][53][69] [9] [11][22] [3] [63] [16] [2] [70] [13] [3] [71] [116][11][13] [1] [27] [16] [2] [33][37] BNU [30]
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Diffusion-VLA[29][43][44] [19]MoRE[46] OTTER[41] [6] RPD[64] SafeVLA[2] [26]UWM[52] [58] Lehigh
Univ.BadVLA

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HULC++[64][90] 交互语言[5] [11]Instruct2Act[12] [2] [3] [4] Act3D[18] RVT[9] RoboCat[5] [7] CCAIVERReLER SkoltechBi-VLA [18]Octo [6] [8] Humanoid-VLA[7] OpenVLA-OFT[8] VLA-Cache[3] [59] VLAS[28] [24]CoT-VLA [8] [9]
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MCIL[5] Transporter Networks[5] [64]HULC[90] 语言成本[24][26][68] [9] Gato[5] [24] [5] [65]UniPi [6] UAlberta [68]RT-RAS JHU 3D 扩散器 Actor[18] [42]PIVOT [5] [8] [30]FAST [6] [8] [11]SpatialVLA[71] TeleAI UP-VLA[1] [27] [1]iRe-VLA [27] [6] ConRFT[4]
2020-2021 (4) 2022 (9) 2023 (21) 2024 (55) 2025 (276)
月

图 9. 2020 年至 2025 年 VLA 模型的时间表。括号内的数字表示相应年份或机构的发表数量。

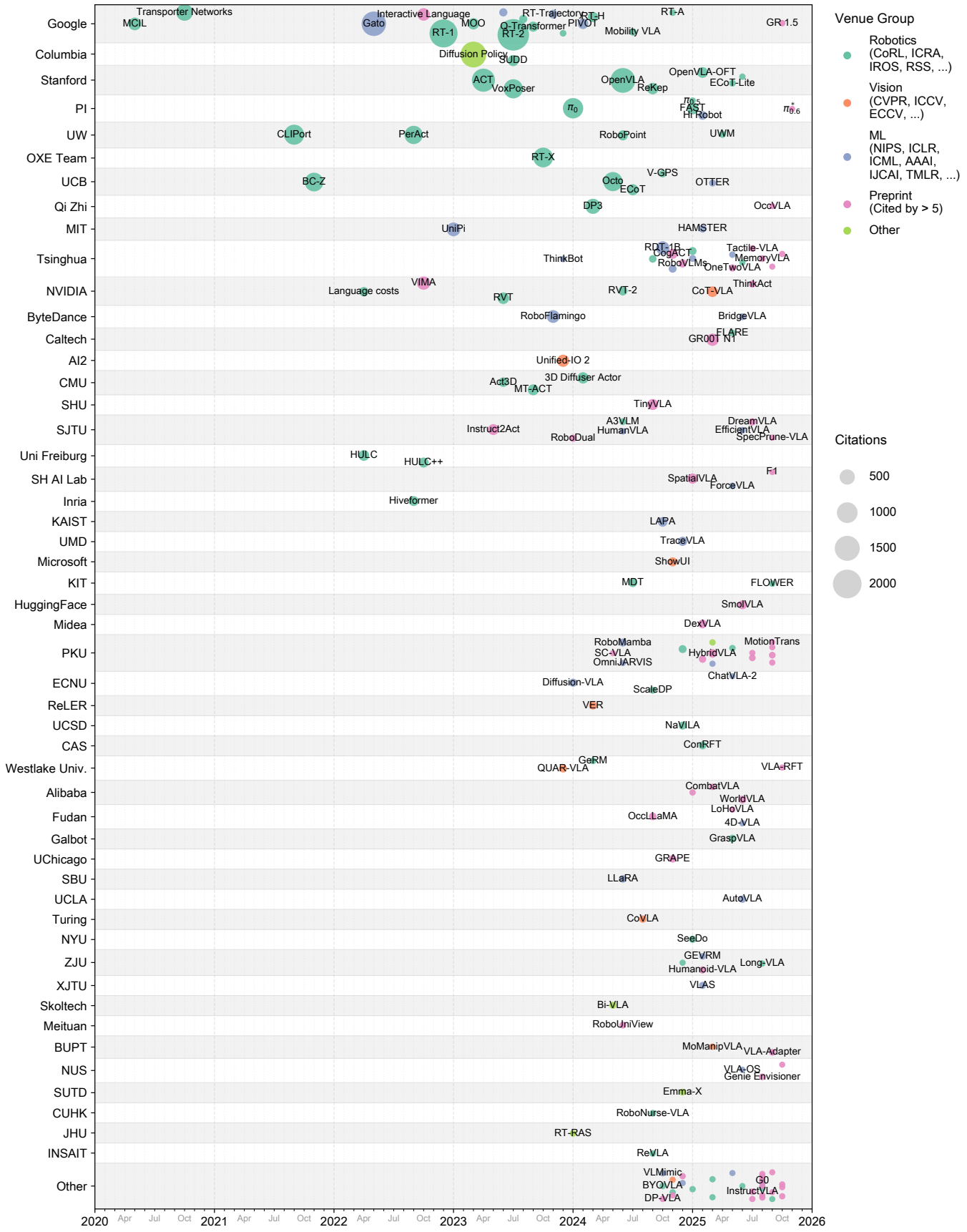


Fig. 10. Visualization of the VLA research landscape.

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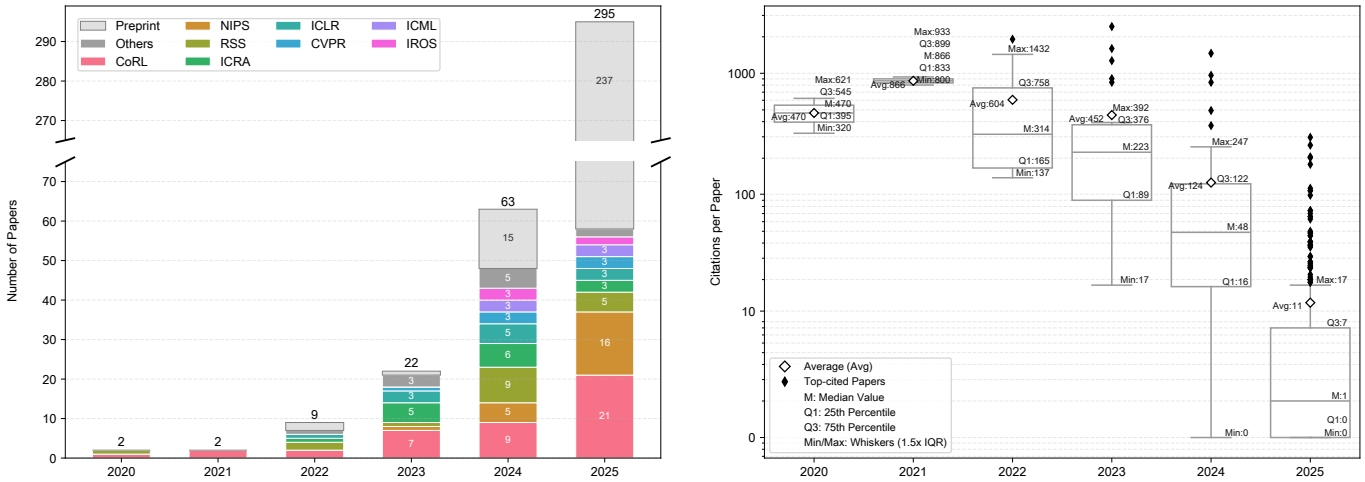
52

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Google MCILTransporter Networks GatoInteractive LanguageRT-1 MOO
Q-TransformerRT-2RT-TrajectoryPIVOTRT-HMobility VLART-A GR 1.5 Venue GroupRobotics
哥伦比亚扩散策略SUDD（CoRL、ICRA、
OpenVLA-OFT IROS、RSS、...）
斯坦福 ACTVoxPoser OpenVLAReKep ECoT-Lite Vision
PI π 0 FASTHi 机器人 π 0.5 π 0.6 *（CVPR、ICCV、ECCV、...）
UW CLIPort PerAct RoboPoint UWM ML
OXE Team RT-X（NIPS、ICLR、ICML、AAAI、
UCB BC-Z Octo V-GPS OTTER IJCAI、TMLR、 ...)
ECOT
预印本
奇智 DP3 OccVLA（引用次数 > 5）
MIT UniPi HAMSTER 其他
清华 ThinkBot RDT-1BCogACTRoboVLMsOneTwoVLATactile-VLAMemoryVLA
VIMA ThinkAct
NVIDIA 语言成本 RVT RVT-2 CoT-VLA
字节跳动 RoboFlamingo BridgeVLA
FLARE
Caltech GR00T N1
AI2 Unified-IO 2
CMU Act3DMT-ACT3D 扩散器 Actor
SHU TinyVLA
A3VLM DreamVLA
SJTU Instruct2Act HumanVLA EfficientVLA
RoboDual SpecPrune-VLA 引文
Uni Freiburg HULC HULC++
SH AI Lab SpatialVLAForceVLA F1 500
Inria Hiveformer
1000
KAIST LAPA
UMD TraceVLA 1500
Microsoft ShowUI
KIT MDT FLOWER 2000
HuggingFace SmolVLA
Midea DexVLA
RoboMamba MotionTrans
北大SC-VLA HybridVLA

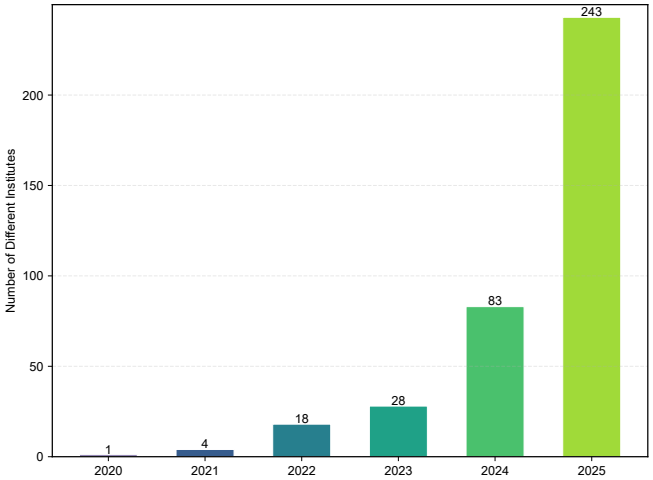
OmniJARVIS
聊天VLA-2
华东师大扩散-VLA量表DP
ReLER VER
UCSD NaVILA
CAS ConRFT
GeRM
西湖大学QUAR-VLA VLA-RFT
CombatVLA
阿里巴巴世界VLA
LoHoVLA
复旦大学OccLLaMA 4D-VLA
Galbot GraspVLA
芝加哥GRAPE
SBU LLaRA
加州大学洛杉矶分校AutoVLA
图灵冠状病毒
NYU SeeDo
GEVRM
ZJU Humanoid-VLA Long-VLA
XJTU VLAS
Skoltech Bi-VLA
美团RoboUniView
北邮MoManipVLA VLA-Adapter
NUS VLA-OSGenie Envisioner
SUTD Emma-X
CUHK RoboNurse-VLA
JHU RT-RAS
INSAIT ReVLA
VLMimic
其他BYOVLA InstructVLAG0
DP-VLA
2020 Apr Jul Oct 2021 Apr Jul Oct 2022 Apr Jul Oct 2023 Apr Jul Oct 2024 Apr Jul Oct 2025 Apr Jul Oct 2026

图 10.VLA 研究景观的可视化。

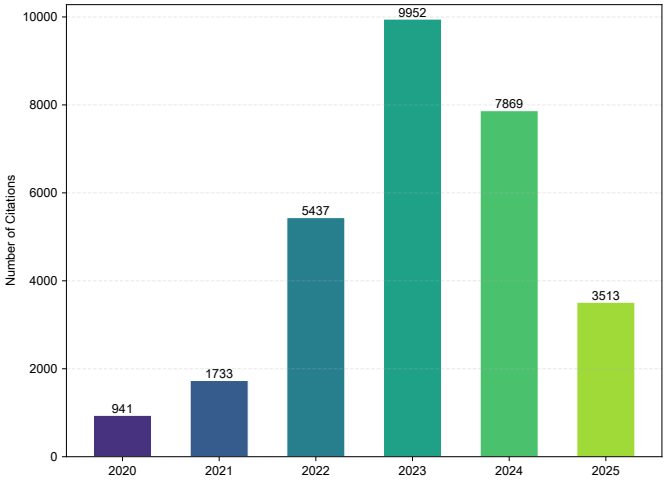


(a) Papers per Year

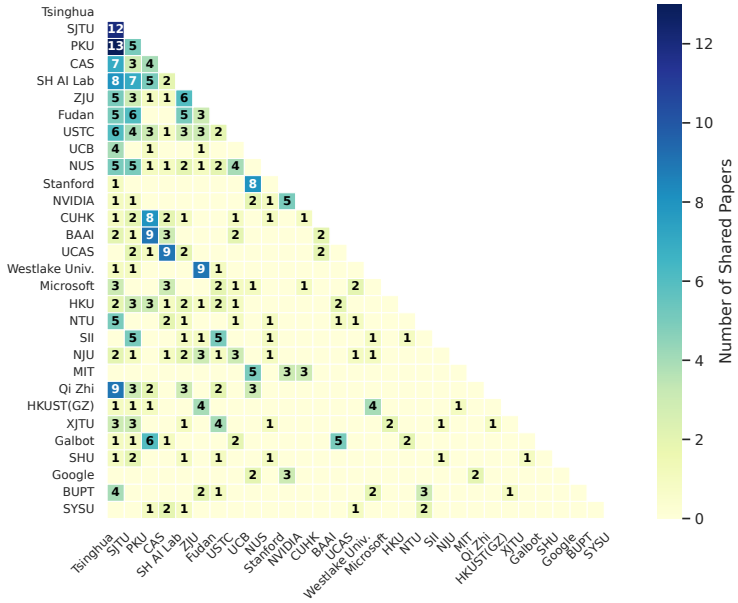
(b) Individual Paper Citations per Year



(c) Institutes per Year



(d) Total Citations per Year



(e) Collaboration Heatmap

Fig. 11. Quantitative analysis of VLA development trends.

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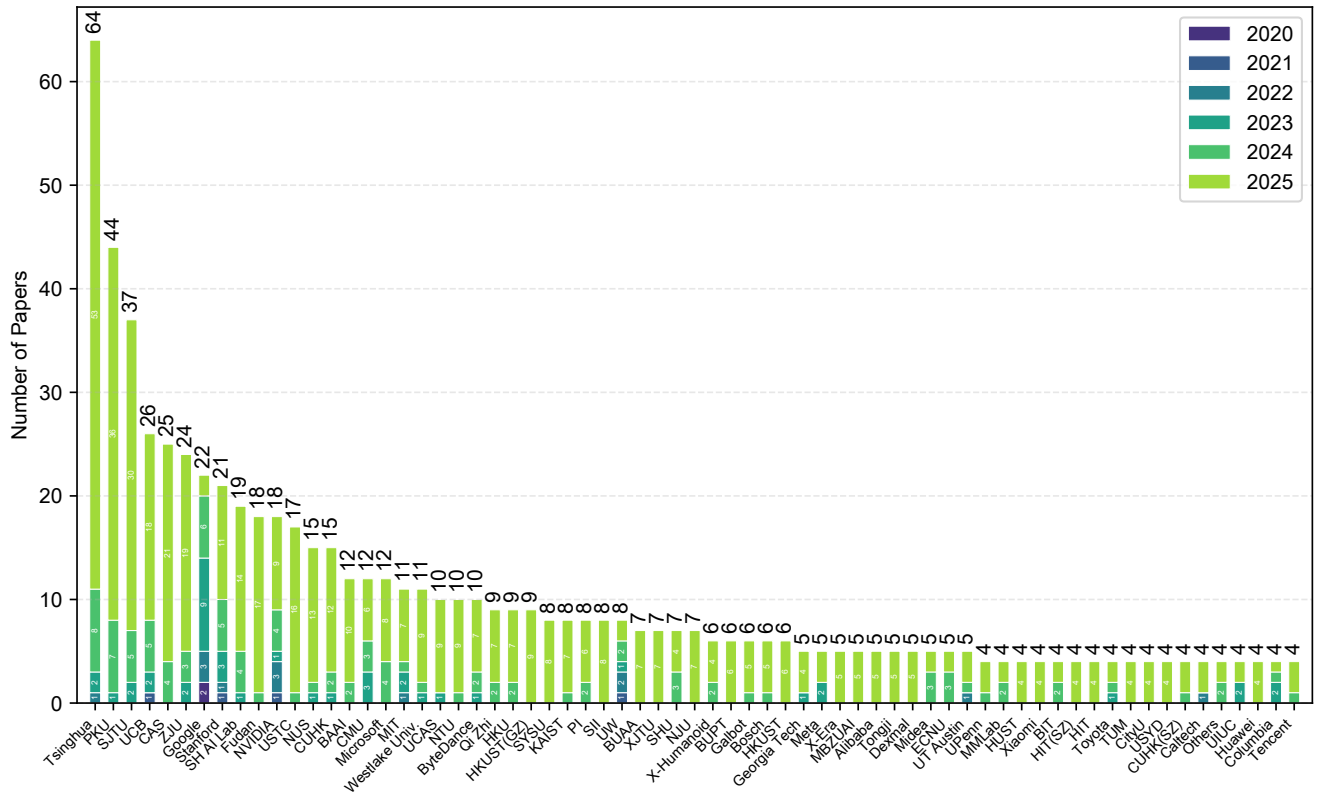
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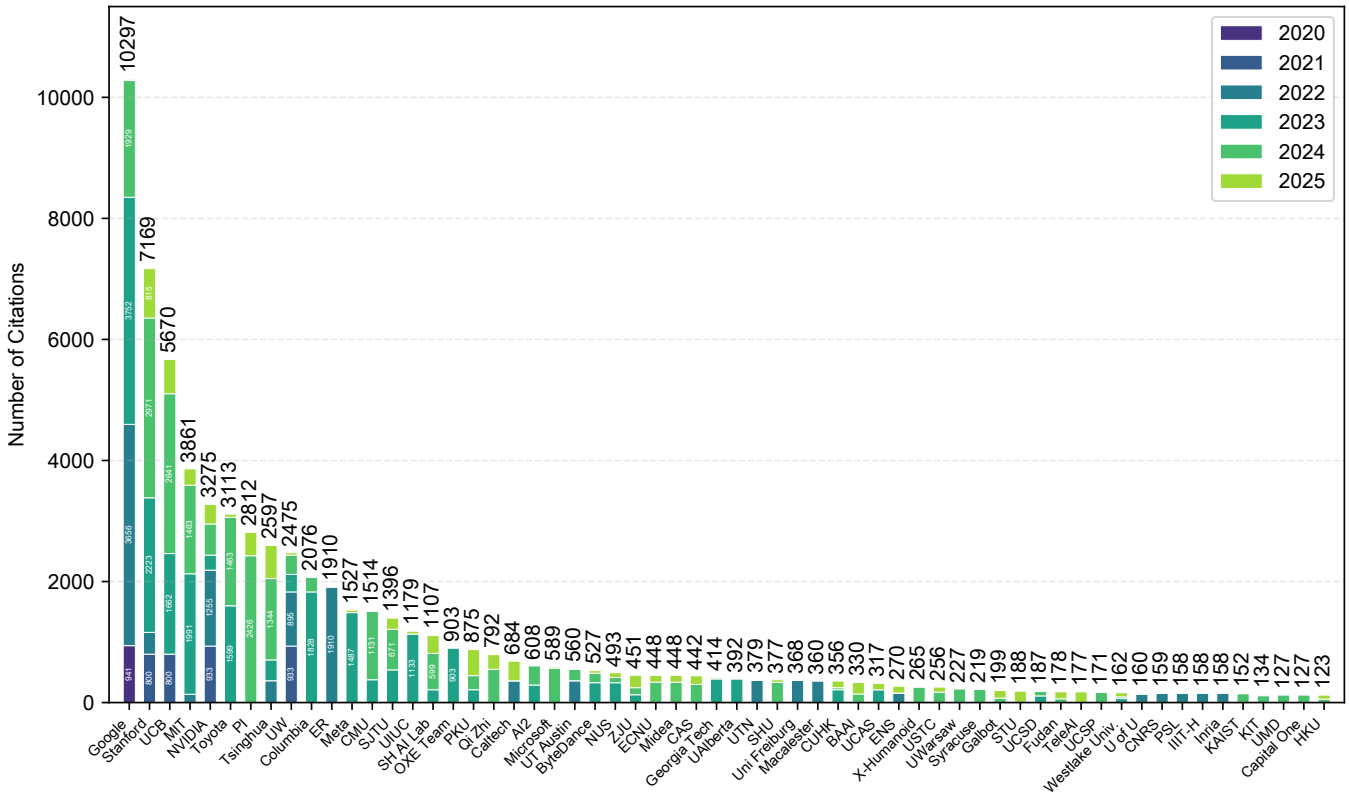
295
预印 NIPS ICLR ICML 最大值:933
290280 其他 CoRL RSS ICRA CVPR IROS 237 1000 最大值:621Q3:545 平均值:866 最小值:800
Q3:899Q1:833M:866 最大值:14 Q3:758 32
M:470平均：604 最大：392
270 平均：470 最小：3 Q1：39520 M：314 平均：452 Q3：376
M：223 最大：2 47
Q1：165
70 63 100 最小：1 37 Q1：89 平均：124 Q3:122
60 M:48
15
3
50 3
40 5 333 33 5 10 最小值:17 Q1:16 平均值:11 最大值 Q3:7:17
30 5 16
22 6 平均值 (Avg)
20 3 9 被引用最多的论文
3 M:中值 M:1
10 2 2 9 57 59 21 0 Q1: 第 25 个百分位 Q3: 第 75 个百分位 最小值/最大值：晶须 (1.5x IQR) M in:0 最小值：0Q1:0
0
2020 2021 2022 2023 2024 2025 2020 2021 2022 2023 2024 2025
(a) 每年论文数量 (b) 每年单篇论文引用次数
243 10000 9952
200
8000 7869
150 6000
5437
100 4000
83 3513
50 2000 1733
28 941
18
0 1 4 0
2020 2021 2022 2023 2024 2025 2020 2021 2022 2023 2024 2025
(c) 每年机构数(d) 每年总引用次数
清华大学

上海交通大学 12
北京大学 13 5 12
CAS 7 3 4
上海人工智能实验室 8 7 5 2
浙江大学 5 3 1 1 6
复旦科技大学 56 64 3 1 53 33 2 10
UCB 4 1 1
新加坡国立大学 5 5 1 1 2 1 2 4
斯坦福 1 8
NVIDIA 1 1 2 1 5 8
香港中文大学 1 2 8 2 1 1 1 1
BAAI 2 1 9 3 2 2
国科大 2 1 9 2 2
西湖大学1 1 9 1
微软 3 3 2 1 1 1 2 6
香港大学 2 3 3 1 2 1 2 1 2
南洋理工大学 5 2 1 1 1 1 1
SII 5 1 1 5 1 1 1
南京大学 2 1 1 2 3 1 3 1 1 1 4
MIT 5 3 3
齐智 9 3 2 3 2 3
香港科技大学（广州） 1 1 1 4 4 1
西安交通大学 3 3 1 4 1 2 1 1
Galbot 1 1 6 1 2 5 2 2
树 1 2 1 1 1 1 1
Google 2 3 2
北京邮电大学 4 2 1 2 3 1
中山大学 1 2 1 1 2
0
(e) 合作热图
清华大学图普库什人工智能实验室中科大中科院浙江大学福旦大学中科院斯坦福大学国立美国大学英伟达大学香港西湖大学BAAIUCAS微软香港图西大学中科大(广州)齐志XJTUGalbotSHUGoogleBUPTSYSU
每篇论文被引次数
论文数量
被引用数量
不同机构数量
共享论文数量

图11 VLA发展趋势定量分析。



(a) Number of Papers per Institute



(b) Number of Citations per Institute

Fig. 12. VLA research output and impact by institution.

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2020
2021
60
2022
2023
2024
50 2025
40
30
20
10
0
(a) 每个研究所的论文数量
2020
2021
10000 2022
2023
2024
2025
8000
6000
4000
2000
0
(b) 每个研究所的引用次数
清华大学北大联合体大学中科院ZJU谷歌斯坦福SH人工智能实验室复旦英伟达中科院中科大香港BAAI微软CMU西湖米图卡斯字节跳动南洋理工大学启智科大(广州)香港大学
SYSUKAISTPISIIUWBUAAXJTUSHUX-HumanoidNJUBUPTGalbot博世佐治亚理工科大
MetaX-EraMBZUAIA阿里巴巴同济德克斯马尔美的华中科技大学奥斯汀分校宾夕法尼亚大学
MMLabHUSTX小米HIT(深圳)BIT HIT丰田TUMCity香港大学（深圳）USY加州理工学院其他UIUC华为哥伦比亚腾讯
Google斯坦福UCBMITNVIDIA丰田清华PI哥伦比亚UW ERMetaCMUSJTUSH AI LabUIUCOXE团队北大齐智加州理工微软AI2UT奥斯汀字节跳动NUSZJUECNUMidea佐治亚理工学院CASU阿尔伯塔UTNUni弗莱堡SHUMacalesterCUHKBAAIUCASX-HumanoidENS USTCU华沙锡拉丘兹
STUUCSDFudanTeleAIWestlake Univ.UCSP U of UCNRSPSLIIIT-HInriaKAISTKITCapital OneUMD
HKU
64

44
53 37
26
论文数量 36 25 24
22
30 21 19
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18 6 11 15 15
21 19
9 12 12 12
14 11 11 10 10 10
82 1 1 7 25 2 1 5 4 32 932 3 11 5 1 4 17 1 3 1 4 16 13 1 122 1 102 633 84 2 1 7 9 1 9 1 9 2 1 7 9 27 9
27 9 9 8 8 8 7 8 62 8 8 8 2 1 2 1 7 7 7 7 7 34 7 7 6 24 6 6 6 5 6 5 6 6 51 4 5 2 5 5 5 5 5 5 5 5 5 5 3 5 3
51 4 4 2 4 4 4 4 4 2 4 4 4 4 41 4 4 4 4 4 4 4 41 4 2 4 2 4 4 4 2 4
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1929
7169
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3752 5670
2971
被引次数
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3113
2812
3656 1463 2597 2475
2223 1463 2076 1910
1662 1255 1344 895 1527 1514 1396 1179 1107
941 800 800 1991 933 1599 2426 933 1828 1910 1487 1131 671 1133 599 903 903 875 792 684 608
589 560 527 493 451 448 448 442 414 392 379 377 368 360 356 330 317 270 265 256 227 219 199
188 187 178 177 171 162 160 159 158 158 158 152 134 127 127 123

图 12. VLA 研究成果和按机构划分的影响。